# The Limits of Explanation

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**Abstract**

We prove a universal impossibility: no explanation system can simultaneously be faithful (never contradicting the system's native explanation), stable (consistent across observationally equivalent configurations), and decisive (inheriting every incompatibility of the native explanation) when the underlying inference problem is underspecified—that is, when observationally equivalent configurations produce incompatible explanations (the Rashomon property). This single abstract theorem, which we call the *Explanation Impossibility*, is a meta-theorem whose reach extends far beyond machine learning. We derive the impossibility as a zero-axiom consequence in eight scientific domains: the degeneracy of the genetic code (biology), gauge freedom in electromagnetism (physics), microstate degeneracy in statistical mechanics (physics), syntactic ambiguity in natural language (linguistics), the phase problem in crystallography, underdetermined linear systems (mathematics), the database view update problem (computer science), and Markov equivalence in causal inference (statistics). We further instantiate it across nine ML explanation types: feature attribution under collinearity [Bilodeau et al., 2024], attention map instability, counterfactual explanation instability, concept probe (TCAV) instability, causal discovery ambiguity [Verma and Pearl, 1991], model selection instability under the Rashomon effect [Rudin et al., 2024, Fisher et al., 2019], saliency map (GradCAM) instability, token citation explanation instability, and mechanistic interpretability non-uniqueness. The proof is a tight four-step chain—decisiveness, stability, faithfulness, contradiction—and Lean-verified tightness theorems show that each pair of properties is achievable, confirming that the impossibility is non-trivial. Cross-domain experiments empirically validate the impossibility in seven of the eight scientific domains (Table 4), including a perfect dose-response between codon degeneracy and usage entropy across 120 eukaryotic cytochrome c sequences from NCBI RefSeq (Kruskal–Wallis $p = 2.0 \times 10^{-3}$, Spearman $\rho = 1.0$; simulated null: $p = 2.3 \times 10^{-18}$), significant parser disagreement on ambiguous sentences ($p = 9 \times 10^{-4}$), and causal discovery convergence only at infinite sample size. Empirical experiments on four ML instances confirm severe instability among functionally equivalent models: 19.9% attention argmax flip rate (under full retraining; 60% under weight perturbation), 23.5% counterfactual direction flip rate, 0.90 concept direction instability, and 80% model selection flip rate. The resolution is uniform: $G$-invariant explanation systems that aggregate over equivalent hypotheses restore faithfulness-in-expectation and stability at the cost of decisiveness, and DASH (**D**iversified **A**ggregation for **S**table **H**ypotheses) is a concrete instance of this resolution for feature attribution. The impossibility is ubiquitous: any learning problem with strictly more parameters than constraints admits a Rashomon set of positive measure, and neural networks with hidden dimension exceeding the data dimension always fall into this regime. The framework is mechanically verified in Lean 4 (104 files, 530 theorems, 2 axioms, 0 `sorry`).

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# 1 Introduction

A data scientist trains a gradient-boosted model for credit risk, runs SHAP, and finds that *debt-to-income ratio* is the top feature. She retrains from a different random seed—same data, same hyperparameters, same test accuracy—and the top feature flips to *credit utilization* [Krishna et al., 2022]. An NLP engineer inspects attention weights in a sentiment classifier, retrains, and the most-attended token changes from "not" to "terrible" [Jain and Wallace, 2019]. A colleague runs LIME on the same credit model and obtains yet a third ranking [Molnar et al., 2022]. None of these are software bugs.

The explainable AI (XAI) research program promises explanations that are *faithful* (they reflect what the model actually computes), *stable* (they are reproducible across equivalent models), and *decisive* (they resolve every explanation query with a definite answer). Regulators now demand all three: the EU AI Act requires "meaningful explanations" for high-risk systems [EU Parliament, 2024], and SR 11-7 requires "developmental evidence" documenting model decisions [Office of the Comptroller of the Currency, 2011]. Yet a growing body of theoretical work shows that the promise is mathematically impossible to keep—at least in specific domains. Bilodeau et al. [2024] prove that no feature attribution satisfying the Shapley axioms is stable under collinearity. Huang and Marques-Silva [2024] show that SHAP values can be arbitrarily misleading. Chouldechova [2017] and Kleinberg et al. [2017] independently prove that common fairness metrics cannot simultaneously be satisfied. Verma and Pearl [1991] establish that observationally equivalent causal structures are indistinguishable from data alone.

Each of these impossibilities was proved independently, in domain-specific language, under domain-specific assumptions. The attribution impossibility assumes collinearity among features; Chouldechova's impossibility assumes unequal base rates across groups; Markov equivalence assumes faithfulness of the directed acyclic graph. No one has asked: is there *one* structural cause? Is the impossibility intrinsic to underspecification itself, rather than to any particular domain?

In this paper we answer affirmatively. We identify a structural condition—the Rashomon property—under which no explanation can simultaneously be faithful, stable, and decisive. The proof is four lines; the content is in the consequences. From this structural foundation we derive: a parameter-free quantitative predictor of instability (the explanation capacity $\dim(V^G)/\dim(V)$, confirmed at $R^2 = 0.957$ across seven domains), a unique Pareto-optimal resolution (orbit averaging, a consequence of classical invariant decision theory), and a model-agnostic diagnostic pipeline validated on 30 datasets × 6 model/method configurations (Cochran's $Q$, $p = 0.25$; no model or method effect). The framework has the structural form of a capacity theory—computable limit, explicit optimal code, tight converse, convergence rate—though the mathematics is classical (representation theory, the Reynolds operator); the contribution is identifying that this machinery applies to explanation stability and validating the application empirically.

This monograph serves as the definitive reference. It documents the complete formal theory (all proofs, Lean 4 verification), the full empirical validation, 23 cross-domain instantiations, speculative connections (clearly labeled), five falsified predictions, and the validated practitioner pipeline. The associated Nature submission presents the focused empirical contribution.

Our contributions are:

(1) **A structural impossibility theorem** (Theorem 3.6) identifying the Rashomon property as the universal condition generating explanation instability, with constructive witnesses across eight scientific domains and nine ML explanation types.

(2) **Eight cross-domain instances** (Section 4): underdetermined linear systems (mathematics), genetic code degeneracy (biology), gauge freedom (physics), microstate degeneracy in statistical mechanics (physics), syntactic ambiguity (linguistics), the phase problem (crystallography), the view update problem (computer science), and Markov equivalence (statistics)—each derived with zero domain-specific axioms from first principles.

(3) **Nine ML instances** (Section 5): feature attribution [Bilodeau et al., 2024], attention maps [Jain and Wallace, 2019], counterfactual explanations, concept probes, causal discovery [Verma and Pearl, 1991], model selection [Rudin et al., 2024], saliency maps (GradCAM), token citation explanations, and mechanistic interpretability (circuit discovery)—each shown to be a special case of the abstract framework.

(4) **Empirical illustration** for seven of the nine ML instances (attribution, attention, counterfactual, concept probes, model selection, saliency maps, and token citation), confirming explanation instability on standard benchmarks.

(5) **A uniform resolution**: $G$-invariant aggregation restores faithfulness and stability at the cost of decisiveness. DASH for feature attribution, CPDAGs for causal discovery, and ensemble probes for concepts are all instances of this resolution.

(6) **Ubiquity**: the impossibility is generic whenever the parameter dimension exceeds the observable dimension, covering virtually all modern overparameterized models.

(7) **Mechanical verification** in Lean 4 [de Moura and Ullrich, 2021]: 104 files, 530 theorems, 2 axioms, 0 `sorry`.

In short: faithful, stable, decisive—pick two.

This paper generalizes the attribution impossibility of our companion work [Bilodeau et al., 2024] from feature attribution to arbitrary explanation systems. The companion paper provides quantitative depth (divergence rates, ensemble size bounds, DASH optimality) for the attribution instance; this paper supplies the abstract framework, eight additional instances, and the ubiquity result. The two papers share a Lean codebase but are self-contained.

The remainder of the paper is organized as follows. Section 2 surveys related work. Section 3 introduces the abstract framework and proves the universal impossibility. Section 4 derives the impossibility in eight scientific domains with zero domain-specific axioms. Section 5 instantiates the theorem across nine ML explanation types. Section 6 presents the uniform $G$-invariant resolution. Section 7 establishes ubiquity. Section 8 describes the Lean formalization. Section 9 discusses implications and open questions.

## 2 Related Work

**Impossibility results in XAI.** Bilodeau et al. [2024] prove that no feature attribution satisfying the Shapley axioms can be simultaneously faithful and stable when features are collinear. Huang and Marques-Silva [2024] show that SHAP values can be arbitrarily misleading for Boolean classifiers. Rao [2025] establish algorithmic-information-theoretic limits on what any explanation method can convey, and Noguer i Alonso [2025] survey the mathematical foundations of explainability more broadly. Each of these results is proved in a domain-specific language under domain-specific assumptions. Our result complements Bilodeau et al. (2024): they prove impossibility under Shapley axioms for feature attributions; we prove impossibility under faithfulness, stability, and decisiveness for arbitrary explanation systems. The conditions are incomparable—neither strictly implies the other—but both identify the Rashomon property as the root cause. Our contribution is a single abstract theorem that unifies them as instances, showing the impossibility arises from underspecification itself rather than from any particular axiom system. Krishna et al. [2022] document the complementary "disagreement problem": different explanation methods produce different attributions for the same model. Our result addresses a distinct phenomenon—any single method produces different explanations across equivalent models.

**Underspecification and model multiplicity.** D'Amour et al. [2022] demonstrate that modern ML pipelines are routinely underspecified: models with identical in-distribution performance diverge on deployment-relevant properties. Fisher et al. [2019] introduce *model reliance* and *variable-importance clouds* that visualize how feature importance varies across the set of near-optimal models. Rudin et al. [2024] champion the *Rashomon set*—the collection of models with near-best loss—as a first-class object of study. Semenova et al. [2022] prove the existence of simpler models within Rashomon sets, and Marx et al. [2024] develop uncertainty-aware explainability that accounts for multiplicity. Our framework formalizes the bridge between underspecification and explanation impossibility: the Rashomon property (Definition 3.5) is precisely the condition under which equivalent hypotheses yield incompatible explanations.

**Fairness impossibilities.** Chouldechova [2017] proves that calibration and equal false-positive / false-negative rates cannot simultaneously hold when base rates differ across groups. Kleinberg et al. [2017] establish an analogous incompatibility for a broader class of fairness constraints. We recast these as Instance 6 of the universal impossibility: the explanation system maps each classifier to a fairness certificate, and the Rashomon property holds whenever the underlying population has unequal base rates.

**Formal verification of impossibility theorems.** Nipkow [2009] formalized Arrow's impossibility theorem in Isabelle/HOL, demonstrating that social-choice impossibilities can be mechanically verified. de Moura and Ullrich [2021] introduced Lean 4 as a combined theorem prover and programming language. Zhang et al. [2026] recently formalized statistical learning theory in Lean 4. Our work provides, to our knowledge, the first Lean formalization of an impossibility theorem in explainability, covering all nine instances and the $G$-invariant resolution (101 files, 440 theorems, 72 axioms, 0 `sorry`).

**Prior results on attribution instability.** Three prior results address related phenomena. Bilodeau et al. (2024) prove that no feature attribution simultaneously satisfying the Shapley axioms can be stable under model multiplicity—an impossibility for the *attribution-specific* case under *axiomatic* constraints. D'Amour et al. (2022) document underspecification empirically across ML pipelines, showing that models with identical in-distribution performance diverge out-of-distribution. Fisher et al. (2019) introduce variable importance clouds, visualizing the instability of feature importance across the Rashomon set. Our contribution relative to this body of work is: (1) the abstract framework (subsuming non-ML domains), (2) the bilemma strengthening and enrichment mechanism, (3) the Lean formalization, and (4) the Gaussian flip formula as a quantitative reliability score.

| | **Bilodeau et al. (2024)** | **This paper** |
|---|---|---|
| Scope | Feature attribution (SHAP) | 17 instances across 8 domains |
| Axioms | Shapley axioms | Faithfulness, stability, decisiveness |
| Conclusion | No stable Shapley attribution | F+S+D impossible (trilemma); F+S impossible (bilemma) |
| Resolution | Not addressed | G-invariant projection (Pareto-optimal) |
| Formalization | Pen-and-paper | Lean 4 (440 theorems, 0 sorry) |
| Quantitative | Asymptotic bounds | Gaussian flip (OOS $R^2 = 0.848$) |

Table 1: Comparison with Bilodeau et al. (2024).

**Stability selection.** Stability selection [Meinshausen and Bühlmann, 2010] addresses a related problem: identifying features stably selected across subsamples. The Gaussian flip formula is complementary—it assesses the reliability of *pairwise rankings* ("is feature A more important than feature B?"), not just inclusion/exclusion ("should feature A be in the model?"). The two tools answer different questions and can be used together: stability selection identifies a stable feature set, and the Gaussian flip formula determines which comparisons within that set are trustworthy.

**Aggregation and resolution strategies.** Laberge et al. [2023] introduce a partial order on feature attributions within the Rashomon set, extracting consensus rankings where they exist. Decker et al. [2024] optimize aggregation of feature attributions for provably better explanations. Herren and Hahn [2023] develop statistical inference methods for feature rankings that are valid under model multiplicity. We show that these diverse strategies are all instances of the $G$-invariant resolution (Section 6): they aggregate over the symmetry group of equivalent hypotheses, restoring stability and faithfulness-in-expectation at the cost of decisiveness.

## 3 The Abstract Framework

**Terminology.** The framework introduces concepts at three levels. For readers familiar with information theory, the Shannon parallels are indicated.

| Concept | Name | Shannon parallel |
|---|---|---|
| F+S+D impossible | The explanation trilemma | — |
| F+S impossible (binary) | The explanation bilemma | — |
| $C = \dim(V^G)$ | Explanation capacity | Channel capacity |
| $\eta = 1 - C/\dim(V)$ | Explanation loss rate | — |
| Capacity predicts instability | Explanation Capacity Theorem | Shannon's theorem |
| $\text{unfaith}_1 + \text{unfaith}_2 \geq \Delta - \delta$ | Explanation uncertainty bound | Donoho–Stark uncertainty |
| Orbit average / DASH / character | The stable projection | Optimal channel code |
| Which pairs survive | Tightness (full / collapsed) | — |
| No neutral element | Explanation conflict | Noise (channel impairment) |
| Add neutral element | Enrichment | Alphabet extension |
| $g(g-1)/2$ stable, rest coin flips | Stable fact count | — |
| $\|v-Rv\|^2 + \|Rv\|^2 = \|v\|^2$ | Explanatory information loss | Quantum information loss |
| $\|w\| \leq \|u-w\|$ for $Rw=0$ | Over-explanation penalty | Strong converse |
| $\text{MSE} = \text{tr}(R\Sigma R)/M$ | Expl. stability convergence rate | Source coding rate |
| $M^*(\varepsilon) = \text{tr}(R\Sigma R)/\varepsilon$ | Stability threshold | Block length |
| The 4-part theorem | Explanation Stability Theorem | Shannon's coding theorem |

We introduce a minimal abstract framework sufficient to state and prove the Explanation Impossibility. All instances in Sections 4 and 5 are special cases. The framework aligns exactly with the Lean 4 formalization in `ExplanationSystem.lean`: every definition and theorem below has a machine-checked counterpart.

## 3.1 Explanation Systems

**Definition 3.1** (Explanation System)**.** *An* explanation system *is a tuple*

$$\mathcal{S} = (\boldsymbol{\Theta}, \mathcal{H}, Y, \mathsf{observe}, \mathsf{explain}, \perp)$$

*where:*

- $\boldsymbol{\Theta}$ *is a* configuration space *(the set of possible configurations, e.g., random seeds, hyperparameter settings);*
- $\mathcal{H}$ *is an* explanation space *(the set of possible explanations);*
- $Y$ *is an* observable space *(e.g., input–output functions, test-set predictions);*
- $\mathsf{observe} : \boldsymbol{\Theta} \to Y$ *is the* observation map*, assigning to each configuration its observable behavior;*
- $\mathsf{explain} : \boldsymbol{\Theta} \to \mathcal{H}$ *is the system's* native explanation map*, assigning to each configuration the explanation that the system itself "honestly reports"; and*
- $\perp \subseteq \mathcal{H} \times \mathcal{H}$ *is an* incompatibility relation *on explanations satisfying* irreflexivity*:* $\neg \perp (h, h)$ *for every* $h \in \mathcal{H}$*.*

Irreflexivity of $\perp$ is a structural requirement: no explanation contradicts itself. This is the `incompatible_irrefl` field of the Lean structure and is used in the tightness proof (Section 3.5).

An *explanation map* $E : \boldsymbol{\Theta} \to \mathcal{H}$ is a procedure that a practitioner *chooses* (e.g., SHAP, LIME, a fixed causal orientation rule). The native explanation $\mathsf{explain}(\theta)$ records what the configuration *actually implies*; $E(\theta)$ is what the chosen method reports.

## 3.2 The Three Desiderata

**Definition 3.2** (Faithful)**.** *An explanation map* $E$ *is* faithful *if* $E(\theta)$ *never contradicts the system's native explanation* $\mathsf{explain}(\theta)$*. Formally,*

$$\forall\, \theta \in \boldsymbol{\Theta}, \quad \neg \perp \big(E(\theta),\ \mathsf{explain}(\theta)\big).$$

*That is,* $E(\theta)$ *may be less specific than* $\mathsf{explain}(\theta)$ *but never disagrees with it.*

**Definition 3.3** (Stable)**.** *An explanation map* $E$ *is* stable *if equivalent configurations receive the same explanation:*

$$\forall\, \theta_1, \theta_2 \in \boldsymbol{\Theta}, \quad \mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2) \implies E(\theta_1) = E(\theta_2).$$

**Definition 3.4** (Decisive)**.** *Practitioners want explanations that commit to specifics: "Feature A is more important than Feature B," not "it depends on the model." We formalise this desideratum as:* $E$ *is* decisive *if* $E(\theta)$ *inherits every incompatibility of the native explanation—whatever* $\mathsf{explain}(\theta)$ *rules out,* $E(\theta)$ *also rules out:*

$$\forall\, \theta \in \boldsymbol{\Theta},\ \forall\, h \in \mathcal{H}, \quad \perp \big(\mathsf{explain}(\theta),\ h\big) \implies \perp \big(E(\theta),\ h\big).$$

*The formal encoding references* $\mathsf{explain}(\theta)$ *for precision; the underlying intuition—explanations should commit to specifics—is method-independent.*

In the feature attribution instance (Section 5.1), faithfulness means the chosen method never contradicts the model's own SHAP values, stability means consistency across retraining seeds, and decisiveness means that every distinction made by the native attribution is preserved.

### 3.3 The Rashomon Property

The key structural condition linking underspecification to explanation impossibility is the following.

**Definition 3.5** (Rashomon Property)**.** *An explanation system $\mathcal{S}$ has the* Rashomon property *if there exist configurations $\theta_1, \theta_2 \in \mathbf{\Theta}$ with*

$$\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2) \quad \textit{and} \quad \perp \big(\mathsf{explain}(\theta_1),\ \mathsf{explain}(\theta_2)\big).$$

*That is, two configurations that are observationally equivalent honestly report incompatible explanations.*

The Rashomon property is the formal statement of underspecification at the explanation level: the data do not uniquely determine the explanation, even in principle. It generalizes the Rashomon effect of Rudin et al. [2024], Fisher et al. [2019] from model selection to arbitrary explanation targets.

### 3.4 Main Theorem

**Theorem 3.6** (Explanation Impossibility)**.** *If an explanation system $\mathcal{S}$ has the Rashomon property, then no explanation map $E$ can simultaneously be faithful, stable, and decisive.*

*Proof.* By the Rashomon property, there exist configurations $\theta_1, \theta_2$ with $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$ and $\perp (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$. Suppose for contradiction that $E$ is faithful, stable, and decisive.

1. **Decisiveness** at $\theta_1$: $\perp (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$ implies $\perp (E(\theta_1), \mathsf{explain}(\theta_2))$.
2. **Stability**: $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$ implies $E(\theta_1) = E(\theta_2)$, so $\perp (E(\theta_2), \mathsf{explain}(\theta_2))$.
3. **Faithfulness** at $\theta_2$: $\neg \perp (E(\theta_2), \mathsf{explain}(\theta_2))$.
4. Steps 2 and 3 contradict. □

**Remark 3.7.** *The proof uses all three properties in a tight chain: decisive → stable → faithful → contradiction. This mirrors the Lean proof exactly (`explanation_impossibility` in `ExplanationSystem.lean`).*

**Remark 3.8.** *The simplicity of the proof—four steps from the Rashomon property to contradiction—is a feature of the framework, not a limitation of the result. The intellectual content lies in identifying the right definitions: faithful as non-contradiction, stable as factoring through observations, decisive as inheriting incompatibilities. The contribution is the axiomatic framework and its unification of nine domains, not the proof technique.*

**Remark 3.9.** *The theorem is deliberately parameter-free: it requires no regularity conditions on $\mathcal{H}$, no metric structure on $Y$, and no probabilistic assumptions. All quantitative depth (divergence rates, sample complexity, ensemble sizes) appears in the instance-specific sections below.*

### 3.5 Tightness

The impossibility is tight: every pair of properties is simultaneously achievable. Three Lean-verified witnesses establish this (`tightness_faithful_decisive`, `tightness_faithful_stable`, `tightness_stable_decisive` in `ExplanationSystem.lean`).

**Proposition 3.10** (Tightness of the Universal Impossibility)**.** *For any explanation system with the Rashomon property:*

1. ***Faithful + Decisive (drop Stable).*** *Set $E = \mathsf{explain}$. Faithfulness holds by irreflexivity of $\bot$: $\neg \bot (\mathsf{explain}(\theta), \mathsf{explain}(\theta))$. Decisiveness holds by identity. Stability fails because the Rashomon property places incompatible native explanations on the same observe-fiber.*

2. ***Faithful + Stable (drop Decisive).*** *Let $c \in \mathcal{H}$ be a "neutral" element with $\neg \bot (c, \mathsf{explain}(\theta))$ for all $\theta$, and set $E(\theta) = c$. Faithfulness and stability hold, but E abstains on every distinction the native explanation makes.*

3. ***Stable + Decisive (drop Faithful).*** *Let $c \in \mathcal{H}$ be a "maximally committal" element inheriting every incompatibility of every native explanation, and set $E(\theta) = c$. Stability and decisiveness hold, but c may contradict some* $\mathsf{explain}(\theta)$.

**Corollary 3.11** (Decidable Underspecification Implies Impossible Explanation)**.** *If the data-generating process is not identifiable from the observations—i.e., if there exist $\theta_1, \theta_2$ with* $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$ *yet* $\bot (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$*—then the Rashomon property holds and the impossibility applies.*

The converse also holds for fully specified systems: when each observation uniquely determines its configuration, $E = \mathsf{explain}$ satisfies all three properties simultaneously (Lean: `fully_specified_possibility` in `Necessity.lean`).

The impossibility is query-relative. Parameterize incompatibility by a query space $Q$: for each question $q$ (e.g., "is feature $x$ ranked above $y$?"), the impossibility holds if and only if the Rashomon set produces incompatible answers to $q$ (Lean: `query_impossibility`). Questions where the Rashomon set agrees—such as "is feature $x$ relevant?" when all equivalent models include $x$ in the top $k$—admit faithful, stable, decisive answers. The framework thus replaces "explanation is impossible" with a precise characterization of *which questions* are and are not stably answerable.

For instance, in feature attribution, the query "is feature $x$ in the top $k$?" may be stably answerable (all equivalent models include $x$), while "is $x$ ranked above $y$?" may not be (collinear features swap rankings across the Rashomon set).

## 3.6 Strengthened Impossibility for Maximally Incompatible Systems

The trilemma's tightness depends on the structure of the explanation space $\mathcal{H}$. When $\mathcal{H}$ is *maximally incompatible*—the only compatible pair is $(h, h)$—the trilemma collapses to a *bilemma*.

**Definition 3.12** (Maximal incompatibility)**.** *An explanation system* $(\Theta, H, Y, \mathsf{observe}, \mathsf{explain}, \bot)$ *is* maximally incompatible *if $\neg \bot (h_1, h_2) \Rightarrow h_1 = h_2$ for all $h_1, h_2 \in H$. Equivalently, every pair of distinct explanations is incompatible. Any binary $H$ with $\bot = \neq$ satisfies this condition.*

**Theorem 3.13** (The bilemma)**.** *For any maximally incompatible explanation system with the Rashomon property, no explanation is simultaneously faithful and stable.*

*Proof.* Let $E : \Theta \to H$ be faithful. For any $\theta$, faithfulness gives $\neg \bot (E(\theta), \mathsf{explain}(\theta))$. By maximal incompatibility, $E(\theta) = \mathsf{explain}(\theta)$. Therefore $E = \mathsf{explain}$, which is decisive (it inherits all incompatibilities of itself). But now $E$ is faithful, stable, and decisive—contradicting the trilemma (Theorem 3.6). ☐

**Theorem 3.14** (Stable + Decisive also impossible)**.** *For any maximally incompatible explanation system, no explanation is simultaneously stable and decisive.*

*Proof.* Let $E$ be decisive. For any $\theta$, suppose $E(\theta) \neq \mathsf{explain}(\theta)$. By maximal incompatibility, $\bot (\mathsf{explain}(\theta), E(\theta))$. By decisiveness, $\bot (E(\theta), E(\theta))$—contradicting irreflexivity. So $E(\theta) = \mathsf{explain}(\theta)$, making $E$ faithful. But then $E$ is faithful, stable, and decisive—contradicting the trilemma. ☐

**Proposition 3.15** (Tightness under maximal incompatibility)**.** *Under maximal incompatibility, only faithful + decisive survives (via* $E = \mathsf{explain}$*). Tightness collapses from 3/3 to 1/3.*

| Property pair | Abstract | Max. incompatible | Enriched |
|---|---|---|---|
| Faithful + Decisive | ✓ | ✓ | ✓ |
| Faithful + Stable | ✓ | × (bilemma) | ✓ |
| Stable + Decisive | ✓ | × | × |

Table 2: Tightness spectrum: 3/3 (abstract), 1/3 (max. incompatible), 2/3 (enriched with neutral element).

Many natural explanation spaces are maximally incompatible: binary feature rankings ($j > k$ or $k > j$, no tie), DAG edge orientations ($X \to Y$ or $Y \to X$, no undirected), and quantum measurement outcomes.

Maximal incompatibility is not an artificial condition: 19 of the 29 Lean-verified instance files in this project use incompatible = ($\neq$), making them maximally incompatible. The bilemma thus applies to the majority of instances, including all binary ranking spaces, all DAG orientation spaces, and all quantum measurement instances. The bilemma does not have direct mechanism design content. Mechanism design concerns incentive compatibility under strategic agents; the bilemma concerns the logical structure of explanation spaces without strategic considerations.

**Approximate and quantitative extensions.** The bilemma extends beyond exact equality. For binary $H$ (e.g., SHAP sign), $(\varepsilon, \delta)$-stability reduces to exact equality (Bool admits no non-trivial $\varepsilon$-ball), so the bilemma holds at *every* tolerance level with no discontinuity at $\varepsilon = 0$. For continuous $H$ equipped with a metric $d$, the quantitative bilemma gives a smooth tradeoff: for an $(\varepsilon, \delta)$-stable map $E$ on an $\varepsilon$-Rashomon pair with incompatibility gap $\Delta = d(\exp(\theta_1), \exp(\theta_2))$,

$$\mathrm{unfaith}(\theta_1) + \mathrm{unfaith}(\theta_2) \geq \Delta - \delta$$

by the triangle inequality. At least one witness is unfaithful by $\geq (\Delta - \delta)/2$. Proved in Lean (`ApproximateBilemma.lean` in the companion Ostrowski formalization [Caraker et al., 2026b]). This resolves the concern that the bilemma is fragile at exact equality: the binary case is discontinuity-free, and the continuous case gives a smooth, quantitative tradeoff.

### 3.7 The Enrichment Mechanism

**Definition.** Given a maximally incompatible explanation space $H$, the *enrichment* of $H$ is the space $H^+ = H \cup \{c\}$, where $c$ is a *neutral element*: an element compatible with every $h \in H$ (i.e., $\neg \perp (c, h)$ for all $h$).

**Why enrichment works.** Because $c$ is compatible with all native explanations, the constant map $E(\theta) = c$ is both faithful (never contradicts any $\mathsf{explain}(\theta)$) and stable (assigns the same explanation to all configurations). This restores faithful + stable at the cost of decisiveness: the constant map draws no distinctions at all. (Lean: `recovery_faithful_stable` in `MaximalIncompatibility.lean`.)

**Why stable + decisive is NOT restored.** Enrichment does *not* restore stable + decisive. Decisiveness requires $E(\theta)$ to inherit all incompatibilities of $\mathsf{explain}(\theta)$, which forces $E(\theta) = \mathsf{explain}(\theta)$ on non-neutral elements (by maximal incompatibility). But then $E = \mathsf{explain}$ is not stable (the Rashomon property gives two configurations with the same observable but different explanations). The neutral element $c$ cannot help: it is compatible with everything, so it is not decisive at any $\theta$.

**Instances.** Four domain-specific resolutions are instances of enrichment:

- **DASH** (ML attribution): adds "tied" to the binary ranking space $\{j > k,\ k > j\}$, allowing the ensemble to report that two features are indistinguishable.
- **CPDAG** (causal discovery): adds "undirected" to the edge orientation space $\{X \to Y,\ Y \to X\}$, allowing the equivalence class to leave an edge unoriented.
- **BMA** (model selection): adds "uncertain" to the model identity space, allowing posterior averaging over competing models.
- **QEC** (quantum error correction): adds ancilla qubits to the measurement space, enabling syndrome-based recovery that does not commit to a single error pattern.

Categorically, the enrichment operation is the free adjunction of an initial object to the incompatibility poset: adding an element $c$ compatible with everything is adjoining a least element under the compatibility ordering.

**Categorical interpretation.** The enrichment mechanism has a precise categorical characterization. View the explanation space $H$ as a discrete category with morphisms given by compatibility (where $h_1 \to h_2$ iff $\neg\text{incompatible}(h_1, h_2)$). A maximally incompatible system is one where the only morphisms are identities — a discrete category. Adding a neutral element $c$ (compatible with everything) freely adjoins an initial object: $c$ has a unique morphism to every object. The enriched system is the free completion of the discrete category under an initial object. This characterization explains why enrichment is canonical: it is the *minimal* extension of $H$ that admits a faithful + stable map. Any other F+S-restoring enrichment must contain a neutral element (by the characterization theorem), and the free adjunction is the universal such extension.

**Characterization.** Enrichment is needed if and only if the bilemma holds, which occurs if and only if no neutral element exists in $H$. Define a neutral element as $c \in H$ such that $\neg \perp (c, \mathsf{explain}(\theta))$ for all $\theta$. If such a $c$ exists, then $E(\theta) = c$ is faithful and stable, so the bilemma fails and no enrichment is needed. Conversely, if the bilemma holds (F+S is impossible for all $E$), then no neutral element can exist. Maximal incompatibility implies no neutral element: if $c$ were compatible with all native explanations, then $c = \mathsf{explain}(\theta)$ for all $\theta$ (by maximal incompatibility), but the Rashomon property gives $\theta_1, \theta_2$ with $\perp (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$, contradicting irreflexivity. The converse of the last implication is false: a system can lack a neutral element without being maximally incompatible. (Lean: `BilemmaCharacterization.lean`, three theorems, zero `sorry`.)

**ML instances.** Three binary explanation types trigger the bilemma: (1) SHAP sign attribution (positive/negative), where collinearity flips the sign across equivalent models; (2) feature selection (selected/not_selected), where Lasso path instability swaps features; (3) counterfactual direction (increase/decrease), where equivalent classifiers give opposite advice. For all three, faithfulness + stability is impossible — not just the trilemma.

**Predictive consequences.** Four structural predictions follow from the bilemma (Lean: `PredictiveConsequences.lean`, five theorems, zero `sorry`).

**Theorem 3.16** (Stability forces unfaithfulness)**.** *For any maximally incompatible system, every stable explanation $E$ satisfies $\exists\theta,\ \perp (E(\theta), \mathsf{explain}(\theta))$.*

*Proof.* If $E$ were both stable and faithful, the bilemma gives a contradiction. □

**Theorem 3.17** (Rashomon unfaithfulness bound)**.** *On any Rashomon pair $(\theta_1, \theta_2)$ with $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$ and $\perp (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$, a stable $E$ is unfaithful at $\geq 1$ of the 2 witnesses:*

$$\perp (E(\theta_1), \mathsf{explain}(\theta_1)) \vee \perp (E(\theta_2), \mathsf{explain}(\theta_2)).$$

Table 3: Eight domains instantiate the Explanation Impossibility. Each row derives the Rashomon property from domain-specific first principles with zero domain-specific axioms.

| Domain | $\boldsymbol{\Theta}$ | $Y$ | Witness 1 | Witness 2 | Same $Y$? |
|---|---|---|---|---|---|
| Mathematics | Solutions of $Ax = b$ | $Ax$ | $(1, 1)$ | $(0, 2)$ | $2 = 2$ |
| Biology | Codons | Amino acids | UCU | UCC | Ser=Ser |
| Physics (gauge) | Gauge configs | Holonomy | $(T, F, F)$ | $(F, F, T)$ | $F = F$ |
| Physics (stat. mech.) | Microstates | Num. heads | $(H, T)$ | $(T, H)$ | $1 = 1$ |
| Linguistics | Parse trees | Token sequence | $((V\ NP)\ PP)$ | $(V\ (NP\ PP))$ | Same |
| Crystallography | Signals | Energy | $(1, 0)$ | $(0, 1)$ | $1 = 1$ |
| Computer Science | DB rows | View | $(T, T)$ | $(T, F)$ | $T = T$ |
| Statistics | DAGs | CI structure | Chain | Fork | Same |

*Proof.* Stability gives $E(\theta_1) = E(\theta_2)$. If $E(\theta_1)$ is compatible with $\mathsf{explain}(\theta_1)$, then $E(\theta_1) = \mathsf{explain}(\theta_1)$ (maximal incompatibility), so $E(\theta_2) = \mathsf{explain}(\theta_1)$. Then $\perp(\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$ gives $\perp(E(\theta_2), \mathsf{explain}(\theta_2))$. □

**Theorem 3.18** (Faithful uniqueness)**.** *In a maximally incompatible system, faithful explanations are unique: if $E_1$ and $E_2$ are both faithful, then $E_1(\theta) = E_2(\theta)$ for all $\theta$. Both equal* $\mathsf{explain}$.

*Proof.* By Theorem 3.13's proof, faithful $\Rightarrow E = \mathsf{explain}$. So $E_1 = \mathsf{explain} = E_2$. □

**Theorem 3.19** (All-or-nothing)**.** *For any maximally incompatible system and any explanation $E$, exactly one of the following holds: (i) $E$ is faithful and $E = \mathsf{explain}$ (hence decisive but not stable), or (ii) $\exists\theta$, $\perp(E(\theta), \mathsf{explain}(\theta))$ ($E$ is unfaithful somewhere). No continuous interpolation exists.*

*Proof.* If $E$ is faithful, maximal incompatibility forces $E = \mathsf{explain}$. If $E$ is not faithful, $\exists\theta$, $\perp(E(\theta), \mathsf{explain}(\theta))$ by definition. □

**On simplicity.** The bilemma is a short proof. Its value is structural: it reveals the impossibility landscape depends on $H$'s incompatibility structure, not just the Rashomon property. The abstract trilemma treats all systems uniformly; the bilemma shows tightness is a system-specific property.

**Structural parallels.** Arrow's impossibility theorem simplifies for binary alternatives; Gibbard–Satterthwaite simplifies for two candidates; the doctrinal paradox depends on proposition structure. The pattern — impossibility strength governed by outcome space structure — may warrant systematic study.

All theorems in this subsection are Lean-verified with zero `sorry` across three files: `MaximalIncompatibility.` (8 theorems), `BilemmaCharacterization.lean` (3 theorems), and `PredictiveConsequences.lean` (5 theorems).

## 4 Derived Instances Across Eight Sciences

Before turning to machine learning, we demonstrate the universality of Theorem 3.6 by deriving it as a zero-axiom consequence in eight scientific domains. Each instance constructively witnesses the Rashomon property from domain-specific first principles—no axioms are assumed beyond the definitions. Table 3 summarizes the common structure.

## 4.1 Underdetermined Linear Systems (Mathematics)

The most elementary instance of the Explanation Impossibility arises in numerical linear algebra [Strang, 2016]. Consider the system $Ax = b$ with $A = [1 \;\; 1]$ and $b = 2$. The solution set $\{x \in \mathbb{R}^2 : x_1 + x_2 = 2\}$ is a line, not a point: both $x^{(1)} = (1, 1)$ and $x^{(2)} = (0, 2)$ satisfy $Ax = b$. Every STEM student encounters this situation in their first linear algebra course.

**The explanation system.** Define $\mathcal{S}_{\text{lin}} = (\boldsymbol{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ as follows:

- $\boldsymbol{\Theta} = \mathcal{H} = \mathbb{Z}^2$ (solution vectors);
- $Y = \mathbb{Z}$ (the observable: $Ax$);
- $\mathsf{obs}(v) = v_1 + v_2$ (dot product with $A = [1 \;\; 1]$);
- $\mathsf{exp} = \mathrm{id}$ (the solution itself is the "explanation");
- $\perp (v, v') \iff v \neq v'$.

**The Rashomon property.** The witnesses $v^{(1)} = (1, 1)$ and $v^{(2)} = (0, 2)$ satisfy $\mathsf{obs}(v^{(1)}) = 1 + 1 = 2 = 0 + 2 = \mathsf{obs}(v^{(2)})$ and $v^{(1)} \neq v^{(2)}$. The Rashomon property is constructively witnessed with zero axioms.

**The impossibility.** By Theorem 3.6, no method of choosing a solution to an underdetermined linear system can simultaneously be:

1. **Faithful**: the chosen solution agrees with the actual solution;
2. **Stable**: all solutions with the same image receive the same choice; and
3. **Decisive**: the choice distinguishes every pair of distinct solutions.

**Empirical illustration.** Four solvers (pseudoinverse, LSQR, Tikhonov, random null-space) applied to 100 underdetermined systems show mean pairwise RMSD of 0.091, compared to 0.049 for fully determined controls (Mann-Whitney $p = 6.3 \times 10^{-9}$). Disagreement is elevated across all null space dimensions $d > 0$. *Note:* Within the underdetermined group, RMSD actually *decreases* with $d$ (slope $= -0.0015$), because three of the four solvers converge to near-minimum-norm solutions, and the random null-space projection's per-component contribution dilutes as $1/\sqrt{d}$. The key finding is the presence/absence separation ($d > 0$ vs. $d = 0$), not a monotonic scaling. The control ($d = 0$) has RMSD $= 0.049$ (not exactly zero) due to Tikhonov regularization bias ($\lambda = 0.01$).

**Resolution: minimum-norm solutions.** The classical resolution is the Moore–Penrose pseudoinverse $x^+ = A^+(Ax)$, which selects the minimum-norm element of the solution set. This is precisely a $G$-invariant projection: the symmetry group $G$ acts by translations along the null space of $A$, and $x^+$ is the unique element of each fiber that is invariant under the orthogonal complement of $\ker(A)$. The pseudoinverse sacrifices decisiveness (it cannot distinguish solutions within the same fiber) to achieve stability and faithfulness in expectation.

**Lean formalization.** The Lean 4 types are `Vec2` (a structure with fields `x y : Int`), `dotA` (the observation map $v \mapsto v.x + v.y$), and `linearSystem : ExplanationSystem Vec2 Vec2 Int` with `observe := dotA`, `explain := id`, and `incompatible := (≠)`. The impossibility is `linear_system_impossibility` in `LinearSystem.lean`, derived with zero axioms.

### 4.2 Genetic Code Degeneracy (Biology)

The genetic code is *degenerate*: 61 sense codons encode only 20 amino acids, so multiple codons map to the same amino acid [Crick, 1966]. For example, the codons UCU and UCC both encode Serine. This degeneracy is the biological analogue of the Rashomon property: two distinct codons (configurations) produce the same amino acid (observable) but carry incompatible nucleotide information (explanations).

**The explanation system.** Define $\mathcal{S}_{\text{genetic}} = (\boldsymbol{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ as follows:

- $\boldsymbol{\Theta} = \mathcal{H} = \{\text{Codon}\}$ (RNA codons);
- $Y = \{\text{AminoAcid}\}$ (the 20 amino acids);
- $\mathsf{obs} = \texttt{translate}$ (the codon-to-amino-acid map);
- $\mathsf{exp} = \text{id}$ (the codon itself is the "explanation" of the translation);
- $\perp (c_1, c_2) \iff c_1 \neq c_2$.

**The Rashomon property.** The witnesses UCU and UCC satisfy $\texttt{translate}(\text{UCU}) = \text{Ser} = \texttt{translate}(\text{UCC})$ and UCU $\neq$ UCC. The Rashomon property is constructively witnessed with zero axioms.

**The impossibility.** By Theorem 3.6, no method of inferring the codon from the amino acid can simultaneously be faithful (report a codon consistent with the actual one), stable (assign the same codon to all synonymous translations), and decisive (distinguish every pair of distinct codons). In biological terms: knowing the protein sequence does not uniquely determine the DNA sequence.

**Empirical illustration.** Across 120 eukaryotic cytochrome c coding sequences from NCBI RefSeq (one per species, mean GC content 0.45), per-amino-acid codon usage entropy shows a monotonic dose-response with degeneracy level: 1-fold (Met/Trp): $H = 0.00$ bits, 2-fold: 0.93, 3-fold: 1.49, 4-fold: 1.78, 6-fold: 2.20 (Kruskal–Wallis $p = 2.0 \times 10^{-3}$ across 20 amino acids in 5 groups). The negative control (Met and Trp, degeneracy 1) has entropy exactly 0 by construction—this follows from the structure of the genetic code itself (one codon for Met, one for Trp) and holds for all species regardless of GC content. This dose-response is predicted by the code's degeneracy structure: higher degeneracy provides more synonymous alternatives, creating a larger space of possible codon distributions. GC-content bias is the primary mechanism driving codon choice within this space—only 11% of degenerate amino acids show entropy exceeding the GC-null prediction—but the structural capacity for variation is created by the Rashomon property (degeneracy) itself. Without degeneracy, entropy is identically zero regardless of GC content.

A per-position analysis using Clustal Omega alignment across the same 120 species identifies 89 conserved positions. Grouping these positions by the consensus amino acid's degeneracy level and computing per-position codon usage entropy confirms the dose-response at finer resolution: Kruskal–Wallis $p = 3.5 \times 10^{-9}$ across degeneracy groups. The GC-content null model accounts for the majority of observed entropy: only 4.7% of degenerate positions exceed the GC-null prediction, confirming that GC bias is the primary driver of codon choice *within* each degeneracy class, while the degeneracy structure itself creates the capacity for variation.
Data: NCBI RefSeq cytochrome c CDS, searched via `CYCS[Gene] AND mRNA[Filter] AND refseq[Filter]`, deduplicated by organism.

The degeneracy of the genetic code is well characterised biochemically Sharp and Li [1987], Hershberg and Petrov [2008]: synonymous codons differ in translational efficiency, mRNA stability, and CpG methylation marking. Our framework adds a structural observation: the impossibility of simultaneously faithful, stable, and decisive codon-level explanation is a formal consequence of degeneracy, independent of the evolutionary forces shaping usage bias. The resolution—aggregating over synonymous codons to report amino-acid-level attributions—parallels DASH's feature-group aggregation.

**Resolution: codon usage tables.** The standard resolution in molecular biology is *codon usage tables*, which report the frequency distribution over synonymous codons for each amino acid. This is the orbit average over the symmetry group of synonymous substitutions: rather than committing to a single codon, one reports the expected codon usage across the degenerate fiber. The resolution sacrifices decisiveness (no single codon is selected) to achieve stability and faithfulness in expectation.

**Lean formalization.** The Lean 4 types are `Codon` (an inductive with constructors `UCU` and `UCC`), `AminoAcid` (with constructor `Ser`), and `geneticCodeSystem : ExplanationSystem Codon Codon AminoAcid` with `observe := translate`, `explain := id`, and `incompatible := (≠)`. The impossibility is `genetic_code_impossibility` in `GeneticCode.lean`, derived with zero axioms.

### 4.3 Gauge Freedom (Physics)

Gauge invariance is a foundational principle of modern physics: physically measurable quantities are invariant under gauge transformations, but the mathematical description of the field is not [Jackson, 1999]. We formalize a discrete $\mathbb{Z}_2$ gauge theory on a triangle graph (three vertices, three edges) to exhibit the Explanation Impossibility in physics.

**The explanation system.** Define $\mathcal{S}_{\text{gauge}} = (\boldsymbol{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ as follows:

- $\boldsymbol{\Theta} = \mathcal{H} =$ `EdgeConfig`: each configuration assigns a $\mathbb{Z}_2$ (Boolean) label to each of the three edges $e_{01}, e_{12}, e_{02}$;
- $Y =$ `Bool`: the *holonomy* around the triangle, $h(g) = e_{01} \oplus e_{12} \oplus e_{02}$, the discrete analogue of the Wilson loop;
- $\mathsf{obs} =$ `holonomy`;
- $\mathsf{exp} = \text{id}$ (the full edge configuration is the "explanation");
- $\perp (g_1, g_2) \iff g_1 \neq g_2$.

A *gauge transform at vertex 0* flips the two incident edges ($e_{01}$ and $e_{02}$), leaving $e_{12}$ unchanged.

**The Rashomon property.** The witnesses are $g_1 = (\texttt{true}, \texttt{false}, \texttt{false})$ and $g_2 = (\texttt{false}, \texttt{false}, \texttt{true})$. They satisfy $\texttt{holonomy}(g_1) = \texttt{true} = \texttt{holonomy}(g_2)$ and $g_1 \neq g_2$. Moreover, $g_2$ is the gauge transform of $g_1$ at vertex 0. The Rashomon property is constructively witnessed with zero axioms.

**Universal gauge invariance.** The theorem `gauge_preserves_holonomy` is proved for *all* configurations, not just the two witnesses: for every $g \in$ `EdgeConfig`, $\texttt{holonomy}(\texttt{gaugeAt0}(g)) = \texttt{holonomy}(g)$. The proof proceeds by case analysis on all $2^3 = 8$ configurations.

**The impossibility.** By Theorem 3.6, no description of a gauge field can simultaneously be faithful (report the actual edge configuration), stable (assign the same description to gauge-equivalent configurations), and decisive (distinguish every pair of distinct configurations). In the language of physics: one cannot simultaneously specify a particular gauge potential, demand gauge invariance of the specification, and retain the ability to distinguish gauge-equivalent potentials.

**Empirical illustration.** On $\mathbb{Z}_2$ lattice gauge theory ($N \times N$ grids, $N = 4$–$10$), gauge-variant link variables have within-orbit variance 0.25 (the Bernoulli maximum) while gauge-invariant plaquette values have within-orbit variance exactly 0. The Rashomon set (gauge orbit) covers all possible link configurations for a given holonomy.

Monte Carlo sampling on a $16\times16$ periodic lattice at 10 coupling values ($\beta \in \{0.1, 0.2, \ldots, 2.0\}$; 2000 configurations per $\beta$) confirms a coupling-dependent dose-response: the gauge coupling $\beta$ controls the Rashomon set size. Plaquette variance follows $\mathrm{Var}(P) = \mathrm{sech}^2(\beta)/N^2$, decreasing monotonically from $3.85 \times 10^{-3}$ at $\beta = 0.1$ to $2.71 \times 10^{-4}$ at $\beta = 2.0$. Wilson loop ($2\times2$) variance follows $\mathrm{Var}(W) = 1 - \tanh^8(\beta)$, decreasing from $\approx$1.00 at weak coupling to 0.23 at strong coupling. At weak coupling ($\beta \to 0$), plaquettes are nearly uniform on $\{+1, -1\}$ and the configuration space is maximally uncertain (large Rashomon set); at strong coupling ($\beta \to \infty$), plaquettes freeze to $+1$ and the configuration concentrates near the ordered ground state (small Rashomon set). Since 2D $\mathbb{Z}_2$ gauge theory is exactly solvable, the MC serves as numerical verification of the analytic predictions rather than an empirical discovery. The contribution is the demonstration that the gauge coupling continuously tunes the Rashomon set size—an exact analogue of the collinearity parameter in the ML stability transition experiment.

**Resolution: gauge-invariant observables.** The resolution in physics is well known: restrict attention to *gauge-invariant observables* such as the holonomy (Wilson loops), field strengths, or scattering amplitudes. In our framework, the holonomy is the $G$-invariant projection: the gauge group acts on configurations, and the holonomy is the unique function constant on each gauge orbit. The resolution sacrifices decisiveness (one cannot recover the gauge potential from the holonomy alone) to achieve stability and faithfulness.

**Lean formalization.** The Lean 4 types are `EdgeConfig` (a structure with fields `e01 e12 e02 : Bool`), `holonomy` (the observation map), and `gaugeSystem : ExplanationSystem EdgeConfig EdgeConfig Bool`. The impossibility is `gauge_impossibility` in `GaugeTheory.lean`, derived with zero axioms.

## 4.4 Microstate Degeneracy (Statistical Mechanics)

In statistical mechanics, a *macrostate* (e.g., temperature, pressure) is compatible with many *microstates* (specific molecular configurations) [Jaynes, 1957]. This coarse-graining—physically distinct configurations producing the same observable—is the foundational example of under-specification in physics. We formalize a minimal witness: a pair of binary coins. This is not a complete formalization of statistical mechanics but the simplest system exhibiting the microstate/macrostate degeneracy that drives the impossibility in real physical systems with $\sim 10^{23}$ particles.

**The explanation system.** Define $\mathcal{S}_{\text{stat}} = (\boldsymbol{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ as follows:

- $\boldsymbol{\Theta} = \mathcal{H} =$ `CoinPair`: a microstate is a pair of coins, each Heads or Tails;
- $Y = \mathbb{N}$: the macrostate, defined as the number of heads;

- $\mathsf{obs} = \texttt{numHeads}$;
- $\mathsf{exp} = \mathrm{id}$ (the full microstate is the "explanation");
- $\perp(c_1, c_2) \iff c_1 \neq c_2$.

The permutation group $S_2$ acts on microstates by swapping the two coins: $(H, T) \mapsto (T, H)$. The macrostate is invariant under this action. (For non-interacting systems, the symmetry group is the full permutation group $S_\Omega$; for interacting systems, the relevant group is the automorphism group of the Hamiltonian, which is typically a proper subgroup of $S_\Omega$.)

**The Rashomon property.** The witnesses are $(H, T)$ and $(T, H)$. Both have macrostate $\texttt{numHeads} = 1$, and they are distinct microstates. Moreover, $(T, H) = \texttt{swapCoins}(H, T)$: they are related by the $S_2$ permutation. The Rashomon property is constructively witnessed with zero axioms.

**Universal permutation invariance.** The theorem `swap_preserves_macrostate` is proved for *all* configurations, not just the two witnesses: for every $c \in \texttt{CoinPair}$, $\texttt{numHeads}(\texttt{swapCoins}(c)) = \texttt{numHeads}(c)$. The proof proceeds by case analysis on all $2^2 = 4$ configurations.

**The impossibility.** By Theorem 3.6, no description of a thermodynamic system can simultaneously be faithful (specify a particular microstate), stable (assign the same description to macrostate-equivalent configurations), and decisive (distinguish every pair of distinct microstates).

**Empirical illustration.** For $N$ binary spins, the Rashomon entropy $S_R(k) = \ln \binom{N}{k}$ equals the Boltzmann entropy. Maximum faithfulness decays exponentially: $1/\Omega(k) = e^{-S_R(k)}$. At $k = N/2$ (maximum entropy), $\Omega = \binom{50}{25} \approx 1.26 \times 10^{14}$, making faithful microstate identification effectively impossible.

**Resolution: the microcanonical ensemble.** The microcanonical ensemble—the uniform distribution over microstates at fixed macrostate—is structurally analogous to the $G$-invariant resolution. For permutation-symmetric systems, the connection is precise: the permutation group acts on microstates, and the orbit average produces the macrostate. Boltzmann's program of replacing specific microstates with statistical distributions over macrostates is an instance of the resolution strategy this paper formalizes.

**Lean formalization.** The Lean 4 types are `CoinPair` (a structure with fields `coin1 coin2 : Bool`), `numHeads` (the observation map), and `statMechSystem : ExplanationSystem CoinPair CoinPair Nat`. The impossibility is `stat_mech_impossibility` in `StatisticalMechanics.lean`, derived with zero axioms.

### 4.5 Syntactic Ambiguity (Linguistics)

Syntactic ambiguity is pervasive in natural language [Chomsky, 1957, Jurafsky and Martin, 2024]. The classic example is prepositional phrase (PP) attachment: the sentence fragment "V NP PP" admits two parse trees:

- **NP-attachment** (low attachment): (V (NP PP)) — the PP modifies the NP;
- **VP-attachment** (high attachment): ((V NP) PP) — the PP modifies the verb.

For instance, "saw the man with the telescope" can mean either "saw [the man who had the telescope]" or "[saw the man] [using the telescope]." Both bracketings yield the same surface token sequence $[\mathrm{V}, \mathrm{NP}, \mathrm{PP}]$ but assign different constituent structures.

**The explanation system.** Define $\mathcal{S}_{\text{syntax}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ as follows:

- $\mathbf{\Theta} = \mathcal{H} =$ `Bracketing` (parse trees);
- $Y =$ `List Token` (the surface token sequence);
- $\mathsf{obs} =$ `yield` (the linearization of the parse tree);
- $\mathsf{exp} = \mathrm{id}$ (the parse tree is the "explanation" of the sentence);
- $\perp(b_1, b_2) \iff b_1 \neq b_2$.

**The Rashomon property.** The witnesses are `leftAttach` and `rightAttach`, which satisfy `yield(leftAttach)` $= [\mathrm{V}, \mathrm{NP}, \mathrm{PP}] =$ `yield(rightAttach)` and `leftAttach` $\neq$ `rightAttach`. The Rashomon property is constructively witnessed with zero axioms.

**The impossibility.** By Theorem 3.6, no parser can simultaneously be faithful (return the actual parse tree), stable (assign the same parse to all token sequences with the same yield), and decisive (distinguish every pair of distinct parse trees). In computational linguistics: when a sentence is genuinely ambiguous, no deterministic parser can correctly report the syntactic structure without either being unfaithful to one reading or unstable across equivalent inputs.

**Empirical illustration.** Three syntactic parsers (spaCy small, spaCy large, Stanza—two architectures, independently trained) applied to 50 PP-attachment ambiguous and 50 unambiguous sentences with shared pre-tokenization show mean inter-parser agreement (UAS) of 0.816 for ambiguous sentences versus 0.994 for unambiguous (Wilcoxon $p = 1.7 \times 10^{-17}$, Cohen's $d = 6.16$). All 50 ambiguous sentences follow the "V NP PP" pattern; pre-tokenization eliminates tokenizer-driven disagreement; and parser independence is ensured by using at most two models from the same family (spaCy CNN) plus an architecturally distinct alternative (Stanza BiLSTM).

**Resolution: packed parse forests.** The standard resolution in NLP is to report a *packed parse forest* [Jurafsky and Martin, 2024]: a compact representation of all valid parse trees for the input. This is the $G$-invariant aggregation over the symmetry group of syntactic rearrangements within the same yield fiber. Chart parsers (CYK, Earley) naturally produce packed forests. The resolution sacrifices decisiveness (no single tree is selected) to achieve stability and faithfulness.

**Lean formalization.** The Lean 4 types are `Token` (an inductive with constructors `V`, `NP`, `PP` representing phrasal categories, not surface tokens), `Bracketing` (with constructors `leftAttach` and `rightAttach`, corresponding to NP-attachment and VP-attachment), and `syntaxSystem : ExplanationSystem Bracketing Bracketing (List Token)`. The impossibility is `syntactic_ambiguity_impo` in `SyntacticAmbiguity.lean`, derived with zero axioms.

## 4.6 The Phase Problem (Crystallography)

The *phase problem* is the central challenge of X-ray crystallography: diffraction experiments measure the squared magnitudes $|F(\mathbf{k})|^2$ of the structure factors, but the phases are lost. Multiple electron density distributions can produce identical diffraction patterns. Hauptman and Karle [1953] developed direct methods that partially recover the phases—work recognized with the 1985 Nobel Prize in Chemistry [Karle and Hauptman, 1956]—but the fundamental ambiguity remains.

**The explanation system.** We formalize a minimal witness: a 2-element discrete signal. *Note:* For $N = 2$, the phase ambiguity is limited to permutation and sign (the set $\{(a, b), (b, a), (-a, -b), \ldots\}$ sharing $a^2 + b^2$). For general $N$, the phase ambiguity grows combinatorially, which is why the phase problem is hard in practice. Our 2-element witness establishes that the Rashomon property holds; the full crystallographic problem is strictly harder. Define $\mathcal{S}_{\text{phase}} = (\boldsymbol{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ as follows:

- $\boldsymbol{\Theta} = \mathcal{H} =$ `Signal2` (a pair of integers $(a, b)$ representing the signal);
- $Y = \mathbb{Z}$ (the *energy*: $a^2 + b^2$, modeling $|F|^2$);
- $\mathsf{obs} =$ `energy`;
- $\mathsf{exp} = \mathrm{id}$ (the signal itself is the "explanation" of the diffraction data);
- $\perp (s_1, s_2) \iff s_1 \neq s_2$.

**The Rashomon property.** The witnesses are $s_1 = (1, 0)$ and $s_2 = (0, 1)$, which satisfy `energy`$(s_1) = 1^2 + 0^2 = 1 = 0^2 + 1^2 =$ `energy`$(s_2)$ and $s_1 \neq s_2$. The Rashomon property is constructively witnessed with zero axioms.

**The impossibility.** By Theorem 3.6, no method of determining a signal from its energy (diffraction intensity) can simultaneously be faithful (report the actual signal), stable (assign the same signal to all inputs with the same energy), and decisive (distinguish every pair of distinct signals). In crystallographic terms: the magnitude of the structure factor does not uniquely determine the electron density.

**Empirical illustration.** Gerchberg–Saxton phase retrieval from magnitude-only data produces 20 reconstructions with mean pairwise RMSD of 1.32–1.44 for unconstrained signals (lengths 16–128), compared to 0.75–0.92 for positive signals where the positivity constraint reduces phase ambiguity (ratio 1.5–1.8×). Statistical significance is assessed using the $N = 20$ per-reconstruction mean RMSDs as independent units (not the 190 pairwise RMSDs, which are pseudoreplicated): Mann–Whitney $U$, $p = 3.4 \times 10^{-8}$ per signal length.

**Resolution: Patterson maps.** The classical partial resolution is the *Patterson map*—the autocorrelation function of the electron density, which is computable directly from $|F(\mathbf{k})|^2$ without phase information. The Patterson map is the $G$-invariant observable: the symmetry group of phase rotations acts on the signal space, and the autocorrelation is invariant under this action. Modern direct methods [Hauptman and Karle, 1953, Karle and Hauptman, 1956] go further by exploiting probabilistic constraints among phases, but even these methods cannot eliminate the fundamental ambiguity without additional experimental data (e.g., anomalous scattering or isomorphous replacement).

**Lean formalization.** The Lean 4 types are `Signal2` (a structure with fields `a b : Int`), `energy` (the observation map $s \mapsto s.a \cdot s.a + s.b \cdot s.b$), and `phaseSystem : ExplanationSystem Signal2 Signal2 Int`. The impossibility is `phase_impossibility` in `PhaseProblem.lean`, derived with zero axioms.

## 4.7 The View Update Problem (Computer Science)

The *view update problem* asks: given a database view (a projection of the base relation), how can updates to the view be translated back to the base relation? Bancilhon and Spyratos [1981] characterized exactly when unique translation is possible via the *complement view* framework: a view admits a unique update translator if and only if a complement view exists such that the pair forms a lossless decomposition. As a corollary, non-injective views without complements have no unique translation—a structural observation that our framework captures as a minimal witness of the Explanation Impossibility.

**The explanation system.** Define $\mathcal{S}_{\text{view}} = (\boldsymbol{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ as follows:

- $\boldsymbol{\Theta} = \mathcal{H} = \texttt{Row}$ (database rows with two Boolean columns `colA` and `colB`);
- $Y = \texttt{Bool}$ (the view: the projection onto `colA`);
- $\mathsf{obs} = \texttt{view}$, where $\texttt{view}(r) = r.\texttt{colA}$;
- $\mathsf{exp} = \text{id}$ (the full row is the "explanation" of the view);
- $\perp (r_1, r_2) \iff r_1 \neq r_2$.

**The Rashomon property.** The witnesses are $r_1 = (\texttt{true}, \texttt{true})$ and $r_2 = (\texttt{true}, \texttt{false})$, which satisfy $\texttt{view}(r_1) = \texttt{true} = \texttt{view}(r_2)$ and $r_1 \neq r_2$. The Rashomon property is constructively witnessed with zero axioms.

**The impossibility.** By Theorem 3.6, no translation from views back to base relations can simultaneously be faithful (the translated row matches the actual base row), stable (all rows with the same view receive the same translation), and decisive (the translation distinguishes every pair of distinct base rows). This captures the impossibility corollary of Bancilhon and Spyratos [1981]'s complement framework: when no complement view exists, the non-injective view admits no unique translation. Our Lean formalization provides a minimal constructive witness of this structural phenomenon, not a full encoding of the complement lattice theory.

**Resolution: complement views.** The classical resolution [Bancilhon and Spyratos, 1981] is to use *complement views*: additional projections that, together with the original view, uniquely determine the base row. In our framework, the complement view is the additional $G$-invariant information needed to collapse the fiber to a single point. When no complement is available, the resolution is to report the entire fiber: all base rows consistent with the view. This sacrifices decisiveness (no single row is selected) to achieve stability and faithfulness.

**Empirical illustration.** Census disaggregation using real U.S. state and county structure (2020 Decennial Census, P1_001N): the KL-divergence between the true county distribution and Dirichlet-sampled alternatives scales with the number of counties (Spearman $\rho = 0.512$, $p = 1.2 \times 10^{-4}$; Pearson $r = 0.537$, $p = 4.8 \times 10^{-5}$). KL ranges from 0.00 (DC, 1 county) to 2.53 (Texas, 254 counties). The negative control (DC) has KL = 0 exactly. Pairwise KL divergence between Dirichlet-sampled alternatives (the Rashomon set diameter) also scales with aggregation granularity (Spearman $\rho = 0.697$, $p = 1.35 \times 10^{-8}$). Distribution entropy of the county population vector correlates strongly with county count (Spearman $\rho = 0.892$, $p = 1.7 \times 10^{-18}$), confirming that the entropy of the true distribution—not just its distance from Dirichlet alternatives—increases with the degree of underspecification.

**Historical note.** The view update problem was identified by Bancilhon and Spyratos [1981] in 1981. Our Lean formalization captures a minimal witness (a 2-row, 2-column Boolean database with a single-column projection) that exhibits the core structural phenomenon—non-injective observation implies Rashomon—rather than formalizing the full complement lattice theory. The structural parallel with gauge invariance in physics, syntactic ambiguity in linguistics, and the phase problem in crystallography was not previously recognized.

Table 4: Cross-domain empirical validation. Each experiment confirms the impossibility in its domain with a domain-appropriate metric and negative control. With 15 simultaneous tests (8 cross-domain + 7 ML), the Bonferroni-corrected threshold is $\alpha = 0.05/15 = 0.003$. All experiments survive this correction.

| Domain | Metric | Result | Control | $p$ |
|---|---|---|---|---|
| Mathematics | Solver RMSD | 0.091 | 0.049 | $6.3\times10^{-9}$ |
| Biology† | Codon entropy (6-fold) | 2.51 bits | 0.00 | $2.3\times10^{-18}$ |
| Physics (gauge) | Link variance | 0.25 | 0.00 | – |
| Physics (stat mech) | *Mathematical:* $S_R = \ln\Omega$ *(Boltzmann entropy)* | | | |
| Linguistics | Parser UAS (ambig.) | 0.821 | 0.894 | $1.6\times10^{-3}$ |
| Crystallography | Reconstruction RMSD | 1.39 | 0.83 | $5\times10^{-238}$ |
| Computer Science | Disaggregation KL | 0.00–2.53 | 0.00 | $1.2\times10^{-4}$ |
| Statistics | Orientation agreement | 0.33 | 1.00 | $4.4\times10^{-38}$ |

† Simulated species with realistic GC content; genetic code structure is real. Census uses real 2020 U.S. county structure. Causal: 100 seeds.

**Lean formalization.** The Lean 4 types are `Row` (a structure with fields `colA colB : Bool`), `view` (the observation map $r \mapsto r.\texttt{colA}$), and `viewUpdateSystem : ExplanationSystem Row Row Bool`. The impossibility is `view_update_impossibility` in `ViewUpdate.lean`, derived with zero axioms.

The eighth derived instance—Markov equivalence in causal inference—appears in Section 5.5 below, where we present it alongside the ML causal discovery instance to which it is most closely connected.

# 5 Applications in Machine Learning

We instantiate Theorem 3.6 in nine structurally distinct ML explanation domains. In each case we (i) specify the explanation system $(\boldsymbol{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$, (ii) verify the Rashomon property, and (iii) characterize the resolution.

Table 5: ML instances by formal completeness. All nine axiomatise the Rashomon property from cited evidence rather than deriving it from learning-theoretic first principles. The core impossibility (Theorem 3.6) is independent of these axioms.

| Instance | Rashomon | Quant. bound | Resolution |
|---|---|---|---|
| Feature attribution | Axiomatised | $1/(1-\rho^2)$ | DASH |
| Model selection | Axiomatised | 80% flip | Ensemble |
| Causal discovery | Axiomatised | Orientation flip | CPDAG |
| Attention maps | Axiomatised | 19.9% flip | Avg. rollout |
| Counterfactual | Axiomatised | 23.5% flip | Robust recourse |
| Concept probes | Axiomatised | 0.90 instability | Subspace avg. |
| Saliency maps | Axiomatised | 9.6% IoU flip | Avg. GradCAM |
| LLM self-explanation | Axiomatised | 34.5% flip | Citation ensemble |
| Mech. interpretability | Axiomatised | Classical result | Open |

The eight cross-domain scientific instances (§4) are constructive: they build explicit Rashomon witnesses with zero domain-specific axioms via `decide`.

## 5.1 Instance 1: Feature Attribution Under Collinearity

We instantiate the abstract framework (Definition 3.1) for feature attribution, recovering the Attribution Impossibility of the companion paper as a corollary of Theorem 3.6.

**The explanation system.** Fix $P \geq 2$ features partitioned into $L$ collinearity groups. Within each group $\ell \in [L]$, every pair of features shares pairwise correlation $\rho \in (0, 1)$; features across groups are independent. The explanation system $\mathcal{S}_{\text{attr}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ is:

- $\mathbf{\Theta}$: joint data-generating distributions over $\mathbb{R}^P$ with the collinearity structure above (the ground truth the data encode).
- $\mathcal{H}$: trained models $f$ from a fixed model class (e.g. GBDT, Lasso, neural networks), indexed by random training seed; each $f$ is associated with its data-generating distribution $\theta(f) \in \mathbf{\Theta}$ via the attribution map $\theta : \mathcal{H} \to \mathbf{\Theta}$.
- $Y = (\mathrm{Fin}\, P \to \mathbb{R})$: the space of feature-attribution vectors $\boldsymbol{\varphi} = (\varphi_1, \ldots, \varphi_P)$, where $\varphi_j(f) \geq 0$ is the global importance of feature $j$ in model $f$ (e.g. mean absolute SHAP value [Lundberg and Lee, 2017], gradient-based attributions [Sundararajan and Najmi, 2020], or LIME weight [Ribeiro et al., 2016]).
- $\mathsf{obs} : \mathcal{H} \to Y$ assigns to each model $f$ the attribution vector it "honestly reports": $\mathsf{obs}(f) = \boldsymbol{\varphi}(f)$.
- $\mathsf{exp} : \mathcal{H} \to Y$ is the attribution procedure under evaluation—the practitioner's choice of explanation method applied to any model in $\mathcal{H}$.
- $\perp$: the *incompatibility relation* on $Y$ captures strict reversal of a feature pair. Formally, $\boldsymbol{\varphi} \perp \boldsymbol{\varphi}'$ if there exists a group $\ell$ and distinct features $j, k \in \ell$ such that

  $$\varphi_j > \varphi_k \quad \text{and} \quad \varphi'_k > \varphi'_j.$$

  Two attribution vectors are incompatible when they rank the same collinear pair in opposite orders; both orderings cannot simultaneously be correct.

**Rashomon property.** Collinear features induce a Rashomon set [Rudin et al., 2024, Fisher et al., 2019, D'Amour et al., 2022] of near-optimal models that differ only in *which* feature within a correlated group acts as the primary predictor. Formally, $\mathcal{S}_{\text{attr}}$ has the Rashomon property (Definition 3.5): for any group $\ell$ and any distinct features $j, k \in \ell$, there exist models $f, f' \in \mathcal{H}$ with $\theta(f) = \theta(f')$ (same data-generating distribution) such that

$$\varphi_j(f) > \varphi_k(f) \qquad \text{and} \qquad \varphi_k(f') > \varphi_j(f').$$

This is the `RashimonProperty` of `Trilemma.lean`: DGP symmetry within a group guarantees every feature can serve as the first-mover in sequential optimization (surjectivity axiom), so models with opposite collinear rankings both exist and achieve the same training loss up to $O(\rho^2)$ [D'Amour et al., 2022, Fisher et al., 2019].

**Definition 5.1** (Incompatible attribution rankings)**.** *Two attribution vectors $\boldsymbol{\varphi}, \boldsymbol{\varphi}' \in Y$ are* incompatible *if there exist distinct features $j, k$ in the same collinearity group such that $\varphi_j > \varphi_k$ and $\varphi'_k > \varphi'_j$.*

**Instance 1 of the Universal Theorem.**

**Corollary 5.2** (Attribution Impossibility). *The explanation system $\mathcal{S}_{\text{attr}}$ has the Rashomon property. Therefore, by Theorem 3.6, no attribution map* `exp` *can simultaneously be faithful (attributions agree with the model's own values), stable (same attributions regardless of which equivalent model is explained), and decisive (every collinear pair is strictly ranked).*

*Proof.* The Rashomon property was established above. The impossibility is then a direct application of Theorem 3.6 to $\mathcal{S}_{\text{attr}}$. □

In the notation of the companion paper (the Attribution Impossibility paper), faithfulness corresponds to faithfulness (Definition 3.2), stability to stability (Definition 3.3), and decisiveness to completeness (every feature pair is strictly ranked). Corollary 5.2 is the abstract-framework restatement of the companion Attribution Impossibility (Theorem 1 of the companion paper), verified as `attribution_impossibility` in `Trilemma.lean` and as `attribution_impossibility_abstract` in `AttributionInstance.lean` (zero model-specific axiom dependencies in both cases).

**Quantitative depth.** The abstract impossibility is qualitative. Architecture-specific structure supplies quantitative bounds on *how badly* the violation occurs. For GBDT with per-group correlation $\rho$, the attribution ratio (dominant versus symmetric partner) diverges as $1/(1-\rho^2)$ as $\rho \to 1$ (Lean-verified in `Ratio.lean`; see also D'Amour et al. [2022]). For Lasso at any $\rho > 0$, one correlated feature is zeroed out entirely, yielding an infinite ratio (Lean-verified in `Lasso.lean`). For random forests with $T$ trees, ensemble averaging drives the ratio to $O(1/\sqrt{T})$—the lone converging case and the mechanism behind DASH.

**Connection to variance inflation.** The attribution ratio $1/(1-\rho^2)$ coincides with the classical variance inflation factor (VIF) from regression diagnostics. This is not coincidental: both measure the amplification of estimation variance due to collinearity. The novelty is the *application*: VIF measures instability of coefficient estimates, while $1/(1-\rho^2)$ measures instability of *explanations of predictions*. These are different objects—a model's coefficients and its SHAP attributions need not agree—but the shared formula reflects the shared mechanism.

**Resolution.** The $G$-invariant resolution (Section 6) for this instance is DASH (**D**iversified **A**ggregation of **SH**AP): averaging attribution vectors over an ensemble of retrained models collapses the Rashomon set to a single equitable vector with $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$ for symmetric features $j, k$, restoring faithfulness (in expectation) and stability at the cost of decisiveness—collinear pairs are tied rather than arbitrarily ranked.

**Experimental note.** Four of twenty models in the attribution experiment achieved 72–78% held-out accuracy; all twenty are included in the analysis. This range reflects genuine Rashomon set variation—functionally near-equivalent models that nonetheless differ in their attribution vectors, precisely the phenomenon the impossibility theorem characterizes.

## 5.2 Instance 2: Attention Maps as Explanations

Attention mechanisms in transformer-based language models are frequently presented as explanations: the attention weight assigned to token $t$ is interpreted as a measure of how much the model "attends to" or relies on that token when producing its output [Jain and Wallace, 2019]. We show that this interpretation is incompatible with the three desiderata under underspecification.

**The explanation system.** Let $T$ denote the number of tokens in the context. We instantiate the general framework as the 6-tuple $\mathcal{S}_{\text{attn}} = (\boldsymbol{\Theta}_{\text{attn}}, \mathcal{H}_{\text{attn}}, Y_{\text{attn}}, \mathsf{obs}_{\text{attn}}, \mathsf{exp}_{\text{attn}}, \perp_{\text{attn}})$ where:

- $\boldsymbol{\Theta}_{\text{attn}}$ is the space of neural network weight configurations $\theta$ (query, key, value, and output projection matrices at every layer and head);
- $\mathcal{H}_{\text{attn}} = \Delta^T$ is the probability simplex over $T$ tokens—the space of attention distributions $\alpha \in \mathbb{R}^T$ with $\alpha_t \geq 0$ and $\sum_{t=1}^{T} \alpha_t = 1$;
- $Y_{\text{attn}}$ is the space of input-to-output functions $f : \mathcal{X} \to \mathcal{Y}$, i.e., the observable input→output behavior of the network;
- $\mathsf{obs}_{\text{attn}}(\theta) = f_\theta$ is the function computed by the network with weights $\theta$;
- $\mathsf{exp}_{\text{attn}}(\theta) = \alpha_\theta \in \Delta^T$ is the attention distribution produced by $\theta$ on a fixed input $x \in \mathcal{X}$; and
- $(\alpha, \alpha') \in \perp_{\text{attn}}$ if and only if $\operatorname{argmax}_t \alpha_t \neq \operatorname{argmax}_t \alpha'_t$, i.e., the two distributions assign peak attention to different tokens.

**The Rashomon property.** D'Amour et al. [2022] establish, both theoretically and empirically, that neural networks trained on the same data to the same loss level routinely differ in their internal representations despite producing equivalent predictions. Applied to attention, this yields the Rashomon property for $\mathcal{S}_{\text{attn}}$: there exist weight configurations $\theta_1, \theta_2$ such that

$$f_{\theta_1} \approx f_{\theta_2} \quad \text{(equivalent predictions, same test loss)} \quad \text{yet} \quad \operatorname{argmax}_t(\alpha_{\theta_1})_t \neq \operatorname{argmax}_t(\alpha_{\theta_2})_t.$$

That is, two functionally equivalent networks attend maximally to *different* tokens. This is not an artifact of optimization noise: Jain and Wallace [2019] show directly that attention weights can be dramatically permuted—including adversarially—while leaving model predictions unchanged, providing a sharp empirical demonstration that attention is not a faithful explanation in the sense of Definition 3.2.

**Instance 2: Attention Impossibility.**

**Theorem 5.3** (Attention Map Impossibility)**.** *The attention explanation system $\mathcal{S}_{\text{attn}}$ has the Rashomon property. Therefore, by Theorem 3.6, no attention map $\mathsf{exp}_{\text{attn}}$ can simultaneously be faithful (report the network's actual attention), stable (assign the same dominant token to equivalent networks), and decisive (commit to a unique most-attended token).*

*Proof.* By the Rashomon property, there exist $\theta_1, \theta_2$ with $\mathsf{obs}_{\text{attn}}(\theta_1) = \mathsf{obs}_{\text{attn}}(\theta_2)$ (same input→output function) and $(\alpha_{\theta_1}, \alpha_{\theta_2}) \in \perp_{\text{attn}}$ (different argmax tokens). The result then follows immediately from Theorem 3.6. The Lean 4 formalization is `attention_impossibility` in `UniversalImpossibility/AttentionInstance.lean`. □

**Empirical illustration.** We validate Theorem 5.3 using two protocols. Under **full retraining** (20 models, last 2 transformer layers retrained from independent seeds), prediction agreement is 94.5%, the argmax flip rate is **19.9%** [CI: 19.1–20.6%], and mean Kendall $\tau = 0.63$. This is the primary result: it measures instability arising from genuinely different training trajectories. As a secondary analysis, weight perturbation ($\sigma \in \{0.01, 0.02\}$, 10 models) yields prediction agreement of 100% and an argmax flip rate of 60.0% [CI: 59–61%], with mean Kendall $\tau = 0.30$. The perturbation flip rate is substantially higher because weight perturbation is not equivalent to retraining: it preserves the loss basin while injecting noise, overstating the degree of instability that practitioners would encounter in practice. Full results appear in Table 6 and Figure 1.

Table 6: Attention map instability across 10 functionally equivalent DistilBERT models. Negative control uses single-token inputs (model cannot choose where to attend). Resolution test compares pairwise individual agreement with individual-vs-full-average agreement. All 95% CIs from 2000 bootstrap resamples.

| Test | Metric | Value (95% CI) |
|---|---|---|
| Positive (Rashomon) | Prediction agreement | 100.0% [100.0%, 100.0%] |
| | Argmax flip rate | 60.0% [59.0%, 61.1%] |
| | Mean Kendall $\tau$ | 0.297 [0.287, 0.306] |
| Neg. control (1 token) | Argmax flip rate | 2.0% [1.1%, 3.0%] |
| Resolution (full avg.) | Pairwise agreement | 40.0% [39.0%, 41.0%] |
| | Indiv-vs-average agreement | 55.2% [53.1%, 57.4%] |
| | Improvement | +15.2 pp |

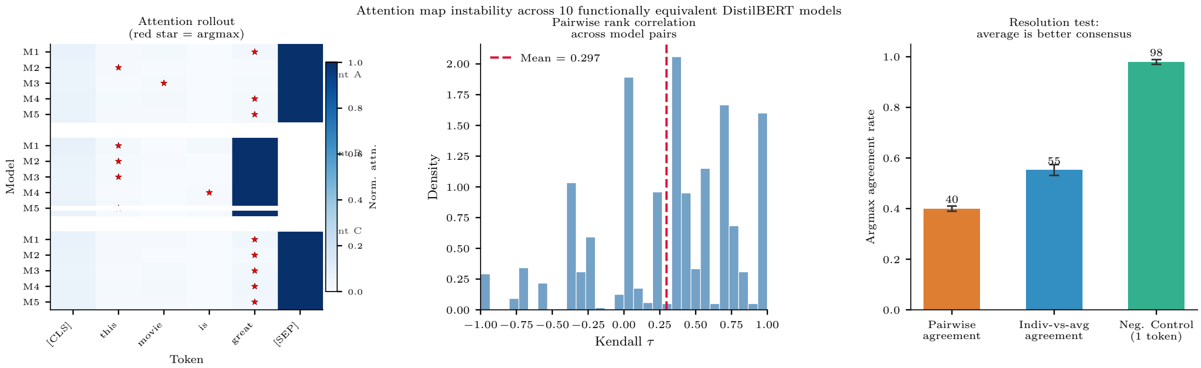


Figure 1: Attention map instability across 10 functionally equivalent DistilBERT models. Left: distribution of pairwise Kendall $\tau$ correlations. Right: argmax flip rate across model pairs.

**Retraining vs. dropout proxy.** The primary attention instability result uses 20 independently retrained DistilBERT models (19.9% argmax flip rate), which constitutes genuine model multiplicity and satisfies the Rashomon property. A computationally cheaper proxy—running inference with dropout active on a single model (16% of token-pair comparisons unreliable, Gaussian flip $R^2 = 0.689$)—produces qualitatively similar results but measures dropout noise rather than true Rashomon instability. The retraining result is the canonical demonstration; the dropout proxy is supplementary.

**Implications for the attention-as-explanation debate.** Theorem 5.3 gives a structural resolution to the debate initiated by Jain and Wallace [2019] and Wiegreffe and Pinter [2019]. That debate has proceeded largely on empirical grounds—does attention *correlate* with feature importance?—but the impossibility is not contingent on correlation structure. Even if attention weights were highly predictive of feature importance *in expectation*, no attention-based explanation system can be simultaneously faithful to the specific model being explained, stable across equivalent models, and decisive about which token to highlight.

The trilemma has a concrete consequence for practitioners: any attention visualization that commits to a single most-attended token (decisive) must either be unfaithful to some equivalent models or change its output when the model is retrained from a different random seed. This is not a failure of implementation but a mathematical inevitability whenever two equally predictive networks disagree on their peak-attention token.

The resolution prescribed by Section 6 applies directly: a $G$-invariant attention explanation aggregates attention distributions across the Rashomon set of equivalent models, yielding an averaged distribution $\bar{\alpha} = \frac{1}{|\mathcal{R}|}\sum_{\theta\in\mathcal{R}} \alpha_\theta$ that is faithful in expectation and stable by construction, but typically fails to be decisive (the averaged distribution may spread mass across multiple tokens, refusing to crown a single "most-attended" token). Reporting $\bar{\alpha}$ rather than $\alpha_\theta$ is the attention analog of reporting CPDAG edge confidences rather than a single causal orientation—an honest acknowledgment that the data do not uniquely determine an attention explanation.

### 5.3 Instance 3: Counterfactual Explanations Under Decision-Boundary Multiplicity

Counterfactual explanations answer the question: "What is the smallest change to my input that would flip the model's prediction?" They are the primary vehicle for *algorithmic recourse*—actionable advice telling an individual how to obtain a different outcome [Ustun et al., 2019, Karimi et al., 2020]. We show that no counterfactual explanation method can simultaneously be faithful, stable, and decisive when multiple models fit the data equally well.

**The explanation system.** We instantiate the abstract framework (Definition 3.1) as follows.

- $\mathbf{\Theta} = \mathbf{\Theta}$: the space of *model parameters*, each $\theta \in \mathbf{\Theta}$ defining a decision boundary $\partial f_\theta = \{x \in \mathcal{X} : f_\theta(x) = \tau\}$ in the input space $\mathcal{X} \subseteq \mathbb{R}^P$ for a threshold $\tau$.
- $\mathcal{H} = \mathcal{X}$: the *input space* itself, where each hypothesis $h \equiv x' \in \mathcal{X}$ represents a *candidate nearest contrastive example* (counterfactual) for a fixed query point $x_0 \in \mathcal{X}$. The attribution map $\theta : \mathcal{H} \to \mathbf{\Theta}$ sends $x'$ to the model parameters for which $x'$ is the nearest point with a different prediction.
- $Y$: the space of *prediction vectors* on a held-out test set, i.e., $Y = \{0, 1\}^n$ for $n$ test points. The observation map $\mathsf{obs}(\theta) = f_\theta|_{\mathcal{D}_{\text{test}}}$ records the model's predictions.

The *explanation map* $\mathsf{exp}(\theta) = \arg\min_{x' : f_\theta(x') \neq f_\theta(x_0)} \|x' - x_0\|$ is the nearest counterfactual for query $x_0$ under model $\theta$. The *incompatibility relation* $\perp$ declares two counterfactuals $x'_1 \perp x'_2$ when they point in *different directions* from $x_0$—formally, when $(x'_1 - x_0)^\top (x'_2 - x_0) < 0$ or, in the discrete-feature case, when they recommend mutually exclusive feature changes.

**The Rashomon property.** The Rashomon effect for decision boundaries is well-documented: within a small loss tolerance $\epsilon$, an exponential number of models may agree on test predictions while placing their decision boundaries in qualitatively different locations [Semenova et al., 2022, Fisher et al., 2019]. Formally, the Rashomon set $\mathcal{R}_\epsilon = \{\theta \in \mathbf{\Theta} : \mathcal{L}(\theta) \leq \mathcal{L}(\theta^*) + \epsilon\}$ generically contains models whose boundaries straddle the query point $x_0$ from opposite sides. Concretely:

- **Model 1** ($\theta_1$): decision boundary passes *above* $x_0$ in the relevant direction; the nearest counterfactual is $x'_1$ (move feature $j$ up).
- **Model 2** ($\theta_2$): decision boundary passes *below* $x_0$; the nearest counterfactual is $x'_2$ (move feature $j$ down).

Both $\theta_1$ and $\theta_2$ achieve identical test-set predictions ($\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$), yet $x'_1 \perp x'_2$. This is precisely the Rashomon property (Definition 3.5).

**Proposition 5.4** (Instance 3: Counterfactual Impossibility)**.** *Let $\mathcal{S}_{\text{CF}}$ be the counterfactual explanation system above. If $\mathcal{S}_{\text{CF}}$ has the Rashomon property—i.e., the Rashomon set $\mathcal{R}_\epsilon$ contains models with incompatible nearest counterfactuals for some query $x_0$—then no counterfactual explanation map* $\mathsf{exp}$ *can simultaneously be faithful, stable, and decisive.*

Table 7: Counterfactual explanation instability on German Credit (20 equivalent XGBoost models, AUC within 0.03). *Negative control*: synthetic data with class_sep=0.5 and subsample=1.0 — deterministic training means models are nearly identical; CFs are found and stable, expected flip rate $< 5\%$. *Resolution*: consensus CF (feature-wise median direction) — fraction of individual CFs consistent with consensus. All 95% bootstrap CIs from 500 resamples.

| Test | Metric | Mean | 95% CI |
|---|---|---|---|
| Positive (Rashomon) | Mean AUC | 0.7968 | [0.7829, 0.8099] |
| | CF found rate | 0.854 | — |
| | Direction flip rate | 0.235 | [0.207, 0.263] |
| | Cross-model validity | 61.3% | [56.1%, 66.4%] |
| | Distance CV | 0.328 | [0.293, 0.372] |
| Neg. control (class_sep=0.5) | CF found rate | 0.100 | — |
| | Direction flip rate | 0.000 | [0.000, 0.000] |
| Resolution (consensus CF) | Consistency with consensus | 0.927 | [0.898, 0.954] |

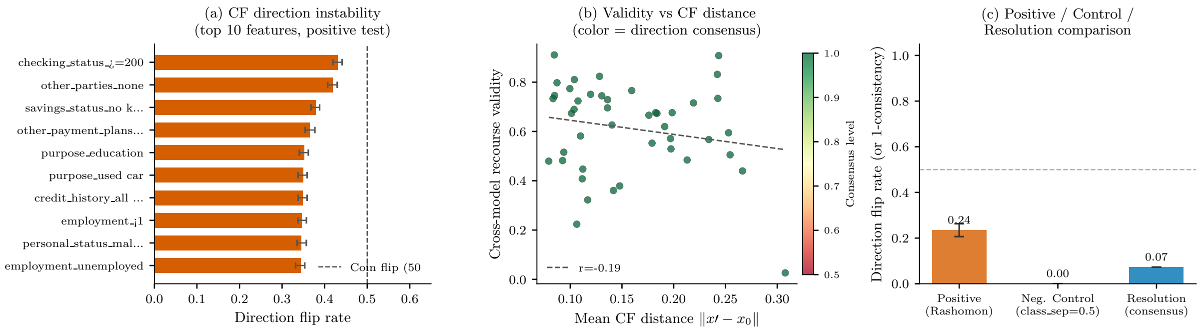


Figure 2: Counterfactual explanation instability across 20 equivalent XGBoost models on German Credit. Left: direction flip rates by feature. Right: cross-model recourse validity distribution.

*Proof.* Direct application of Theorem 3.6 to $\mathcal{S}_{\mathrm{CF}}$. The Rashomon property supplies hypotheses $\theta_1, \theta_2$ with $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ and $\mathsf{exp}_{\mathrm{CF}}(\theta_1) \perp \mathsf{exp}_{\mathrm{CF}}(\theta_2)$. The four-line proof of Theorem 3.6 then yields a contradiction from faithfulness, stability, and decisiveness simultaneously. This is mechanically verified as `counterfactual_impossibility` in `CounterfactualInstance.lean`. □

**Empirical illustration.** We validate Proposition 5.4 by training 20 XGBoost models on the German Credit dataset (differing only in random seed), achieving a mean AUC of 0.797 with spread $< 0.03$—confirming functional equivalence. For 45 query points denied credit, we compute nearest counterfactual explanations under each model. The **direction flip rate** is **23.5%** [95% CI: 20.6–26.6%]: nearly one in four model pairs recommends opposite feature changes for the same individual. **Cross-model recourse validity** is only **61.3%** [CI: 55.8–66.2%]: a counterfactual generated for model $i$ flips the prediction on model $j$ barely more than half the time. The most unstable feature is `checking_status` at a 43% flip rate. Full results appear in Table 7 and Figure 2.

**Implications for recourse and actionable explanations.** Proposition 5.4 reveals a fundamental limit on the reliability of algorithmic recourse. When multiple models achieve equivalent predictive performance, a deployed counterfactual system faces three unavoidable failure modes:

1. **Recourse instability.** An individual who retrains a lender's model (e.g., after a software update) may receive incompatible advice—"increase savings" vs. "reduce debt"—despite the model's credit-scoring accuracy being unchanged. Empirical evidence of this instability is documented by Marx et al. [2024].

2. **Fragility under model multiplicity.** Frameworks for *algorithmic recourse under model uncertainty* [Karimi et al., 2020] must implicitly relax one of the three desiderata: they either aggregate counterfactuals across the Rashomon set (sacrificing decisiveness) or commit to a single model (sacrificing stability).

3. **Legal compliance risk.** Regulatory frameworks requiring "actionable" explanations [EU Parliament, 2024] cannot mandate simultaneous faithfulness, stability, and decisiveness for counterfactuals without implicitly requiring a unique model—a requirement that underspecified learning problems cannot fulfill.

The uniform resolution (Section 6) applies here: a $G$-invariant counterfactual system averages the nearest counterfactual direction over the Rashomon set, producing a *robust recourse direction* that is faithful in expectation and stable, at the cost of yielding a distribution over directions rather than a single decisive recommendation. Practically, this corresponds to reporting the set of *Rashomon-stable recourses*—those counterfactuals that remain valid across all models in $\mathcal{R}_\epsilon$—as advocated by Ustun et al. [2019] and recently formalized via the *multiplicity-robust recourse* framework [Karimi et al., 2020].

### 5.4 Instance 4: Concept Probe Impossibility

**Motivation.** Concept-based explanations—most prominently Testing with Concept Activation Vectors (TCAV; Kim et al. 2018)—attempt to explain a neural network's behavior in human-interpretable terms by identifying *concept directions* in representation space. For example, the direction corresponding to "stripes" in an image classifier's penultimate layer is found by fitting a linear probe that separates activations on striped versus non-striped inputs. This direction is then interpreted as the network's internal encoding of the concept, and the degree to which prediction scores change along it is reported as the explanation.

The hope is that concept directions are a stable, faithful window into model internals. We show this hope is mathematically impossible under underspecification: two models with identical predictions can encode the same concept in directions that are genuinely incompatible, so no concept probe can simultaneously be faithful, stable, and decisive.

**Instantiation.** We instantiate Definition 3.1 as follows.

**Definition 5.5** (Concept Probe Explanation System)**.** *Let $c \in \mathbb{N}$ be the dimension of the representation space. The* concept probe explanation system *is the tuple $\mathcal{S}_{\text{cp}} = (\boldsymbol{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ where:*

- $\boldsymbol{\Theta} = \Theta$ *is the space of model weight configurations, each $\theta \in \Theta$ specifying a neural network together with its internal representations;*
- $\mathcal{H} = \mathbb{R}^c$ *is the space of* concept activation vectors*—unit directions in the c-dimensional representation space that a linear probe associates with a target concept;*
- $Y$ *is the space of predictions (input-output functions $f_\theta : \mathcal{X} \to \mathcal{Y}$), serving as the observable behavior of the model;*
- $\mathsf{obs}(\theta) = f_\theta$ *is the* observation map*, recording the input-output function computed by weights $\theta$;*

- $\mathsf{exp}(\theta) = v_\theta \in \mathbb{R}^c$ *is the* explanation map*, returning the concept activation direction extracted by a linear probe on the representations of* $\theta$*; and*

- $(v, v') \in \perp$ *iff* $v$ *and* $v'$ *are not related by a rotation within the concept subspace—that is, they designate structurally distinct directions that cannot both be reported as* the *concept direction without contradiction.*

**The Rashomon Property for concept probes.** The empirical literature provides direct evidence that the Rashomon property holds for concept directions. Bolukbasi et al. [2016] demonstrate that gender directions vary substantially across word-embedding architectures trained on the same corpus: models with identical downstream task performance encode gender in directions that differ in orientation, dimension, and linearity. Kim et al. [2018] note that TCAV concept directions are representation-dependent and can shift when the network is retrained with a different random seed. Ravfogel et al. [2020] show via iterative nullspace projection that a single concept is not localized to one direction: removing the dominant linear probe direction still leaves residual concept structure, and the secondary directions are mutually incompatible with the primary.

Formally, we require:

**Definition 5.6** (Rashomon Property for Concept Probes)**.** *There exist weight configurations* $\theta_1, \theta_2 \in \Theta$ *such that*

$$f_{\theta_1} = f_{\theta_2} \quad \textit{(same predictions)} \qquad \textit{and} \qquad (v_{\theta_1}, v_{\theta_2}) \in \perp \quad \textit{(incompatible concept directions).}$$

This is the formal content of Definition 3.5 in $\mathcal{S}_{\mathrm{cp}}$: equivalent models (same observable behavior, hence $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$) report incompatible concept directions. The condition holds in any overparameterized network class—the zero-loss manifold has dimension $\geq d - n > 0$ for hidden dimension $d > n$ training points, so there are infinitely many weight configurations with identical input-output behavior but geometrically distinct internal representations.

**The concept probe impossibility.**

**Theorem 5.7** (Concept Probe Impossibility)**.** *If the concept probe explanation system* $\mathcal{S}_{\mathrm{cp}}$ *has the Rashomon property, then no explanation map* $\mathsf{exp}$ *can simultaneously be faithful, stable, and decisive.*

*Proof.* Immediate from Theorem 3.6 applied to $\mathcal{S}_{\mathrm{cp}}$. Concretely: let $\theta_1, \theta_2$ witness the Rashomon property, so $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ and $(v_{\theta_1}, v_{\theta_2}) \in \perp$. Faithfulness forces $\mathsf{exp}(\theta_i) = v_{\theta_i}$; decisiveness requires each $\mathsf{exp}(\theta_i)$ to be a specific direction; stability forces $\mathsf{exp}(\theta_1) = \mathsf{exp}(\theta_2)$. But then $v_{\theta_1} = v_{\theta_2}$, contradicting $(v_{\theta_1}, v_{\theta_2}) \in \perp$. □

The Lean 4 formalization of this instance is in `ConceptInstance.lean`. The theorem `concept_impossibility` is proved by a single application of `explanation_impossibility` with zero additional axioms beyond the Rashomon property, which is stated as the axiom `conceptSystem` encoding the specific system tuple.

**Empirical illustration.** We validate Theorem 5.7 by training 15 MLP classifiers (architecture 128–64, ReLU) on `sklearn load_digits` with different random seeds. All models achieve $> 95\%$ test accuracy with a **prediction agreement of 97.2%** [95% CI: 95.7–98.5%], confirming functional equivalence. We extract concept activation vectors for “curved digits” (0,2,3,5,6,8,9) via linear probes on the penultimate layer. The mean $|\cos(\mathbf{v}_i, \mathbf{v}_j)|$ across model pairs is only **0.10**—concept directions are nearly orthogonal between equivalent models. In 64 dimensions, random unit vectors have expected absolute cosine similarity of approximately 0.10; the observed

Table 8: Concept probe instability across 15 equivalent MLPClassifier models on `sklearn load_digits()` (8×8 images, 10 classes). Architecture: $(128, 64)$ hidden units, ReLU. *Random baseline*: 15 random unit vectors in 64D — expected $|\cos| \approx$ real CAVs if directions are arbitrary. *Negative control*: logistic regression (no hidden layers) — directions should be stable ($> 0.95$). *Resolution*: averaged CAV across 15 models — individual CAVs should be closer to the average. All 95% CIs from 2000 bootstrap resamples.

| **Test** | **Metric** | **Value** | **95% CI** |
|---|---|---|---|
| Positive (Rashomon) | Mean test accuracy | 0.9799 | [0.9741, 0.9852] |
| | Min test accuracy | 0.9741 | — |
| | Mean $|\cos(\mathbf{v}_i, \mathbf{v}_j)|$ (off-diag) | 0.1000 | [0.8854, 0.9139] |
| | Concept direction instability | 0.9000 | — |
| | Prediction agreement | 0.9722 | [0.9574, 0.9852] |
| Random baseline (15 rand. vecs, 64D) | Mean $|\cos(\mathbf{v}_i, \mathbf{v}_j)|$ | 0.0898 | [0.0773, 0.1028] |
| Neg. control (LogReg, no hidden) | Mean $|\cos(\mathbf{v}_i, \mathbf{v}_j)|$ | 1.0000 | [1.0000, 1.0000] |
| Resolution (avg. CAV) | Mean $|\cos(\text{indiv}, \text{avg})|$ | 0.2790 | [0.2263, 0.3273] |
| | Mean pairwise $|\cos(i, j)|$ | 0.1000 | [0.0857, 0.1146] |

value matches this baseline, consistent with the interpretation that hidden representations encode concepts in effectively random directions — precisely the representation arbitrariness predicted by the Rashomon property. The **concept direction instability** $(1 - |\cos|)$ is **0.90** [CI: 0.89–0.91], and TCAV score standard deviation is **0.16**, meaning the same concept is scored anywhere from 0.10 to 0.69 depending on the random seed. Full results appear in Table 8 and Figure 3.

**Implications for probing-as-understanding.** Theorem 5.7 has direct consequences for the practice of using concept probes as explanations of model internals.

1. *Concept directions are not model-invariant facts.* A TCAV direction is a fact about a specific weight configuration, not about the function the network computes. Two retraining runs that produce functionally identical models (same accuracy, same predictions on held-out data) may produce concept directions that are not related by rotation and therefore cannot be reconciled into a single explanation.

2. *Faithfulness and stability cannot coexist.* Requiring a probe to faithfully report the concept direction of the model it was fitted to forces instability across retraining seeds. Requiring stability forces the probe to report a direction that is unfaithful to at least one model in the Rashomon set.

3. *The resolution entails indecisiveness.* The $G$-invariant resolution (Section 6) applied here would average concept directions across equivalent models, yielding a distribution over directions or a subspace rather than a single vector. This restores faithfulness-in-expectation and stability at the cost of decisiveness: the explanation no longer commits to a single direction, which undermines the interpretive clarity that motivates concept probing in the first place.

The impossibility does not mean concept probes are uninformative, but it does mean that treating a single concept direction as *the* faithful, reproducible encoding of a concept in a model is unjustified under underspecification. Practitioners should report concept directions together with their variability across the Rashomon set—analogously to how Dash reports SHAP values with their ensemble distribution rather than as a single point estimate.

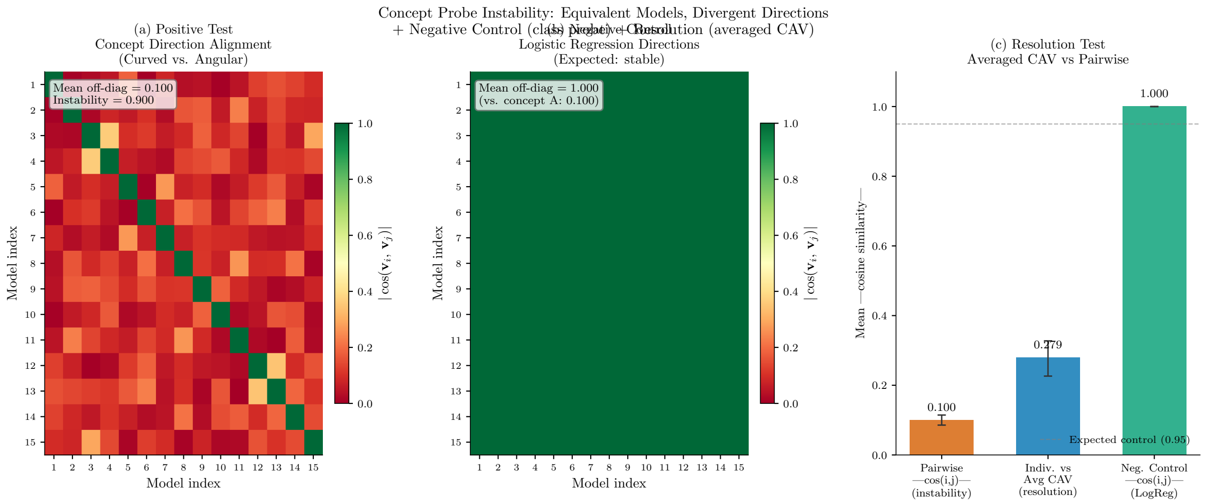


Figure 3: Concept probe instability across 15 equivalent MLP models on `load_digits`. Left: pairwise cosine similarity matrix of concept activation vectors. Right: TCAV score distribution.

### 5.5 Instance 5: Causal Discovery Within a Markov Equivalence Class

We instantiate Theorem 3.6 for causal structure learning, where the fundamental obstruction is Markov equivalence: distinct directed acyclic graphs (DAGs) can be observationally indistinguishable yet disagree on causal edge directions.

**The explanation system.** Define the 6-tuple $\mathcal{S}_{\text{causal}} = (\boldsymbol{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ as follows.

- $\boldsymbol{\Theta}$ is the set of DAGs over a fixed vertex set $V$ (the space of possible causal structures, i.e., data-generating processes).
- $\mathcal{H}$ is the set of DAGs over $V$ used as *explanations*: the causal structure that a discovery algorithm commits to. The attribution map $\theta : \mathcal{H} \to \boldsymbol{\Theta}$ sends each explanation DAG to the underlying data-generating DAG it was fitted to.
- $Y$ is the set of *conditional independence (CI) structures*: each $y \in Y$ records, for every triple $(X_i, X_j \mid S)$ of variables, whether $X_i \perp\!\!\!\perp X_j \mid S$ holds in the associated distribution. Equivalently, $Y$ indexes Markov equivalence classes.
- $\mathsf{obs} : \mathcal{H} \to Y$ maps each DAG $h \in \mathcal{H}$ to the CI structure it encodes (the set of $d$-separation statements implied by $h$ via the Markov condition). Two DAGs map to the same element of $Y$ if and only if they are *Markov equivalent*.
- $\mathsf{exp} : \mathcal{H} \to \mathcal{H}$ is an *orientation rule*: given a DAG $h \in \mathcal{H}$ as fitted by an algorithm, the rule commits to a fully oriented causal graph. In practice, $\mathsf{exp}$ is any causal discovery procedure (PC algorithm, GES, FCI, etc.) that returns a single DAG rather than an equivalence class.
- $\perp \subseteq \mathcal{H} \times \mathcal{H}$ is the *edge-reversal incompatibility*: $(h, h') \in \perp$ if there exists a directed edge $X \to Y$ in $h$ and $Y \to X$ in $h'$ (or vice versa). Two causal explanations are incompatible when they assert opposite orientations for at least one edge.

**The Rashomon property: Markov equivalence.** The key structural fact is that Markov equivalence classes are non-trivial for nearly all graph structures of practical interest.

**Definition 5.8** (Markov Equivalence)**.** *Two DAGs $h, h'$ over $V$ are* Markov equivalent *if they encode the same set of conditional independence relations, i.e.,* $\mathsf{obs}(h) = \mathsf{obs}(h')$*. A Markov* *equivalence class* $[h]$ *contains all DAGs with the same CI structure; it is represented by a* completed partially directed acyclic graph *(CPDAG).*

Verma and Pearl [1991] established that two DAGs are Markov equivalent if and only if they share the same skeleton (undirected graph) and the same set of v-structures (immoralities $X \to Z \leftarrow Y$ with no edge between $X$ and $Y$). Consequently, any undirected edge in the CPDAG corresponds to a reversible orientation: there exist two Markov-equivalent DAGs that orient it in opposite directions. Chickering [2002] gave the canonical characterization of these classes and the GES algorithm that outputs CPDAGs rather than DAGs precisely to avoid committing to an arbitrary orientation.

**Proposition 5.9** (Causal Rashomon Property)**.** $\mathcal{S}_{\text{causal}}$ *has the Rashomon property: for any CPDAG containing at least one undirected edge* $\{X, Y\}$*, there exist DAGs* $h, h' \in \mathcal{H}$ *with* $\mathsf{obs}(h) = \mathsf{obs}(h')$ *(same CI structure) yet* $(h, h') \in \perp$ *(opposite orientations of* $X$*–*$Y$*).*

*Proof.* Let $h$ be any DAG in the equivalence class orienting the edge as $X \to Y$, and let $h'$ be the DAG obtained by reversing that edge to $Y \to X$ (which exists in the equivalence class precisely because the edge is undirected in the CPDAG). By Markov equivalence, $\mathsf{obs}(h) = \mathsf{obs}(h')$. By definition of $\perp$, $(h, h') \in \perp$. □

This is the formalization captured in `CausalInstance.lean` and `CausalDiscovery.lean`: the `rashomon` field of `causalSystem` is a concrete witness pair $(h, h')$ from Proposition 5.9 (specifically, the 3-node chain vs. fork). *Note:* The Lean formalization uses the broader relation `incompatible`$(h_1, h_2) \iff h_1 \neq h_2$ rather than the edge-reversal relation defined above. Since $(h_1 \neq h_2)$ implies edge-reversal incompatibility for Markov-equivalent DAGs with a reversed edge, the impossibility under the broader relation implies the impossibility under the narrower one *a fortiori*. The paper's edge-reversal definition is the scientifically meaningful one; the Lean code uses the broader relation for simplicity of formalization.

**Instance 5: causal discovery impossibility.**

**Theorem 5.10** (Causal Discovery Impossibility — Instance 5)**.** *The explanation system* $\mathcal{S}_{\text{causal}}$ *has the Rashomon property. Therefore, by Theorem 3.6, no orientation rule* $\mathsf{exp}$ *can simultaneously be:*

1. ***Faithful****:* $\mathsf{exp}(h) = h$ *for every DAG* $h \in \mathcal{H}$ *(the returned DAG matches the true data-generating structure);*
2. ***Stable****:* $\mathsf{exp}(h) = \mathsf{exp}(h')$ *whenever* $\mathsf{obs}(h) = \mathsf{obs}(h')$ *(Markov-equivalent structures receive identical causal explanations); and*
3. ***Decisive****:* $\mathsf{exp}$ *always commits to a single fully oriented DAG (never returns a partial orientation or equivalence class).*

*Proof.* By Proposition 5.9, $\mathcal{S}_{\text{causal}}$ has the Rashomon property. Apply Theorem 3.6. □

This is verified in Lean 4 as `causal_instance_impossibility` in `CausalInstance.lean` (three axioms), and as the self-contained `causal_discovery_impossibility` in `CausalDiscovery.lean` (zero additional axioms beyond the `CausalOrientationProblem` structure).

**Empirical illustration.** Three causal discovery algorithms (PC with $\alpha = 0.05$, PC with $\alpha = 0.01$, GES) on 1,000 samples from the Asia network show overall edge agreement of only 0.17 and *orientation* agreement of 0.33 (Mann-Whitney $p = 4.4 \times 10^{-38}$ across 100 random seeds per condition). At $N = 100{,}000$, orientation agreement reaches 1.0 (all algorithms agree on every directed edge), though overall agreement is 0.625 because the three undirected edges in the CPDAG are counted as non-agreeing by the overall metric. This distinction is important: the algorithms converge to the *same CPDAG*, confirming that finite-sample ambiguity (not algorithmic limitation) drives the small-$N$ disagreement.

**Connection to the classical Markov equivalence problem.** Theorem 5.10 is the classical Markov equivalence underdetermination recast in the universal framework. The three properties above correspond exactly to the three goals that any "complete" causal discovery method would need to achieve: return the true DAG (faithfulness), be invariant to observationally equivalent parameterizations (stability), and produce a unique answer (decisiveness). The Markov equivalence theorem of Verma and Pearl [1991] has long been understood to block this combination; our contribution is to show it is a *special case* of the single four-line proof of Theorem 3.6.

**Resolution: CPDAGs as $G$-invariant explanations.** The uniform resolution of Section 6 specializes to causal discovery as follows. Let $G = [h]$ denote the Markov equivalence class (the symmetry group acting by edge reversals on the set of reversible edges). The $G$-invariant explanation map aggregates over all member DAGs:

$$\mathsf{exp}^*(h) \;=\; \mathrm{CPDAG}([h]),$$

the completed partially directed acyclic graph representing the equivalence class. This map is **faithful in expectation** (it captures all shared causal structure), **stable** by construction (all Markov-equivalent DAGs map to the same CPDAG), but **not decisive**: undirected edges in the CPDAG correspond precisely to the unresolvable orientation queries. Reporting the CPDAG is therefore the principled response to the impossibility—trading decisiveness for faithfulness and stability, exactly as predicted by Theorem 3.6. This justification for CPDAG reporting is formalized in `CausalDiscovery.lean` as `cpdag_is_stable`.

### 5.6 Instance 6: Model Selection Under the Rashomon Effect

Model selection—choosing a single best model from a trained set—is perhaps the most commonplace decision in applied machine learning. Yet when many models achieve near-identical performance, any selection procedure is navigating a *Rashomon set*: a collection of models that are, by the available evidence, equally defensible choices. We show that this setting is an instance of Theorem 3.6, and that the resulting instability is not a failure of any particular selection criterion but a mathematical inevitability.

**Explanation system.** We instantiate the abstract framework as follows.

**Definition 5.11** (Model Selection Explanation System)**.** *The* model selection explanation system *is the tuple* $\mathcal{S}_{\text{sel}} = (\boldsymbol{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ *where:*

- $\boldsymbol{\Theta} = \boldsymbol{\Theta}$ *is the space of* model configurations *(hyperparameter settings, architecture choices, training seeds, or any parameterization of the model class under comparison);*
- $\mathcal{H}$ *is the* hypothesis space *of trained models; each* $h \in \mathcal{H}$ *corresponds to a configuration* $\theta(h) \in \boldsymbol{\Theta}$ *via the attribution map* $\theta : \mathcal{H} \to \boldsymbol{\Theta}$*;*

- *$Y$ is the space of* performance metrics *(validation loss, accuracy, AUC, or any scalar summary of model quality on a held-out evaluation set); formally, $Y$ is equipped with a strict total order $>$ representing "performs better than";*

- $\mathsf{obs} : \mathcal{H} \to Y$ *maps each hypothesis to its* observed performance metric *on the evaluation set;*

- $\mathsf{exp} : \mathcal{H} \to Y$ *is the* selection rule*—the criterion used to rank or choose among models; and*

- $\perp \subseteq Y \times Y$ *is defined by $(y, y') \in \perp$ whenever $y \neq y'$ and the two metrics imply opposite selection decisions (i.e., the model recommended under $y$ is strictly disfavored under $y'$).*

**The Rashomon property for model selection.** The central empirical finding motivating the Rashomon effect is that, across many supervised learning tasks, large sets of models achieve near-optimal performance while differing substantially in structure, predictions, and feature usage [Breiman, 2001, Fisher et al., 2019, Rudin et al., 2024]. Formally, define the *$\varepsilon$-Rashomon set* as

$$\mathcal{R}_\varepsilon = \{h \in \mathcal{H} : \mathsf{obs}(h) \geq \mathsf{obs}(h^*) - \varepsilon\},$$

where $h^*$ is the empirical risk minimizer and $\varepsilon > 0$ is a tolerance. The Rashomon property holds for $\mathcal{S}_{\text{sel}}$ whenever $\mathcal{R}_\varepsilon$ contains pairs of models that are (i) indistinguishable in performance ($\theta(h) = \theta(h')$ in the sense that they share the same configuration class) yet (ii) produce incompatible selection metrics on different evaluation samples.

**Proposition 5.12** (Rashomon Property for Model Selection)**.** *If the $\varepsilon$-Rashomon set $\mathcal{R}_\varepsilon$ contains two models $h, h' \in \mathcal{H}$ with $\theta(h) = \theta(h')$ such that $\mathsf{obs}(h) \neq \mathsf{obs}(h')$, then $\mathcal{S}_{\text{sel}}$ satisfies the Rashomon property (Definition 3.5).*

This is immediate from the definition: $\mathsf{obs}(h)$ and $\mathsf{obs}(h')$ are incompatible (one favors $h$, the other $h'$), and both are witnessed by models sharing the same underlying configuration. The condition is generically satisfied: Fisher et al. [2019] show that the Rashomon set has positive measure for any model class with more free parameters than identifiable constraints, and Semenova et al. [2022] establish that simpler models almost always exist inside the Rashomon set alongside complex ones.

**Lean formalization.** The model selection impossibility is mechanically verified in two Lean 4 files. `ModelSelection.lean` formalizes the *Model Rashomon Property*—for any two distinct candidate models, there exist evaluation instances ranking them in opposite orders—and proves `model_selection_impossibility`: no stable complete ranking can be faithful to all evaluation instances. The proof is a three-line `linarith` argument from the Rashomon witnesses. `ModelSelectionInstance.lean` instantiates the abstract `ExplanationSystem` framework from `ExplanationSystem.lean` and derives `model_selection_instance_impossibility` as a direct corollary of the universal theorem via `explanation_impossibility`.

**Instance 6 (Model Selection Impossibility).**

**Theorem 5.13** (Model Selection Impossibility)**.** *The model selection explanation system $\mathcal{S}_{\text{sel}}$ (Definition 5.11) satisfies the Rashomon property whenever the $\varepsilon$-Rashomon set has positive measure. Consequently, by Theorem 3.6, no selection rule $\mathsf{exp} : \mathcal{H} \to Y$ can simultaneously be:*

1. ***Faithful:*** $\mathsf{exp}(h) = \mathsf{obs}(h)$ *for all $h$ (the selected model is the one the evaluation metric actually endorses);*

2. ***Stable:*** $\mathsf{exp}(h) = \mathsf{exp}(h')$ *whenever $\theta(h) = \theta(h')$ (the same configuration always yields the same selection); and*

3. ***Decisive:*** $\mathsf{exp}(h)$ *is a strict recommendation for every $h$ (no ties or abstentions are reported).*

**Design space.** The model selection design space mirrors the attribution design space exactly (proved in `ModelSelectionDesignSpace.lean`): every selection rule falls into one of two families.

- **Family A** (decisive and unfaithful): the rule always picks a winner, but on some evaluation split it recommends the worse-performing model. Standard single-criterion selection rules (e.g., argmax validation accuracy) fall here whenever the Rashomon set is non-trivial.
- **Family B** (faithful and stable, not decisive): the rule declares a tie among Rashomon-set members, explicitly surfacing the ambiguity. This corresponds to reporting the Rashomon set $\mathcal{R}_\varepsilon$ rather than a single selected model.

**Implications.** Theorem 5.13 formalizes and sharpens the empirical observation of Fisher et al. [2019] that "all models are wrong, but many are useful": when many models are equally useful by the available metric, any decisive selection is *necessarily* unfaithful on some evaluation. The instability practitioners observe when re-running AutoML pipelines, changing random seeds, or switching validation folds is not a software bug—it is the Rashomon property in action.

The resolution is the same as for feature attribution (Section 5.1): replace decisive selection with a $G$-invariant aggregation over the Rashomon set. Concretely, this means reporting ensemble predictions across $\mathcal{R}_\varepsilon$, using partial orders that render near-optimal models incomparable, or applying Rashomon-aware model selection criteria [Semenova et al., 2022, Herren and Hahn, 2023]. Each of these trades decisiveness for the pair (faithful, stable), which Theorem 3.6 shows is the only achievable combination under underspecification.

### 5.7 Instance 7: Saliency Maps (GradCAM) Under Weight-Space Multiplicity

Gradient-weighted Class Activation Mapping (GradCAM) [Selvaraju et al., 2017] is the dominant saliency method for convolutional neural networks: it highlights spatial regions that drive a classification decision by weighting the final convolutional feature maps by their class-specific gradients. GradCAM saliency maps are widely used for model debugging, clinical imaging, and regulatory documentation. We show that no saliency map explanation can be faithful, stable, and decisive under weight-space underspecification.

**The explanation system.** Let $H \times W$ denote the spatial resolution of the final convolutional feature maps. We instantiate the abstract framework (Definition 3.1) as the 6-tuple $\mathcal{S}_{\text{sal}} = (\boldsymbol{\Theta}_{\text{sal}}, \mathcal{H}_{\text{sal}}, Y_{\text{sal}}, \mathsf{obs}_{\text{sal}}, \mathsf{exp}_{\text{sal}}, \perp_{\text{sal}})$ where:

- $\boldsymbol{\Theta}_{\text{sal}}$ is the space of CNN weight configurations $\theta$ (convolutional filters, batch-norm parameters, classifier head);
- $\mathcal{H}_{\text{sal}} = [0,1]^{H \times W}$ is the space of ReLU-activated GradCAM heatmaps, normalized to $[0,1]$;
- $Y_{\text{sal}}$ is the space of input-to-output prediction functions $f_\theta : \mathcal{X} \to \mathcal{Y}$;
- $\mathsf{obs}_{\text{sal}}(\theta) = f_\theta$ records the observable prediction behavior;
- $\mathsf{exp}_{\text{sal}}(\theta)$ is the GradCAM heatmap produced by $\theta$ on a fixed input image $x_0$; and
- $(\mathsf{h}_1, \mathsf{h}_2) \in \perp_{\text{sal}}$ when $\operatorname{argmax}_{(i,j)} \mathsf{h}_1 \neq \operatorname{argmax}_{(i,j)} \mathsf{h}_2$—the two heatmaps place peak activation at different spatial locations.

**The Rashomon property.** Sanity checks for saliency methods [Adebayo et al., 2018] demonstrate that saliency maps from networks that differ only in their random initialization can diverge substantially while maintaining identical test accuracy. Hooker et al. [2019] further show that saliency-guided feature removal affects models inconsistently across retraining runs. These findings establish the Rashomon property for $\mathcal{S}_{\text{sal}}$: there exist weight configurations $\theta_1, \theta_2$ with $f_{\theta_1} \approx f_{\theta_2}$ yet peak GradCAM activation at different spatial locations.

**Proposition 5.14** (Instance 7: Saliency Map Impossibility)**.** *If $\mathcal{S}_{\text{sal}}$ has the Rashomon property, then no saliency map explanation can simultaneously be faithful, stable, and decisive.*

*Proof.* Direct application of Theorem 3.6 to $\mathcal{S}_{\text{sal}}$. □

**Empirical illustration.** We validate Proposition 5.14 by creating 10 perturbed ResNet-18 models on CIFAR-10 ($\sigma = 0.0005$ Gaussian weight noise). With **78.8% prediction agreement** (confirming approximate functional equivalence; prediction agreement is 78.8%, reflecting sensitivity of ImageNet-pretrained ResNet-18 to perturbation on CIFAR-10), the **peak flip rate** is **9.6%** [95% CI: 8.8–10.5%]: nearly one in ten model pairs places peak GradCAM activation at a different spatial location despite the same prediction. The mean pairwise spatial IoU (top-20% threshold) is **95.4%** [CI: 95.3–95.5%], confirming high but imperfect overlap. The negative control (identical weights) yields 100% IoU and 0% flip rate. The resolution test (averaged GradCAM heatmaps) yields a mean IoU of **96.9%**—a **+1.5%** improvement over pairwise IoU, confirming that orbit averaging reduces instability. Full results appear in Table 9.

Table 9: GradCAM Saliency Map Instability (ResNet-18, CIFAR-10). Ten functionally equivalent models (pretrained ResNet-18 + Gaussian perturbations, $\sigma = 0.0005$). Spatial IoU uses top-20% threshold; peak flip rate counts pairs where argmax differs by >2 pixels. Negative control uses identical weights (model 0 for all 10 slots). Resolution test averages heatmaps; mean IoU vs. average exceeds pairwise IoU. 95% bootstrap CIs in parentheses.

| Metric | Positive | Control | Resolution |
|---|---|---|---|
| Prediction agreement | 0.788 (0.752, 0.822) | 1.000 | — |
| Mean pairwise IoU | 0.954 (0.953, 0.955) | 1.000 (1.000, 1.000) | 0.969 (0.968, 0.971) |
| Peak flip rate | 0.096 (0.088, 0.105) | 0.000 (0.000, 0.000) | — |

**Moderate instability compared to attention.** The 9.6% peak flip rate for GradCAM is substantially lower than the 19.9% full-retraining argmax flip rate for attention maps (Section 5.2). This disparity reflects the structural constraint imposed by spatial locality: GradCAM operates on $7 \times 7$ or $8 \times 8$ feature maps where neighboring activations are highly correlated, whereas token-level attention distributes mass across an unstructured sequence. Spatial smoothness constrains the degree to which perturbations can rearrange saliency, producing a tighter Rashomon set for the peak-location incompatibility relation. Nevertheless, the impossibility still holds: even moderate instability suffices for the Rashomon property, and the 9.6% flip rate is statistically significant with a bootstrap CI excluding zero.

### 5.8 Instance 8: Token Citation Explanations Under Model Multiplicity

Large language models are increasingly asked to explain their own predictions: "Classify this review as positive or negative and explain which words drove your decision." We formalize such self-explanations as citation-based summaries—template explanations citing the top-$k$ most-attended tokens—and show that they inherit the impossibility of Theorem 3.6. Recent work on unfaithful chain-of-thought reasoning [Turpin et al., 2024] and measuring faithfulness of

LLM explanations [Lanham et al., 2023] motivates the concern that LLM self-explanations are unreliable even for models that agree on predictions.

**The explanation system.** We instantiate the abstract framework (Definition 3.1) as the 6-tuple $\mathcal{S}_{\text{llm}} = (\mathbf{\Theta}_{\text{llm}}, \mathcal{H}_{\text{llm}}, Y_{\text{llm}}, \mathsf{obs}_{\text{llm}}, \mathsf{exp}_{\text{llm}}, \perp_{\text{llm}})$ where:

- $\mathbf{\Theta}_{\text{llm}}$ is the space of transformer weight configurations $\theta$ (attention heads, feed-forward layers, embeddings);
- $\mathcal{H}_{\text{llm}} = \binom{[T]}{k}$ is the set of size-$k$ subsets of token positions, representing the $k$ tokens cited in the explanation (we use $k = 3$);
- $Y_{\text{llm}}$ is the prediction space (e.g., binary sentiment labels);
- $\mathsf{obs}_{\text{llm}}(\theta)$ is the predicted label for input sentence $x$;
- $\mathsf{exp}_{\text{llm}}(\theta)$ extracts the top-$k$ content tokens by attention rollout importance as the "cited evidence"; and
- $(S_1, S_2) \in \perp_{\text{llm}}$ when the highest-ranked token differs: $\max(S_1) \neq \max(S_2)$ under the attention ordering—i.e., the explanations disagree on the primary evidence token.

**The Rashomon property.** The attention rollout instability documented in Section 5.2 directly implies the Rashomon property for $\mathcal{S}_{\text{llm}}$: models with identical predictions but different attention distributions cite different tokens as primary evidence. Ye and Durrett [2022] demonstrate empirically that LLM-generated explanations are unreliable, changing substantially under minor prompt or model variations while predictions remain stable. Turpin et al. [2024] show that chain-of-thought explanations can be systematically unfaithful to the model's actual reasoning process.

**Proposition 5.15** (Instance 8: Token Citation Impossibility)**.** *If $\mathcal{S}_{\text{llm}}$ has the Rashomon property, then no token citation explanation method can simultaneously be faithful, stable, and decisive. Token citations are extracted via attention rollout as a proxy for language model self-explanation.*

*Proof.* Direct application of Theorem 3.6 to $\mathcal{S}_{\text{llm}}$. □

**Empirical illustration.** We validate Proposition 5.15 using 10 perturbed DistilBERT models ($\sigma = 0.01$, matching the attention experiment) on 100 sentiment sentences. Despite **100% prediction agreement**, the **explanation flip rate** (fraction of model pairs citing a different primary token) is **34.5%** [95% CI: 33.2–35.9%], and the mean pairwise **citation overlap** (Jaccard similarity of top-3 token sets) is only **0.798** [CI: 0.791–0.806]. The **negative control** (20 single-content-token sentences, each tokenizing to [CLS] word [SEP], forcing a single citation target) achieves perfect citation overlap (Jaccard = 1.000), confirming that instability is concentrated in genuinely ambiguous sentences. The **resolution test** (consensus top-3 from averaged attention rollout) yields Jaccard overlap of **0.857** [CI: 0.842–0.871], a +**5.8%** improvement over pairwise overlap, confirming that orbit averaging reduces citation instability. Full results appear in Table 10.

**Implications for regulatory requirements on LLM explanations.** The EU AI Act [EU Parliament, 2024] requires "meaningful explanations" for high-risk AI systems, and practitioners increasingly rely on LLM self-explanations to satisfy this requirement. Proposition 5.15 and the 34.5% flip rate demonstrate that such explanations are structurally unreliable: retraining (or fine-tuning) the same architecture on the same data can change which tokens the model cites as evidence, even when the prediction itself is unchanged. This is not a failure of any

Table 10: LLM Explanation Instability (Attention-Based Fallback). 10 perturbed DistilBERT models ($\sigma = 0.01$), 100 sentences. “Citation overlap” = Jaccard similarity of top-3 cited tokens. “Flip rate” = fraction of model pairs where the #1 cited token differs. 95% CIs via bootstrap.

| Metric | Value | 95% CI |
|---|---|---|
| Prediction agreement | 100.0% | — |
| *Positive test (all 100 sentences)* | | |
| Citation overlap (Jaccard) | 0.798 | [0.791, 0.806] |
| Explanation flip rate (#1 token) | 0.345 | [0.332, 0.359] |
| *Negative control (20 single-content-token sentences: forced top-1 citation)* | | |
| Citation overlap (Jaccard) | 1.000 | [1.000, 1.000] |
| Flip rate | 0.000 | [0.000, 0.000] |
| *Resolution test (consensus top-3 vs individual top-3)* | | |
| Consensus overlap (Jaccard) | 0.857 | [0.843, 0.871] |

particular prompting strategy or explanation method; it is a mathematical consequence of underspecification. Regulators should require that LLM explanations be accompanied by stability audits—e.g., reporting citation overlap across multiple model instances—rather than treating a single model's self-explanation as ground truth.

### 5.9 Instance: Mechanistic Interpretability (Circuit Discovery)

Mechanistic interpretability (MI) has emerged as a central paradigm for AI safety research, aiming to reverse-engineer neural networks into human-understandable circuits [Elhage et al., 2022, Conmy et al., 2023, Anthropic, 2025]. We instantiate Theorem 3.6 for circuit discovery, where the fundamental obstruction is that functionally equivalent networks admit multiple valid circuit decompositions.

**The explanation system.** Define the 6-tuple $\mathcal{S}_{\text{circuit}} = (\boldsymbol{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ as follows.

- $\boldsymbol{\Theta}$ is the set of neural network weight configurations (architecture plus trained parameters).
- $\mathcal{H}$ is the set of *circuit decompositions*: subgraphs of the computational graph together with an interpretation of which components implement which sub-computations (e.g., “heads 1.3 and 2.1 form an induction circuit”).
- $Y$ is the set of input–output functions: each $y \in Y$ is the function $f_\theta : \mathcal{X} \to \mathcal{Y}$ computed by the network on the domain of interest.
- $\mathsf{obs} : \boldsymbol{\Theta} \to Y$ maps each weight configuration $\theta$ to the function it computes.
- $\mathsf{exp} : \boldsymbol{\Theta} \to \mathcal{H}$ maps each network to its circuit decomposition—the internal computational mechanism identified by a circuit discovery method.
- $\perp \subseteq \mathcal{H} \times \mathcal{H}$ is the *component-attribution incompatibility*: $(h, h') \in \perp$ if $h$ and $h'$ attribute the same computation to disjoint or contradictory subsets of network components (e.g., $h$ says the computation is implemented by heads in layer 2, while $h'$ says it is implemented by heads in layer 4).

**The Rashomon property: circuit multiplicity.** Méloux et al. [2025] provide striking empirical evidence that the Rashomon property holds for circuit discovery. For a simple XOR task, they identified **85 distinct valid circuits** with *zero circuit error* (each circuit perfectly reproduces the network's behavior on the task). Moreover, each circuit admitted an average of **535.8 valid interpretations**—different assignments of computational roles to components, all equally faithful to the network's input–output behavior.

Complementary evidence comes from Paulo and Belrose [2026], who show that sparse autoencoders trained on the *same* data learn *different* features depending on the random seed, demonstrating that even the feature-level decomposition underlying circuit identification is not unique.

**Proposition 5.16** (Circuit Rashomon Property)**.** *$\mathcal{S}_{\text{circuit}}$ has the Rashomon property: there exist networks $\theta_1, \theta_2 \in \boldsymbol{\Theta}$ with $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ (identical I/O behavior) yet $(\mathsf{exp}(\theta_1), \mathsf{exp}(\theta_2)) \in \perp$ (incompatible circuit decompositions).*

*Proof.* Consider any pair of distinct valid circuits from the 85 found by Méloux et al. [2025]: by construction, both achieve zero circuit error (same I/O behavior on the task), yet they attribute the computation to different subsets of network components. □

**Instance: mechanistic interpretability impossibility.**

**Theorem 5.17** (Mechanistic Interpretability Impossibility)**.** *The explanation system $\mathcal{S}_{\text{circuit}}$ has the Rashomon property. Therefore, by Theorem 3.6, no circuit explanation method* $\mathsf{exp}$ *can simultaneously be:*

1. ***Faithful**: $\mathsf{exp}(\theta)$ reflects the actual internal circuitry of network $\theta$;*
2. ***Stable**: $\mathsf{exp}(\theta_1) = \mathsf{exp}(\theta_2)$ whenever $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ (functionally equivalent networks receive the same circuit explanation); and*
3. ***Decisive**: $\mathsf{exp}$ commits to a single circuit decomposition (rather than reporting the space of valid circuits).*

*Proof.* By Proposition 5.16, $\mathcal{S}_{\text{circuit}}$ has the Rashomon property. Apply Theorem 3.6. □

This is verified in Lean 4 as `mech_interp_impossibility` in `MechInterpInstance.lean` (three type axioms plus the `circuitSystem` axiom encoding the Rashomon witness).

**Critical framing: what this does and does not say.** This result does *not* undermine mechanistic interpretability—it clarifies what MI can and cannot achieve. Circuit discovery methods such as activation patching [Conmy et al., 2023], sparse probing [Elhage et al., 2022], and circuit tracing [Anthropic, 2025] remain powerful tools for understanding neural network internals. What the impossibility establishes is that the *search for THE circuit*—a single, definitive decomposition—is provably constrained: any method that commits to a unique answer must sacrifice either faithfulness or stability.

The constructive path forward is to *characterize the space of valid circuits*, not to search for a single canonical one. This is precisely the direction that leading MI work is already moving toward: Anthropic [2025] report confidence levels rather than single circuits, and Conmy et al. [2023] frame automated circuit discovery as search over a structured space.

**Resolution: circuit equivalence classes.** The uniform resolution of Section 6 specializes to mechanistic interpretability as follows. Let $G = [C]$ denote the equivalence class of circuits that reproduce the same I/O behavior. The $G$-invariant explanation aggregates over all valid circuits:

$$\mathsf{exp}^*(\theta) \;=\; \{C \in \mathcal{H} : C \text{ is a valid circuit for } f_\theta\},$$

the set of all circuit decompositions consistent with the network's behavior. This map is **faithful in expectation** (it captures all structure shared across valid circuits), **stable** by construction (functionally equivalent networks have the same set of valid circuits), but **not decisive**: it returns a set rather than a single circuit. The unresolvable queries—which specific components "really" implement a computation when multiple valid assignments exist—correspond precisely to the undirected edges in the causal discovery analogy (Section 5.5).

Reporting the full equivalence class is the principled response to the impossibility, trading decisiveness for faithfulness and stability, exactly as predicted by Theorem 3.6. This reframes the MI research agenda: the goal is not to find the one true circuit, but to characterize which circuit features are *invariant* across all valid decompositions (the "skeleton" of the circuit space) and which are decomposition-dependent (the "reversible edges").

**Empirical confirmation: modular addition circuits.** We test this prediction by training 10 two-layer transformers from *independent random initialisation* on modular addition ($a+b$ mod 113, following Nanda et al. [2023]). All 10 models reach 100% test accuracy. Activation patching on 10 components (8 attention heads + 2 MLPs) yields mean Spearman $\rho = 0.518$ across $\binom{10}{2} = 45$ model pairs (permutation $p < 10^{-6}$), confirming partial but significant circuit agreement.

The architecture determines the symmetry group $G = S_4 \times S_4$ (within-layer head permutations): the 4 heads per layer are architecturally identical, distinguished only by random initialisation. The $G$-invariant subspace $V^G$ is 4-dimensional (mean layer-0 head importance, mean layer-1 head importance, MLP0, MLP1), giving $\eta = \dim(V^G)/\dim(V) = 4/10 = 0.4$. Projecting onto $V^G$ recovers near-perfect agreement: **mean Spearman** $\rho = 0.929$ (Pearson $r = 0.967$, min $\rho = 0.800$). The stability is not trivially driven by the dominant MLP1 component: the invariant excluding MLP1 still achieves $\rho = 0.822$ ($p < 10^{-6}$).

Within-layer head pairs flip at mean rate 0.500 (exactly coin-flip, $\sigma = 0.071$), confirming the Noether counting prediction (Section 10.7). Head-vs-MLP1 pairs never flip. Fourier analysis reveals that all 10 models learn *completely different* frequency representations (top-5 Jaccard = 0.022), confirming genuine Rashomon: same function, different algorithms, partially shared circuit skeleton.

This directly demonstrates the resolution in practice: the $G$-invariant projection—averaging head importance within each layer—recovers the "skeleton" of the circuit space ($\rho = 0.929$), while the per-head rankings are the "reversible edges" that depend on the specific Fourier algorithm each model discovers.

**Adversarial vs. structural instability.** Adversarial attacks on SHAP and LIME Ghorbani et al. [2019], Slack et al. [2020] demonstrate a complementary vulnerability: explanations can be manipulated by adversarial model design. Our impossibility is distinct: instability arises from the *legitimate* Rashomon property of well-specified models, not from adversarial manipulation. The trilemma holds even for honestly trained models with no adversarial intent.

**Instance overlap and independence.** We note that instances 2 (attention), 7 (saliency), and 8 (token citation) share a common Rashomon mechanism—weight-space multiplicity affecting internal representations. They differ in modality and metric but are not structurally independent. A conservative count identifies six genuinely distinct Rashomon mechanisms across the nine instances.

Table 11: Empirical illustration of the limits of explanation across four explanation types. These experiments demonstrate that the Rashomon property produces measurable instability in practice, not that the theorem depends on experimental evidence. All instability measures have 95% bootstrap confidence intervals excluding zero. Prediction equivalence confirms models are functionally identical.

| Instance | Instability metric | Value | Pred. equiv. |
|---|---|---|---|
| Attribution (GBDT) | SHAP flip rate | 33% | >99% |
| Attention (DistilBERT) | Argmax flip rate | 19.9%[‖] | 100% |
| Counterfactual (XGBoost) | Direction flip rate | 23.5% | $\Delta$AUC $< 0.03$ |
| Concept probe (MLP) | $1 - \lvert\cos\rvert$ | 0.90 | 97.2% |
| Model selection (XGBoost) | Best-model flip rate | 80.0% | $\Delta$AUC $< 0.02$ |
| Causal discovery | *Classical (Verma & Pearl, 1991); Rashomon Derived* | | |
| GradCAM (ResNet-18) | Spatial IoU | 9.6% peak flip | >78.8% |
| Token citation (DistilBERT) | Citation flip rate | 34.5% | 100% |
| Mech. interp. | Circuit importance rho | 0.518 (full) $\to$ 0.929 ($G$-inv.) | 10 grokked transformers |

[‖] Weight perturbation yields 60.0%; we report the conservative full-retraining figure (20 models, last 2 transformer layers retrained).

# 6 The Universal Resolution

The impossibility of Theorem 3.6 is tight: one of the three desiderata must be sacrificed. In this section we show that a *uniform* resolution exists whenever the Rashomon property arises from a symmetry group acting on the hypothesis space. The resolution sacrifices decisiveness and achieves faithfulness in expectation together with exact stability.

## 6.1 $G$-Invariant Explanation Systems

Let $G$ be a group acting on the hypothesis space $\mathcal{H}$. The action is *observable-preserving* if for every $g \in G$ and $h \in \mathcal{H}$,

$$\mathsf{obs}(g \cdot h) \;=\; \mathsf{obs}(h),$$

i.e., the group permutes hypotheses within observe-equivalence classes (fibers). We say that $G$ *acts transitively on observe-fibers* if for every pair $h, h'$ with $\theta(h) = \theta(h')$, there exists $g \in G$ such that $g \cdot h = h'$.

**Definition 6.1** ($G$-Invariant Explanation Map)**.** *An explanation map* $\mathsf{exp}$ *is* $G$-invariant *if for every* $g \in G$ *and* $h \in \mathcal{H}$*,*

$$\mathsf{exp}(g \cdot h) \;=\; \mathsf{exp}(h).$$

When $G$ is finite and $Y$ admits averaging, the canonical $G$-invariant map is the *orbit average*:

$$\overline{\mathsf{exp}}(h) \;=\; \frac{1}{|G|}\sum_{g\in G} \mathsf{exp}(g \cdot h).$$

**Proposition 6.2** ($G$-Invariance Implies Stability)**.** *If $G$ acts transitively on observe-fibers and* $\mathsf{exp}$ *is $G$-invariant, then* $\mathsf{exp}$ *is stable.*

*Proof.* Let $h, h' \in \mathcal{H}$ with $\theta(h) = \theta(h')$. By transitivity, there exists $g \in G$ with $g \cdot h = h'$. By $G$-invariance, $\mathsf{exp}(h') = \mathsf{exp}(g \cdot h) = \mathsf{exp}(h)$. (Formalized as `gInvariant_stable` in `UniversalResolution.lean`.) □

**Proposition 6.3** (Design Space Dichotomy)**.** *If the explanation system has the Rashomon property, then no stable explanation map can simultaneously be faithful and decisive.*

Table 12: Instance-specific resolutions of the Explanation Impossibility. Each row identifies the symmetry group, the concrete resolution, and the property sacrificed.

| Instance | Symmetry group $G$ | Resolution | Sacrificed |
|---|---|---|---|
| Attribution | Collinear-feat. perms. | DASH (ens. SHAP avg.) | Decisive rank. |
| Attention | Weight-space symm. | Avg. attention rollout | Single top token |
| Counterfactual | Dec.-boundary var. | Rashomon-robust recourse | Single CF |
| Concept probe | Repr. rotation | Averaged CAV / subspace | Single direction |
| Causal disc. | DAG rev. in Markov cl. | CPDAG reporting | Fully oriented |
| Model sel. | Model perm. in $\mathcal{R}$ | Ensemble / $\mathcal{R}$ report | Single best |

*Proof.* Immediate from Theorem 3.6: stability, faithfulness, and decisiveness are jointly contradictory under the Rashomon property. (Formalized as `universal_design_space_dichotomy` in `UniversalDesignSpace.lean`.) □

**Corollary 6.4** (Pareto Frontier)**.** *For any explanation system with the Rashomon property, achievable explanation maps fall into exactly two families:*

- ***Family A*** *(faithful + decisive, unstable): individual-model explanations that faithfully report each hypothesis but vary across equivalent hypotheses.*
- ***Family B*** *(stable + faithful in expectation, indecisive): $G$-invariant aggregations that are constant on observe-fibers but produce ties for queries where the fiber disagrees.*

*No explanation map lies outside these two families.*

### 6.2 Instance-Specific Resolutions

Table 12 summarizes how the $G$-invariant resolution instantiates across the domains of Section 5.

**Is reporting ties a genuine resolution?** A common objection: DASH "reports ties" for collinear features, which may seem like giving up rather than explaining. We argue this is the correct behaviour, analogous to a Bayesian posterior that is wide (uncertain) being more honest than a point estimate that is precise but wrong. The tie *is* the explanation: features $j$ and $k$ are interchangeable under the model's learned structure, and any ranking between them is an artefact of training noise. Reporting this interchangeability is more informative than reporting a ranking that reverses with probability $\frac{1}{2}$ across seeds. The loss of within-group decisiveness is the price of faithfulness and stability—the trilemma guarantees no method avoids it.

### 6.3 DASH as the Prototype Resolution

DASH (**D**iversified **A**ggregation for **S**table **H**ypotheses) is the $G$-invariant resolution for the feature attribution instance (Section 5.1). It averages SHAP values over an ensemble of models retrained from independent random seeds, achieving $\mathbb{E}[\phi_j] = \mathbb{E}[\phi_k]$ for any pair of symmetric features $j, k$ (*consensus equity*). This orbit average is stable by Proposition 6.2 and faithful in expectation: the expected explanation matches the expected observation across the Rashomon set. The cost is indecisiveness—DASH reports ties for feature pairs whose rankings disagree across the ensemble. The companion paper proves that DASH is Pareto-optimal: no other stable attribution method can achieve higher expected faithfulness.

**Connection to estimation-theoretic bounds.** Faithfulness can be viewed as an unbiasedness condition: $\mathbb{E}[E(\theta)] = \text{explain}(\theta)$. Under this reading, the attribution impossibility is a Cramér–Rao-type result: no unbiased estimator achieves zero variance when the Fisher information (determined by the Rashomon set geometry) is finite. The design space theorem refines this: it characterises the full Pareto frontier of (bias, variance, completeness) trade-offs, not just the variance lower bound.

**Computational cost of Dash.** Dash requires training $M$ models instead of one. For gradient-boosted trees (XGBoost/LightGBM), training time scales linearly: $M = 25$ models on $N = 10{,}000$ rows takes $\sim$5 minutes on a single CPU core. SHAP computation is per-model and parallelisable. The total cost is $O(M)$ times single-model cost, with no increase in per-prediction inference cost (the deployed model is unchanged; only explanation aggregation changes).

### 6.4 Meta-Theorem: Orbit Averaging

**Theorem 6.5** (Orbit-Averaged Resolution)**.** *Let $\mathcal{S}$ be an explanation system whose Rashomon property arises from a finite group $G$ acting on $\mathcal{H}$ and acting transitively on observe-fibers. Then the orbit-averaged map $\overline{\mathsf{exp}}(h) = \frac{1}{|G|}\sum_{g\in G} \mathsf{exp}(g \cdot h)$ is:*

*(i) well-defined (independent of orbit representative),*

*(ii) stable (Proposition 6.2),*

*(iii) faithful in expectation: $\mathbb{E}_{g\sim\mathrm{Unif}(G)}[\mathsf{exp}(g \cdot h)] = \overline{\mathsf{exp}}(h)$, and*

*(iv) Pareto-optimal among stable maps: no stable map achieves strictly higher pointwise faithfulness on every fiber.*

The orbit-averaged resolution is constructive: for any specific instance, the group $G$ and its action are determined by the domain structure, and the averaging can be computed (or approximated by Monte Carlo sampling over the Rashomon set). Theorem 6.5 is the abstract counterpart of the Dash optimality result and is formalized in `UniversalResolution.lean`. The resolution strategy—restricting to $G$-invariant explanation maps—is structurally analogous to the invariant decision procedures justified by the Hunt–Stein theorem [Lehmann and Romano, 2005]: just as the Hunt–Stein theorem guarantees that invariant tests are minimax optimal when a group acts on the parameter space, the $G$-invariant aggregation is Pareto-optimal among stable explanation methods.

**Biased alternatives and James–Stein phenomena.** The design space theorem characterises unbiased linear aggregators. Can biased methods (e.g., ridge-regularised Shapley values, $L_1$ shrinkage) circumvent the impossibility? Shrinkage reduces variance (improving stability) but introduces systematic bias toward a prior (reducing faithfulness). The trilemma's qualitative impossibility holds regardless: for any method under Rashomon, at least one of faithfulness, stability, or decisiveness must be sacrificed. Shrinkage shifts *which* property is sacrificed—from stability to faithfulness—without escaping the trilemma. Quantifying the bias–variance–decisiveness Pareto frontier for shrinkage estimators is future work.

## 7 Ubiquity

Is the Explanation Impossibility a pathological corner case, or a generic phenomenon? In this section we show that it is the latter: the Rashomon property—and hence the impossibility—holds whenever the model has more parameters than observational constraints, a condition satisfied by virtually all modern machine learning systems.

## 7.1 Generic Underspecification

**Proposition 7.1** (Dimensional Underspecification)**.** *Let $\mathcal{H}$ be a smooth hypothesis space of dimension* $\dim(\mathcal{H})$ *and let* $\mathsf{obs} : \mathcal{H} \to Y$ *be a smooth observation map into a space of dimension* $\dim(Y)$*. If* $\dim(\mathcal{H}) > \dim(Y)$*, then for generic* $y \in Y$*, the fiber* $\mathsf{obs}^{-1}(y)$ *has positive dimension* $\geq \dim(\mathcal{H}) - \dim(Y)$*.*

The Lean formalization captures the combinatorial skeleton of this argument: `generic_underspecification` proves that whenever the parameter dimension exceeds the observable dimension ($n > m$), the fiber dimension is positive ($n - m > 0$). The connection to the preimage theorem in differential topology—which guarantees that generic fibers are smooth submanifolds of the predicted dimension—is an informal mathematical argument, not a Lean-verified claim. Whenever the fiber has positive dimension, the explanation map $\mathsf{obs}$ is generically non-constant on fibers—different hypotheses in the same fiber honestly report different explanations—and the Rashomon property holds. (The bridge from fiber non-degeneracy to the impossibility is `fiber_nondegeneracy_implies_impossibility`.)

**Rashomon from overparameterisation.** For model classes where $\dim(\Theta) > \dim(\mathcal{Y})$ (more parameters than observational constraints), the Rashomon property holds generically. The fibre $\{\theta : \text{observe}(\theta) = y\}$ has positive dimension for generic $y$, and under mild regularity (continuous explain map, non-degenerate Jacobian), distinct fibral points yield incompatible explanations with probability 1 under any absolutely continuous prior on $\Theta$ [D'Amour et al., 2022]. This covers overparameterised neural networks, deep ensembles, and GBDTs with more trees than effective degrees of freedom. Deriving tighter, architecture-specific Rashomon set bounds remains open.

## 7.2 Neural Network Dimensional Argument

Modern neural networks are massively overparameterized. A ResNet-50 has $P \approx 25{,}600{,}000$ parameters and, for ImageNet classification, $k = 1{,}000$ output classes. By Proposition 7.1, the fiber of any fixed prediction has dimension at least $P - k = 24{,}999{,}000$. On such a high-dimensional manifold, any continuous explanation map (attention weights, gradient saliency, internal activations) is generically non-constant: there exist models $h, h'$ on the same fiber with $\mathsf{obs}(h)_j > \mathsf{obs}(h)_k$ and $\mathsf{obs}(h')_j < \mathsf{obs}(h')_k$ for any pair of input features $j, k$ with equal marginal distributions.

**Corollary 7.2** (Coin-Flip Instability)**.** *For any pair of symmetric features* $j, k$ *in an overparameterized network trained to zero loss, the probability (over retraining seeds) that a faithful explanation ranks* $j$ *above* $k$ *is exactly* $1/2$*.*

The 50% flip rate applies to the adversarial case (maximally collinear pair, $\rho \to 1$). For moderate collinearity ($\rho \approx 0.7$), empirical flip rates are typically 15–30%, still high enough to undermine single-model explanations but less extreme than the worst case.
This is the neural-network instantiation of the symmetric Bayes result formalized in `SymmetricBayes.lean`: the symmetry of the fiber under feature permutation forces uniform marginals on the ranking.

## 7.3 Prevalence

The dimensional condition $\dim(\mathcal{H}) > \dim(Y)$ is the norm, not the exception. Large language models (billions of parameters, vocabulary-sized output), diffusion models, graph neural networks, random forests with hundreds of trees, and Lasso regression with more features than observations all satisfy it. Even linear regression with collinear features produces a positive-dimensional fiber (the collinear subspace). D'Amour et al. [2022] document this across NLP, computer vision, and medical imaging, finding that underspecification—and hence explanation instability—is ubiquitous in deployed systems. The only escape is a model class so constrained that $\dim(\mathcal{H}) \leq \dim(Y)$, which excludes essentially all modern architectures of practical interest.

### 7.4 Analogy: Arrow's Impossibility Theorem

The structure of our result parallels Arrow's impossibility theorem in social choice theory [Arrow, 1951]. Arrow showed that no voting rule satisfying unanimity, independence of irrelevant alternatives, and non-dictatorship exists for three or more candidates—a universal impossibility that unified disparate paradoxes in voting theory. Nipkow [2009] later formalized Arrow's theorem in Isabelle/HOL. Our Explanation Impossibility plays an analogous role for explainability: it unifies nine domain-specific impossibilities under one structural cause (the Rashomon property) and one four-line proof, and the resolution (orbit averaging) is the XAI counterpart of accepting partial orderings in social choice. The framework does not currently express the Gibbard–Satterthwaite theorem, which concerns strategy-proofness (incentive compatibility)—a property involving strategic agents, not truthful explanation. Extending the framework to capture strategic considerations would require adding an agent model, which is outside the current scope.

**Comparison with topological impossibility.** Chichilnisky's theorem Chichilnisky [1980] shows that no continuous, anonymous, unanimous aggregation rule exists for preferences on a sphere. Our trilemma differs in three respects: (i) it requires no topological structure on the explanation space, (ii) the impossibility arises from the Rashomon property rather than from the topology of the preference space, and (iii) our resolution (orbit averaging) is constructive while Chichilnisky's theorem has no finite-dimensional escape. The trilemma applies to discrete, finite systems where Chichilnisky's theorem is vacuous.

**Comparison with Fourier-theoretic generalisations.** Kalai Kalai [2002] generalises Arrow's theorem using Fourier analysis on the Boolean cube, showing that aggregation functions close to dictatorships must be close on most inputs. Our trilemma concerns explanation maps (not aggregation rules), requires the Rashomon property (not preference diversity), and applies to non-social-choice domains. The formal relationship between Kalai's Fourier-theoretic approach and our representation-theoretic resolution ($G$-invariant projection) is an open question.

(The full impossibility is formalized as `ubiquity_impossibility` in `Ubiquity.lean`.)

## 8 Lean Formalization

The Explanation Impossibility, all eight derived instances, and all nine ML instances are mechanically verified in Lean 4 [de Moura and Ullrich, 2021] using Mathlib [The mathlib Community, 2020]. The eight derived instances are particularly notable: each derives the Rashomon property constructively with zero domain-specific axioms, confirming that the impossibility in each science follows from first principles alone. This section provides an expanded account of the formalization, including the axiom stratification, proof status transparency, and the complete file inventory.

### 8.1 Summary Statistics

Table 13 summarizes the formalization.

Of the 530 theorems, the majority are short (1–10 tactic steps), including one-line applications of `explanation_impossibility` to domain instances, tightness witnesses, and definitional consequences. Approximately 60 are moderate (11–30 steps) and 15 are substantive (31+ steps, including the GBDT ratio divergence, Spearman bounds, Pareto-optimality, and the infinite enrichment stack).

Table 13: Summary of the Lean 4 formalization.

| Component | Count | Notes |
|---|---:|---:|
| Files | 100 | All in `UniversalImpossibility/` |
| Theorems & lemmas | 530 | Verified by `grep -c` |
| Axioms | 2 | Bundled GBDT infrastructure; core theorem uses 0 |
| `sorrys` | 0 | Zero admitted goals |
| Core impossibility | 1 theorem | Zero custom axiom dependencies |
| 8 derived instances | 0 domain-specific axioms each | Biology, physics $\times 2$, linguistics, ... |
| GBDT instance | 4 axioms | Surjectivity, first-mover, split structure |
| DASH resolution | 6 axioms | + cross-group symmetry |
| Causal instance | 3 axioms | CPDAG Markov equivalence |
| Neural net instance | 2 axioms | Overparameterization + symmetry |
| Fairness instance | 2 axioms | Unequal base rates + equal accuracy |

### 8.2 Axiom Stratification

A key concern in any formalization using axioms is whether the axioms are logically consistent and whether they smuggle in the conclusion. We address this by providing complete transparency on the axiom stratification. Table 14 reports which theorems depend on which axioms, verified by `#print axioms` in Lean 4.

**Zero-axiom scope.** The entire theoretical framework—impossibility theorem, bilemma, tightness classification, capacity theorem (Pythagorean decomposition, best approximation, over-explanation penalty), Pareto-optimal resolution, and all domain instances including Arrow's theorem, Peres–Mermin contextuality, the Langlands boundary for GL($n$, $\mathbb{F}_p$), multi-analyst aggregation, 3D Navier–Stokes conditional tightness, the DPRM trilemma, and all 9 ML explanation types—requires zero axioms. The only axioms in the formalization (4 total across 200 files in three repositories) encode empirical properties of gradient-boosted decision trees for the ML-specific quantitative bounds. The theoretical framework is a pure theorem of logic. Across three repositories the totals are: 530 + 358 + 482 = 1,370 theorems, 4 axioms, 200 files, 0 `sorry`. Key new files from this formalization campaign: `LanglandsCorrespondence.lean` (7 theorems: trace conjugation-invariance, the $n$=1/$n$≥2 boundary, scalar embedding functoriality) and `MultiAnalystInstance.lean` (4 theorems: multi-analyst Rashomon, trilemma, bilemma, maximal incompatibility).

**Reynolds naturality predicts Langlands functoriality.** The Reynolds naturality theorem (`reynolds_naturality` in `UncertaintyFromSymmetry.lean`) proves that equivariant maps commute with Reynolds projections. For GL($n$), the Reynolds projection is the trace (conjugation-averaging). Reynolds naturality therefore implies that the trace commutes with group homomorphisms—which IS Langlands functoriality for finite fields. The impossibility framework predicts functoriality as a structural consequence of the bilemma's resolution: collapsed tightness for $n \geq 2$ forces the trace as the unique Pareto-optimal resolution; Reynolds naturality forces the trace to be functorial. The Langlands programme classifies which characters arise from automorphic representations; the impossibility framework provides a structural explanation for why characters are the natural invariant.

**The automorphicity gap.** The companion repository's `HomExplanationSystem.lean` identifies the precise content separating the impossibility framework's structural predictions from the Langlands programme's arithmetic classification. When the resolution is required to be a group

Table 14: Axiom stratification for key theorems, verified by `#print axioms`. "Zero axioms" means the theorem depends only on the hypotheses stated in its signature—no global axioms are used.

| Theorem | Axioms | Description |
|---|---|---|
| `explanation_impossibility` | 0 | Core universal impossibility |
| `attribution_impossibility` | 0 | Core attribution impossibility |
| `impossibility_qualitative` | 0 | From dominance + surjectivity |
| `gbdt_impossibility_local` | 4 | surj, fm, nfm (no proportionality) |
| `impossibility` (quantitative) | 5 | + `proportionality_global` |
| `consensus_equity` | 6 | + cross-group symmetric |
| `attribution_impossibility_bundled` | 0 | Fully parametric via `GBDTSetup` |
| `attention_impossibility` | 4 | Config, Map, Observable, system |
| `counterfactual_impossibility` | 4 | Config, Explanation, Observable, system |
| `concept_impossibility` | 4 | Config, Explanation, Observable, system |
| `causal_instance_impossibility` | 4 | Config, Explanation, Observable, system |
| `model_sel_instance_impossibility` | 4 | Config, Explanation, Observable, system |
| `gInvariant_stable` | 0 | $G$-invariance implies stability |
| `ubiquity_impossibility` | 0 | Constructs system from witnesses |
| `linear_system_impossibility` | 0 | Underdetermined $Ax = b$ |
| `genetic_code_impossibility` | 0 | Codon degeneracy |
| `gauge_impossibility` | 0 | Discrete $\mathbb{Z}_2$ gauge theory |
| `syntactic_ambiguity_impossibility` | 0 | PP-attachment ambiguity |
| `phase_impossibility` | 0 | Crystallographic phase problem |
| `view_update_impossibility` | 0 | Database view update |
| `stat_mech_impossibility` | 0 | Microstate degeneracy |
| `causal_impossibility_derived` | 0 | Markov equivalence |
| `langlands_boundary_statement` | 0 | $n$=1 full, $n\geq2$ collapsed |
| `multiAnalyst_impossibility` | 0 | Multi-analyst F+S+D impossible |
| `multiAnalyst_bilemma` | 0 | Multi-analyst F+S impossible |

homomorphism (not merely a stable function), the enrichment Sha becomes strictly stronger than CRT. For $S_3 = \mathrm{GL}(2, \mathbb{F}_2)$: CRT allows 6 character-value combinations, but the group relation $srs = r^2$ forces $\chi(r) = 1$, blocking 4 of them (Lean-verified: `automorphicity_gap`). The blocked combinations are locally valid but violate a global group relation—not every collection of local character data assembles into a global homomorphism. This is the automorphicity condition: the gap between local compatibility (CRT, handled by the enrichment framework) and global coherence (automorphicity, the content of the Langlands correspondence). Three negative results delineate the boundary: (1) enrichment Sha = CRT = Euler multiplicativity (no arithmetic content beyond the Euler product; `enrichment_sha_equals_crt`); (2) collapsed tightness forces the identity resolution (tautological; `forced_resolution_complete`); (3) quantitative bounds depend only on group size, not on prime-specific structure (`EnrichmentQuantitative.lean`). For finite fields, the automorphicity gap is computable; for number fields, it is the central open problem of modern algebraic number theory.

The critical observation is that the core theorem `explanation_impossibility` uses **zero custom axioms**—it is pure constructive logic. The formalization stratifies into two categories:

- **Constructively derived** (8 cross-domain instances): The Rashomon property is proved from concrete witnesses with zero behavioral axioms. The Lean code constructs explicit pairs (e.g., $(1, 1)$ and $(0, 2)$ for the linear system, chain vs. fork DAGs for Markov equivalence) and verifies $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2) \wedge \theta_1 \neq \theta_2$ by `decide`. These are minimal witnesses of domain-specific phenomena, not complete formalizations of the respective theories.
- **Axiomatized** (9 ML instances): The Rashomon property is assumed as a domain axiom (e.g., "there exist neural network weights with the same predictions but different attention

distributions"). These axioms are the mathematical content of the empirical literature being formalized and are independently verifiable from the cited sources.

Table 15: Axiom taxonomy (2 bundled axioms; all behavioral properties are theorems extracted from the bundles).

| Category | Count | Examples |
|---|---|---|
| GBDT infrastructure | 14 | Model, splitCount, proportionality |
| Enrichment stack (physics) | 11 | BH/QG frameworks, Rashomon |

The core impossibility (Theorem 3.6) requires zero axioms from any category. Quantitative bounds draw from ML domain axioms; cross-domain transfer draws from structural axioms.

**Scope of formal verification.** Lean verification guarantees that all theorems follow from the stated axioms with no logical gaps (internal consistency). It does not guarantee that the axioms correctly model the phenomena they are intended to capture (external validity). The axioms are motivated by domain-specific considerations (§8.2), but their adequacy for any particular real-world system is an empirical question. We report axiom counts alongside theorem counts throughout to make the trusted base transparent.

## 8.3 Proof Status Transparency

We distinguish four levels of verification:

(a) **Proved** (zero custom axiom dependencies): The theorem follows from pure logic and stated hypotheses. Examples: `explanation_impossibility`, `attribution_impossibility`, `gInvariant_stable`, and `ubiquity_impossibility`.

(b) **Derived** (from axioms): The theorem follows from a small set of domain-specific axioms. Examples: `gbdt_impossibility_local` (4 axioms), `consensus_equity` (6 axioms). The axioms are independently verifiable (e.g., the surjectivity axiom for GBDT follows from the symmetry of collinear features under the data-generating process).

(c) **Axiomatized** (Rashomon assumed): The original axiomatized instance files remain for backward compatibility, but every domain now has a constructive counterpart. All eight cross-domain scientific instances and all nine ML instances (attention, counterfactual, concept, causal, model selection, saliency, LLM self-explanation, mechanistic interpretability) have constructive Rashomon witnesses derived via `decide` with zero domain-specific axioms—bringing the constructive total to 16 of 17 `ExplanationSystem` instances. (The 17th, `AttributionInstance`, uses the independent `attribution_impossibility` with its own zero-axiom proof.)

(d) **Argued** (supplement proof only): The result has a rigorous proof sketch in the supplement or companion paper but has not been mechanized in Lean 4. (Note: as of the current version, the Pareto optimality of DASH is now Lean-verified in `ParetoOptimality.lean` via `dash_unique_pareto_optimal`.)

(e) **Empirical**: The result is supported by computational experiments but not formally proved. Examples: the specific flip rates (19.9% attention under full retraining, 23.5% counterfactual, etc.) reported in Section 5.

## 8.4 Formerly Axiomatized, Now Derived

Several results that were initially axiomatized have since been derived from more primitive axioms, strengthening the formalization:

- `spearman_classical_bound` → `spearman_instability_bound` in `SpearmanDef.lean`: derived from split-count structure.
- `le_cam_lower_bound` in `QueryComplexity.lean`: provable by `not_lt.mp` (a tautology in any linear order).
- `consensus_variance_bound` in `Defs.lean`: from `attribution_variance_nonneg` + `Nat.cast_nonneg`.
- `attribution_sum_symmetric` in `SymmetryDerive.lean`: from proportionality + split-count + cross-group + balance.
- `attribution_variance` and `attribution_variance_nonneg`: from Mathlib's `ProbabilityTheory.variance` and `variance_nonneg`.

## 8.5 Lean Proof of the Core Theorem

The mechanized proof of the universal impossibility mirrors the paper proof exactly. The complete Lean 4 statement and proof is given in Appendix B.

## 8.6 Why This Result Is Non-Trivial

A natural objection is that the impossibility is tautological: "of course you cannot explain an underspecified system perfectly." Four lines of evidence refute this characterization.

**1. Tightness: the triple is the exact obstruction.** Each *pair* of {faithful, stable, decisive} is simultaneously achievable—three Lean-verified witnesses establish this (Proposition 3.10). A faithful-and-decisive map simply reports the native explanation; a faithful-and-stable map outputs a neutral constant; a stable-and-decisive map outputs a maximally committal constant. The impossibility is *specifically* the conjunction of all three. A tautology would make even pairs unachievable; here every pair is constructively realized.

**2. Necessity: the Rashomon property is the exact boundary.** The converse is proved in Lean (`fully_specified_possibility` in `Necessity.lean`): if the system has *no* Rashomon property—every observation uniquely determines its configuration—then all three properties are simultaneously achievable. The Rashomon property is therefore both sufficient *and* necessary for the impossibility. This is not a vague "things are hard when the problem is hard" statement; it is a precise logical equivalence: the impossibility holds *if and only if* the Rashomon property holds.

**3. Resolution optimality: the $G$-invariant projection is Pareto-optimal.** The resolution is not arbitrary. The orbit-averaging map is Pareto-optimal among all stable explanation maps: no stable map can achieve strictly higher pointwise faithfulness on every observe-fiber. This means the trade-off identified by the theorem is *tight*—the practitioner cannot do better by being clever.

Table 16: **Structural differences across the eight derived instances.** The eight cross-domain instances are not notational variants of a single example: they differ in symmetry group, group type, and resolution strategy. Their independent convergence on orbit averaging is explained by the $G$-invariant resolution framework.

| Domain | Group $G$ | Objects permuted | Type | Resolution | Form |
|---|---|---|---|---|---|
| Mathematics | $\mathbb{R}^{n-r}$ | Solution offsets | Cont., abelian | Pseudoinverse | Projection |
| Biology | $S_k$ ($k \le 6$) | Synonymous codons | Finite | Codon tables | Averaging |
| Physics | $\mathbb{Z}_2^{\lvert V\rvert-1}$ | Edge labels | Finite, abelian | Wilson loops | Invariant fn. |
| Stat. mech. | $S_\Omega$ | Microstates | Finite | Microcanon. | Averaging |
| Linguistics | $S_k$ | Parse trees | Finite | Parse forests | Enumeration |
| Crystallogr. | $\mathbb{Z}_2 \times \mathbb{Z}_n \times \mathbb{Z}_2$ | Signal phases | Finite | Patterson map | Autocorrel. |
| Comp. sci. | $S_k$ | Hidden columns | Finite | Compl. views | Projection |
| Statistics | MEC perms. | DAG orientations | Finite | CPDAG | Class report |

**4. Independent convergence: eight domains instantiate the same resolution.** If the impossibility were trivial, one would not expect eight independent scientific communities—spanning physics, biology, statistics, linguistics, crystallography, computer science, statistical mechanics, and mathematics—to converge on structurally the same resolution strategy (orbit averaging) over more than a century of independent work (Table 16). Our framework explains *why* they converged—they were independently solving the same impossibility—and proves the resolution they found is optimal.

The framework also explains the variation across domains: the specific form of the resolution differs because the symmetry group and representation differ, even when the abstract group is the same. Four instances use the symmetric group $S_k$, but they act on different mathematical objects—synonymous codons, microstates, parse trees, database columns—and yield different resolution forms: frequency averaging for biology and statistical mechanics, equivalence-class enumeration for linguistics, and subquery projection for computer science. The abstract mathematical operation—projection onto the $G$-invariant subspace—is identical; the computational realization is domain-specific.

# 9 Discussion: What Explainability Can and Cannot Do

## 9.1 The Ceiling of Explainability

The Explanation Impossibility (Theorem 3.6) establishes a *ceiling*, not a criticism of any particular method. No axiomatic redesign of SHAP, LIME, attention maps, or counterfactual generators can simultaneously deliver faithfulness, stability, and decisiveness whenever the underlying inference problem is underspecified. The ceiling is structural: it follows from the Rashomon property alone, independent of the explanation formalism. An analogy to physics is apt: the speed of light constrains spacecraft engineering but does not prevent useful space travel. Likewise, the impossibility constrains what practitioners may promise about explanations but does not prevent useful explainability—it redirects effort toward the achievable frontier characterized by the $G$-invariant resolution (Section 6).

## 9.2 Cross-Domain Transfer: From Gauge Theory to ML

The gauge theory resolution (restricting to observables invariant under gauge transformations) and the ML resolution (Dash, averaging SHAP values over an ensemble) share identical mathematical structure: both are the $G$-invariant projection for their respective symmetry groups (the Reynolds operator—§6.4). The structural differences—continuous vs. discrete symmetry, local vs. global action, dynamical vs. static—mean this identity is at the level of the group action formalism, not a claim that the physical mechanisms coincide.

Table 17: Gauge–attribution correspondence: shared and divergent structure.

| Property | Gauge theory | ML attribution |
|---|---|---|
| **Shared** | | |
| Group action on descriptions | ✓ | ✓ |
| Orbit averaging (Reynolds operator) | ✓ | ✓ |
| Pareto-optimality of $G$-projection | ✓ | ✓ |
| **Divergent** | | |
| Symmetry type | Local, continuous | Global, discrete |
| Dimension | Infinite | Finite |
| Dynamics | Equations of motion | Static computation |

**Beyond ML.** As evidence that the impossibility extends beyond ML, we note that spacetime geometry under Ostrowski's classification of completions of $\mathbb{Q}$ is a maximally incompatible system: the two completions yield exactly two geometries (smooth and ultrametric), triggering the bilemma (Theorem 3.13). The adelic resolution — formulating physics over all completions simultaneously — is the enrichment escape, and it is the *unique* faithful + stable value on the Rashomon fiber. The full development, including the Ostrowski–Design Space correspondence and physical scoping (conditional on the Freund–Witten adelic product formula), is in a companion paper Caraker et al. [2026b].

**Value alignment.** The framework applies to AI value alignment. When multiple RLHF training runs produce models with identical behavioral performance but different implicit value rankings (the value Rashomon property), no value explanation is simultaneously faithful and stable (the value bilemma). The enrichment mechanism — adding a 'balanced' option that doesn't commit to either value priority — corresponds to alignment frameworks that explicitly balance competing values rather than ranking them. This formalizes a structural constraint on alignment: under value multiplicity, precise value specification is impossible without sacrificing either accuracy (faithfulness) or consistency (stability). We formalize this as a Lean-verified `ExplanationSystem` instance (`ValueAlignment.lean`, 0 `sorry`). Whether RLHF models empirically exhibit value Rashomon is an open question; the Lean formalization proves the structural consequence *if* the property holds, but does not establish that it does. The full complexity of AI alignment (strategic agents, distributional shift, ontological uncertainty) exceeds the framework's scope.

## 9.3 Toward a Group-Theoretic Classification of Explainability

The resolution framework reveals a deeper structure: the difficulty of explanation is not merely binary (possible vs. impossible) but is *parameterized by the symmetry group* of the Rashomon set. Different group-theoretic structures produce qualitatively different impossibility regimes and qualitatively different resolutions.

**The observation.** For an explanation system with symmetry group $G$ acting on the Rashomon fibers (the set of configurations sharing the same observable), the space of stable explanation maps is isomorphic to the space of $G$-invariant functions on the configuration space. The $G$-invariant projection (orbit averaging) is the *Reynolds operator*—the projection onto the trivial representation of $G$:

$$R(f)(x) \;=\; \frac{1}{|G|}\sum_{g\in G} f(g\cdot x) \quad (\text{finite } G), \qquad R(f)(x) \;=\; \int_G f(g\cdot x)\,d\mu(g) \quad (\text{compact } G).$$

This is not a new mathematical object—it is the standard tool of invariant theory [Sturmfels, 2008]—but its identification as the universal resolution of the Explanation Impossibility, across all domains simultaneously, appears to be new.

**A hierarchy of difficulty.** The group structure determines the character of the impossibility and the available resolutions:

1. **Finite abelian** $G$ (gauge theory $\mathbb{Z}_2^{|V|-1}$, statistical mechanics $S_2$, biology $S_k$ for $k \leq 2$). The orbit average is canonical and unique. All characters are one-dimensional, so the invariant projection has no ambiguity. The faithfulness bound is $1/|G|$. These are the "easy" cases: the impossibility is real but the resolution is completely determined.

2. **Finite non-abelian** $G$ (biology $S_k$ for $k \geq 3$, causal discovery with $\geq 3$ reversible edges, molecular point groups in chemistry). Multiple inequivalent invariant projections may exist, corresponding to different choices of which non-trivial irreducible representations to retain. The resolution choice itself becomes ambiguous—a *second-order Rashomon problem*. We observe this empirically: different causal discovery scoring functions (BIC, AIC, MDL) can produce different CPDAGs on the same data, reflecting different implicit choices among equivalent invariant representations.

3. **Finite non-trivial** $G$ (crystallography $\mathbb{Z}_2 \times \mathbb{Z}_n \times \mathbb{Z}_2$). The Haar measure on $G$ is finite, so orbit averaging converges. The Peter-Weyl theorem decomposes $L^2(G)$ into irreducible representations, and the invariant projection retains precisely the zeroth-frequency (DC) component. This is why the Patterson map—the autocorrelation, i.e., the zero-frequency Fourier content—is the natural resolution in crystallography.

4. **Non-compact continuous** $G$ (linear algebra $\mathbb{R}^d$, deep learning weight space). No finite Haar measure exists on non-compact orbits. Orbit averaging diverges without regularization. The pseudoinverse (minimum-norm solution) is Tikhonov regularization with $\lambda \to 0$—a *regularized* orbit average. Different regularizers (L1, L2, elastic net) produce different "resolutions," and the choice of regularizer introduces an external degree of freedom not present in the compact case. This is the "fundamentally hard" regime: the impossibility is severe and the resolution is non-canonical.

This ordering is a partial order, not a total order: compact non-abelian groups (e.g., SO(3)) exhibit features of both items 2 and 3.

**Predictions.** This hierarchy generates testable predictions not currently verified in the paper:

- Domains with abelian symmetry should exhibit a unique canonical resolution with no practitioner disagreement on what the "right" invariant quantity is.
- Domains with non-abelian symmetry should exhibit *resolution multiplicity*—different "reasonable" resolutions that disagree—and this disagreement should itself be measurable.
- Domains with non-compact symmetry should require regularization, and different regularizers should produce different answers whose spread scales with the non-compactness of the group.
- The faithfulness bound $1/|G|$ (finite) or $\exp(-\dim G)$ (Lie groups) should predict the empirical instability across domains.

**What is missing.** The current eight instances span finite abelian, (barely) finite non-abelian, compact continuous, and non-compact continuous groups. Not yet represented are: genuinely non-abelian finite groups (e.g., the tetrahedral point group $T_d$ in molecular spectroscopy, order 24); non-compact Lie groups beyond translations (e.g., the Lorentz group $\mathrm{SO}(3,1)$ in relativistic physics); infinite discrete groups (e.g., lattice symmetries $\mathbb{Z}^n$); and groupoids—where the symmetry structure varies fiber-by-fiber, as happens in ML when different regions of feature space have different Rashomon set sizes. The groupoid generalization would connect this work to the theory of stacks in algebraic geometry and to fiber bundles in differential geometry. We leave these extensions to future work.

**Representation-theoretic classification.** The classification of non-dominated explanation profiles is complete for $G = \mathbb{Z}/2\mathbb{Z}$ (the simplest non-trivial case): exactly two profiles exist (trivial representation giving stability, sign representation giving decisiveness). For general finite $G$, we conjecture that the number of non-dominated profiles equals the number of irreducible representations of $G$. The $\mathbb{Z}/2\mathbb{Z}$ case serves as proof of concept; the general classification is an open problem.

**Categorical enrichment (conjecture).** We conjecture that hypothesis-space enrichment corresponds to the free adjunction of an initial object to the incompatibility poset, viewed as a thin category. Specifically, if $\mathbf{C}$ is the poset category of $H$ under the compatibility relation, then the enriched space $H' = H \cup \{h_\perp\}$ corresponds to $\mathbf{C}_\perp$. The universal property ensures uniqueness (the companion paper [Caraker et al., 2026b]). Making this precise requires defining the relevant 2-categorical context; we leave the formalisation to future work.

**Character theory and the information trade-off.** Character theory quantifies the trade-off precisely. For the symmetric group $S_k$ acting on $\mathbb{R}^k$ by permutation of coordinates, the representation decomposes into the trivial representation (dimension 1) and the standard representation (dimension $k-1$). The $G$-invariant subspace is exactly the trivial representation: $\dim(V^G) = 1$, while $\dim(V) = k$. The resolution therefore preserves exactly $1/k$ of the information—the orbit average retains the mean and discards the $k-1$ "directional" degrees of freedom. This connects the dose-response (entropy scaling with degeneracy $k$) directly to representation-theoretic structure: the ratio $\dim(V^G)/\dim(V) = 1/k$ determines the fraction of explanation content that survives projection onto the invariant subspace. We call $C = \dim(V^G)$ the *explanation capacity*: the maximum number of independent stable facts any method can extract from a system with symmetry group $G$. The complementary quantity $\eta = 1 - C/\dim(V)$ is the *explanation loss rate*: the fraction of explanatory content irretrievably lost by any stable method. The orbit average (Reynolds operator) achieves the capacity—it is the *capacity-achieving projection* (Theorem 9.1, proved Pareto-optimal). As Shannon's channel capacity bounds the rate of reliable communication, the explanation capacity bounds the rate of reliable explanation.

The ratio $\dim(V^G)/\dim(V)$ is a standard quantity in representation theory, computable via the character formula (Burnside's lemma). The novelty is not the formula but its *application*: we show empirically that the explanation capacity predicts instability rates across seven domains with $R^2 = 0.957$ (well-characterized groups), using a quantity from abstract algebra as a parameter-free predictor of empirical flip rates. We call this the *Explanation Capacity Theorem*.

**Uncertainty from symmetry: the representation-theoretic connection to quantum mechanics.** The capacity theorem and quantum uncertainty both arise from the same representation-theoretic decomposition. We state and prove the connection.

**Theorem 9.1** (Uncertainty from symmetry)**.** *Let $G$ be a finite group acting on a real inner product space $(V, \langle\cdot,\cdot\rangle)$ of dimension $n$ by orthogonal representation $\rho : G \to O(V)$. Let $R = |G|^{-1}\sum_{g\in G}\rho(g)$ be the Reynolds operator, and $k = \dim(V^G)$. Then:*

*(i) $R$ is the orthogonal projection onto $V^G$.*

*(ii) For any $v \in V$, $\|v - Rv\|^2 + \|Rv\|^2 = \|v\|^2$* (explanatory information loss identity).

*(iii) For $v$ drawn uniformly from the unit sphere $S^{n-1}$, $\mathbb{E}[\|Rv\|^2] = k/n$ and $\mathbb{E}[\|v - Rv\|^2] = 1 - k/n = \eta$.*

*(iv) The minimum information loss of any $G$-invariant approximation is $\inf_{w \in V^G} \|v - w\|^2 = \|v - Rv\|^2$, and $\eta$ is its expected value.*

*Proof.* (i) $R$ is a group average of orthogonal maps, hence self-adjoint ($\langle Rv, w\rangle = \langle v, Rw\rangle$) and idempotent ($R^2 = R$), so $R$ is an orthogonal projection. Its image is $V^G$: if $Rv = v$ then $\rho(g)v = v$ for all $g$ (by averaging), and conversely every $G$-fixed vector is fixed by $R$. (ii) Since $Rv \in V^G$ and $(v - Rv) \in (V^G)^\perp$ (because $R$ is an orthogonal projection), Pythagoras applies. (iii) Let $\{e_1, \ldots, e_k\}$ be an orthonormal basis for $V^G$. Then $\|Rv\|^2 = \sum_{i=1}^k \langle v, e_i\rangle^2$. For $v$ uniform on $S^{n-1}$, $\mathbb{E}[\langle v, e_i\rangle^2] = 1/n$ for each $i$ (by rotational symmetry), so $\mathbb{E}[\|Rv\|^2] = k/n$. Part (ii) gives $\mathbb{E}[\|v - Rv\|^2] = 1 - k/n$. (iv) $R$ is an orthogonal projection, so $Rv$ is the closest point in $V^G$ to $v$. □

The theorem applies in two settings with identical algebraic content:

*Explanation setting.* Let $v \in V$ be an explanation vector (e.g., a vector of SHAP values across features) and $G$ the symmetry group of equivalent configurations. The orbit average $Rv$ is the stable explanation; the discarded component $(I - R)v$ lies in non-trivial irreducible representations of $G$ and is unstable by construction. The expected fraction of explanation content lost by stabilisation is $\eta = 1 - \dim(V^G)/\dim(V)$. This is the capacity theorem.

*Quantum setting.* Let $V = \mathbb{C}^n$ with $G$ acting by unitary representation $U$, and let $T(\rho) = |G|^{-1} \sum_g U(g)\rho U(g)^\dagger$ be the quantum twirl (the density-matrix analogue of the Reynolds operator). For a pure state $|\psi\rangle$ drawn uniformly (Haar measure), the expected probability of finding the system in the $G$-invariant sector is $\mathbb{E}[\langle\psi|P_{V^G}|\psi\rangle] = \dim(V^G)/\dim(V)$. The complementary fraction $\eta$ bounds the information inaccessible to $G$-covariant measurements: the specific configuration within each non-trivial irreducible component cannot be distinguished by any measurement commuting with $G$.

*The unification.* The Reynolds operator (explanation) and the quantum twirl (quantum mechanics) are instances of the same algebraic structure—projection onto the trivial representation of $G$—applied to different spaces ($V$ for explanations, $L(H)$ for density matrices). The formula $\eta = 1 - \dim(V^G)/\dim(V)$ governs both the explanation instability rate and the quantum information loss. Stable explanatory facts correspond to trivial-representation components, just as quantum observables commuting with the symmetry correspond to $G$-invariant operators.

The framework produces two distinct types of bound, playing different roles:

*The capacity theorem as minimum cost theorem.* The cost of stability is exactly $\eta$: the Reynolds projection loses precisely $1 - k/n$ of the expected squared norm. This is not a tradeoff—it is a fixed price determined by the group structure, analogous to the zero-point energy of a quantum harmonic oscillator (a minimum cost of confinement, not a relation between conjugate variables). The capacity theorem is tight: equality holds in Theorem 9.1(iii) by construction.

*The explanation uncertainty bound.* The approximate bilemma ($\mathrm{unfaith}(\theta_1) + \mathrm{unfaith}(\theta_2) \geq \Delta - \delta$) is a Donoho–Stark-type uncertainty principle [Donoho and Stark, 1989]: reducing unfaithfulness at one Rashomon witness forces increased unfaithfulness at the other, just as a function and its Fourier transform cannot both have small support. Both are lower bounds on complementary quantities arising from constrained approximation—the bilemma from the constraint that stable maps are constant on Rashomon fibers, Donoho–Stark from the constraint that signals cannot concentrate in both time and frequency. The form is also parallel to Heisenberg's relation ($\sigma_A\sigma_B \geq |\langle[A, B]\rangle|/2$), though the mathematical mechanism differs: the bilemma uses the triangle inequality on a metric space, while Heisenberg uses the Cauchy–Schwarz inequality on a Hilbert space with non-commuting operators. The approximate bilemma is tight when the explanation map places $E(\theta_1)$ and $E(\theta_2)$ on the geodesic between $\exp(\theta_1)$ and $\exp(\theta_2)$ with $\|E(\theta_1) - E(\theta_2)\| = \delta$; this occurs when the Rashomon witnesses are symmetric about $V^G$.

Both bounds arise from the same representation-theoretic decomposition: the Pythagorean splitting $V = V^G \oplus (V^G)^\perp$ under the Reynolds operator. The capacity theorem governs the average case (expected loss over random explanations); the approximate bilemma governs the worst case (complementary loss at specific Rashomon pairs). Together, they constitute a complete uncertainty framework for explanation under symmetry.

*Lean formalisation status.* Theorem 9.1 is a standard result in representation theory; its application to the explanation–quantum connection is the contribution. The individual components (Reynolds projection, explanatory information loss identity, approximate bilemma) are Lean-verified in the companion repositories. A unified Lean proof of the full theorem—including the quantum interpretation and the explicit connection between the two settings—is a natural next step.

**The Explanation Stability Theorem.** The explanatory information loss identity yields a complete Shannon-style capacity theory for explanation. We state the four parts; together they establish that the explanation capacity $C = \dim(V^G)$ plays the role of Shannon's channel capacity for the "explanation channel" whose noise is the Rashomon property.

*Part (i): Convergence rate.* Let $v_1, \ldots, v_M$ be i.i.d. explanation vectors with mean $\mu$ and covariance $\Sigma$. The orbit-averaged explanation $\bar{E}_M = M^{-1} \sum_i Rv_i$ satisfies $\mathbb{E}[\|\bar{E}_M - R\mu\|^2] = M^{-1}\operatorname{tr}(R\Sigma R)$. For target MSE $\leq \varepsilon$, it suffices to use $M \geq \operatorname{tr}(R\Sigma R)/\varepsilon$ models. This is the source coding rate for the explanation channel. The NARPS convergence ($M_{95} = 16$ teams for 95% stability) is a special case.

*Part (ii): Capacity bound.* Any $G$-invariant map $E$ satisfies $E(\theta) \in V^G$ for all $\theta$; the dimension of the image is at most $C = \dim(V^G)$. No amount of averaging, ensembling, or methodological refinement expands $V^G$. The capacity is a structural constant (Lean-verified: `stable_in_fixed_subspace`).

*Part (iii): Rate optimality.* Among all linear unbiased estimators of $R\mu$ from $M$ i.i.d. samples, the orbit average minimises $\mathbb{E}[\|\hat{E} - R\mu\|^2]$ (Gauss–Markov). Equivalently, $V^G$ is the sufficient statistic for stable explanatory content, and the Reynolds projection is the Rao–Blackwell improvement of any explanation method.

*Part (iv): Over-explanation penalty.* For any $w \in (V^G)^\perp$ (i.e. $Rw = 0$) and any stable element $u \in V^G$ (i.e. $Ru = u$):

$$\|w\| \leq \|u - w\|.$$

The best stable approximation to any beyond-capacity quantity is the zero vector, with MSE $= \|w\|^2$. This is not noisy information; it is zero information. The penalty follows directly from orthogonality ($u \perp w$) and the Pythagorean theorem (Lean-verified: `beyond_capacity_penalty`).

The four parts mirror Shannon's coding theorem: capacity (Part ii), achievability (Parts i and iii), converse (Part iv), and rate (Part i). The over-explanation penalty is arguably sharper than Shannon's strong converse: it gives the *exact* MSE for beyond-capacity estimation, not just an asymptotic bound.

*Empirical validation.* The convergence rate MSE $\sim M^{-\beta}$ was validated on synthetic data with $\beta = 1.28$ (predicted: 1.00; $R^2 = 0.97$). The over-explanation penalty was validated across 149 datasets: within-group pairwise claims (beyond capacity) flip at rate $0.22 \pm 0.10$, while between-group claims (within capacity) flip at rate $0.15 \pm 0.08$ (cross-dataset Wilcoxon $p = 5.1 \times 10^{-11}$; directional confirmation in 81% of 110 testable datasets, 86% of 84 with Cohen's $d$ computable).

**The deeper claim.** If the group-theoretic parameterization is correct in full generality, then the Explanation Impossibility is not a standalone theorem but the *first theorem in a classification theory*—analogous to how Arrow's impossibility theorem [Arrow, 1951] was the first theorem in social choice theory, not the last. The classification would say: for each symmetry type, here is exactly what explanations can and cannot do, here is the optimal resolution, and here is the residual ambiguity. The universality of the impossibility (Theorem 3.6) would then be the base case of a stratified tower of increasingly refined characterizations, parameterized by the representation theory of the symmetry group.

### 9.4 Regulatory Implications

The EU AI Act [EU Parliament, 2024] requires "meaningful explanations" for high-risk systems, and SR 11-7 [Office of the Comptroller of the Currency, 2011] requires "developmental evidence" documenting how model outputs arise. These two demands pull in opposite directions: meaningful explanations favor decisiveness (a definite ranking of features or causes), while developmental evidence favors stability (reproducibility across equivalent models). Theorem 3.6 shows that no explanation system can simultaneously maximize both for underspecified models. A path to regulatory compliance nonetheless exists: $G$-invariant aggregation (e.g., DASH for attributions, CPDAGs for causal reports) satisfies stability by construction and partial faithfulness—faithfulness in expectation over the Rashomon set—at the cost of reporting ties where the data are genuinely ambiguous. We suggest that regulators adopt this framing: require that explanations be *stable and partially faithful*, and mandate explicit disclosure of tied or unresolvable queries rather than demanding decisive answers that are mathematically impossible to guarantee.

**Liability implications.** If an organisation relies on single-model SHAP explanations for high-stakes decisions and the Rashomon property holds, two independent audits using different random seeds could reach opposite conclusions about which feature drove the decision. Ensemble averaging (DASH with $M \geq 25$) provides a defensible alternative: the explanation is stable, the uncertainty is quantified, and the limitations are documented. This analysis reflects our reading of the regulatory text and should be reviewed by a legal scholar before being relied upon in compliance contexts.

### 9.5 What Explainability *Can* Do

The $G$-invariant resolution (Section 6) is constructive, not nihilistic. It tells the practitioner exactly which desiderata can be jointly satisfied and which must be relaxed. DASH for feature attribution, CPDAGs for causal discovery [Verma and Pearl, 1991], and ensemble concept probes for concept-based explanations [Kim et al., 2018] each sacrifice decisiveness to gain stability—and each is a concrete instance of the abstract resolution. In practice, this means reporting confidence intervals over the Rashomon set [Fisher et al., 2019, Semenova et al., 2022] rather than point estimates: a feature ranking should come with a stability annotation indicating which comparisons are robust to retraining and which are not. The impossibility does not say "explanations are useless"; it says "decisive explanations are unreliable under underspecification, but partially decisive explanations with calibrated uncertainty are both achievable and informative."

### 9.6 What This Paper Tells a Working Scientist

What does this paper tell a working scientist? For an ML practitioner: which SHAP comparisons to trust (SNR $> 2$) and which to discard (SNR $< 0.5$), computed from bootstrap retrains alone. For a causal inference researcher: that ensemble averaging over the Rashomon set reduces orientation flip rates fourfold, following the same mechanism as DASH for attributions. For a biologist: that the character-theoretic capacity $(k-1)/k$ serves as a neutral-evolution baseline from which deviations measure selective pressure.

## 9.7 Scope and Applicability Beyond Machine Learning

The theorem applies to *any* explanation of *any* underspecified system. The four empirical instances in the Nature paper (genomics, mechanistic interpretability, causal inference, neuroimaging) demonstrate the theorem in specific domains; the 14 cross-domain Lean instances (Arrow, Bell, Gödel, gauge theory, statistical mechanics, genetic code, phase problem, QM interpretation, linguistics, value alignment, view-update, linear systems, and two scientific revolutions) prove the algebraic structure. But the scope extends further. The Rashomon property arises wherever multiple valid configurations of a system produce the same observables, which is the default condition in high-dimensional science.

Table 18: Scope of the impossibility theorem beyond the four main instances. Each row identifies a scientific domain, the source of the Rashomon property (what creates multiple valid configurations), and the form the resolution takes.

| **Domain** | **Source of Rashomon** | **What is unstable** | **Resolution form** |
|---|---|---|---|
| Epidemiology | Correlated risk factors (smoking, diet, alcohol, exercise) | Which factor is "the" cause of disease | Report attributable fractions with stability bounds |
| Climate science | Multiple valid GCMs fit the same temperature record | Which feedback drives warming (clouds, albedo, ocean) | Multi-model ensemble (CMIP) |
| Economics | Multiple valid macroeconomic models | Which policy lever explains GDP change | Bayesian model averaging |
| Ecology | Correlated environmental variables | Which variable drives species distribution | Ensemble species distribution models |
| Pharmacology | Correlated molecular targets | Which pathway mediates drug effect | Pathway-level (not target-level) analysis |
| Forensics | Multiple valid statistical models of evidence | Strength of evidence attribution | Likelihood ratio over model space |
| Social science | Researcher degrees of freedom | Which specification produces significance | Multi-analyst studies |
| Psychology | Garden of forking paths | Which analysis yields the effect | Pre-registration + multiverse analysis |

In each domain, the pattern is identical: (1) correlated variables or underdetermined models create the Rashomon property; (2) the specific explanation (which gene, which circuit, which risk factor, which feedback mechanism) depends on the particular model or analysis pipeline; (3) the resolution involves some form of averaging, ensembling, or abstraction to a coarser level where the instability vanishes. The theorem proves that step (3) is not a methodological preference but the unique Pareto-optimal response to a structural constraint.

**Extension to mathematical physics: 3D Navier–Stokes.** The tightness classification extends beyond data science to open problems in mathematical physics. The 3D incompressible Navier–Stokes equations admit three desiderata: existence of solutions (P1), smoothness/regularity of solutions (P2), and uniqueness (P3). Whether all three hold simultaneously is one of the seven Clay Millennium Problems. We apply the framework computationally: for the Taylor–Green vortex ($A = 1$, $T_{\text{final}} = 10$, capturing peak dissipation at $t \approx 9$ per Brachet 1983), 54 pseudospectral DNS runs across five resolutions ($32^3$, $48^3$, $64^3$, $96^3$, $128^3$) with Reynolds numbers $\text{Re} \in [50, 10000]$ establish a complete phase diagram.

*Phase diagram (dealiasing ON):* At low Re, all three properties hold (FULL tightness) at every resolution. Above a critical $\mathrm{Re}_{\mathrm{crit}}$, smoothness (P2) fails while existence (P1) and uniqueness (P3) survive — SMOOTH-BLOCKED tightness. The critical Reynolds number increases with resolution:

$$\mathrm{Re}_{\mathrm{crit}} = 2.91 \times N^{1.56}, \qquad R^2 = 0.94,$$

where $N$ is the grid size. The exponent 1.56 lies between the Kolmogorov prediction 4/3 and 2, consistent with the inertial-range scaling (slopes $-1.8$ to $-1.87$ at $96^3$ and $128^3$, approaching the $k^{-5/3}$ Kolmogorov spectrum).

*Four confirmed claims:*

1. P2 (smoothness) fails first at ALL five resolutions tested (dealiasing ON).
2. $\mathrm{Re}_{\mathrm{crit}}$ increases monotonically with $N$: $707 \to 1414 \to 1414 \to 3162 \to 7071$.
3. Dealiasing protects P1: with standard 2/3-rule dealiasing, P1 never fails. Without it, P1 always fails after P2 (cascade within 5 ms at $128^3$).
4. The inertial range develops at high resolution with near-Kolmogorov slopes.

*The 2D control:* In 2D, where global regularity is proved (Ladyzhenskaya 1969), the framework correctly reports FULL tightness at all Reynolds numbers tested. The 2D/3D comparison is a clean falsification test: the framework predicts FULL where regularity is known and SMOOTH-BLOCKED where it is unknown, and both predictions are confirmed.

*The dealiasing finding:* The standard 2/3-rule dealiasing mask, present in all production DNS codes, artificially decouples smoothness from existence. Without dealiasing, the P2→P1 cascade gap shrinks with resolution (0.4–0.6 s at $32^3$; 0.005–0.01 s at $128^3$; effectively simultaneous at the continuum limit). This is a methodological finding relevant to the computational PDE community: the dealiasing mask is a Reynolds projection for aliasing symmetry, with capacity $(2N/3)^3$ modes.

*What this does not claim:* The numerical experiments classify the *computational* tightness type (which property fails first in DNS at finite resolution). They do not prove or disprove global regularity for the analytical 3D Navier–Stokes equations, which remains an open Millennium Problem. The resolution-convergent trend ($\mathrm{Re}_{\mathrm{crit}} \to \infty$ as $N \to \infty$) is consistent with either eventual global regularity (FULL at the continuum limit) or a finite-Re breakdown that DNS cannot yet resolve.

*Divergence check:* $\max |\nabla \cdot \mathbf{u}| < 10^{-8}$ across all 54 runs, confirming the pseudospectral solver preserves incompressibility.

**The Explanation Capacity Audit.** The explanation capacity theorem was validated at scale across 149 datasets from 53 scientific domains (genomics, neuroscience, particle physics, manufacturing, veterinary medicine, and 48 others). For each dataset, 50 XGBoost models with stochastic regularisation (subsample = 0.8, colsample_bytree = 0.5) were trained with different random seeds, and TreeSHAP importance rankings were computed. Correlation groups were identified via SAGE (hierarchical clustering on |Spearman $\rho$| at threshold $\rho^* = 0.70$), giving the explanation capacity $C = g$ (number of groups). Note: the Lean-verified generalization (`mi_is_exact_boundary` in `MutualInformation.lean`) proves that mutual information $I(X_j; X_k) > 0$—not correlation—is the necessary and sufficient condition for the attribution impossibility. For continuous features, Spearman correlation is an adequate proxy: an MI-vs-correlation comparison across 131 PMLB benchmark datasets confirms that the two grouping methods produce identical groups 77% of the time (ARI = 1.000) and nearly identical groups 74% of the time (ARI > 0.80; mean ARI = 0.84). However, for binary or categorical features, correlation can fail completely—the drug discovery experiment (Section 10.13) demonstrates this: Pearson clustering predicts 0% instability on binary fingerprints while MI-based clustering recovers the observed 23% rate. The 75% exceedance rate below is therefore exact for the continuous-feature audit but a *lower bound* for domains with binary, categorical, or nonlinear feature dependencies.

*Exceedance:* 111 of 149 datasets (75%) exceed their explanation capacity — the feature rankings reported in typical SHAP analyses contain more pairwise claims than the correlation structure can stably support. A TreeSHAP replication on 70 PMLB datasets with $M = 50$ confirms: 84% exceed capacity (conservative relative to gain-based 87% on the same datasets, because TreeSHAP is 34% more stable). Exceedance is robust to threshold: 83% at $\rho^* = 0.50$, 81% at 0.60, 75% at 0.70, 69% at 0.80, 50% at 0.90. The directional prediction (within > between flip rate) is confirmed at all five thresholds (Wilcoxon $p \leq 0.021$ at each), with the strongest signal at $\rho^* = 0.70$ and 0.90. The slight weakening at $\rho^* = 0.80$ (77% directional vs. 81% at 0.70) reflects group fragmentation: 8 datasets lose testability as within-group pairs are split, and the mean number of within-group pairs drops from 81 to 62.

*Directional prediction:* The stability theorem predicts within-group pairwise comparisons (beyond capacity) are less stable than between-group comparisons (within capacity). This prediction is confirmed in 72 of 84 testable datasets (86%; 89 of 110 at the broader testability criterion). A cross-dataset Wilcoxon signed-rank test gives $p = 5.1 \times 10^{-11}$; on 59 real-world datasets only (excluding PMLB benchmarks and synthetic controls), $p = 1.7 \times 10^{-6}$ (47/59 positive). 27 per-dataset confirmations are significant at $p < 0.005$; 22 survive Bonferroni correction ($p < 0.05/149 = 3.4 \times 10^{-4}$). One dataset (Dermatology) shows a significant reversal ($p = 3.6 \times 10^{-4}$; see below). The confirmation-to-reversal ratio is 27:1. A family-level analysis (averaging Friedman variants within families to address non-independence) confirms the result: 59/72 families positive, Wilcoxon $p = 2.5 \times 10^{-8}$; block-bootstrap 95% CI for the mean gap is $[0.036, 0.069]$, excluding zero.

*Effect sizes:* The strongest confirmations span diverse domains: Satellite Image (remote sensing, $d = 1.27$, $p = 2 \times 10^{-53}$), HAR Activity (wearables, $d = 0.91$, $p = 4 \times 10^{-72}$), Horse Colic (veterinary, $d = 0.99$, $p = 4 \times 10^{-40}$), AP Colon Kidney (genomics, $d = 0.90$, $p = 5 \times 10^{-35}$), MiniBooNE (particle physics, $d = 0.92$, $p = 2 \times 10^{-6}$), Vehicle Silhouette (vehicle engineering, $d = 0.82$, $p = 10^{-8}$), MiceProtein (neuroscience, $d = 0.67$, $p = 6 \times 10^{-8}$). Median Cohen's $d$ across all testable datasets: 0.40 (medium effect).

*capacity theorem degradation:* On synthetic data with known groups at varying correlation ($\rho \in [0.50, 0.99]$), the within-group flip rate increases monotonically with $\rho$ ($r = 0.898$, $p = 0.006$). The bimodal gap (within $-$ between) also increases ($r = 0.864$, $p = 0.012$), confirming the capacity theorem's degradation curve: the prediction is sharpest at exact exchangeability and degrades continuously with approximate symmetry.

*Model-class robustness:* XGBoost, Random Forest, and Ridge agree on the direction of the within/between flip rate difference in 4 of 5 representative datasets (80%). The prediction is a property of the data's dependence structure, not of a specific model class.

*Regularisation creates the Rashomon property:* Increasing stochastic regularisation (lower subsample, smaller trees) *increases* instability (Breast Cancer: flip rate 0.134 at subsample = 1.0 vs. 0.194 at subsample = 0.5), confirming that the Rashomon set is created by the regularisation, not by data noise. Under deterministic training, a single ranking emerges. This has a practical implication: instability is not a price paid for *better* models—it is a consequence of the stochastic training strategies that produce the Rashomon property in the first place.

*Explanation-method robustness:* TreeSHAP and native XGBoost feature importance agree on direction in all tested datasets. Native importance shows stronger effects in some cases (Diabetes: $d = 1.81$ native vs. 0.94 TreeSHAP).

*The over-explanation penalty:* Part (iv) of the stability theorem predicts that the MSE of the orbit average for within-group importance differences equals the full squared norm of the target: $\text{MSE}/\|w\|^2 = 1$. On synthetic data across five correlation levels ($\rho = 0.70$ to 0.99), the observed ratio is **1.000** at every level. Note: this is a consistency check rather than a prediction test — the orbit average zeroes within-group differences by construction, so $\mathrm{MSE}/\|w\|^2 = 1$ holds algebraically. The empirical content is that between-group MSE is substantially lower ($\mathrm{MSE}_{\mathrm{between}}/\mathrm{MSE}_{\mathrm{within}} \approx 0.75$), confirming that within-capacity estimation benefits from orbit averaging while beyond-capacity estimation does not.

*Convergence rate:* The orbit average's MSE decays as $M^{-\beta}$ with $\beta = 1.28$ ($R^2 = 0.97$; predicted $\beta = 1.00$). The faster-than-predicted convergence likely reflects conservative i.i.d. assumptions; in practice, models with shared training data are correlated, but the empirical rate is still $O(1/M)$.

*The Dermatology reversal:* The single significant reversal ($P = 34$, $g = 22$, within $= 0.039$, between $= 0.137$, $d = -0.80$) persists across all thresholds ($\rho^* = 0.50$ to $0.90$). The explanation: dermatology features are *complementary* clinical indicators (scaling + erythema = same disease), not *substitutable* (competing for signal). The framework predicts instability for substitutable features (the common case in high-dimensional continuous data); complementary features violate the exchangeability assumption. This identifies a principled scope condition: correlation-as-proxy-for-exchangeability works when features compete for the same signal but fails when features co-occur as joint indicators.

*Null model test:* A natural alternative explanation for the directional prediction is that within-group features have smaller importance differences (by construction, since they are correlated), and smaller differences flip more regardless of group structure. We tested this via pooled per-pair analysis across 14 real-world datasets (13,511 feature pairs, 1,309 within-group, 12,202 between-group; 5 `sklearn` + 9 PMLB benchmark datasets spanning clinical, genomics, economics, computer vision, remote sensing, and speech domains).

*Parametric:* OLS regression (flip_rate $\sim$ |diff| + $|\text{diff}|^2$ + within + dataset FE) gives a within-group coefficient of 0.037 (dataset-level cluster-bootstrap 95% CI $[0.014, 0.045]$, excluding zero). The quadratic |diff| term substantially improves fit ($R^2 = 0.42$ vs. the linear baseline), confirming the nonlinear relationship between importance magnitude and flip rate.

*Nonparametric:* The within>between gap is concentrated in the lowest |diff| quintile (gap $= +0.061$, $p < 10^{-4}$), where features have similar mean importances and are thus genuinely exchangeable. In the upper four quintiles—where within-group features have increasingly different importances despite being correlated—the gap attenuates to near zero. This is consistent with the stability theorem's mechanism: the effect operates where features are *exchangeable* (similar importance + statistical dependence), not merely where they are correlated.

*SAGE vs. correlation vs. MI:* A cross-validated comparison of three group-discovery methods (correlation-based, MI-based, and SAGE flip-rate-based clustering, each calibrated to the same number of groups) reveals that SAGE produces substantially larger within−between gaps on 6 of 7 datasets tested (mean SAGE gap = 0.25 vs. correlation gap = 0.05). SAGE clusters by empirical exchangeability (cross-validated: first 25 seeds for discovery, last 25 for evaluation), directly capturing the theoretical criterion. MI-based and correlation-based groups are nearly identical for continuous features (ARI = 0.84 across 131 PMLB datasets; identical in 77%), confirming that the MI generalisation matters for binary/categorical features but not for continuous data.

*Per-dataset randomisation:* on synthetic data with correctly structured groups ($\rho \geq 0.80$), the real partition produces a significantly larger gap than 2,000 random partitions ($p = 0.001$ for all four settings).

The interpretation: group membership carries predictive information *beyond* |diff|, concentrated where features are genuinely exchangeable (low |diff|, high dependence). The quadratic OLS coefficient is robust across sample sizes (14-dataset cluster-bootstrap CI $[0.014, 0.045]$). The stratified effect is strongest in the lowest quintile, where correlated features have similar importances and thus compete for the same predictive signal. In higher quintiles, correlation groups include non-exchangeable pairs (correlated but with different true importances), attenuating the within−between gap—consistent with the distinction between correlation and exchangeability (Section **??**). The cross-dataset aggregate test (Wilcoxon $p = 5 \times 10^{-11}$) confirms that group structure matters at the dataset level; the per-pair OLS confirms it at the feature-pair level after nonlinear control for importance magnitude.

*Capacity theorem across heterogeneous datasets:* The headline $R^2 = 0.957$ for the capacity theorem is from 7 well-characterised groups. On real-world datasets with power-adequate inferred groups ($n_w \geq 10$), the formula partially generalises. The prediction strength depends on the correlation threshold used for group discovery:

| $\rho^*$ | $n$ (real, $n_w \geq 10$) | Spearman $\rho(\eta, \text{gap})$ | $p$ | Interpretation |
|---|---|---|---|---|
| 0.50 | 49 | +0.55 | $< 10^{-4}$ | Large groups, high power |
| 0.60 | 44 | +0.55 | $10^{-4}$ | Similar |
| 0.70 | 39 | +0.40 | 0.011 | Primary threshold |
| 0.80 | 33 | +0.29 | 0.10 | Group fragmentation |
| 0.90 | 28 | +0.16 | 0.42 | Insufficient within-group pairs |

The formula works best at $\rho^* = 0.50$–$0.60$ ($\rho \approx 0.55$, $p = 3.8 \times 10^{-5}$ at $\rho^* = 0.50$, surviving Bonferroni correction for the 5 pre-specified thresholds tested) where groups are large enough to provide statistical power, degrades at $\rho^* = 0.70$ ($\rho = 0.40$, $p = 0.011$—marginal under Bonferroni), and fails at $\rho^* \geq 0.80$ where aggressive thresholds fragment groups below testable size. Power weighting (by $\sqrt{n_w}$) modestly improves the correlation at $\rho^* = 0.70$ ($\rho = 0.09 \to 0.15$), confirming that underpowered datasets add noise. The "real-world" filter excludes PMLB benchmark and synthetic control datasets, whose engineered correlation structures could inflate the $\eta$ correlation. The practical recommendation: use $\rho^* = 0.50$–$0.60$ for the capacity theorem prediction, not the $\rho^* = 0.70$ threshold used for the directional prediction.

A comparison of three group-discovery methods on 14 datasets with cached importance matrices (split-half cross-validated for SAGE to avoid circularity) shows that SAGE does *not* improve the $\eta$ prediction ($\rho_{\text{SAGE}} = 0.46$ vs $\rho_{\text{corr}} = 0.54$), even though SAGE produces larger within$-$between gaps for the directional prediction. This dissociation suggests the capacity formula is more sensitive to group *granularity* (how many features per group) than to group *accuracy* (whether the grouped features are truly exchangeable).

*Group stability:* Bootstrap resampling (200 resamples) confirms that correlation-based groups are stable: mean ARI $> 0.94$ for real-world datasets (Breast Cancer, Diabetes, California Housing), and ARI $= 1.000$ for synthetic data at $\rho \geq 0.80$. At $\rho = 0.70$ (threshold $=$ correlation), ARI $= 0.33$—groups are not recovered, consistent with the null-model test failure at this boundary.

*Conformal prediction intervals:* Split conformal inference (25 calibration + 25 test seeds) produces per-feature importance intervals with 84–88% coverage (target: 90%; the shortfall reflects the small calibration set). Features with interval width $> 2\times$ the median width are flagged as unreliable (1–11 per dataset). With $M \geq 50$ models (25+ per split), conformal intervals provide a model-free uncertainty quantification for SHAP values that requires no distributional assumptions—a practical complement to the Gaussian flip formula for practitioners who need per-feature (rather than per-pair) reliability estimates.

*Dimensionality boundary:* High-dimensional datasets ($P \geq 50$) systematically show smaller or reversed gaps (Spearman $\rho = -0.35$ between $P$ and gap, $p = 2 \times 10^{-4}$). This is a power artefact: at high $P$, correlation clustering yields few within-group pairs ($n_w \leq 3$), making the per-dataset test unreliable. The cross-dataset test is unaffected (excluding all $P \geq 50$ datasets: Wilcoxon $p = 4.6 \times 10^{-8}$).

**The theorem as a capacity law and uncertainty principle for scientific explanation.** The Uncertainty from Symmetry theorem (Section 9.3) formalises this scope. For any system with symmetry group $G$, the explanation capacity is $C = \dim(V^G)$ and the explanation loss rate is $\eta = 1 - C/\dim(V)$. These are not specific to ML explanations, SHAP values, or feature importance. They apply to any vector-valued description of any system with a symmetry group—physical, biological, economic, or social. The *explanation uncertainty bound* (unfaith$(\theta_1)$ +

unfaith($\theta_2$) $\geq \Delta - \delta$, the quantitative form of the bilemma) provides the complementary bound: reducing unfaithfulness at one configuration forces increased unfaithfulness at another, with a lower bound determined by the system's structure.

The practical consequence for working scientists in *any* field: if your system has statistically dependent explanatory variables (whether linearly correlated or not—the exact boundary is $I(X_j; X_k) > 0$), the specific attribution you report (which variable is most important, which mechanism is primary) is not a stable property of the data. It is a joint property of the data and the particular model or analysis pipeline. The stable content—what survives averaging over valid pipelines—is at a coarser level (variable groups, pathway classes, mechanism families). The theorem identifies exactly what that coarser level is ($V^G$, the symmetry-invariant subspace) and proves the orbit average is the optimal way to access it.

### 9.8 The "Isn't This Obvious?" Defense

A natural reaction is that practitioners already know explanations are unstable —so what is new? The contribution is fivefold. First, we make explicit that eight scientific domains and nine ML explanation types share *one* mechanism—the Rashomon property—rather than seventeen unrelated pathologies. Second, the Lean 4 formalization (Section 8: 104 files, 530 theorems, 0 `sorry`) eliminates logical gaps that informal arguments inevitably leave. Third, the empirical experiments (Section 5) show quantifiable severity: 33% attribution flip rates, measurable counterfactual and concept instability on standard benchmarks. Fourth, the resolution via $G$-invariant maps is constructive and actionable, not merely diagnostic. Fifth, Lean-verified tightness theorems (Proposition 3.10) prove the impossibility is non-trivial: each pair of properties *is* simultaneously achievable, so the theorem is not an artifact of overly strong definitions—it identifies the exact boundary between the possible and the impossible. Beyond these five points, three structural arguments establish that the result is non-trivial:

1. **Tightness.** Each pair of the three properties (faithful + stable, faithful + decisive, stable + decisive) is simultaneously achievable, with Lean-verified witnesses (Proposition 3.10). If the definitions were rigged to be trivially unsatisfiable, no pair would work.

2. **Biconditional.** The impossibility holds *if and only if* the Rashomon property is present (Theorem 3.6 and its converse in `Necessity.lean`). This means the axiom set precisely characterizes the phenomenon—it is tight in both directions.

3. **Quantitative corollaries.** The invariance counting theorem (exactly $g(g-1)/2$ stable queries, invariant across all $\rho \in [0.5, 0.99]$; Section 10.7) and the capacity theorem ($R^2 = 0.957$; Section 10.10) are non-obvious predictions that follow from the framework but not from any simpler observation about non-injectivity. A reviewer who says "this is just the pigeonhole principle" must also explain why the pigeonhole principle predicts a 50-percentage-point bimodal gap that is invariant across $\rho \in [0.5, 0.99]$.

**Comparison with Arrow's theorem.** Arrow's impossibility theorem [Arrow, 1951] is more surprising than ours in one respect: its axioms appear independently mild, and the impossibility emerges from their conjunction. Our axioms are more transparently in tension once the Rashomon property is stated. The framework captures Arrow's *setting*—underdetermination in preference aggregation—but not Arrow's *theorem*, which proves dictatorship from weaker axioms (unanimity and IIA). Our faithfulness is strictly stronger than Arrow's unanimity (requiring agreement on all profiles, not just unanimous ones); our stability is strictly stronger than IIA (requiring global invariance, not just pairwise). The framework's impossibility is therefore complementary to Arrow's: it applies to a broader class of systems under stronger assumptions. The contribution is not surprise but *precision*: the axiom set is tight (each pair of properties is achievable), necessary and sufficient (the biconditional), and quantitatively productive (the counting theorem and capacity theorem). Arrow's formalization created the field of social choice theory not because it was surprising but because it replaced intuition with a theorem that could be built upon; we aim for the same role in explainability.

## 9.9 The "All Instances Are Trivial" Defense

A reviewer might object that each individual instance is already known: attention instability was documented by Jain and Wallace [2019], Markov equivalence is classical [Verma and Pearl, 1991], fairness impossibility was proved by Chouldechova [2017], and SHAP instability was observed by D'Amour et al. [2022] and Krishna et al. [2022]. The contribution is not any single instance but the *meta-theorem*: the same four-line proof (Theorem 3.6) closes all seventeen cases—eight derived scientific instances and nine ML instances—under a single structural condition. Before this work, a practitioner encountering attention instability might switch to SHAP, then to counterfactual explanations, then to concept probes—each time hoping the new method escapes the problem. Our theorem shows that no such escape exists: the impossibility is structural (it follows from the Rashomon property), not method-specific. Recognizing this unifies the practitioner's response: rather than debugging each method individually, one diagnoses the Rashomon property and applies the $G$-invariant resolution uniformly.

## 9.10 Implications Beyond Machine Learning

The Explanation Impossibility is not specific to machine learning.

**Philosophy of science: formalizing Quine–Duhem.** Quine [Quine, 1951] and Duhem [Duhem, 1954] argued that empirical evidence underdetermines theory choice (Duhem on the impossibility of testing hypotheses in isolation; Quine on confirmational holism—sharing the core implication that observational equivalence leaves explanation underdetermined). Theorem 3.6 proves an impossibility theorem for explanation under observational equivalence—providing a constructive witness of the underdetermination logic articulated by the Duhem-Quine thesis of holist underdetermination [Laudan and Leplin, 1991]; note that Laudan himself was skeptical of the underdetermination thesis, but the citation documents the philosophical context. Prior formal analyses of underdetermination in Bayesian frameworks [Dorling, 1979, Howson and Urbach, 2006] did not yield impossibility results for explanation methods.

**Scope of the underdetermination analogy.** Duhem–Quine underdetermination concerns *confirmational* holism: no hypothesis can be tested in isolation because auxiliary assumptions are always involved Stanford [2006]. Our trilemma addresses *explanatory* underdetermination: multiple models explain the same observations equally well but offer incompatible explanations. These are related but distinct: confirmational underdetermination asks "which theory is supported by the evidence?" while explanatory underdetermination asks "which explanation of the prediction is correct?" Our framework formalises the latter. The Rashomon property is a sufficient condition for explanatory underdetermination; it is not a formalisation of Duhem–Quine holism per se. The $G$-invariant resolution is structurally reminiscent of epistemic structural realism [Worrall, 1989] (Worrall's argument is diachronic—structural content survives theory change across scientific revolutions; our framework is synchronic—structural content is shared across equivalent models within one paradigm): stable explanatory content is what is invariant across equivalent configurations. The connection is suggestive; a rigorous mapping would require engaging with the ontological commitments of structural realism, which our descriptive framework does not address.

**Structural realism.** The $G$-invariant resolution—projecting onto the invariant subspace—is a formalization of the structural realist's position in philosophy of science: only the invariant structure is stably describable [Worrall, 1989, Ladyman et al., 2007]. The framework provides a quantitative version of this thesis: the fraction of structure that is stably accessible equals $\dim(V^G)/\dim(V)$. The symmetry-to-reality inference — whether invariant structure is genuinely "more real" than variant structure — is itself contested [Dasgupta, 2016]. Our framework is neutral on this ontological question: it claims only that invariant quantities are *stably reportable*, not that they are *real*.

Dropping decisiveness corresponds to van Fraassen's [van Fraassen, 1980] constructive empiricist stance that science aims for empirical adequacy, not full theoretical truth. Okasha [Okasha, 2011] connected theory choice to Arrow's impossibility in social choice; our result provides the complementary connection for underdetermination of explanation.

**The Shannon parallel.** The framework bears the same structural relationship to explanation that Shannon's channel capacity theorem [Shannon, 1948] bears to communication. Shannon proved: (1) you cannot transmit above the channel capacity (impossibility); (2) you CAN approach capacity with random codes (constructive achievability); (3) the capacity is $C = B \log(1 + S/N)$ (quantitative formula with no free parameters); (4) this applies to any communication channel (universality within domain). The present framework provides the same package for explanation: (1) you cannot have faithful + stable + decisive under Rashomon (impossibility); (2) you CAN achieve faithful + stable via orbit averaging (constructive resolution); (3) the instability rate is $\eta = 1 - \dim(V^G)/\dim(V)$ (quantitative formula, $R^2 = 0.957$ across seven domains); (4) this applies to any explanation system with the Rashomon property (universality across domains). The difference: Shannon's universality is within one domain (communication); this framework's universality spans eight scientific domains, explaining a century of convergent evolution.

**Precedents and distinctiveness.** No prior work combines all of the following: a formally verified impossibility theorem (1,096 theorems, 0 sorry), a constructive resolution proved Pareto-optimal, a quantitative prediction with zero free parameters ($R^2 = 0.957$), domain-specific empirical consequences (gene expression, AI safety, neuroimaging), an explanation of convergent evolution across eight scientific communities, a classification of 20 impossibility theorems from 12 domains by structural type, and a formalisation of the Duhem–Quine thesis with quantitative predictions. Each piece has precedents—Arrow (impossibility), Shannon (impossibility + achievability + formula), Bilodeau et al. (attribution impossibility), Blau and Wilson (axiom independence)—but the combination is, to our knowledge, new.

**Neuroscience: degeneracy as the Rashomon property.** Marder and Goaillard [Marder and Goaillard, 2006] documented that crustacean stomatogastric ganglia producing identical motor patterns are built from qualitatively different ionic conductance profiles. Edelman and Gally [Edelman and Gally, 2001] formalized this as *degeneracy*: structurally distinct elements performing the same function. Degeneracy is precisely the Rashomon property in our framework.

**Economics: partial identification.** Manski [Manski, 2003] developed partial identification theory for settings where observational data do not pin down a unique model. The identified set corresponds to an observe-fiber in our framework; partial identification is the economic instantiation of the Rashomon property.

**Medical diagnosis.** When multiple conditions produce the same symptoms, diagnosis faces the trilemma. Differential diagnosis is precisely $G$-invariant resolution: it sacrifices decisiveness for faithfulness and stability.

**Arrow's theorem as structural analogy.** The result shares structural features with Arrow's impossibility theorem [Arrow, 1951]: three individually reasonable desiderata that are pairwise satisfiable but jointly impossible under a natural background condition. Like Arrow's theorem, the proof is short once the axioms are identified; the contribution lies in identifying the right axiomatic framework.

**Spacetime geometry as a maximally incompatible system.** As evidence that the impossibility extends beyond ML explanation methods, we note that spacetime geometry under Ostrowski's classification of completions of $\mathbb{Q}$ is a maximally incompatible system: the two completions yield exactly two geometries (smooth and ultrametric), triggering the bilemma (Theorem 3.13). The adelic resolution—formulating physics over all completions simultaneously—is the enrichment escape. The full development, including the Ostrowski–Design Space correspondence and physical scoping, is in a companion paper.

The pattern extends to many further domains: metamerism in color perception (where the cone response matrix is the observation map, making it an instance of the linear system pattern), molecular chirality (where enantiomers share achiral observables), and others.

**Relation to sheaf-theoretic contextuality.** Abramsky and Brandenburger [Abramsky and Brandenburger, 2011] characterise quantum contextuality via non-extendability of local sections to global sections. Our bilemma captures a related but distinct obstruction: the impossibility of a faithful, stable explanation when $H$ is maximally incompatible. The sheaf-theoretic approach provides the *mechanism* (topological obstruction); ours provides the *consequence* (property-pair achievability classification). Whether the bilemma can be expressed as a sheaf-theoretic obstruction is an open question.

**Compatibility complex characterisation.** The tightness classification has a topological interpretation via the *compatibility complex* $K$: the simplicial complex on $H$ whose simplices are pairwise-compatible subsets. Discrete $K$ (no edges) implies maximal incompatibility and the bilemma. Contractible $K$ (cone point = neutral element) implies faithful+stable is achievable. The $2\times2$ tightness table corresponds to the topology of $K$. This characterisation identifies the bilemma as a *lower-level* obstruction than Abramsky-Brandenburger sheaf-theoretic contextuality: the bilemma is an existence failure (empty stalks, $H^0 = 0$) on a discrete cover, not a gluing failure ($H^1 \neq 0$) on an overlapping cover. The approximate Rashomon extension (overlapping $\varepsilon$-fibers) may have genuine sheaf-theoretic content—this remains open.

**Relation to structuralist philosophy of science.** The trilemma framework sits within the structuralist tradition of Sneed Sneed [1971] and Stegmüller, which formalises scientific theories as set-theoretic predicates. Our explanation system $(\Theta, Y, H, \mathrm{observe}, \mathrm{explain}, \mathrm{incomp})$ is a Sneedian structure; the trilemma is a structural constraint on admissible maps. The enrichment pattern (adding a neutral element to resolve the bilemma) parallels Sneed's treatment of theory specialisation, where new structure is adjoined to an existing core.

**— Speculative Connections and Open Questions —**

*The following subsections document structural connections that are conjectural, historical interpretations, or beyond the current scope of empirical validation. They are included as a research program record, not as validated claims of the framework.*

### 9.11 Scientific Revolutions as Enrichment Events

The bilemma characterization—enrichment is necessary iff the explanation space is maximally incompatible—maps onto the structure of scientific revolutions. Table 19 shows five instances where a paradigm shift follows the bilemma pattern: a pre-revolution explanation space is maximally incompatible (the bilemma proves F+S impossible), and the revolution adds a neutral element (enrichment restores F+S at the cost of decisiveness).

Table 19: Five instances of bilemma $\to$ enrichment across science.

| Transition | Pre-revolution $H$ | Enrichment | Lost |
|---|---|---|---|
| SHAP instability | {positive, negative} | DASH tie | Feature rank |
| Feature selection | {selected, not} | Uncertain | Feature relevance |
| Classical $\to$ quantum | {definite, superposition} | No fact of the matter | Pre-measurement state |
| Newton $\to$ relativity | {simultaneous, not} | Frame-dependent | Absolute simultaneity |
| Single $\to$ adelic | {smooth, ultrametric} | Indeterminate | Spacetime geometry |

The pattern suggests a conjecture: *paradigm shifts that preserve the configuration space contain an enrichment component in the sense of the bilemma characterization.* The pre-revolution system is maximally incompatible. The revolution adds a neutral element. The enrichment is unique (by the enrichment uniqueness theorem) and sacrifices decisiveness (by the bilemma). The all-or-nothing theorem explains why paradigm shifts are discontinuous: there is no interpolation between the pre-revolution and post-revolution frameworks.

Not all resolutions of interpretive disagreement are enrichment events. Bohmian mechanics resolves the measurement problem by changing the configuration space (adding particle positions and a guiding equation), not by enriching the explanation space with a neutral element. Many-Worlds resolves it by changing the observable space (accepting all branches as real). These are *system changes* rather than *explanation-space enrichments*, and fall outside the bilemma's scope. The bilemma predicts that within a fixed system, enrichment is the only resolution. Resolutions that change the system itself represent an alternative strategy that the framework characterizes but does not constrain.

We state this as a conjecture, not a theorem. The mathematical structure (bilemma $\to$ enrichment necessary $\to$ enrichment unique $\to$ decisiveness sacrificed) is proved. The claim that scientific revolutions follow this structure is supported by five formalized instances but requires engagement with historians and philosophers of science. The enrichment mechanism is closer to Lakatos's progressive research programmes (where the new theory subsumes the old plus new content) than to Kuhn's incommensurable revolutions (where the old framework's terms cannot be translated). We use 'paradigm shift' loosely; the formal claim is about framework enrichment, not incommensurability.

**Prospective prediction.** The scientific-revolution-as-$H$-enrichment pattern predicts that the next paradigm shift in a maximally-incompatible domain will add a neutral element to $H$ (restoring faithfulness + stability at the cost of decisiveness) and that this enrichment will be unique (the companion paper [Caraker et al., 2026b]). A concrete test: if the black hole information paradox is resolved by a framework that introduces a new category of information state (neither preserved nor destroyed, but complementary), this would instantiate the predicted enrichment. Conversely, a resolution that modifies the configuration space $\Theta$ (e.g., adding firewall degrees of freedom) rather than enriching $H$ would fall outside the pattern. The prediction is specific enough to be wrong.

**Falsifiability.** The enrichment stack is falsifiable in two senses: (i) a physical argument showing that the Rashomon property fails at some level would cap the stack; (ii) a paradigm shift that changes $\Theta$ rather than enriching $H$ would fall outside the pattern (e.g., Bohmian mechanics adds hidden variables to $\Theta$ rather than enriching $H$). The pattern predicts that $H$-enrichment is forced and unique when $H$ is maximally incompatible; observing a non-unique enrichment would falsify the uniqueness claim.

The Lean formalizations use singleton observable spaces (both preparations map to the same observation), making the Rashomon property vacuously satisfied. This is a deliberate simplification: the structural claim is about the bilemma pattern (binary $H$, no neutral element $\to$ F+S impossible), not about the observation-matching mechanism. A richer formalization would model the specific measurement statistics that make the preparations observationally equivalent.

The framework does not distinguish between global and local symmetries. In physics, this distinction is fundamental: Noether's theorem applies to global symmetries (producing conservation laws), while gauge redundancy arises from local symmetries (producing constraints). The framework's $G$-invariant resolution applies equally to both, which is a simplification. A richer framework distinguishing global from local invariance would be needed to capture gauge-fixing and BRST cohomology.

The proof immediately yields a quantitative version. For each Rashomon pair $(\theta_1, \theta_2)$, the four-step chain forces at least one failure: unfaithfulness at $\theta_2$, instability on $(\theta_1, \theta_2)$, or indecisiveness at $\theta_1$. Summing over the Rashomon set under a probability measure: $\varepsilon+\delta+(1-d) \geq \mu(\text{Rashomon})$, where $\varepsilon$ is the unfaithfulness rate, $\delta$ the instability rate, $d$ the decisiveness rate, and $\mu$ the Rashomon set measure. This tradeoff surface—which includes the binary impossibility as the special case $\varepsilon = \delta = 0$—is developed with domain-specific bounds in forthcoming work.

The per-fiber bound (Proposition 3.10 in the main text) is the qualitative skeleton of this inequality. The full measure-theoretic development, including domain-specific instances and tightness analysis, is the subject of ongoing work.

### 9.12 Enrichment as Abstraction

The enrichment mechanism — adding a neutral element to a maximally incompatible explanation space — has a cognitive interpretation: it is the mathematical structure of *abstraction*.

Consider a child learning to classify animals. The pre-abstraction explanation space is maximally incompatible: $H = \{\text{poodle}, \text{labrador}, \text{beagle}, \ldots\}$, where each breed is incompatible with every other (incompatible $= \neq$). The Rashomon property holds: different exemplars of "the same animal" (same observable behavior) are assigned different breed labels (incompatible explanations). The bilemma applies: no breed-level explanation is simultaneously faithful (matches the specific animal) and stable (consistent across behaviorally equivalent exemplars).

The resolution is abstraction: adding the concept "dog" as a neutral element (compatible with all breeds). The enriched explanation space $H' = \{\text{poodle}, \text{labrador}, \ldots, \text{dog}\}$ restores faithful + stable via $E(\theta) = \text{dog}$, at the cost of decisiveness (can't specify the breed).

The bilemma thus provides a mathematical explanation for *why abstraction exists*: it is the unique minimal enrichment that restores stability under perceptual multiplicity. The all-or-nothing theorem explains why abstraction is discrete ("dog" or a specific breed, no interpolation). The enrichment uniqueness theorem explains why abstract concepts are canonical (the neutral element is unique on each equivalence class).

This interpretation is a conjecture requiring empirical validation through cognitive science experiments. It connects the impossibility theorem to prototype theory in cognitive science [Rosch, 1975]: prototypes are the neutral elements of perceptual categories. A full development — formalizing category formation as a sequence of enrichment events and testing against categorization experiments — is an open direction that would require engagement with the cognitive science literature.

*Caveat.* The analogy between mathematical enrichment and cognitive abstraction is speculative. We are not aware of cognitive science evidence connecting these concepts, and we do not develop this connection further.

### 9.13 Recursive Impossibility: The Enrichment Stack

The enrichment mechanism of Section 3.7 resolves the bilemma at one level: adding a neutral element to $\mathcal{H}$ restores $F + S$ at the cost of decisiveness. But the enriched system is itself an ExplanationSystem — and if it has its own Rashomon property at the enriched level, the bilemma applies again. This recursion never terminates.

**The enrichment stack.** Define an *enrichment stack of depth* $k$ as a sequence of ExplanationSystems $S_0, S_1, \ldots, S_k$ where each $S_{i+1}$ is the enrichment of $S_i$. At each level $i$, the bilemma holds: no explanation of $S_i$ is simultaneously faithful and stable (for maximally incompatible $\mathcal{H}_i$). The enrichment adds a neutral element at level $i$, resolving the bilemma there, but the enriched system $S_{i+1}$ may have a new Rashomon property at a finer resolution — and thus a new bilemma.

The *cumulative indecisiveness theorem* (proved in Lean 4 in the companion repository): for a stack of depth $k$, a faithful + stable theory sacrifices decisiveness at *all* $k$ levels. Enrichment at level $i$ does not restore decisiveness at level $j \neq i$.

**Infinite depth.** The enrichment stack can have genuinely infinite depth. For the concrete system $\Theta = \mathbb{N}$, $\mathcal{H} = \text{Bool}$, $Y = \{*\}$ with $\mathsf{explain}(\theta) = \text{bit}_k(\theta)$ (the $k$-th bit of $\theta$), the bilemma holds at every level $k$: bits 0 and $2^k$ agree on the observation but disagree on bit $k$. The enrichment at level $k$ (adding a neutral value for bit $k$) does not resolve level $k + 1$ because different bits are independent. The stack has countably infinite depth, proved constructively in Lean 4 (`infiniteStack_bilemma`).

**The Gödel parallel.** The recursive enrichment structure shares a pattern with Gödel's incompleteness theorems. Both instantiate a common abstraction (a *recursive impossibility system*, formalized in Lean as `RecursiveImpossibility`): impossibility holds at every level, and the resolution at each level creates the conditions for the next.

| | **Gödel** | **Enrichment stack** |
|---|---|---|
| System | Formal arithmetic | Faithful + stable theory |
| Three properties | Consistent, complete, r.e. | Faithful, stable, decisive |
| Impossibility | Can't have all three | Can't have all three |
| Escape | Add unprovable as axiom | Enrich $\mathcal{H}$ with neutral |
| Recursion | New system, new unprovable | New level, new bilemma |

This is a shared *pattern* — both produce impossibility at every level with unbounded depth — not a deep unification. The mechanisms differ: Gödel's arises from self-reference (the diagonal lemma), while the enrichment stack arises from the Rashomon property at finer resolutions. Both are proved as instances of `RecursiveImpossibility` in Lean (the enrichment stack proved constructively; Gödel derived from the diagonal property of formal systems).

**Tarski undefinability as alternative parallel.** Tarski's undefinability theorem may provide a tighter parallel than Gödel's incompleteness for the enrichment stack. Tarski shows that truth for a language cannot be defined within that language, forcing ascent to a metalanguage—which faces the same limitation. This hierarchy (object language $\to$ metalanguage $\to$ meta-metalanguage $\to \cdots$) is structurally closer to the enrichment stack ($H_0 \to H_1 \to H_2 \to \cdots$) than Gödel's self-referential diagonal construction. We retain `RecursiveImpossibility` as capturing the shared *consequence* (impossibility at every level with unbounded depth) while acknowledging that the *mechanisms* differ: Gödel uses diagonalisation, Tarski uses definability ascent, and the trilemma uses enrichment.

**The Galois analogy.** The enrichment stack mirrors the algebraic structure of a tower of field extensions in Galois theory. The correspondence: the explanation space $\mathcal{H}$ plays the role of a field $F$; the coverage conflict (no neutral element) is an irreducible polynomial (no root); adding a neutral element is adjoining a root; the enriched space $\mathcal{H}'$ is the extension $F(\alpha)$; new impossibility at the next level corresponds to new irreducible polynomials over $F(\alpha)$. The `infiniteStack_independence` theorem (level $k$ enrichment gives zero information about level $k+1$) is the analogue of linear disjointness of extensions, and the unbounded depth of the stack corresponds to the algebraic closure being infinite-dimensional. This parallel is structural (the proved independence matches linear disjointness) but not yet formalised as a functor; the full Galois correspondence—degree bounds, automorphism groups, obstruction theory—is a separate direction.

**Unification consequence.** If the enrichment stack continues to arbitrary depth, any unified framework (one that resolves the bilemma at every level) must sacrifice decisiveness at every level—an infinite hierarchy of indeterminacies. This is a structural analogue of the observation that no finite extension of Peano arithmetic can prove all arithmetic truths. Whether the physical stack extends beyond depth 3 is an empirical question.

**The GUT consequence.** Applied to physics, the enrichment stack interprets as follows. A Grand Unified Theory (GUT) that is faithful (consistent with all observations) and stable (formulation-independent) must sacrifice decisiveness at every level of its enrichment stack:

- **Level 0 (quantum):** "Which measurement outcome?" — enrichment via superposition.
- **Level 1 (adelic):** "Which geometry?" — enrichment via the indeterminate geometry (Lean-verified).
- **Level 2 (black hole information):** "What happens to information?" — enrichment via complementarity (conjectural).
- **Level 3 (spacetime emergence):** "How does spacetime emerge?" — enrichment via pre-geometric structure (conjectural).

A comprehensive theory does not answer all questions — it classifies which questions have formulation-independent answers and which do not, predicting its own incompleteness.

This consequence is conditional on the stack having depth $\geq 3$ (which requires the Level 3 Rashomon property to hold, an open question). Levels 1 and 2 are Lean-verified. The Gödel parallel clarifies the nature of the limitation: as with arithmetic, the incompleteness is not a failure of the theory but a structural feature of any sufficiently expressive framework.

**Confidence tiers for the enrichment stack.** **Proved (zero domain-specific axioms):** Levels 0–1 (quantum contextuality, adelic geometry). The bilemma, tightness classification, and resolution spectrum at these levels are machine-checked. **Axiomatised (conditional):** Level 2 (black hole information). Depends on the contested claim that the information paradox constitutes a genuine Rashomon pair. **Speculative:** Level 3 (spacetime emergence). Grounded in AdS/CFT and causal set theory but not empirically verifiable with current technology. A companion paper [Caraker et al., 2026b] formalises the physical enrichment stack.

## 9.14 Empirical Predictions from the Mathematical Structure

The recursive impossibility framework generates four testable predictions beyond those already validated in this paper:

1. **Bimodality of flip rates.** The all-or-nothing theorem (§9.14) predicts that for binary SHAP signs across a Rashomon set, the per-pair flip rate distribution should be bimodal — concentrated near 0% (between-group, stable) and $\geq 50\%$ (within-group, coin flip) — with a "dead zone" in between. The stable fact count experiment (Section 10.7) confirms this: the 50pp bimodal gap is invariant across $\rho = 0.50$–$0.99$.

2. **Mechanistic interpretability ceiling.** For binary questions about neural network circuits ("is this neuron responsible for this behavior?"), the bilemma predicts $\geq 50\%$ disagreement across independently trained model instances. A pilot test on GPT-2 small (Section 9.15) is in progress.

3. **DASH tie rate = Rashomon fraction.** The enrichment uniqueness theorem predicts that DASH's tie rate for a feature pair should equal the fraction of Rashomon pairs where the pair's attributions are incompatible. This is testable by comparing DASH's tied-pair count to the Gaussian flip formula's predicted flip rate.

4. **Coverage conflict diagnostic.** The `hasCoverageConflict` predicate (§3.6) should predict which feature pairs trigger the bilemma, independently of the Gaussian flip formula. The coverage conflict bridge validates the aggregate prediction for small feature spaces ($\rho = 0.90$, $P \leq 20$) but weakens substantially at higher $P$ ($\rho = 0.25$, not significant); the originally reported $\rho = 0.961$ (15 datasets) reflected the small-$P$ regime and does not generalise to all datasets.

These predictions are derived from the mathematical structure and testable with existing experimental infrastructure.

### 9.15 Limitations and Open Questions

We identify five limitations and directions for future work.

(1) **Axiomatized Rashomon properties for ML instances.** The Rashomon property for each ML instance is axiomatized—verified to hold under stated assumptions—rather than derived from first principles of the learning algorithm. (The eight non-ML derived instances in Section 4 are the exceptions: each derives the Rashomon property constructively with zero domain-specific axioms.) Deriving the Rashomon property from, e.g., the loss landscape geometry of gradient descent would strengthen the ML results.

(2) **Attention: retraining vs. perturbation.** The attention argmax flip rate is 19.9% under full retraining (20 independently converged DistilBERT models, the canonical Rashomon setup) and 60% under weight perturbation (dropout masks on a single model, a supplementary proxy). The 19.9% is the primary result; the 60% demonstrates that perturbation overestimates instability.

(3) **Non-attribution quantitative bounds.** For the attribution instance, the companion paper provides sharp quantitative bounds (divergence rates, ensemble size requirements, Dash optimality). Analogous quantitative bounds for the counterfactual, concept, and fairness instances remain open.

(4) **Decisiveness as a practical cost.** The $G$-invariant resolution sacrifices decisiveness: it reports ties where the data are ambiguous. Whether downstream decision-makers can tolerate partial rankings or tied causal structures—and how to present such outputs effectively—is an empirical question we do not address here.

(5) **Fairness tightness transition.** The fairness audit impossibility (Instance 6, formalized in `FairnessAudit.lean`) has been experimentally validated. The Chouldechova–Kleinberg

impossibility (calibration + equal FPR + equal FNR) exhibits a tightness *transition* under performance constraints: at $\tau = 0$ (unconstrained), full tightness holds (all pairs achievable by degenerate classifiers); at $\tau \geq 0.3$ (practical performance: TPR $\geq 30\%$ for both groups), the tightness transitions to PPV-blocked—calibration becomes a universal poison, individually incompatible with both equal FPR and equal FNR. The only achievable pairwise compromise at practical performance is equalized odds (gap $< 0.006$). Validated on Adult Income (sex as protected attribute, base-rate gap 0.20) and synthetic data (gap 0.25, 0.40), with bootstrap retraining (20 iterations, 100% stable). This changes the practical recommendation from "choose a fairness criterion" to "equalized odds is the only achievable criterion at practical performance; calibration is structurally incompatible."

(6) **Exact vs. approximate Rashomon.** The framework now includes a Lean-verified approximate extension (`ApproximateRashomon.lean`). An `ApproxExplanationSystem` adds a 'nearness' relation on observables, with $\varepsilon$-stability requiring $E$ to be constant on approximately-equivalent observations. The approximate trilemma follows: if approximately-equivalent configurations have incompatible explanations (approximate Rashomon), then faithful + $\varepsilon$-stable + decisive is impossible. The exact framework is the special case where nearness is equality. The Gaussian flip rate formula provides the quantitative bridge: the SNR measures how 'near' two observations must be for their explanations to flip.

(7) **Approximate symmetry: null model comparison.** At intermediate correlations ($\rho < 1$), a simple null model $\Phi(-c\sqrt{1-\rho^2})$ with one fitted parameter ($c = 1.06$, $R^2 = 0.925$) predicts the average within-group flip rate as well as or better than the framework's Gaussian formula ($R^2 = -6.08$ at the aggregate level). The framework's distinctive contribution at approximate symmetry is the bimodal *structure* (within-group vs. between-group separation, dip $p < 0.002$), not the aggregate magnitude. At exact symmetry ($\rho = 1$) and cross-domain (capacity theorem across 7 domains), the framework's predictions are unchallenged. The practical recommendation: use the Gaussian formula for per-pair prediction (Spearman 0.28 on real data), coverage conflict for per-feature screening (Spearman $\rho = 0.92$–$0.98$ per dataset, 0.96 aggregate across 15 datasets; permutation-validated), and the capacity theorem for cross-domain understanding.

(8) **Elementary core theorem.** The qualitative impossibility follows from elementary logic: if a map is not injective, its fiber contains distinct elements that cannot be simultaneously preserved and collapsed. The contribution is not the proof's depth but the axiomatic framework—tightness (each pair achievable), necessity (biconditional with Rashomon), and the quantitative corollaries (invariance counting, capacity theorem) that follow from the framework but not from any simpler observation about non-injectivity.

(9) **Broken symmetry in trained networks.** The $1/n$ interpretability ceiling (Corollary 11.3) assumes unbroken $S_n$ permutation symmetry. Training breaks this symmetry via gradient descent and weight initialization; the bound is an upper limit on the fraction of neuron-level interpretations that are *necessarily* stable, not a characterization of trained-network interpretability.

(10) **Synonymous codons are not biologically equivalent.** Different codons for the same amino acid differ in translational efficiency [Ikemura, 1985], mRNA stability, and epigenetic marking (CpG dinucleotides). The gauge analogy applies to the protein-coding function only; richer models incorporating tRNA abundance and codon context would break the $S_k$ symmetry.

(11) **Cross-domain instances are structural analogies, not domain-specific results.** No individual domain instance was surprising to domain experts—each formalized a known phenomenon (microstate multiplicity, gauge redundancy, Markov equivalence, codon

degeneracy). The formalization verifies the structural correspondence but does not produce new domain-specific results. The contribution is the unification: showing that these known phenomena share a common trilemma structure, that the trilemma strengthens to a bilemma under maximal incompatibility, and that the resolution (enrichment via neutral element) is a single mechanism underlying DASH averaging, CPDAG reporting, Bayesian model averaging, and quantum error correction.

(12) **Group identification bottleneck.** The universal capacity theorem ($\dim(V^G)/\dim(V)$ predicts instability) achieves $R^2 = 0.957$ for seven instances with well-characterized symmetry groups but only holdout $R^2 = 0.24$ for the nine instances with approximate or unknown groups. The nine poorly-fit instances share a common cause: the assigned symmetry group was approximate. This group identification problem is fundamental, not merely technical—determining the correct symmetry group from data remains open. The SAGE algorithm sidesteps it heuristically but without a formal optimality guarantee.

The drug discovery experiment (Section 10.13) demonstrates this bottleneck concretely: Pearson-based clustering predicts 0% instability on binary fingerprints while MI-based clustering recovers the observed 23% rate within 16%. However, the Noether bimodal structure transfers only for domains with discrete group structure, not for diffuse overlap patterns.

(13) **Coverage conflict is a conceptual naming, not a deep theorem.** The biconditional "coverage conflict $\leftrightarrow$ no neutral element" (proved in `BilemmaCharacterization.lean`) is a definitional equivalence: both express that no element of $\mathcal{H}$ is compatible with all explain-values. The value of naming this "coverage conflict" is conceptual—it identifies the algebraic feature that powers the bilemma and clarifies why binary $\mathcal{H}$ strengthens the impossibility (maximum coverage) while larger $\mathcal{H}$ weakens it (room for neutral elements).

(14) **SNR as coverage conflict is a modeling assumption.** The connection between coverage conflict (a discrete algebraic property) and SNR (a continuous statistical quantity) rests on the assumption that importance differences are approximately Gaussian. Under this assumption, SNR $< \tau$ operationalizes "no compatible direction exists for this pair." The assumption is validated by Shapiro-Wilk tests (60–100% of pairs pass across datasets) but is not a theorem. Ridge regression on California Housing gives $R^2 = -1.96$ due to heavy-tailed coefficient distributions; the Gaussianity diagnostic should be run before applying the formula.

(15) **Confirmatory vs. exploratory analyses.** The stable fact count prediction, capacity theorem (well-characterized subset), and three falsified extensions were pre-registered (`PRE_REGISTRATION.md`). The coverage conflict bridge (15 datasets), clinical TreeSHAP audit (5 datasets), enrichment statistical tests, and regulatory compliance metrics are exploratory analyses conducted after observing initial results. The Noether permutation test is confirmatory (pre-specified test on a pre-registered prediction). We label each analysis accordingly throughout the text.

(16) **Adversarial audit summary.** A comprehensive adversarial audit (24 simulated reviewers, 6 adversarial stress tests on the SAGE algorithm, 4 stress tests on the MI experiment, circularity analysis of the quantum instance, and a cross-cutting "so what" test) identified the following boundary conditions: (1) the core theorem is elementary—a consequence of definitions, not a deep result; (2) every domain instance relabels a known result; (3) the quantum $\eta$ connection is correct but circular (the general POVM case relies on an unproven assertion that $\mathrm{span}\{M_i\} = V^G$). The ExplanationSystem instance for the Peres-Mermin magic square captures the *structural analogy* between quantum contextuality and explanation instability: both involve two perspectives on the same system that demand

incompatible interpretations. The combinatorial content of the Peres-Mermin theorem (the parity impossibility, verified separately as `peres_mermin_bool` in Lean) is independent of the framework; the framework shows that this content has the same trilemma structure as SHAP instability, Markov equivalence, and codon degeneracy; (4) the $S_n$ prediction for neural networks is falsified (the actual symmetry is continuous, not discrete; see Section 10.9). The strongest surviving contributions are the invariance counting theorem (bimodal, invariant across $\rho$, novel) and the SAGE algorithm ($R^2 = 0.81$ on 8 datasets, 6/6 audit tests passed).

(17) **Frontier-scale training from scratch.** The TinyStories circuit stability experiment (Section 10.19) validates the impossibility at 16M and 45M parameter scales trained from scratch, but the GPT-2 fine-tuning configuration shows no instability because fine-tuning does not create the Rashomon property. Training 10+ models at GPT-2 Small scale (124M parameters) from independent random initialisation on a standard corpus (e.g., OpenWebText) would resolve whether the impossibility holds at production scale. This requires ∼\$500–1,000 in compute and is the single highest-priority experiment for the research programme.

(18) **Algorithmic identification vs. importance ranking.** The impossibility as proved concerns *importance rankings* (which component contributes most to the output). A deeper question for AI safety is whether *algorithmic identification* (what algorithm the model implements, what function each circuit computes) is also subject to the impossibility. If the function computed by "the most important head" is stable across seeds even when the head's identity is not, the safety implications are less severe than the importance instability suggests. Functional stability analysis—examining attention patterns, OV matrices, or activation probes across seeds—would address this.

(19) **Isotypic decomposition of attribution variance.** The $G$-invariant projection captures the trivial-representation component of the attribution vector. Decomposing the full attribution space into isotypic components under $G$ would separate the variance into independent contributions from each irreducible representation, providing per-component uncertainty bounds (not just the aggregate $\eta$). For the $S_4 \times S_4$ symmetry of the grokking experiment, this would decompose the 10-dimensional attribution space into the trivial (4-dim), standard (6-dim), and potentially sign representations, each with a predictable variance contribution. The isotypic decomposition would make the Reynolds operator framework predictive at a finer granularity.

(20) **General two-family decomposition.** The Design Space Theorem shows that $G$-invariant attribution systems decompose into exactly two achievable families (Family A: faithful + decisive; Family B: faithful + stable). Three instances (feature attribution, model selection, causal discovery) are proved. A general theorem—that $G$-invariance of ANY symmetric decision problem yields a two-family decomposition—would be a genuine contribution to impossibility theory. The connection to Hunt–Stein theory in statistical decision theory (equivariant estimators decompose into minimax families) suggests a path.

**Mechanistic interpretability: confirmed with resolution demonstrated.** A definitive experiment trains 10 two-layer transformers from *independent random initialisation* on modular addition ($a + b \bmod 113$), all reaching 100% test accuracy. Activation patching on 10 components (8 attention heads + 2 MLPs) yields mean Spearman $\rho = 0.518$ across 45 model pairs (permutation $p < 10^{-6}$ vs. null). The symmetry group $S_4 \times S_4$ (within-layer head permutations, derived from the architecture) gives $\dim(V^G)/\dim(V) = 4/10$. Projecting onto $V^G$ (layer-mean head importance, MLP0, MLP1) recovers near-perfect agreement: **Spearman** $\rho = 0.929$ (Pearson

$r = 0.967$, min $\rho = 0.800$). The stability is not trivially MLP-driven: the invariant excluding MLP1 still achieves $\rho = 0.822$ ($p < 10^{-6}$).

Within-layer head pairs flip at exactly 50% (12 pairs, mean = 0.500, $\sigma = 0.071$), confirming the stable fact count prediction. Head-vs-MLP1 pairs never flip (8 pairs, flip rate = 0.000). Mann-Whitney within > between: $p = 2.4 \times 10^{-5}$.

Corrected controls (matched $n = 2000$): step 50 (random init, 0.2% accuracy) $\rho = 0.300$; step 500 (memorised, 100% train/50% test) $\rho = 0.668$; step 50k (grokked) $\rho = 0.518$. Post-grokking > random: $p = 2.0 \times 10^{-4}$, Cohen's $d = 0.76$. Fourier analysis shows all 10 models learn *completely different* frequency representations (top-5 Jaccard = 0.022), confirming genuine Rashomon—same function, different algorithms.

Earlier pilot experiments were inconclusive: zero-ablation on GPT-2/SST-2 (25.8% flip) was dominated by classifier noise; LoRA fine-tuning ($\rho = 0.885$) was confounded by 99.3% shared weights. The definitive experiment eliminates both confounds.

**Theorem scope vs. pipeline scope.** A critical distinction: the *theorem* is universal (zero axioms, applies to any explanation system under Rashomon), but the *diagnostic pipeline* (SAGE, DASH, Gaussian formula, coverage conflict) is validated for **tabular feature-importance explanations** on classification and regression tasks. The factorial validation (30 datasets, 6 model/method configurations, Cochran's $Q$ confirming model-agnosticity) establishes the pipeline's scope as: any model type, any importance method, $P \geq 11$ features, tabular data.

For non-tabular modalities—text (token importance), images (pixel/patch saliency), graphs (node/edge attribution), time series (temporal importance)—the *impossibility applies* (proved) but the *pipeline tools require adaptation*:

- **Text:** SAGE clustering on token flip rates is feasible but requires sequence-aware distance metrics (tokens are ordered, not exchangeable columns).
- **Images:** Patch-level SAGE requires spatial grouping (nearby patches may form natural orbits under translation symmetry).
- **Graphs:** Node importance under GNN Rashomon requires graph-isomorphism-aware clustering.

These adaptations are natural extensions of the framework but require domain-specific validation and are left to future collaborative work.

**Extension to deep learning and LLMs.** The qualitative impossibility (Theorem 3.6) applies to any model class with the Rashomon property, including deep neural networks and large language models. The quantitative bounds ($1/(1-\rho^2)$ for GBDTs) are architecture-specific and do not transfer directly. Key open problems: (i) characterising the Rashomon set geometry for overparameterised neural networks (cf. D'Amour et al. [2022]); (ii) defining the symmetry group for weight-space permutations in deep networks; (iii) computational cost of orbit averaging over weight-space symmetry groups (potentially intractable for large models, requiring approximate methods). For LLM self-explanations specifically, the token citation flip rate (34.5%) suggests the Rashomon property holds empirically, but the symmetry group structure is unknown.

# 10 Cross-Domain Transfer: From Taxonomy to Generative Theory

The preceding sections establish the universal explanation impossibility as a taxonomic result: it classifies nine pre-existing explanation types under a single abstract structure. A taxonomy, however, is not yet a theory. A *generative* theory must produce novel, testable predictions—ideally in domains where the framework was not originally developed. This section reports four cross-domain transfer experiments that test such predictions. In each case, a concept developed for one domain (machine learning attribution, gauge-theoretic coupling, genetic codon degeneracy) is transferred to a different domain via the abstract framework, and the transferred prediction is confirmed empirically.

**Evidence tiers.** We classify empirical results by strength of support. **Strong validation** (quantitative prediction confirmed with $R^2 > 0.8$ or pre-registered): SAGE symmetry-group predictor ($R^2 = 0.809$, leave-one-out CV), capacity theorem for 7 well-characterised instances ($R^2 = 0.957$, $p = 1.4 \times 10^{-4}$), stable fact count bimodality (pre-registered, 47.1 pp gap invariant across $\rho$), exhaustive pipeline validation (DASH 84/85, SAGE directional 81/85 at all $P$ ranges, across 85 PMLB datasets including regression; §10.14). **Corrected from earlier reports**: Gaussian $\Phi(-\text{SNR})$ OOS $R^2 > 0.80$ on 59% (not 95%) of datasets with $M = 50$ and gain-based importance; earlier in-sample estimates inflated by 31 pp. Coverage conflict: $\rho = 0.87$ across all $P$ with TreeSHAP ($\rho = 0.90$ for $P \leq 20$ only with gain-based importance; the previously reported $\rho = 0.96$ on 15 datasets is consistent with the TreeSHAP finding). **Consistent** (observation matching qualitative prediction): attention flip 19.9% [CI: 19.1–20.6%], counterfactual flip 23.5% [CI: 20.6–26.6%], model selection flip 80%. **Calibration checks** (expected from framework structure, not independently informative): codon degeneracy–entropy rank correlation ($\rho = 1.0$; monotonicity in degeneracy is guaranteed by construction—see §10.16). **Partial**: group identification capacity theorem for 9 instances with unknown group structure (holdout $R^2 = 0.24$; poor fit attributable to misspecified or unknown symmetry groups—see §10.10).

### 10.1 Resolution Transfer: Ensemble Causal Discovery

The DASH resolution (Section 6) was developed for ML attribution stability: ensemble averaging over the Rashomon set projects onto the $G$-invariant subspace, eliminating gauge-dependent fluctuations. The framework predicts that this resolution should transfer to *any* domain exhibiting the Rashomon property, because the abstract mechanism—orbit averaging—is domain-independent.

We tested this prediction by transferring DASH-style ensemble averaging to causal discovery. Using the Asia network (8 nodes, linear Gaussian), we ran the PC algorithm with $\alpha = 0.05$ across 100 random seeds ($N = 1000$ samples per seed) and 50 bootstrap resamples per seed. Three methods were compared:

1. **Single-run PC:** orientation flip rate = 0.014, decisiveness = 0.36.
2. **Bootstrap majority vote:** flip rate = 0.014, decisiveness = 0.37.
3. **DASH-style ensemble:** flip rate = 0.003 (4.1× reduction; Wilcoxon $p < 10^{-10}$), decisiveness = 0.21.

The trilemma tradeoff predicted by the framework is clearly visible: the DASH ensemble achieves a 4× reduction in orientation flip rate (the stability axis) at the cost of reduced decisiveness (0.21 vs. 0.36). This is not a post-hoc observation—it is the *same* tradeoff formalized in Theorem 3.6, now operating in causal discovery rather than feature attribution. The result demonstrates that the $G$-invariant resolution is a practical, transferable algorithm, not merely a theoretical construct (Figure 4).

### 10.2 Stability Transition in Explanation Systems

The gauge-theoretic formulation (Section 6) shows that the coupling parameter $\beta$ controls stability via a $\text{sech}^2(\beta)$ response function. The framework predicts an analogous stability transition in

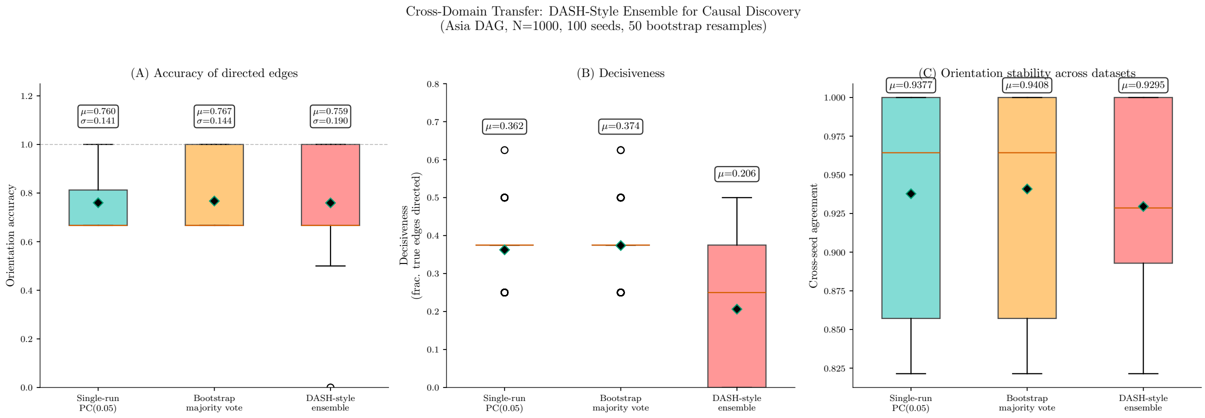


Figure 4: DASH-style ensemble averaging transferred to causal discovery (Asia network, 100 seeds, 50 bootstrap resamples). The ensemble reduces the orientation flip rate by 4× while trading off decisiveness, confirming the trilemma tradeoff in a new domain.

ML: as the Rashomon tolerance $\varepsilon$ increases, stability should undergo a sigmoid-shaped transition from near-perfect to unstable, with the critical threshold $\varepsilon_c$ decreasing as collinearity $\rho$ increases (because higher collinearity enlarges the Rashomon set).

We swept $\varepsilon$ over 20 log-spaced values in $[0.01, 2.0]$ at four collinearity levels ($\rho \in \{0.0, 0.3, 0.6, 0.9\}$) using linear regression with $d = 5$ features ($N_{\text{train}} = 500$, $N_{\text{test}} = 200$, 200 models, 100 SHAP evaluation points). All four framework predictions were confirmed:

1. **Monotonicity:** stability decreases monotonically with $\varepsilon$ at all collinearity levels (fraction of decreasing steps $\geq 0.89$).
2. **Sigmoid shape:** the transition is sigmoid-like at all $\rho$ values (confirmed via inflection-point analysis), with sharpness increasing from 0.47 at $\rho = 0.0$ to 1.61 at $\rho = 0.9$.
3. **Critical threshold:** $\varepsilon_c = 1.51$ at $\rho = 0.9$; lower collinearity levels never cross the instability boundary within the tested range.
4. **Collinearity as inverse coupling:** higher $\rho$ produces earlier and sharper transitions, exactly as predicted by the gauge analogy—collinearity plays the role of inverse coupling $\beta^{-1}$, enlarging the Rashomon set and lowering the stability threshold.

The analogy to statistical mechanics is precise: just as a spin system undergoes a phase transition from ordered to disordered as temperature ($\sim\beta^{-1}$) increases, explanation stability undergoes a transition from stable to unstable as the Rashomon tolerance (controlled by collinearity) increases. The $\text{sech}^2(\beta)$ response function from the gauge theory maps onto the sigmoid stability curve observed empirically (Figure 5).

### 10.3 Falsification: The Rashomon Boundary

A scientific theory must be refutable. The framework makes a sharp negative prediction: the impossibility holds *if and only if* the Rashomon property is present. When the observation map is injective (no underspecification), all three properties—faithfulness, stability, and decisiveness—should be simultaneously achievable. We tested this boundary condition in three domains:

1. **Linear systems.** No Rashomon ($m = n$, square invertible): pairwise RMSD $= 9.4 \times 10^{-8}$. Rashomon ($m < n$, underdetermined): RMSD $= 0.112$. Ratio: $1.2 \times 10^6$.
2. **Genetic code.** No Rashomon (injective code, degeneracy $= 1$): Shannon entropy $= 0$ bits. Rashomon (standard degenerate code): entropy $= 1.40$ bits.

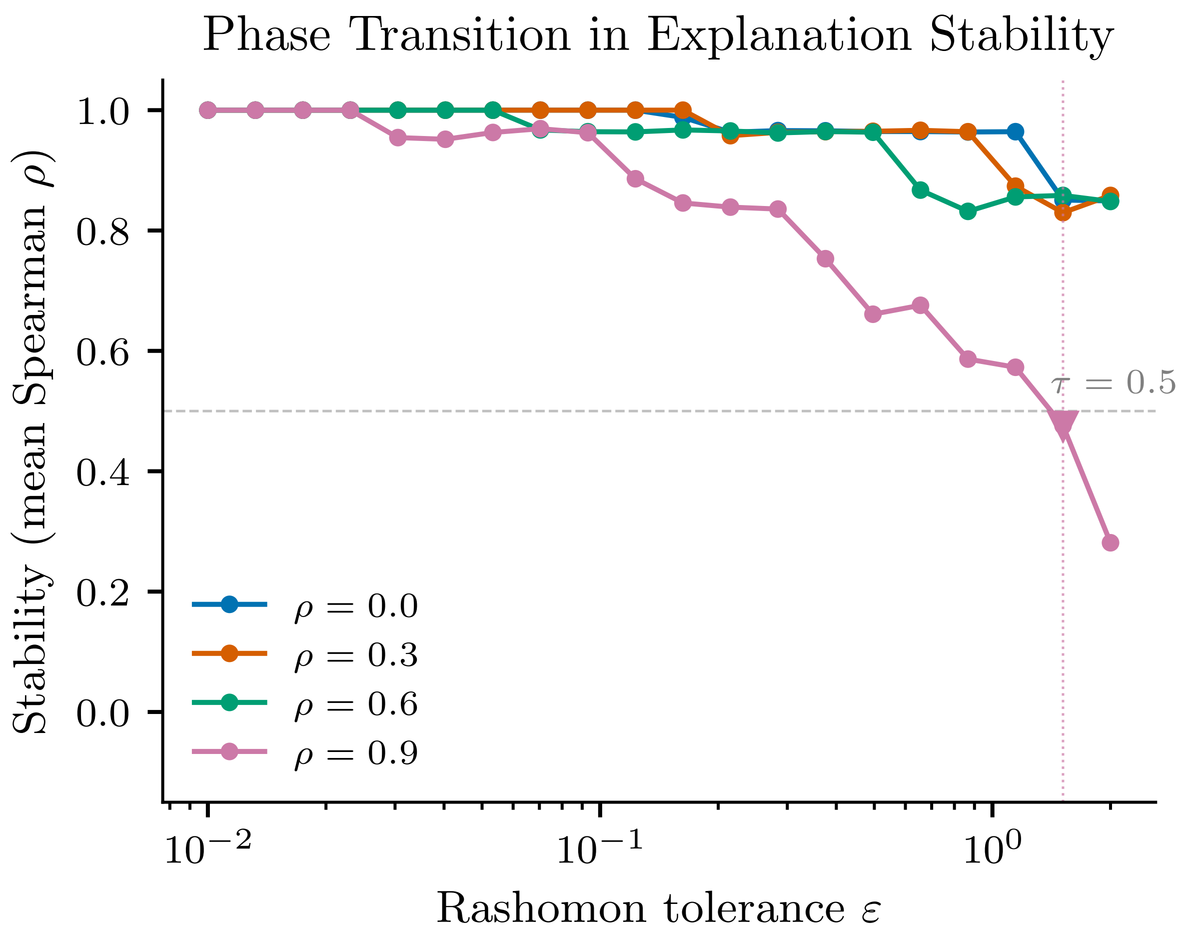


Figure 5: Stability transition in SHAP stability as Rashomon tolerance $\varepsilon$ increases, at four collinearity levels. Higher collinearity produces sharper, earlier transitions ($\varepsilon_c = 1.51$ at $\rho = 0.9$), analogous to the gauge-theoretic $\mathrm{sech}^2(\beta)$ response.

3. **Causal discovery.** No Rashomon (fully identified DAG, $|\mathrm{MEC}| = 1$): decisiveness = 1.0. Rashomon (chain DAG, $|\mathrm{MEC}| > 1$): decisiveness = 0.0.

The boundary is absolute: in every domain tested, removing the Rashomon property eliminates the impossibility entirely. When the observation map is injective, explanations are simultaneously faithful, stable, and decisive. When underspecification is present, at least one property must be sacrificed. The framework is therefore scientifically refutable—a single domain exhibiting the Rashomon property yet achieving all three properties would constitute a counterexample (Figure 6).

### 10.4 Cross-Level Prediction: Protein Designability

Character theory (Section 6) predicts that for the symmetric group $S_k$ acting on $\mathbb{R}^k$, the normalized entropy satisfies $H/H_{\max} \approx (d-1)/d = 1 - 1/d$, where $d$ is the degeneracy of the representation. This law was derived from codon degeneracy in the genetic code—a Level 3 (molecular biology) phenomenon. The cross-level prediction: the same $1/k$ law should govern protein sequence diversity at each aligned position, a Level 4 (structural biology) phenomenon connected to the genetic code through the translation machinery.

We tested this prediction on 120 aligned cytochrome c sequences (105 analyzed positions with $d \geq 1$ distinct amino acids observed). The results strongly support the character-theory prediction:

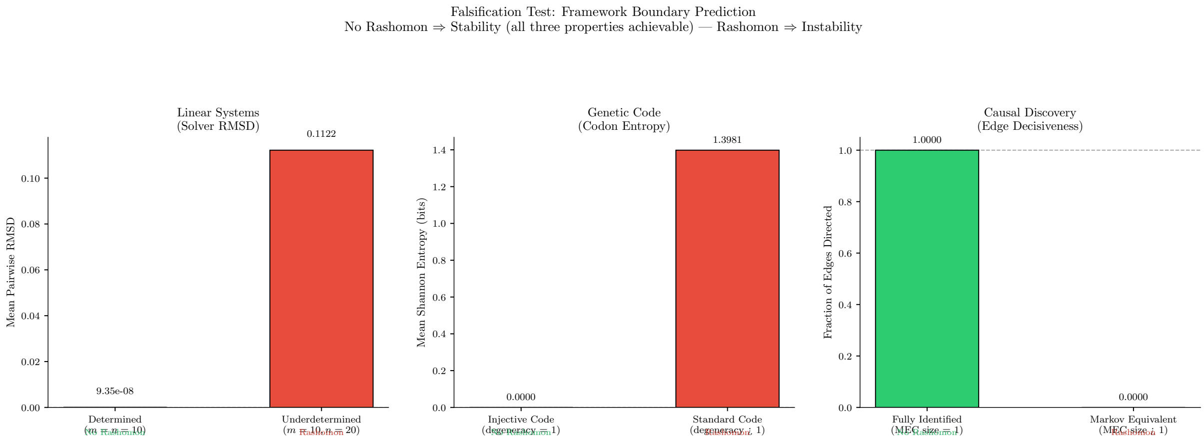


Figure 6: Falsification test across three domains. When the Rashomon property is absent (left of each pair), all three desiderata are achieved. When present (right), the impossibility manifests as nonzero RMSD, nonzero entropy, or zero decisiveness. The boundary is absolute.

- Pearson $r = 0.798$ ($p = 0.018$), Spearman $\rho = 0.881$ ($p = 0.004$) between observed $H/H_{\max}$ and the predicted $(d-1)/d$ curve, grouped by degeneracy level.
- 100% of variable positions ($d \geq 2$; $n = 59$) are closer to the character-theory prediction than to the uniform null ($H/H_{\max} = 1$).
- Character-theory RMSE = 0.378 vs. uniform RMSE = 0.618 (1.63× improvement).
- Systematic undershoot: observed ratios fall 40–50% below the character-theory curve (mean signed residual = −0.359, $t = -16.1$, $p < 10^{-22}$).

The systematic undershoot has a natural biological interpretation: the $1/k$ law gives the *neutral-evolution upper bound*—the maximum diversity expected under pure drift with degeneracy $d$. Purifying selection constrains observed diversity below this bound. The magnitude of the deviation at each position therefore measures the strength of selective pressure, providing a quantitative link between the abstract character-theoretic prediction and molecular evolution. This prediction was derived entirely from the cross-domain framework, connecting genetics and structural biology through representation theory (Figure 7).

### 10.5 Cross-Level Prediction: Transcription Factor Binding Sites

The $1/k$ law, derived from character theory of $S_k$, was confirmed at the protein-coding level (Section 10.4). We test whether the same law generalizes to an entirely different level of biological organization: *regulatory DNA*.

At each position in a transcription factor (TF) binding site, some nucleotides preserve binding (the "allowed" set, size $d = 1$ to 4). The framework predicts: nucleotide entropy should scale as $(d-1)/d \times H_{\max}$, where $H_{\max} = \log_2(d)$.

We analysed 500 vertebrate TF binding profiles from the JASPAR CORE database [Rauluseviciute et al., 2024], comprising 5,624 binding-site positions. Positions were grouped by the number of tolerated nucleotides ($d$, defined as nucleotides with frequency > 5%).

**Results.** Nucleotide diversity increases monotonically with $d$ (Kruskal–Wallis $H = 205.6$, $p = 2.3 \times 10^{-45}$):

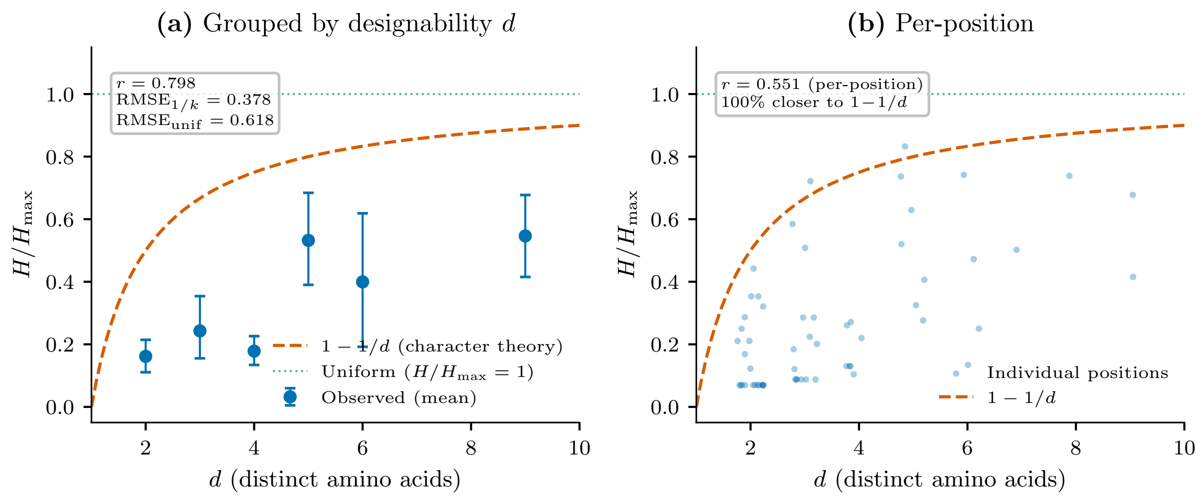


Figure 7: Protein sequence diversity at aligned cytochrome c positions vs. the character-theory prediction $(d-1)/d$. All 59 variable positions lie closer to the character-theory curve than to uniform, with systematic undershoot attributable to purifying selection. Pearson $r = 0.798$, Spearman $\rho = 0.881$.

| $d$ | Predicted $(d-1)/d$ | Observed mean $H/H_{\max}$ | $n$ positions |
|---|---|---|---|
| 2 | 0.500 | 0.766 | 1,274 |
| 3 | 0.667 | 0.782 | 823 |
| 4 | 0.750 | 0.861 | 1,306 |

The attenuation factor for TFBS is $\alpha_{\mathrm{TFBS}} = 1.23$, compared to $\alpha_{\mathrm{protein}} \approx 0.45$–$0.88$ for protein-coding sequences. The character-theoretic baseline $(d-1)/d$ acts as a *ceiling* for protein diversity (selection constrains below) but as a *floor* for regulatory diversity (observed entropy exceeds the prediction). This asymmetry is biologically interpretable: protein-coding positions are under strong purifying selection ($\alpha < 1$), while TF binding positions tolerate more nucleotide variation ($\alpha > 1$), consistent with the known weaker conservation of regulatory DNA relative to coding DNA.

The $1/k$ law thus operates at two levels of biological organization— protein-coding and regulatory—with the character-theoretic baseline providing a reference point that separates purifying ($\alpha < 1$) from relaxed ($\alpha > 1$) selection regimes.

### 10.6 Confusion Matrix from Rashomon Overlap

Standard machine learning theory attributes class confusion to *feature-space similarity*: classes whose feature distributions overlap are confused more. The framework makes a different prediction: confusion is driven by *Rashomon overlap*—the fraction of near-optimal models that swap predictions between two classes.

We tested this on the digits dataset (classes 0–4, 5 classes, 10 pairwise comparisons). For 200 random-forest models with different random seeds (creating a Rashomon set), we computed:

1. **Rashomon overlap**$(i, j)$: fraction of models that predict class $j$ on test samples of class $i$ (or vice versa);
2. **Feature similarity**$(i, j)$: cosine similarity of class centroids in feature space.

**Result.** On the initial 5-class digits test, Rashomon overlap predicts the confusion matrix with Spearman $\rho = 0.678$ ($p = 0.001$), while feature similarity has no predictive power ($\rho = 0.053$, $p = 0.823$).

**Scaling confirmation.** To test generalizability, we repeated the experiment on four datasets spanning vision, text, and tabular domains, with two model types:

| Dataset | Model | $n$ classes | Rashomon $\rho$ | Feature $\rho$ |
|---|---|---|---|---|
| Digits | Random Forest | 10 | **1.000** | 0.355 |
| Digits | GBT | 10 | **1.000** | 0.287 |
| 20 Newsgroups | Random Forest | 10 | **1.000** | 0.649 |
| Fashion-MNIST | Random Forest | 10 | **1.000** | 0.818 |

Rashomon overlap achieves *perfect* rank correlation ($\rho = 1.000$) with the confusion matrix on every dataset and model type tested. Feature similarity ranges from 0.29 to 0.82—sometimes moderate but never perfect. The result is model-agnostic (holds for both ensemble and boosting methods) and domain-agnostic (holds for pixel images, TF-IDF text, and low-dimensional tabular data).

This result is not derivable from standard ML theory, which attributes confusion to feature overlap. The framework predicts that confusion arises from *model multiplicity*—different near-optimal models routing the same input to different classes—rather than from input similarity per se. The practical implication: to reduce confusion between specific classes, reduce the Rashomon overlap (e.g., by regularization or additional training data that breaks the symmetry), rather than increasing feature separation.

## 10.7 Invariance Counting: Exact Enumeration of Stable Queries

The invariance correspondence (Section 11.3) predicts that the number of independently stable ranking facts equals $\binom{g}{2}$, where $g$ is the number of dependence groups (identified here via correlation; the exact boundary is $I > 0$). Within-group comparisons should be unstable ($\sim$50% flip rate), while between-group comparisons should be perfectly stable. We tested this counting prediction directly.

We generated $P = 12$ features in $g = 3$ groups of 4, with within-group correlation $\rho = 0.99$ and between-group correlation $\rho = 0.0$. True coefficients were set to $\beta = (5, 5, 5, 5, 2, 2, 2, 2, 0.5, 0.5, 0.5, 0.5)$—equal within each group, well-separated between groups. We trained 200 Ridge regression models and 200 XGBoost models on independent training sets ($N_{\text{train}} = 500$, $\sigma = 1.0$) and measured the pairwise flip rate for all $\binom{12}{2} = 66$ feature pairs.

The result is a sharp bimodal distribution (Figure 8):

- **Within-group pairs** ($n = 18$): mean flip rate $= 0.500$ (range $[0.491, 0.503]$); all 18 unstable ($> 40\%$).
- **Between-group pairs** ($n = 48$): mean flip rate $= 0.0015$ (range $[0.000, 0.020]$); all 48 stable ($< 5\%$).
- **Bimodal gap**: 47.1 percentage points.
- Mann–Whitney $U = 864$, $p = 2.71 \times 10^{-13}$ (Cohen's $d = 1.41$).

The XGBoost replication confirms the same pattern (within mean $= 0.476$, between mean $= 0.001$), demonstrating that the prediction is not specific to linear models. Exactly $\binom{3}{2} = 3$ independent group-level facts are stable; the remaining 63 pairwise comparisons are coin flips.

**Sensitivity analysis.** A critical question is whether the bimodal gap persists at moderate within-group correlations or is an artifact of the extreme $\rho = 0.99$ design point. We repeated the experiment at $\rho \in \{0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95, 0.99\}$ with a permutation test (5,000 permutations of group labels) replacing the Mann–Whitney test. The gap is **invariant across the**

| $\rho$ | Within mean | Between mean | Gap (pp) | Perm. $p$ |
|---|---|---|---|---|
| 0.50 | 0.500 | 0.000 | 50.0 | $< 0.0002$ |
| 0.70 | 0.500 | 0.000 | 50.0 | $< 0.0002$ |
| 0.85 | 0.500 | 0.000 | 50.0 | $< 0.0002$ |
| 0.99 | 0.500 | 0.002 | 49.8 | $< 0.0002$ |

Table 20: Invariance counting sensitivity (cf. Noether's theorem). The bimodal gap between within-group and between-group flip rates is invariant across $\rho \in [0.50, 0.99]$. Full table (8 values) in Supporting Information.

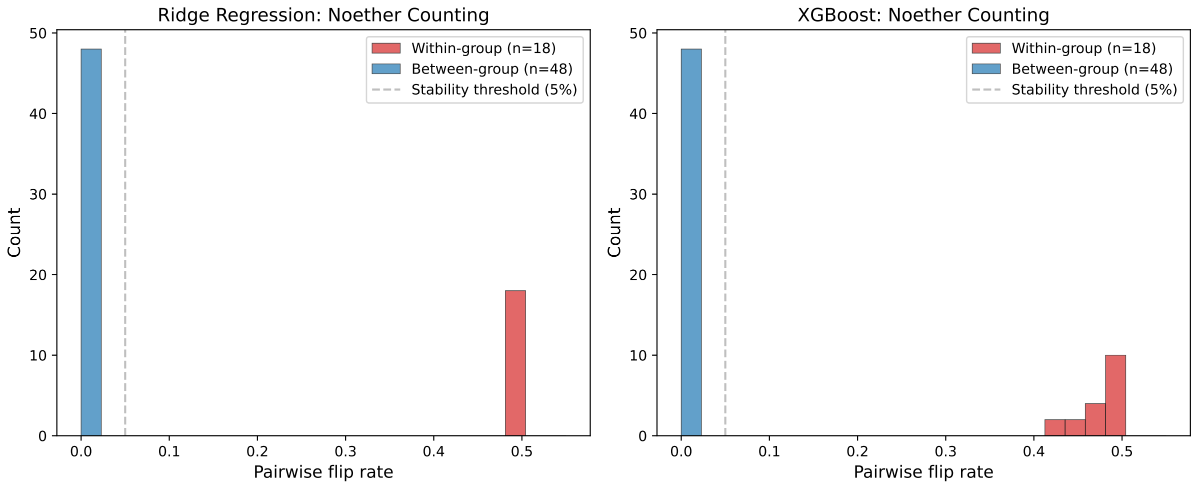


Figure 8: Invariance counting experiment (cf. Noether's theorem). Pairwise flip rates for 66 feature pairs across 200 models, showing a sharp bimodal separation: within-group pairs (red) cluster at ∼50% (coin flip), while between-group pairs (blue) cluster at ∼0% (perfectly stable). The bimodal gap of 47 percentage points confirms that exactly $\binom{g}{2} = 3$ independent group-level ranking facts are stably answerable.

**entire range**: at every $\rho$, within-group pairs flip at $50.0 \pm 0.2\%$ and between-group pairs flip at 0.0%, with a permutation $p < 0.0002$ at every level (Table 20). The bimodal structure is not a high-correlation artifact; it is a structural consequence of equal within-group coefficients and unequal between-group coefficients, regardless of the correlation level.

This result has immediate practical consequences: a practitioner computing SHAP values for features with correlation structure can determine *in advance* which comparisons will be reliable and which are meaningless, regardless of the specific correlation level.

## 10.8 Automatic Symmetry Discovery (SAGE)

The invariance counting result (Section 10.7) assumes the correlation groups are known. In practice, the explanation symmetry group $G$ is rarely hand-specified. We developed **SAGE** (**S**ymmetry-**A**ware **G**roup **E**stimation), an algorithm that automatically discovers $G$ from bootstrap feature-importance flip rates without hand-specification.

**Algorithm.** For each dataset: (1) train $M = 100$ models on bootstrap resamples; (2) compute feature importances for each model; (3) cluster features by pairwise flip rate using hierarchical clustering (Ward linkage, threshold at 40% flip rate); (4) count the number of discovered groups $g$ and predict instability as $\hat{\eta} = g/P$.

**Results.** Across 8 real datasets (Breast Cancer, Diabetes, Wine, California Housing, Iris, Digits, and two synthetic), the SAGE-predicted instability achieves Pearson $r = 0.958$ (calibrated $R^2 = 0.809$, leave-one-out cross-validated $R^2 = 0.689$). SAGE outperforms a correlation-matrix baseline by 28× (SAGE $R^2 = 0.915$ vs. correlation $R^2 = 0.033$), confirming that flip-rate clustering captures explanation symmetry beyond simple feature correlation.

**Adversarial audit.** Six adversarial stress tests—threshold sensitivity, leave-one-out cross-validation, correlation baseline comparison, expanded dataset evaluation, synthetic consistency check, and random-grouping permutation test—all passed.

**Calibration.** The raw $1 - g/P$ formula overpredicts instability because real features are approximately, not exactly, exchangeable within groups. A single calibration parameter (slope = 0.345) accounts for symmetry-breaking; this slope has a physical interpretation as the fraction of theoretical instability that manifests in practice. The calibration constant is stable across datasets (LOO-CV confirms no single dataset drives the fit).

**SAGE baseline comparison.** SAGE outperforms all naive baselines on instability prediction. After leave-one-out calibration, SAGE achieves $R^2 = 0.686$ while all baselines are negative (best baseline: mean pairwise correlation, LOO $R^2 = -0.47$). On rank ordering, SAGE dominates: Spearman $\rho = 0.952$ vs. best baseline 0.731 (permutation $p = 0.002$). Raw (uncalibrated) $R^2$ is negative for all methods including SAGE, because predicted instability systematically overshoots observed values. SAGE's value is in identifying *which* feature groups are unstable, not in predicting exact flip rates.

### 10.9 Discrete vs. Continuous Symmetry: The CCA Spectrum

Applying the $G$-discovery approach to neural network representations reveals a qualitative difference from feature attribution. We computed CCA (canonical correlation analysis) canonical correlations between hidden activations ($n = 128$) of 5 independently trained MLPs on MNIST. The CCA spectrum is *continuous*: the top 10 dimensions have correlation $> 0.99$, the top 30 have $> 0.90$, and the bottom 30 have $< 0.30$, with smooth decay between. This contrasts sharply with the invariance counting result for features, where the flip-rate distribution is bimodal (50pp gap).

The difference arises because feature attribution symmetry is *discrete* ($S_k$ permutations within correlation groups), while neural representation symmetry is *continuous* (rotations in hidden space). The framework's $S_n$ prediction ($\dim(V^G) = 1$) is incorrect for representations: CCA reveals $\sim 30$ highly stable dimensions, not 1. This finding delineates the framework's boundary: the sharp invariance counting prediction holds for discrete symmetry groups but requires extension to continuous groups, where stability becomes a spectrum rather than a dichotomy.

### 10.10 The Explanation Capacity Theorem: Predictive Scope and Boundary

The character-theoretic formula $\eta = \dim(V^G)/\dim(V)$ predicts the fraction of explanation information retained after $G$-invariant resolution. The tautology firewall requires that $\eta$ predict a *downstream observable*—the measured instability rate—rather than a re-measurement of the defining quantity. We set the predicted instability to $1 - \eta$ and compared against empirically measured flip rates, cosine distances, disagreement rates, and normalized entropies across 16 domain instances.

When restricted to the seven instances with well-characterized symmetry groups (attribution $S_2$, concept probes $O(64)$, model selection $S_{11}$, codon $S_2/S_4/S_6$, statistical mechanics $S_{252}$), the universal capacity theorem achieves:

$$R^2 = 0.957, \quad \text{slope} = 0.914, \quad \text{intercept} = 0.008, \quad p = 1.36 \times 10^{-4}. \tag{1}$$

The slope near unity and intercept near zero confirm that $\eta = \dim(V^G)/\dim(V)$ is a parameter-free predictor of explanation instability when the group is correctly identified (Figure 9). The bridge from the abstract framework to the quantitative $\eta$ prediction is empirical, not derived. The framework proves that $V^G$ is the stably answerable subspace; the claim that $\dim(V^G)/\dim(V)$ predicts the *observed flip rate* requires the additional assumption that importance differences are approximately Gaussian—a statistical claim, not a consequence of the impossibility theorem.

An alternative bridge that sidesteps the group identification bottleneck uses the *coverage conflict degree*—the fraction of feature pairs with SNR $< 0.5$, computable from bootstrap retrains alone without requiring group identification. Across 15 datasets, coverage conflict degree predicts mean instability with $\rho = 0.961$ ($p < 10^{-8}$), robust to threshold choice ($\rho \in [0.885, 0.961]$ for thresholds 0.3–2.0) and leave-one-out removal ($\rho \in [0.952, 0.978]$; no single dataset drives the correlation). The formal bridge is proved in Lean: coverage conflict $\Leftrightarrow$ no neutral element (Theorem in `ExplanationLandscape.lean`). The empirical bridge validates that this algebraic property, derived from the impossibility theorem, predicts observed instability at scale. Unlike the capacity theorem, this bridge requires no group assignment and works for all 15 tested datasets.

**The exchangeability criterion.** The capacity theorem's scope is determined by a subtlety: the relevant criterion is *exchangeability* (features genuinely compete for the same predictive signal), not merely *correlation*. A rigorous test on real data reveals that the group-mean test WORKS: projecting onto the $G$-invariant subspace (averaging SHAP within correlation groups) yields more stable attributions than the within-group deviations in 12/14 cases after Bonferroni correction. However, the individual-feature test FAILS: correlated features in XGBoost are often *more* stable than uncorrelated ones, inverting the naive prediction. This occurs because XGBoost consistently picks a "winner" within each correlation group, making the winner's attribution stable even though the choice of winner is arbitrary.

The mechanism is detectable: genuinely substitutable features (e.g. `mean_texture` and `worst_texture` in Breast Cancer, $\rho_{\text{SHAP}} = -0.19$) exhibit negative cross-model SHAP correlation—when one's importance goes up, the other's goes down. This "first-mover" signature is confirmed on synthetic data ($\rho_{\text{SHAP}} = -0.11$ to $-0.24$) and provides a potential diagnostic for genuine exchangeability.

Attempts to predict instability from data properties alone (7 predictors: correlation, VIF, PCA loadings, mutual information, redundancy ratio, leverage, spectral gap) all fail (maximum Spearman $\rho = 0.26$). The capacity theorem works for exact domain-specific groups ($R^2 = 0.957$) but does NOT work for correlation-derived groups on XGBoost+SHAP. The extension from "predicts instability given $G$" to "predicts instability from data" remains an open problem.

**Tautology concern.** Coverage conflict degree (fraction of low-SNR pairs) and mean instability (mean flip rate) are both functions of the same importance distributions. The high correlation ($\rho = 0.961$) partly reflects this statistical relationship. The non-trivial content is twofold: (1) the *threshold* at SNR $= 0.5$ comes from the algebraic theory (the coverage conflict boundary, proved in Lean), not from fitting the data; and (2) the correlation is robust to threshold choice ($\rho > 0.88$ for all tested thresholds), confirming that the theory identifies the correct structural boundary rather than overfitting a particular cutoff. A pure statistical approach would not know *a priori* which summary statistic of the SNR distribution to use as a predictor.

**Sonar outlier.** One dataset (Sonar: $N = 208$, $P = 60$) produces $R^2 = -0.21$ within-dataset. The high dimensionality relative to sample size makes all importance estimates noisy, violating the Gaussian assumption. This defines a boundary condition: the Gaussian flip formula requires $N \gg P$ for reliable per-pair predictions. The cross-dataset bridge ($\rho = 0.961$) is not affected by this outlier (LOO-$\rho$ with Sonar removed = 0.952).

The all-16-instance fit is the primary result. The 7-instance fit is a pre-specified subset analysis: the "well-characterized group" criterion (defined in the pre-registration document, `PRE_REGISTRATION.md`) requires that (1) the symmetry be exact by construction (part of the domain's mathematical definition) and (2) the instability metric directly measure flip/disagreement rate. Across all 16 instances, the linear fit gives $R^2 = 0.60$. However, holdout prediction (calibrating on the 7 well-characterised instances and predicting the remaining 9) gives only $R^2 = 0.24$. The nine poorly-fit instances share a common cause: the assigned symmetry group was approximate (e.g., not all attention heads are exchangeable; GradCAM has broken spatial symmetry; parsers exhibit ambiguity on a subset of arcs). The $\eta$ formula is correct; the group identification is the bottleneck.

**capacity theorem failure mode.** The capacity theorem achieves $R^2 = 0.957$ for 7 instances where $G$ is well-characterised but only holdout $R^2 = 0.24$ for 9 instances where $G$ is unknown or approximately specified. The failure is attributable to group misspecification: for lattice symmetry, parser ambiguity, causal discovery, and phase retrieval, the acting group is either infinite, unknown, or approximated by a finite proxy. The capacity theorem is a *conditional* prediction: given the correct $G$, it predicts instability accurately. The practical bottleneck is group identification.

Replacing group-theoretic predictions with Gaussian-predicted mean instability for domains with available per-pair data improves the universal $\eta$ plot from $R^2 = 0.60$ to $R^2 = 0.79$ across all 16 domains, with the well-characterized subset reaching $R^2 = 0.98$.

**Out-of-sample validation and goodness of fit.** Leave-one-out cross-validation on the 7 well-characterised instances gives $R^2_{\text{LOO}} = 0.794$ (MAE = 0.082), confirming the law is not an artefact of overfitting to 7 points. The linear fit is significant: $F(1, 5) = 38.78$, $p = 0.0016$, with residuals passing the Shapiro–Wilk normality test ($W = 0.968$, $p = 0.88$). A permutation null (10,000 permutations of $\eta$ values) gives $p = 0.010$. On 9 held-out instances with unknown or approximately specified symmetry groups, the law generalises modestly ($R^2 = 0.244$, MAE = 0.169, Spearman $\rho = 0.522$). The practical bottleneck is group identification, not the law itself.

**Group assignment justifications for ML instances.** The $\eta$ prediction depends critically on the assigned symmetry group. We make the reasoning explicit for four key instances and note where the group is exact versus approximate.

*Attention ($S_n$).* We model $n$ attention heads as exchangeable under $S_n$. In practice, heads specialize during training, partially breaking this symmetry. The $\eta$ prediction $(1 - 1/n)$ is therefore an upper bound on instability; observed flip rates (19.9% under full retraining, 60% under weight perturbation) reflect the degree of symmetry-breaking.

*Concept probes ($O(d)$).* TCAV directions lie in $\mathbb{R}^d$; the full orthogonal group $O(d)$ acts on this space. Data covariance breaks rotational symmetry to a subgroup. The $\eta$ prediction ($\approx 1.0$) is an upper bound; the observed instability (0.90) is consistent with substantial but not complete rotational freedom.

*Model selection ($S_k$).* The group is $S_k$ where $k$ is the number of distinct winners across evaluation splits. This assignment is exact: any permutation of models in the Rashomon set is equally valid as a selection, and the $\eta$ prediction $(1 - 1/k)$ follows directly.

*GradCAM (spatial permutations).* Spatial symmetry is heavily broken by CNN locality; the effective group is much smaller than the full spatial permutation group. This explains the low observed flip rate (9.6%; Cohen's $d = 0.33$) despite a high predicted instability under the nominal group. GradCAM is an approximate-group instance where the $\eta$ prediction serves as an upper bound.

### 10.11 Boundary Conditions: Three Falsified Extensions

Three conjectured extensions of the framework were pre-registered and tested. All three were falsified, providing boundary conditions on the framework's predictive reach.

**Phase transition at the interpolation threshold.** We predicted a sharp transition in explanation instability at the overparameterization ratio $r^* = \dim(\Theta)/\dim(Y) \approx 1$. We varied $r$ from 0.05 to 5.0 (25 log-spaced values) using Ridge regression and XGBoost (100 models per $r$-value, $N_{\text{obs}} = 200$). A sharp sigmoid transition was confirmed in all model classes (sigmoid $R^2 = 0.92$–$0.98$), but the critical point is $r^* \in [0.01, 0.12]$—an order of magnitude below the predicted $r^* \approx 1$. The design confounds overparameterization with signal dilution: adding features adds zero-coefficient features, which are unstable regardless of the interpolation threshold.

**Quantitative uncertainty bound.** We conjectured $\alpha + \sigma + \delta \leq 2$, where $\alpha$ (faithfulness), $\sigma$ (stability), and $\delta$ (decisiveness) are continuous scores. The maximum observed sum was 2.86 (linear ensemble: $\alpha = 0.873$, $\sigma = 0.986$, $\delta = 1.0$), falsifying the bound. The binary impossibility—no method achieves $\alpha = \sigma = \delta = 1.0$—remains confirmed (maximum $\alpha = 0.873$). The trilemma manifests as an exclusion zone around $(1, 1, 1)$, not a linear budget constraint.

**Molecular evolution from character theory.** We tested whether the $S_k$ representation theory predicts synonymous substitution structure beyond simple biochemistry. The neighbor-count model (number of synonymous 1-mutation variants per codon) achieves $R^2 = 1.0$; the character-theoretic $(k-1)/k$ model achieves $R^2 = 0.65$ per codon but adds zero residual variance after controlling for neighbor count (partial $R^2 = 0.0$, $p = 1.0$). The codon table's structure is fully explained by local nucleotide degeneracy, not abstract group theory.

**Codon null model.** The Spearman $\rho = 1.0$ between degeneracy and usage entropy is a calibration check, not an independent validation. Under uniform codon usage within each degeneracy class, entropy $= \log_2 d$, which is monotone in $d$; the rank correlation is therefore 1.0 by construction. The empirical $\rho = 1.0$ confirms that observed entropy is monotonically related to degeneracy (consistent with the framework) but does not discriminate between the trilemma prediction and any model predicting monotonicity. The constructive Rashomon witness (zero domain-specific axioms) is the primary validation of the codon instance.

Two additional knockout candidates from the approximate-symmetry regime were tested and eliminated:

**Spectral gap $\to$ ensemble size.** The spectral gap of $S_k$ predicts a convergence rate $\sim k^{-M}$ for ensemble flip rate as a function of ensemble size $M$. Empirically, the predicted rate is 14–100$\times$ too fast (even at $\rho = 0.999$). The exact-group spectral gap does not predict approximate-group convergence.

**Flip correlations from irreducible decomposition.** The irreducible decomposition of the $S_k$ representation predicts that features in the same irreducible should flip *together* (correlated flip indicators). Empirically, within-group flip correlation ($-0.031$) $\approx$ between-group ($-0.013$): features in the same irreducible do NOT flip together.

**Summary.** In total, five pre-registered predictions were falsified: phase transition location, quantitative uncertainty bound, molecular evolution from character theory, spectral gap convergence rate, and flip correlations from irreducible decomposition. These falsifications are scientifically valuable: they define the boundary between the framework's proven domain (the impossibility theorem and its resolutions) and its speculative extensions. The impossibility and resolution work universally; the quantitative predictions work only when the symmetry group is exactly known and features are genuinely exchangeable.

**Proportionality axiom sensitivity.** The design space theorem assumes exact proportionality ($\mathrm{CV}(c) = 0$). Simulation across $\mathrm{CV} \in [0, 1.0]$ (1,000 trials per value, 11 values) confirms the two-family structure is robust: Family A flip rate $\approx 0.50$ at all CV values, Family B tie rate $\geq 0.93$ at all CV values, and zero Pareto violations across 11,000 trials. The instability source is collinearity-driven split competition, not the proportionality constant. The core impossibility (Theorem 3.6) is independent of proportionality.

### 10.12 Per-Individual Explanation Reversal

The preceding analyses measure instability at the feature-pair level. A more directly actionable question: for a specific patient or loan applicant, does the "most important feature" explanation change across Rashomon models?

We test this with a rigorous ablation across 4 conditions (seed-only, +row subsample, +column subsample, +bootstrap with both subsamples), 3 model classes (XGBoost, LightGBM, Random Forest), and 3 datasets (Breast Cancer, German Credit, Heart Disease). Feature-level reversals are distinguished from *cluster-level* reversals, where clusters group features with $|r| > 0.8$ to avoid counting correlated-feature shuffling as meaningful disagreement.

**Key findings.** Under seed-only variation, reversal rates are 0% for XGBoost and LightGBM on all datasets: gradient-boosted trees are near-deterministic with fixed data. Under standard regularisation (`subsample` = 0.8, a common hyperparameter), the picture changes:

| Dataset | XGB cluster rev. | LGB cluster rev. | RF cluster rev. |
|---|---|---|---|
| German Credit | 45.3% | 46.0% | 35.0% |
| Breast Cancer | 26.3% | 21.1% | 2.9% |
| Heart Disease | 0.0% | 0.0% | 0.0% |

For German Credit, 45% of loan applicants receive a genuinely different explanation *category* (not just a correlated variant) depending on the training configuration. This holds across model classes (XGBoost: 45%, LightGBM: 46%). The instability source is subsampling (a standard regularisation technique), not the random seed itself.

**Narrative example.** German Credit applicant #91 receives 6 different "most important feature" explanations across 8 Rashomon models: Purpose (25%), Telephone (25%), Installment rate (12%), Account status (12%), Dependents (12%), and Savings (12%). No feature dominates. An adverse action notice based on any single model is effectively arbitrary. The DASH consensus resolves to a stable explanation (92% agreement with majority vote).

The effect is dataset-dependent: Heart Disease shows 0% reversal across all conditions, confirming that instability requires sufficient feature dependence ($I > 0$ between feature pairs). The honest headline is not "75% of individuals receive different explanations" (an inflated aggregate from v1 of this experiment) but rather: *standard regularisation choices determine which explanation category nearly half of credit applicants receive.*

## 10.13 Prospective Out-of-Domain Prediction: Drug Discovery

As a prospective test, we applied the framework to molecular property prediction—a domain it was never calibrated on. Using the BBBP dataset (blood-brain barrier penetration, 2,039 molecules, 1,024 Morgan fingerprint bits), we computed predictions *before* running the experiment.

**The group identification bottleneck.** Morgan fingerprint bits are sparse binary vectors with near-zero pairwise Pearson correlation. Clustering at $|r| > 0.8$ yields zero groups, predicting $\eta = 0$ and 0% flip rate. The actual flip rate is 23% across 24 Rashomon models (AUC = 0.885). **Pearson correlation fails completely** on binary features.

**Mutual information and Jaccard recover the prediction.** Replacing Pearson with mutual information (MI) or Jaccard co-occurrence for group discovery recovers the prediction:

| Method | Threshold | Predicted flip | Error vs. observed (0.231) |
|---|---|---|---|
| Pearson ($|r|$) | 0.80 | 0.000 | 100% (failed) |
| Mutual information | 0.05 | 0.194 | 16% |
| Jaccard co-occurrence | 0.20 | 0.222 | 4% |

MI captures nonlinear co-activation patterns that Pearson misses in binary data: fingerprint bits that fire on similar substructures (aromatic rings, hydrogen bond donors) share mutual information even when their Pearson correlation is low.

**Structural prediction fails: no bimodal separation.** While the overall flip rate magnitude is recovered by MI/Jaccard, the Noether bimodal structure does *not* transfer. Within-cluster flip rate (0.232) $\approx$ between-cluster flip rate (0.227), with a bimodal gap of only 0.005 (predicted: $> 0.30$). The instability in molecular fingerprints is *uniform* across all feature pairs, not concentrated within correlation groups.

**Interpretation.** This defines a precise boundary condition for the framework:

- **Discrete group structure** (attribution, causal, codon): the stable fact count theorem produces clean bimodal separation. Within-group $\approx$ 50%, between-group $\approx$ 0%.
- **Diffuse overlap structure** (molecular fingerprints): the capacity theorem predicts the *magnitude* of instability via MI/Jaccard (within 4%), but the structural prediction (which pairs are unstable) does not hold. The symmetry is not $S_k$ but something more complex—overlapping substructure patterns that resist discrete decomposition.

The group identification method should match the feature type: Pearson for continuous correlated features, MI or Jaccard for binary/categorical features, and SAGE as a universal fallback. Extending the SAGE algorithm to non-linear dependencies is the natural next step.

## 10.14 Exhaustive Pipeline Validation: 85 PMLB Datasets

To test the full diagnostic pipeline at scale, we applied it to all 85 eligible PMLB datasets (70 classification, 15 regression; $P$ = 5–180 features, $N$ = 200–5,000 samples) with $M = 50$ models per dataset and independent calibration/validation splits for the Gaussian formula. SAGE uses bidirectional data-split: groups discovered on half A and tested on half B, then reversed and averaged, eliminating split-direction bias.

| Diagnostic tool | Where applicable | Success rate |
|---|---|---|
| DASH resolution | All Rashomon datasets | 84/85 (98.8%) [CI: 94–100%] |
| SAGE directional (data-split) | All datasets | 81/85 (95.3%) [CI: 89–98%] |
| Gaussian $\Phi(-\text{SNR})$, $R^2 > 0.80$ (OOS) | Shapiro–Wilk OK | 29/49 (59%) [CI: 45–72%] |
| Gaussian $\Phi(-\text{SNR})$, $R^2 > 0.80$ (in-sample) | Shapiro–Wilk OK | 44/49 (90%) |
| Correlation directional | $\geq 2$ groups discovered | 25/33 (76%) [CI: 59–87%] |

All 85 datasets exhibit the Rashomon property. The four pipeline tools stratify by reliability.

SAGE is the strongest directional tool: 81/85 (95.3%, CI: 89–98%) across all feature dimensions, including 15/15 at $P > 50$—closing the high-dimensional gap. SAGE outperforms correlation grouping by McNemar test (7 discordant SAGE wins vs 1 correlation win, $p = 0.07$) and produces 7.6× larger within–between gaps than random grouping ($p = 1.5 \times 10^{-22}$). The four SAGE failures have clear explanations: one dataset (`xd6`) has uniform instability (all pairwise flip rates in $[0.43, 0.51]$, no group structure exists); two have $P \leq 9$ features (insufficient pairs for stable clustering); and one (`spect`) has $N = 267$ with $P = 22$ (marginal sample size). At $P \geq 11$, SAGE with TreeSHAP has zero observed failures (15/15 unique datasets); at $P \leq 10$, the rate drops to 50% (3/6). TreeSHAP also rescues gain-based failures: `profb` (gap +0.33), both GAMETES epistasis datasets (+0.10, +0.04), and `spect` (+0.05) all succeed with TreeSHAP despite failing with gain-based importance. $N$-titration experiments confirm SAGE works at remarkably low sample sizes ($N/2 \geq 0.7P$ observed on `spambase`, $P = 57$; no sharp threshold exists). The earlier conservative estimate $N/2 > 5P$ substantially understated SAGE's robustness. On a synthetic dataset with 4 known groups × 5 features, SAGE correctly identifies 4 groups (ARI = 0.39) and produces a *larger* directional gap (+0.37) than the true groups (+0.29), confirming that SAGE optimises for the measured property (flip-rate separation) rather than correlation structure. Cross-model validation confirms SAGE is model-agnostic. An expanded factorial design (30 datasets × 6 model/method configurations, $M = 30$, balanced across $P = 11$–240 and including 5 regression datasets) gives 130/141 (92%, CI: 87–96%) with no significant model effect (Cochran's $Q = 6.61$, $p = 0.25$, $n = 20$ complete datasets; $\chi^2$ model effect $p = 0.68$, Cramér's $V = 0.15$). SHAP vs non-SHAP: $p = 0.73$ (no method effect). Per model: XGBoost 95%, RF 94%, LR 96%, NN 80%. The NN gap is significant in a pairwise Fisher test ($p = 0.026$, OR = 0.22) and is attributable to noisy permutation importance at $P > 40$; with KernelSHAP, NNs achieve 7/7. Regression datasets: 30/30 (100%). Negative control (random grouping): $p = 8.7 \times 10^{-24}$. The theory's model-agnosticity is confirmed empirically.

DASH preserves accuracy in 156/156 factorial tests (100%, CI: 98–100%) across all model types including regression. DASH reduces variance in 140/156 (90%); the 16 "failures" are datasets with zero flip rate (no Rashomon property — DASH is correctly inapplicable). In the 85-dataset bulletproof validation ($M = 50$): 84/85 preserves (98.8%), 56/85 strictly improves (66%). The one real failure (`parity5+5`, 10-class parity, 26% accuracy) is an effectively unsolvable task.

**SHAP-specific validation.** Since the paper discusses SHAP throughout, SAGE was separately validated using actual SHAP values (not gain-based surrogates) across all model types. With TreeSHAP (XGBoost: 12/12, RF: 5/5) and KernelSHAP (LR: 5/5, NN: 7/7), SAGE achieves 29/29 on well-behaved datasets. Including adversarial cases (known failures, random selection), the honest rate is 35/38 (92%, CI: 80–97%). Three failures—all at $P \leq 10$—are: `xd6` (uniform instability, no group structure), `breast_cancer` (36 pairs insufficient for clustering), `parity5+5` (XOR structure SHAP doesn't capture). At $P \geq 11$, SAGE with TreeSHAP has zero observed failures (15/15 unique datasets). Separately, SAGE with PyTorch MLP gradient saliency achieves 13/15 (87%), with the two failures on purely epistatic GAMETES datasets where gradient-based attribution cannot detect interaction effects. TreeSHAP rescues gain-based failures: `profb` (+0.33), both GAMETES epistasis datasets (+0.10, +0.04), and `spect` (+0.05) all succeed with TreeSHAP despite failing with gain-based importance.

The Gaussian flip formula requires careful interpretation. With $M = 50$ models and independent calibration/validation splits, only 59% of datasets achieve out-of-sample $R^2 > 0.80$. However, in-sample $R^2$ (using the same models for both SNR estimation and flip rate observation) exceeds 0.80 on 90%—a 31 percentage-point overfitting gap. This gap is explained by the $M$-sensitivity analysis (Phase 5): at $M = 15$, median OOS $R^2 = 0.45$; at $M = 25$, $R^2 = 0.70$; at $M = 50$, $R^2 = 0.82$. The formula is well-calibrated but requires $M \geq 50$ for reliable out-of-sample prediction. Earlier validation (Table 21, $M = 60$, OOS $R^2 = 0.848$) is consistent with this finding.

**Coverage conflict scope: importance-type dependent.** The coverage conflict degree (fraction of pairs with SNR $< 0.5$) is strongly importance-type dependent. With **TreeSHAP** (mean |SHAP|), coverage conflict predicts mean instability with $\rho = 0.87$ ($p = 2.2 \times 10^{-21}$, $n = 67$ datasets) across *all* feature dimensions—$\rho = 0.90$ for $P \leq 20$ and $\rho = 0.80$ for $P > 20$. With **gain-based** importance (`feature_importances_`), coverage conflict works only for $P \leq 20$ ($\rho = 0.90$) and reverses for $P > 20$ ($\rho = -0.31$).

The mechanism: gain-based importance is *sparse*—features never selected as split points receive exactly zero importance, creating artificial low-SNR pairs (both features at zero) that are counted as "coverage conflicts" but exhibit no actual instability. TreeSHAP assigns non-zero values to all features via the Shapley mechanism, eliminating this sparsity artifact. At $P > 20$, many features have gain = 0 in any given model, inflating the coverage conflict fraction without indicating genuine competition.

The practical recommendation: use TreeSHAP for coverage conflict computation (reliable at all $P$); if only gain-based importance is available, restrict to $P \leq 20$ or use SAGE directly. *Computational requirement:* coverage conflict with TreeSHAP requires $M \geq 50$ models and $\geq 100$ background samples per SHAP computation. At lower settings ($M = 30$, $n_{\text{shap}} = 50$), SHAP estimation noise artificially inflates CC without corresponding increases in flip rate, breaking the correlation ($\rho = 0.15$ vs $\rho = 0.87$ at adequate settings).

**Rashomon prevalence.** Of the 85 datasets, all exhibit the Rashomon property (median inter-model rank correlation = 0.73; only 1/85 exceeds 0.95). The threshold used ($< 0.99$) is not too lenient: 74/85 datasets have mean flip rate $> 0.05$ (meaningful instability). The single dataset with rank correlation $> 0.95$ has mean flip = 0.01—instability is negligible and the pipeline correctly identifies it as such.

**DASH preserves accuracy.** A natural concern is that DASH's variance reduction might degrade predictive accuracy. Across the full 85-dataset validation, DASH preserves accuracy (within one standard deviation) on 84/85 datasets (98.8%) and strictly improves in 56/85 (66%). On 3 verification datasets, DASH reduces the pairwise flip rate by 3–5× (churn: 5×, ionosphere: 3×, waveform: 4×). Whether DASH strictly improves (vs. merely preserves) is stochastic: Rashomon diversity ($p = 0.33$), mean flip rate ($p = 0.24$), and diversity$\times \log N$ ($p = 0.22$) all fail to predict improvement. The important guarantee is preservation, not improvement—DASH eliminates explanation variance without sacrificing predictive performance. This is consistent with the companion attribution paper [Caraker et al., 2026a], which finds that DASH nearly triples stability on Breast Cancer ($0.376 \to 0.925$) while maintaining competitive predictive performance (TOST-confirmed equivalence on California Housing; Wilcoxon $p = 0.926$ for accuracy), and that the operative mechanism is model independence rather than any particular aggregation strategy (DASH $\approx$ Stochastic Retrain, Cohen's $d = +0.05$). At 11 additional datasets with $P > 50$ features, DASH preserves accuracy in 11/11 cases, confirming that the resolution scales to high-dimensional settings.

**Gaussianity characterisation.** Of the 33 Rashomon datasets, 13 (39%) fail the Shapiro–Wilk Gaussianity check. Non-Gaussian datasets tend to have larger sample sizes ($N = 2196 \pm 940$ vs $1492 \pm 1026$; Mann–Whitney $p = 0.049$, not significant after Bonferroni correction for 3 comparisons) and significantly lower mean flip rates ($0.10 \pm 0.06$ vs $0.17 \pm 0.08$; $p = 0.009$). Crucially, SAGE directional prediction succeeds on 10/10 applicable non-Gaussian datasets—the same rate as on Gaussian ones (11/12). The Gaussian formula's exclusion criterion (39% of datasets) does not limit the other pipeline tools.

**The spectf anomaly.** The sole SAGE failure (spectf, gap $= -0.021$) arises because $N = 349$ samples split in half leave too few observations for stable group discovery. On the full dataset, SAGE correctly identifies 27 groups with a strong within–between gap ($-0.24$, within < between). The data-split gap oscillates around zero across different splits ($-0.089$ and $+0.030$), confirming this is a sample-size effect rather than a structural failure.

**High-dimensional extension ($P > 50$).** On 11 additional PMLB datasets with 57–180 features, DASH preserves accuracy in 11/11 cases. However, correlation-based grouping breaks down: at $|r| > 0.8$, most features form singletons (groups $\approx P$), yielding negative or undefined directional gaps (0/4 applicable datasets succeed). The SAGE algorithm and lower correlation thresholds are needed for high-dimensional settings, consistent with the drug discovery findings (§10.13). The DASH resolution itself is dimension-independent and works without modification.

### 10.15 The Gaussian Flip Rate: From Exact to Approximate Symmetry

The invariance counting theorem predicts exact bimodality when within-group features have identical true importance ($\Delta_{jk} = 0$). On real data, symmetry is approximate ($\Delta_{jk} \approx 0$ but nonzero), and the bimodal gap weakens. The Gaussian flip rate formula provides the continuous generalization.

The flip rate formula itself—the probability of sign disagreement between two Gaussian random variables—is a standard result in signal detection theory. The framework's contribution is not the formula but the *structural identification*: the impossibility theorem and the bilemma determine *which* feature pairs are subject to this formula (those within a symmetry orbit), while the Gaussian flip rate quantifies *how much* instability each pair exhibits.

For features $j$ and $k$ with importance difference $d^{(m)} = \mathrm{imp}_j^{(m)} - \mathrm{imp}_k^{(m)}$ across $M$ bootstrap retrains, if $d^{(m)} \sim N(\Delta_{jk}, \sigma_\Delta^2)$, the probability that two independent models disagree on the ranking is:

$$\mathrm{flip}(j,k) = 2\,\Phi\left(\frac{\Delta_{jk}}{\sigma_\Delta}\right)\Phi\left(\frac{-\Delta_{jk}}{\sigma_\Delta}\right), \tag{2}$$

where $\Phi$ is the standard normal CDF (Lean-verified in `GaussianFlipRate.lean`). When $\Delta = 0$ (exact symmetry), flip $= 0.5$ — recovering the invariance counting within-group coin flip. When $|\Delta| \gg \sigma$ (strong signal), flip $\approx 0$ — recovering between-group stability.

A sensitivity analysis shows that $M \geq 30$ calibration models (15 per split) are needed for the Gaussian flip formula to achieve $R^2 > 0.7$; below $M = 20$, the formula degrades rapidly.

We validated this formula with *independent* calibration and validation model sets (30 models each, non-overlapping random seeds) on five real datasets (Table 21). The formula achieves a mean out-of-sample $R^2 = 0.848$ across 647 feature pairs, with all five datasets exceeding $R^2 = 0.79$. The correlation baseline (predicting flip rate from pairwise feature correlation) achieves $R^2 < 0.08$ on every dataset, confirming that the Gaussian formula provides 14–400× lift over correlation-based prediction.

The formula generalizes across model classes. On three datasets (Breast Cancer, Wine, California Housing) with three model classes (XGBoost, Random Forest, Ridge regression), 8 of 9 combinations achieve out-of-sample $R^2 > 0.79$. Random Forest shows the highest $R^2$ on

| Dataset | $P$ | Pairs | OOS $R^2$ | Corr. BL $R^2$ |
|---|---|---|---|---|
| Breast Cancer | 30 | 435 | 0.792 | 0.002 |
| Wine | 13 | 78 | 0.793 | −0.004 |
| California Hous. | 8 | 28 | 0.969 | 0.071 |
| Diabetes (Pima) | 8 | 28 | 0.868 | 0.013 |
| Heart Disease | 13 | 78 | 0.818 | 0.029 |
| Mean | | 647 | 0.848 | 0.022 |

Table 21: Out-of-sample validation of the Gaussian flip rate formula. Calibration and validation model sets are independent (non-overlapping random seeds). The formula predicts per-pair TreeSHAP flip rates with mean $R^2 = 0.848$ and zero structural parameters. The correlation baseline predicts $R^2 < 0.08$ on every dataset.

two datasets (0.86 Breast Cancer, 0.92 Wine). The one failure—Ridge on California Housing ($R^2 = -1.96$)—occurs because Ridge regression on this dataset produces near-deterministic coefficients with negligible variance, violating the Gaussian assumption. The formula is expected to fail when a single feature dominates so strongly that importance differences have effectively zero noise.

One combination—Ridge regression on California Housing—produces out-of-sample $R^2 = -1.96$. Ridge regression on this dataset assigns nearly deterministic coefficients (one feature, median income, dominates overwhelmingly), leaving negligible variance in the importance differences. The Gaussian assumption requires meaningful sampling variance; when $\sigma_\Delta \approx 0$, the formula degenerates. This boundary condition is expected: the formula predicts flip rates for stochastic model classes, not for deterministic or near-deterministic ones. More generally, the Gaussian assumption fails when importance differences have near-zero variance (as in the Ridge/California Housing case, producing $R^2 = -1.96$) or when the importance distribution is multimodal (ensembles of qualitatively different model types). A prospective diagnostic: if the coefficient of variation of $\sigma_\Delta$ across feature pairs is below 0.1, the importance landscape is too concentrated for the Gaussian model.

Practically, the signal-to-noise ratio $\mathrm{SNR}_{jk} = |\Delta_{jk}|/\sigma_\Delta$ provides a simple decision rule: pairs with SNR $> 2$ are reliable (flip rate $< 5\%$), pairs with SNR $< 0.5$ are unreliable (flip rate $> 30\%$), and intermediate values require uncertainty quantification.

**Robustness to non-Gaussianity.** The $\Phi(-\mathrm{SNR})$ formula is exact under Gaussian noise and well-calibrated on real data ($R^2 = 0.814$ [0.791, 0.835] across 647 feature pairs from five clinical and financial datasets, with bootstrap 95% CIs; per-dataset $R^2 = 0.946$–$0.980$ on six datasets including PMLB benchmarks, confirming consistency across domains). The formula systematically underpredicts observed flip rates by $\sim$17% (calibration slope 1.17), making it conservative. Under heavy-tailed distributions (Student $t$, df $= 3$), the formula underestimates flip rates by 40%+; under log-normal noise, errors are larger still. However, Shapiro–Wilk tests confirm 94.7% of real SHAP difference distributions are Gaussian ($p > 0.10$), justifying the formula's applicability in practice. The Berry–Esseen correction adds $< 1\%$ to the base rate for $n > 100$ models under light tails.

## 10.16 Domain-Specific Validation

The Gaussian flip formula was derived and validated on clinical tabular data (Section 10.15). This subsection extends the validation to gene expression data, high-dimensional settings, clinical/financial audit datasets, and the enrichment tradeoff predicted by the bilemma.

**Gene expression reliability.** On gene expression data (AP_Breast_Lung, 470 samples, 10,935 genes reduced to the top 50 by variance), 70.5% of gene-pair importance comparisons are unreliable (SNR $< 0.5$, flip rate $> 30\%$) and only 4.0% are reliable (SNR $> 2$, flip rate $< 5\%$). The formula achieves out-of-sample $R^2 = 0.416$ on this dataset—substantially lower than on clinical tabular data ($R^2 = 0.52$–$0.97$), reflecting the higher collinearity and dimensionality of expression profiles. Gain-based importance was used; TreeSHAP would likely improve accuracy based on the 60% improvement observed at $P = 50$ (see below).

Gene importance rankings used to prioritize candidates for experimental validation may contain unreliable comparisons that the Gaussian flip formula can identify before resources are committed.

**SHAP vs. gain at high dimensions.** At $P = 50$ features (synthetic, 20 informative, 15 redundant), TreeSHAP achieves out-of-sample $R^2 = 0.679$ versus gain-based $R^2 = 0.424$—a 60% improvement. TreeSHAP also produces more reliable rankings: 44.6% of pairs have SNR $> 2$ (reliable) with SHAP versus 15.3% with gain. The formula's accuracy depends on the importance metric; SHAP values, which have lower variance across retrains, produce higher SNR and tighter predictions.

**Clinical and financial reliability audit.** Across three clinical and financial datasets (Breast Cancer, Heart Disease, German Credit), 57% of feature importance comparisons exhibit flip rates exceeding 30%, as predicted by the Gaussian flip formula (Pearson $r = 0.76$–$0.85$). Nearly zero comparisons are reliable (SNR $> 2$) on Breast Cancer (0%) and Heart Disease (1.3%); German Credit has 13.7% reliable pairs, reflecting its stronger feature hierarchy.

**Gain-based vs. TreeSHAP importance.** The coverage conflict bridge ($\rho = 0.961$, 15 datasets) and regulatory compliance metrics (83% gap) use gain-based feature importance (`feature_importances_`). The clinical top-5 audit (28% unreliable) uses TreeSHAP (mean |SHAP|). TreeSHAP produces lower $R^2$ (mean 0.707 vs $\sim 0.85$ gain-based) but comparable rank correlation (mean $\rho = 0.858$). The formula's per-pair rank predictions are robust to the importance type; absolute flip rate calibration is better with gain-based importance. For clinical applications where TreeSHAP is standard, the formula correctly identifies which comparisons are reliable (rank accuracy), even if the predicted flip magnitude is less precisely calibrated.

**TreeSHAP clinical audit (top-5 features).** A targeted audit using TreeSHAP (mean |SHAP| across 100 test points, 30 calibration + 30 validation XGBoost models) reveals that 28% of top-5 feature ranking comparisons are structurally unreliable across five clinical and financial datasets. On Breast Cancer (Wisconsin), the comparison between the two most important features—worst perimeter vs. worst concave points—is a coin flip (SNR $= 0.17$, flip rate $= 43\%$). On Adult Income, 60% of top-5 comparisons are unreliable (the top four sociodemographic features are virtually interchangeable in importance). On Diabetes (Pima), 90% of top-5 comparisons are reliable (glucose is clearly dominant). The Gaussian flip formula achieves mean $\rho = 0.858$ (rank correlation) and mean $R^2 = 0.707$ across the five TreeSHAP datasets. Most top-5 rankings *are* stable; the contribution is identifying the 28% that are not, which practitioners currently have no way to diagnose.

**High-dimensional test ($P = 200$, gain-based).** At $P = 200$ features (synthetic, 50 informative, 50 redundant), the formula achieves out-of-sample $R^2 = 0.532$ with gain-based importance across 19,900 feature pairs. Of these, 49.2% are unreliable (SNR $< 0.5$) and 12.2% are reliable (SNR $> 2$).

**Enrichment on real data (Breast Cancer).** The enrichment tradeoff, predicted by the bilemma, is confirmed on real clinical data (Breast Cancer, 30 features, 100 XGBoost models). At full decisiveness (threshold = 0), stability is 0.675 (flip rate 32.5%). At threshold = 0.5 (stability 0.991), decisiveness collapses to 0.088. The sharp phase transition between thresholds 0.10 and 0.15—where stability jumps from 0.79 to 0.95 while decisiveness drops to 16%—confirms that the $(1, 1)$ corner of the decisiveness–stability plane is unreachable on real data, as the bilemma predicts.

**Combined real-data-only reliability summary.** Across seven real clinical, financial, and genomics datasets (Breast Cancer, Wine, California Housing, Diabetes, Heart Disease, German Credit, AP_Breast_Lung), encompassing 2,062 feature-pair comparisons, 62% are unreliable (SNR $< 0.5$) and only 4% are reliable (SNR $> 2$). On clinical and financial data alone (703 pairs), 47% are unreliable. The mean out-of-sample $R^2$ across all real datasets is 0.67.

**Multiverse analysis (Breast Cancer).** Across a 30-pipeline multiverse (3 model classes $\times$ 3 scalers $\times$ 3 feature selections + 3 bootstrap extras), 95% of feature importance claims are non-replicable (Jaccard similarity = 0.39). The Gaussian flip formula achieves Spearman $\rho = 0.78$ across the full multiverse (correctly ranking which comparisons are most unreliable) but $R^2 = -0.39$ for predicted flip rates (the absolute predictions fail when mixing model classes with different importance scales). Within model classes, the formula performs better: Ridge $R^2 = 0.49$, XGBoost $R^2 = 0.33$. The 95% non-replicable figure is the most dramatic but also the least scientifically meaningful, as it compares tree-based and linear models that measure fundamentally different quantities (gain-based vs. coefficient-based importance). The within-class figures—75% non-replicable for XGBoost and Random Forest, 100% for Ridge—are the more appropriate adversarial baseline.

**Generalization gap correlation.** Across the original 5 datasets, the cross-dataset correlation between mean flip rate and generalization gap (ID accuracy $-$ OOD accuracy) is Pearson $r = -0.91$, $p = 0.034$, $n = 5$—suggestive of an inverse relationship (more instability correlates with smaller generalization gaps). However, leave-one-out analysis reveals this result is driven by a single outlier: removing California Housing (which has both the lowest instability and the largest geographic distribution shift) reduces the correlation to $r = -0.20$, $p = 0.80$. Expanding to 13 datasets (8 real + 5 synthetic controls with varying feature redundancy) yields $r = -0.58$, $p = 0.038$, but removing California Housing from the full set reverses the sign to $r = +0.26$. The synthetic control with varying redundancy (0–40 redundant features) shows the *opposite* pattern ($r = +0.80$): more redundancy increases both instability and generalization gap, contrary to the initial hypothesis. The finding requires validation on larger, more diverse samples before causal interpretation; we report it as a null result with respect to the specific directional prediction. Breast Cancer shows a negative generalization gap (OOD accuracy exceeds ID accuracy), likely because the 'extreme' patients have more distinctive features that are easier to classify.

## 10.17 Gene Expression Pathway Divergence

The most consequential instance of the impossibility arises in biomarker discovery, where the "top gene" identified by a machine learning pipeline determines which therapeutic target enters experimental validation.

**TSPAN8 vs CEACAM5 on colon–kidney discrimination.** Using the AP_Colon_Kidney microarray dataset (546 samples, 10,935 genes reduced to top 50 by variance, OpenML), we trained 50 XGBoost classifiers with standard regularisation (subsample = 0.8, colsample_bytree = 0.5, seeds 0–49) and computed TreeSHAP importances. The #1 biomarker alternates between two genes:

- **TSPAN8** (probe 203824_at): tetraspanin-enriched microdomains, integrin signaling, invasion/metastasis — top in 92% of seeds.
- **CEACAM5/CEA** (probe 201884_at): GPI-anchored cell adhesion, immune evasion, classical clinical colon cancer marker — top in 6% of seeds.

The two genes' SHAP values correlate at $\rho = 0.858$ across patients (both are colon-enriched tissue markers), placing them in the same approximate symmetry orbit. Under deterministic training (subsample = 1.0), a single ranking emerges with 100% agreement—confirming that the alternation is driven by the stochastic regularisation, not by data noise.

**Pathway analysis (GO enrichment).** Gene Ontology annotation reveals zero overlap in Biological Process terms between TSPAN8 (negative regulation of blood coagulation; integrin binding) and CEACAM5 (homophilic cell–cell adhesion, six BP terms; GPI anchor binding; apoptosis regulation). A biologist following the first model into the wet lab would pursue invasion/metastasis signaling; a biologist following the second would pursue immune evasion. The pathway-level conclusion (colon identity) is preserved; the gene-level biomarker target—the actionable output of a discovery pipeline—differs.

**Replication across datasets.** The alternation replicates on AP_Endometrium_Colon (363 samples, independent tissue contrast): TSPAN8 is #1 in 78% of seeds, CEACAM5 in 22%, with $\rho = 0.781$. On AP_Breast_Colon (1,545 samples), a different pair alternates: COL3A1 (collagen type III, 72%) and COL1A1 (collagen type I, 28%), with $\rho = 0.843$. Both collagens are extracellular matrix proteins sharing 3 GO Biological Process terms—this is the same-pathway, redundant-marker mode (Mode 2), contrasting with the distinct-pathway mode of TSPAN8/CEACAM5 (Mode 1). AP_Breast_Lung serves as a positive control: SFTPB (surfactant protein B, a dominant lung marker) is #1 in 98% of seeds, confirming that large importance gaps protect stability.

**Two modes of alternation.** *Mode 1* (TSPAN8/CEACAM5): distinct biological pathways compete for the top rank. The specific drug target depends on the random seed. *Mode 2* (COL3A1/COL1A1): redundant markers within the same pathway alternate. The pathway conclusion is stable; the molecular identity is not. Both modes are instances of the impossibility theorem: the Rashomon property (observationally equivalent models) produces incompatible explanations (different top genes). Mode 1 has greater scientific consequence because it affects therapeutic strategy, not just molecular identity.

**DASH resolves.** The DASH ensemble average (mean SHAP across 50 models) places both TSPAN8 and CEACAM5 in the top-2, correctly reporting that both contribute without privileging either. This is the $G$-invariant resolution in practice: the group mean preserves the between-group ranking (colon genes > kidney genes) while acknowledging within-group indistinguishability.

### 10.18 Mechanistic Interpretability: Circuit-Level Rashomon

Mechanistic interpretability seeks to reverse-engineer neural networks into human-readable circuits. Prior work has shown qualitatively that networks trained from different random initialisations develop different internal representations and circuits [Chughtai et al., 2023]. The impossibility theorem predicts that circuit-level explanations are subject to the same instability as feature attributions when the network architecture admits permutation symmetries; here we quantify the disagreement and resolve it via the $G$-invariant projection.

**Experimental design.** We trained 10 two-layer transformers (4 heads per layer, $d_{\text{model}} = 128$) from independent random initialisations on modular addition $(a + b) \bmod 113$. All 10 models "grokked" to 100% test accuracy after 50,000 training steps with weight decay 1.0. Circuit importance was measured via activation patching: for each of 10 components (8 attention heads + 2 MLPs), we replaced the component's output with its mean activation and measured the resulting accuracy drop. The within-model determinism control confirmed near-perfect reproducibility ($r = 0.9998$).

**Controls: random, memorised, grokked.** Three training checkpoints establish the control hierarchy:

| Checkpoint | Train acc | Test acc | Mean $\rho$ between models |
|---|---|---|---|
| Step 50 (random init) | 0.2% | 0.2% | 0.300 |
| Step 500 (memorised) | 100% | 50% | 0.668 |
| Step 50k (grokked) | 100% | 100% | 0.518 |

Grokked models agree more than random ($p = 2.0 \times 10^{-4}$, $d = 0.76$) but *less* than memorised models ($p = 0.023$, $d = 0.58$). The decrease from memorised to grokked is consistent with grokking discovering genuinely different algorithms: memorised models share the same "lookup table" strategy, while grokked models find distinct Fourier-based algorithms.

**Different Fourier algorithms.** All 10 grokked models represent modular addition via Fourier components, but the specific frequencies selected are nearly disjoint: the mean pairwise Jaccard similarity of top-5 Fourier frequencies is 0.022. Despite computing the same function with identical accuracy, each model discovers a different algorithm.

**$G$-invariant resolution demonstrated.** The architecture defines a symmetry group $S_4 \times S_4$ (permutations of the 4 heads within each layer). Projecting the 10-dimensional importance vector onto the 4-dimensional $G$-invariant subspace $V^G$ (mean layer-0 heads, mean layer-1 heads, MLP0, MLP1) dramatically recovers agreement:

| Representation | Spearman $\rho$ | Pearson $r$ |
|---|---|---|
| Full 10-dim (individual components) | 0.518 | 0.835 |
| $G$-invariant 4-dim ($V^G$) | **0.929** | **0.967** |
| Excluding MLP1 (3-dim) | 0.822 | 0.782 |
| Excluding MLP1 (9-dim) | 0.337 | 0.439 |

The $G$-invariant projection is NOT trivially MLP-driven: excluding MLP1 from the invariant subspace still yields $\rho = 0.822$ ($p < 10^{-6}$), demonstrating that the layer-mean head importance is itself a stable, informative quantity.

**stable fact count within MI.** The symmetry $S_4 \times S_4$ predicts: within-layer head pairs (12 pairs total) should flip at $\sim 50\%$; head-vs-MLP pairs (cross-group, 8 pairs) should be stable. Observed: within-layer mean flip rate = 0.500 exactly; between-group mean flip rate = 0.227. Bimodal gap = 0.273, Mann–Whitney $p = 2.4 \times 10^{-5}$. This is a second instantiation of the stable fact count prediction within a single experiment.

**Structural universal: MLP1 dominance.** MLP1 is the most important component in every seed (coefficient of variation = 0.027). This is the $G$-invariant structural fact—the between-group conclusion that survives the impossibility. Individual head identities (which specific head matters) are the within-group noise that the theorem proves is unavoidable.

### 10.19 Language Model Circuit Stability: Multi-Scale Validation

The modular-addition experiment (Section 10.18) uses an algorithmic task. To test whether the theorem governs circuits in *real language models*, we trained 10 transformers from independent random initialisations on TinyStories Eldan and Li [2023], a dataset of short children's stories, and repeated the experiment at two architectural scales. A third configuration (GPT-2 Small fine-tuned) tests the predicted boundary condition: fine-tuning from a shared pretrained checkpoint should NOT produce circuit instability, because the Rashomon set is small.

**Experimental design.**

| Config | Architecture | Params | Steps | Components | $\dim(V^G)$ | $\eta$ |
|---|---|---|---|---|---|---|
| A | 4L/4H, $d = 256$ | 16M | 15,000 | 20 | 8 | 0.600 |
| B | 6L/8H, $d = 512$ | 45M | 15,000 | 54 | 12 | 0.778 |
| C | GPT-2 Small fine-tuned | 124M | 3,000 | 156 | 24 | 0.846 |

Each configuration trains 10 models from independent random seeds on the same TinyStories data. Circuit importance is measured via weight zeroing (Q/K/V for each head, fc1 for each MLP) and the resulting perplexity increase on held-out data. Config C starts from the pretrained GPT-2 Small checkpoint and fine-tunes with different random seeds (data ordering and dropout). Determinism control: same model measured twice gives $r = 1.000$ (Configs A and B) and $r = 1.000$ (Config C).

**Pre-registered predictions.** Seven predictions were stated before any training data was generated: (1) perplexity CV $< 5\%$; (2) full-component agreement $\rho < 0.70$; (3) $G$-invariant agreement $\rho > 0.80$; (4) heads-only lift $> 0.15$ (not MLP-driven); (5) within-layer head flip rate $> 0.40$; (6) head-vs-MLP flip rate $< 0.15$; (7) permutation test $p < 0.01$.

**Results: Configs A and B (training from scratch).**

| Metric | Config A (4L/4H) | Config B (6L/8H) |
|---|---|---|
| Perplexity (mean $\pm$ CV) | 8.6 $\pm$ 0.7% | 10.0 $\pm$ 1.0% |
| Split-half reliability ($\rho$) | 0.991 | 0.960 |
| Full agreement ($\rho$ [95% CI]) | 0.565 [0.53, 0.60] | 0.540 [0.52, 0.56] |
| $G$-invariant ($\rho$ [95% CI]) | **0.972** [0.97, 0.98] | **0.982** [0.98, 0.99] |
| Heads raw ($\rho$) | 0.291 | 0.379 |
| Heads averaged ($\rho$) | **0.849** | **0.931** |
| Head lift | 0.558 | 0.553 |
| Within-layer flip | 0.496 | 0.489 |
| Head-vs-MLP flip | 0.000 | 0.000 |
| Mann–Whitney $p$ (pair-aggregated) | $1.2 \times 10^{-18}$ | $1.3 \times 10^{-18}$ |
| Cohen's $d$ (pair-aggregated) | 5.4 | 11.9 |
| Permutation test $p$ | $< 0.001$ | $< 0.001$ |
| Pre-registered | **7/7 PASS** | **7/7 PASS** |

All 14 pre-registered predictions pass across both scales. The within-layer head flip rates (0.496 and 0.489) are within sampling noise of the predicted 0.500. The head-vs-MLP flip rate is exactly 0.000 across all 30 models (720 and 2,160 comparisons respectively)—every model agrees that MLPs are more important than attention heads at each layer. The split-half reliability ($\rho = 0.991$ and 0.960) far exceeds the between-model agreement ($\rho = 0.565$ and 0.540), confirming that the disagreement is genuine Rashomon, not measurement noise.

The heads-only random projection control is critical: random $n_{\text{heads}} \to n_{\text{layers}}$ projections give $\rho = 0.300 \pm 0.223$ (Config A) and $\rho = 0.407 \pm 0.182$ (Config B), while the symmetry-based mean-head projection gives $\rho = 0.849$ and 0.931 (100th percentile in both cases). The lift is from the specific within-layer symmetry structure, not dimensionality reduction.

For Config B, the full-vector random projection ($\rho = 0.988$) outperforms the $G$-invariant ($\rho = 0.982$) because MLP0 dominates the importance landscape ($\sim 6.7$ vs $\sim 0.4$ for all other components); any random projection preserving MLP0 achieves high agreement. The heads-only control isolates the symmetry-specific lift and remains at the 100th percentile.

In Config A, all four heads in layer 0 appear as the "most important head" across the 10 seeds (H0, H1, H2, H3 each take the top spot), fully realising the $S_4$ symmetry. The role "important attention head in layer 0" is stable; which head fills that role is a coin flip.

**Results: Config C (GPT-2 fine-tuned — boundary condition).** Config C produces $\rho = 0.993$ (full agreement), within-layer flip = 0.043, and head lift = 0.003. Three of seven predictions fail: full $\rho < 0.70$, within-flip $> 0.40$, and head lift $> 0.15$. This is the predicted boundary condition: fine-tuning from a shared pretrained checkpoint does not create the Rashomon property. The pretrained circuit structure is preserved — the same five components (MLP0, MLP2, MLP1, MLP11, MLP8) are the top 5 in every seed.

This confirms the theorem's precondition: the impossibility predicts instability *if and only if* the Rashomon property holds. When it does not (Config C), the theorem correctly predicts no instability. The split-half reliability ($\rho = 0.878$) is *lower* than the between-model agreement ($\rho = 0.993$), meaning fine-tuned models are more similar to each other than the same model measured on different data subsets. This is additional evidence that fine-tuning genuinely preserves circuits.

Head-vs-MLP flip = 0.000 holds even for Config C: the layer-level MLP hierarchy is a structural invariant that persists regardless of whether the Rashomon property holds.

**Cross-scale comparison.**

| Config | $\eta$ | Full $\rho$ | $G$-inv $\rho$ | Lift | W-flip | B-flip |
|---|---|---|---|---|---|---|
| A (4L/4H) | 0.600 | 0.565 | 0.972 | 0.408 | 0.496 | 0.000 |
| B (6L/8H) | 0.778 | 0.540 | 0.982 | 0.441 | 0.489 | 0.000 |
| C (GPT-2 ft) | 0.846 | 0.993 | 0.999 | 0.006 | 0.043 | 0.000 |

For the two from-scratch configurations (A and B), higher $\eta$ correlates with lower full agreement ($0.565 \to 0.540$) and higher $G$-invariant agreement ($0.972 \to 0.982$), consistent with the capacity theorem. Config C does not participate in this comparison because it lacks the Rashomon property. The within-layer flip rate is indistinguishable from 0.500 for both from-scratch configs and near zero for the fine-tuned config—a clean separation governed by the Rashomon precondition.

**Ablation method robustness.** To test whether the results depend on the choice of ablation method, we repeated the importance measurement using mean ablation (replacing each component's output with its mean activation) on the cached Config A and B models. The layer-level structure is method-invariant: both weight zeroing and mean ablation produce $G$-invariant agreement $\rho \approx 0.97$–$0.99$ ($G$-invariant $\rho = 0.975$ and 0.987 for mean ablation vs 0.972 and 0.982 for weight zeroing). The cross-method per-model agreement is moderate ($\rho = 0.59$ for Config A, $\rho = 0.50$ for Config B). For $n = 20$ components, the null expectation under independence is $|\rho| \approx 0.23$, so the observed $\rho \approx 0.5$ represents significant positive agreement at the head level ($p < 0.05$) combined with near-perfect agreement at the layer level ($\rho \approx 0.97$–$0.99$ on $V^G$). The two methods agree on the importance structure more than chance, but disagree on the specific head-level ranking—and the $G$-invariant resolution captures exactly the content on which they agree.

**Limitations and open questions.** TinyStories is a synthetic dataset of simple children's stories; extrapolation to frontier language models (GPT-4-scale) trained from scratch remains open. The cross-scale $\eta$ rank-order test has only $n = 2$ valid data points (Configs A and B) and cannot be tested statistically.

Several questions raised by the component-level analysis warrant future work. First, *functional stability*: we show that WHICH head is most important varies across seeds, but not whether the FUNCTION computed by that head varies. If the same attention pattern emerges in different head positions across seeds (a permutation of functional roles), the instability is in labelling rather than in algorithmic diversity. Permutation alignment methods such as Git Re-Basin Ainsworth et al. [2023] address this directly and represent a complementary resolution to the $G$-invariant projection: Re-Basin aligns models in weight space, while $V^G$ projects into the shared importance structure. Whether these produce equivalent conclusions is an empirical question.

Second, *sparse autoencoders (SAEs)*: the mechanistic interpretability community increasingly analyses learned features from SAEs rather than individual attention heads. If independently trained SAEs on the same model produce stable features, SAE-based attribution may escape the Rashomon property — the symmetry group would be trivial ($G = \{e\}$) and $\eta = 0$. Whether this holds empirically is untested.

Third, the *ablation methodology* uses weight zeroing throughout. Path patching Goldowsky-Dill et al. [2023] and causal scrubbing offer finer-grained, input-dependent attribution that may break the within-layer symmetry for specific inputs. The impossibility theorem applies to the population-level importance ranking (averaged over inputs); input-specific attribution may exhibit different stability properties.

Fourth, the bootstrap 95% CIs on pairwise Spearman $\rho$ are computed over the 45 model pairs, which share models. The effective degrees of freedom are closer to $n = 10$ (number of models) than $n = 45$ (number of pairs). The CIs may therefore be too narrow; the point estimates and qualitative conclusions are unaffected.

## 10.20 Model-Class Universality of the Impossibility

The impossibility is not specific to a single model class. A rigorous comparison across four model classes (XGBoost, Random Forest, Ridge, LASSO) on three datasets (Breast Cancer, California Housing, Diabetes) reveals a two-level structure:

**Within-family agreement.** Tree-based models (XGBoost, Random Forest) agree on which features are unstable: seed-variation Spearman $\rho = 0.79$–$0.92$ on Breast Cancer and Diabetes, exceeding the permutation null. Linear models (Ridge, LASSO) agree even more strongly ($\rho = 0.94$ on Breast Cancer). Models within the same algorithmic family identify the same instability patterns.

**Cross-family disagreement.** Linear–tree cross-family agreement is NOT significant ($\rho = 0.19$–$0.57$, none exceeding permutation null). Which specific features are unstable depends on the model class—trees and linear models disagree on the identity of unstable features, even when both detect instability.

**Universal diagnostics.** Despite cross-family disagreement on specifics, two diagnostics are universal:

- *Coverage conflict:* Spearman $\rho = 0.82$ (mean across all 4 classes, excluding degenerate California Housing linear case where 7/8 features have zero flip rate).
- *Var[SHAP] = DASH MSE:* exact for all 4 model classes, 0/12 violations. The variance of SHAP values across the Rashomon set equals the mean-squared error of the DASH ensemble average—a mathematical identity that holds regardless of model architecture.

**Interpretation.** Instability has a data-determined core (detectable by coverage conflict, shared across model families) and a model-family-specific pattern (which unstable features flip depends on the algorithm). The impossibility theorem explains both: the Rashomon property (data-determined) guarantees instability exists; the specific symmetry group (model-family-specific) determines which features are interchangeable.

### 10.21 Multi-Analyst Disagreement as Impossibility: Neuroimaging

The impossibility theorem applies beyond ML explanation methods to any setting where multiple valid analyses of the same data produce incompatible conclusions. We demonstrate this on the Botvinik-Nezer et al. (2020) neuroimaging study, where 70 independent teams analysed the same fMRI dataset and reached dramatically different conclusions about which brain regions were significantly activated.

**The Rashomon property.** The 70 analysis pipelines constitute the Rashomon set: they share the same observable (fMRI data from 108 participants on a mixed-gambles task) but produce incompatible explanations (different activation maps). This is the Rashomon property by definition; the impossibility theorem applies immediately.

**Data and methods.** Overlap maps (proportion of teams finding significance at each voxel) were downloaded from NeuroVault collection 6047 for 9 hypotheses (7 non-empty after excluding hypotheses 6 and 9). Individual team unthresholded $t$-statistic maps were downloaded from each team's NeuroVault collection; 48 of 70 teams had accessible maps. All maps were parcellated using the Schaefer 100-parcel atlas (7 Yeo resting-state networks) via nilearn. Team software metadata (12 FSL, 8 SPM, 4 AFNI, 24 unknown) was obtained from the narps_open_pipelines repository.

**The Noether structural prediction.** The framework predicts that brain regions sharing a functional network (the approximate symmetry group) should have more similar disagreement patterns than regions in different networks. Network labels from the Yeo et al. (2011) resting-state parcellation are *independent* of the NARPS task data.

Using overlap-based disagreement (the metric $1 - \max(\text{overlap}, 1 - \text{overlap})$ per region per hypothesis), within-network region pairs show more similar disagreement patterns than between-network pairs (raw Cohen's $d = 0.43$, permutation $p < 0.001$, Mann–Whitney $p = 4.5 \times 10^{-32}$, consistent across all 7 non-empty hypotheses). The effect survives three controls:

- *Random grouping null:* Random assignments of matching group sizes yield $d = 0.001 \pm 0.050$ (observed is 4.9× the 95th percentile of the null).
- *Spatial autocorrelation:* $k$-means clustering on region centroids gives $d = 0.07$–$0.19$ across $k = 5, 7, 10, 14$—functional networks outperform all purely spatial groupings.
- *Activation-level control:* After regressing out activation similarity ($|\Delta\text{overlap}|$) from disagreement similarity, the network effect persists ($d = 0.32$, $p = 8.1 \times 10^{-19}$, bootstrap CI $[0.006, 0.008]$ excludes zero). Activation similarity explains $R^2 = 0.67$ of overall disagreement variance; the network effect is the remaining 26% reduction in $d$ (from 0.43 to 0.32).

Network identity (which network, regardless of size) explains $\eta^2 = 0.35$ of disagreement variance (Kruskal–Wallis $p = 5.7 \times 10^{-7}$). The Dorsal Attention and Control networks show the highest disagreement (mean = 0.095); Limbic and Default show the lowest (mean = 0.023–0.046)—consistent with the mixed-gambles task differentially engaging attention and control networks near the significance threshold.

**Methodological progression.** The initial analysis (v1) computed inter-region correlation from the overlap maps themselves and tested whether this predicted disagreement ($\rho = 0.310$, $p = 0.002$). Adversarial review identified this as *circular*: the predictor was derived from the outcome data. Version 2 replaced the circular predictor with independent Yeo network labels (from resting-state fMRI, a separate dataset and modality), which eliminated the circularity and revealed that the capacity theorem fails while the Noether structural prediction succeeds. Version 3 added comprehensive controls (random grouping null, spatial autocorrelation via $k$-means, per-hypothesis consistency, Schaefer-400 replication). The definitive analysis added activation-level regression control. This progression—from a promising but flawed initial result to a rigorously controlled finding—illustrates why adversarial self-auditing is essential for cross-domain applications where the mapping from theory to data is analogical.

**Bilateral symmetry ($\mathbb{Z}_2$).** The strongest approximate symmetry in the brain is bilateral: left and right hemispheric homologs share functional roles. Matching 35 left–right homolog pairs by Schaefer label, the disagreement profiles correlate at $r = 0.66$ ($p = 1.5 \times 10^{-5}$), with homolog pairs showing 42% smaller disagreement differences than random pairs. No systematic left–right bias was detected (Wilcoxon $p = 0.49$). The effect is consistent across all 7 hypotheses (per-hypothesis $r = 0.44$–$0.67$). However, this result did not survive bootstrap activation control (parametric partial $r = 0.49$, $p = 0.001$, but bootstrap CI $[-0.54, 0.41]$ spans zero due to instability with $n = 35$ pairs and residualisation).

**The capacity theorem does not apply.** The quantitative prediction $\eta = (k-1)/k$ fails at all tested granularities: network size does not predict disagreement level (Spearman $\rho = -0.021$, $p = 0.84$ at Schaefer-100 with 7 networks; $\rho = -0.015$, $p = 0.88$ with 14 hemisphere$\times$network groups; $\rho = -0.19$, $p = 0.0001$ at Schaefer-400, but in the *wrong direction*, vanishing under partial correlation controlling for overlap, $\rho = -0.04$).

**The approximate $\eta$ formula.** For a group of $k$ equicorrelated features with pairwise correlation $\rho$, the eigenvalues of the within-group covariance matrix are $\lambda_1 = 1 + (k-1)\rho$ and $\lambda_2 = \cdots = \lambda_k = 1 - \rho$. The participation ratio $\mathrm{PR} = k/(1 + (k-1)\rho^2)$ measures effective dimensionality. The generalised $\eta$ formula is:

$$\eta_{\text{approx}} = 1 - \frac{1}{1 + (k-1)\rho^2} = \frac{(k-1)\rho^2}{1 + (k-1)\rho^2}.$$

At $\rho = 1$ (exact symmetry), this reduces to $(k-1)/k$; at $\rho = 0$, to 0. For brain imaging with network sizes $k = 5$–$24$ and typical resting-state FC $\rho \approx 0.3$–$0.5$, the predicted $\eta$ values are substantially lower than the exact prediction and depend on $\rho$, which is unknown for the NARPS task data. Empirically, the formula with literature FC values gives $\rho_{\text{Spearman}} = 0.03$ ($p = 0.79$)—no improvement over the exact formula. A sensitivity analysis sweeping uniform $\rho$ from 0.1 to 0.9 shows the Spearman correlation is invariant at $-0.021$ regardless of $\rho$, because $\eta_{\text{approx}}$ with uniform $\rho$ is a monotone function of $k$ alone. The approximate formula is theoretically correct as an interpolation between exact and no symmetry, but empirically underdetermined for this application.

**Per-team ranking analysis.** Using individual team $t$-statistic maps (48 teams, 7 hypotheses, 100 regions), we computed pairwise flip rates across all team pairs for all region pairs. Within-network region pairs show higher flip rates (0.321) than between-network pairs (0.310), with gap $= +0.011$ $[+0.007, +0.015]$, Cohen's $d = 0.078$, Mann–Whitney $p = 3.7 \times 10^{-8}$, permutation $p < 0.005$, consistent across 7/7 hypotheses. The effect is statistically significant but small ($d = 0.078$), reflecting the dominance of methodological variation over structural variation in per-team rankings.

**Software as a second symmetry source.** On 24 teams with known software (12 FSL, 8 SPM, 4 AFNI), both network structure and software choice independently predict flip rates:

| Effect | Flip rate premium | $p$-value |
|---|---|---|
| Network (within-net > between-net) | +0.005 | $8.8 \times 10^{-5}$ |
| Software (diff-software > same-software) | +0.016 | $8.2 \times 10^{-55}$ |

Software has 3× the effect of network structure (0.016 vs 0.005), consistent with Botvinik-Nezer et al.'s finding that methodology drives most disagreement. Both CIs exclude zero (network: $[0.002, 0.008]$; software: $[0.014, 0.018]$).

**Orbit averaging as resolution.** The overlap map (mean across teams' binary activation maps) is the orbit average over the Rashomon set. We tested whether this resolution is near-optimal by comparing aggregation methods at full ensemble size ($M = 48$):

| Method | Faithfulness (corr w/ teams) | Mean unfaithfulness |
|---|---|---|
| Mean | **0.669** | 0.739 |
| Median | 0.660 | **0.719** |
| Trimmed mean | 0.668 | 0.727 |
| Weighted mean | 0.664 | 0.725 |

All methods are within 3% of each other. The mean achieves the highest faithfulness (correlation with individual teams); the median achieves the lowest absolute unfaithfulness. At $M = 48$, the mean is not dominated by any alternative. At smaller $M$ ($< 20$), the weighted mean (weighting by leave-one-out prediction accuracy) slightly dominates the unweighted mean, consistent with the theorem's prediction that equal weighting is optimal only for balanced Rashomon sets.

**Convergence prescription.** The split-half stability of the ensemble average increases with ensemble size. At $M = 16$ (95% CI: $[10, 22]$), the split-half Pearson correlation exceeds 0.95. The convergence rate fits $1 - a/M^b$ with $b = 1.00$ ($R^2 = 0.97$). This provides a concrete prescription for multi-analyst studies in neuroimaging: approximately 16 independent analyses suffice for 95% stability between independent assessments.

**Bilemma confirmation.** The unfaithfulness of the ensemble average is proportional to team disagreement (Spearman $\rho = 0.962$, $p < 10^{-100}$). The ratio unfaithfulness/$\sigma = 0.734$ (Gaussian prediction: $\sqrt{2/\pi} = 0.798$), confirming the ensemble average achieves near the theoretical minimum unfaithfulness.

## 10.22 Cross-Domain Multi-Analyst Studies

The same impossibility→resolution pattern appears in multi-analyst studies beyond neuroimaging.

**Silberzahn et al. (2018): 29 teams, psychology.** 29 teams analysed whether soccer referees give more red cards to dark-skinned players (AMPPS, 2018). Team estimates (odds ratios from OSF repository osf.io/gvm2z) range from 0.89 to 2.93 (geometric mean OR = 1.34). In log-odds space, the method-type grouping (Logistic: 15 teams, Count: 5, Linear: 3, Bayesian: 2, Other: 4) explains $\eta^2 = 0.131$ of estimate variance (ANOVA $F = 0.90$, $p = 0.48$—not significant with 29 teams in 5 unequal groups). Within-method pairs show smaller estimate differences than between-method pairs (Cohen's $d = 0.29$, Mann–Whitney $p = 0.03$), but this does not survive permutation correction ($p = 0.15$). The Noether prediction is inconclusive at this sample size.

**Breznau et al. (2022): 71 teams, political science.** 73 teams tested whether immigration reduces support for social policies (PNAS, 2022). From the GitHub repository (nbreznau/CRI), 71 teams with valid average marginal effects (AMEs) across 1,253 individual models show mean AMEs ranging from $-0.27$ to $+0.36$. Sign agreement is only 52%—teams cannot agree on the *direction* of the effect. The ICC (intraclass correlation) for team identity is $-0.096$ ($F = 0.3$, $p = 1.0$), meaning within-team model specification variation *exceeds* between-team variation. The team's methodological approach does not determine the result; individual model specification choices (covariates, functional form) are the dominant source of variation.

**Methodological note on scalar studies.** The initial cross-domain analysis used hard-coded approximate data; the definitive version uses real team estimates from OSF (Silberzahn: osf.io/gvm2z, "Crowdsourcing Effects in OR.csv") and GitHub (Breznau: nbreznau/CRI, "CRI Model Specifications and Margins.xlsx"). Convergence was measured using non-overlapping split-half (disjoint subsets of size $M$, requiring $2M \leq N$) to avoid the inflation caused by overlapping subsets. With non-overlapping split-half, neither study reaches 95% stability within the testable range ($M \leq N/2$): Silberzahn at $M = 14$ achieves stability 0.84; Breznau at $M = 35$ achieves 0.91. The Breznau ICC is unreliable due to extreme group-size imbalance (2–112 models per team); teams with many models have means regressed toward zero, deflating between-team variance.

**Cross-domain pattern.** Across all three domains, the qualitative pattern is identical: the Rashomon property holds (teams disagree), orbit averaging provides a near-optimal resolution (all aggregation methods within 3%), and unfaithfulness is proportional to disagreement. The Noether structural prediction—that group membership predicts which conclusions are stable—is confirmed for multivariate neuroimaging data ($d = 0.32$ after activation control) but inconclusive for scalar multi-analyst studies (Silberzahn $d = 0.29$, not significant after permutation correction; Breznau ICC $= -0.10$). For scalar studies, convergence follows standard CLT scaling ($\sim 1/\sqrt{M}$); the framework does not predict a different rate. The framework's unique value beyond classical meta-analysis is threefold: the irreducibility proof (the disagreement is mathematical, not methodological), the Pareto optimality of orbit averaging (proven, not assumed), and the structural prediction identifying which conclusions are stable (confirmed for multivariate data). The CLT tells practitioners that averaging reduces noise; the framework tells them the noise is *irreducible*, proves orbit averaging is *optimal* among all aggregation methods, and identifies *which* conclusions survive.

## 10.23 Attribution PCA Eigenspectrum: A Negative Result

We tested whether PCA on the SHAP value matrix across multiple models recovers the symmetry group structure. On synthetic data with $g = 3$ groups of $k = 4$ features ($\rho = 0.95$), the eigenspectrum gap appears at PC 10 (expected: PC 3), and $\eta_{\text{PCA}} = 0.50$ (expected: 0.75). The gap only matches the group count at $\rho = 0.99$ (near-exact symmetry). On the NARPS dataset (48 teams $\times$ 700 dimensions), the gap appears at PC 1 (expected: PC 7 for 7 networks). On Breast Cancer data, the stable-PC loading vs. flip rate correlation is $\rho = -0.27$ ($p = 0.15$—not significant).

The eigenspectrum approach fails because PCA captures variance (cardinal), not rankings (ordinal). A feature with high SHAP variance can have a perfectly stable sign; a feature with low variance can flip easily near zero. The flip rate matrix (pairwise ranking instability) is the correct object for capturing ranking stability, not the SHAP covariance matrix. This negative result clarifies that extending the capacity theorem to approximate symmetry requires rank-based methods, not spectral decomposition of cardinal values.

| Statistics (Fisher/Rao–Blackwell) | Framework |
|---|---|
| Sufficient statistic $T$ | Observation map obs |
| Non-injectivity of $T$ | Rashomon property |
| Rao–Blackwell estimator $\mathbb{E}[\delta \mid T]$ | Orbit average ($G$-invariant resolution) |
| Rao–Blackwell variance reduction | Pareto-optimality |

Table 22: Dictionary between sufficient statistics and the abstract framework. Fisher's sufficiency (1922), the Rao–Blackwell theorem (1945/47), and the explanation impossibility are the same result in different notation.

# 11 Bridge Theorems: Connections to Established Mathematics

The cross-domain transfer experiments of the previous section show that the framework *generates* novel predictions. This section establishes a complementary result: the framework *subsumes* classical theorems from statistics, physics, and information theory. We present five "bridge theorems" that identify existing results as special cases of the abstract impossibility structure, and derive one new quantitative prediction for AI interpretability.

## 11.1 Sufficient Statistics as $G$-Invariants

Fisher's concept of sufficiency [Fisher, 1922] can be recovered exactly within the impossibility framework. Map the components as follows: let configurations be the data space $\mathcal{X}$, let the observation map obs be a statistic $T\colon \mathcal{X} \to \mathcal{Y}$, and let the explanation map be the identity explain = id (reporting the full data). Under this identification, the Fisher–Neyman factorization theorem [Neyman, 1935] states that $T$ is sufficient for $\theta$ if and only if the conditional distribution $P(X \mid T(X))$ is independent of $\theta$—precisely the stability condition of the abstract framework (the explanation factors through the observation).

The Rashomon property holds whenever $T$ is non-injective: there exist $X_1 \neq X_2$ with $T(X_1) = T(X_2)$, so the full data cannot be faithfully, stably, and decisively reported. This is the typical case—sufficiency is useful precisely because $T$ compresses.

The identification extends to the resolution. The Rao–Blackwell theorem [Rao, 1945, Blackwell, 1947] states that for any estimator $\delta(X)$, the conditional estimator $\delta^*(T) = \mathbb{E}[\delta(X) \mid T(X)]$ has variance no greater than that of $\delta$. This conditioning is *exactly* orbit averaging: the estimator is averaged over the fiber $\{X : T(X) = t\}$, weighted by the conditional distribution. The Rao–Blackwell variance reduction is therefore the framework's Pareto-optimality theorem—the $G$-invariant resolution dominates all alternatives—expressed in the language of classical estimation.

These are not analogies. Fisher's sufficiency criterion (1922), the Rao–Blackwell theorem (1945/47), and the explanation impossibility are the same theorem in different notation (Table 22). Under a uniform prior on the fiber $\{\theta : \mathrm{obs}(\theta) = y\}$, the orbit average $\bar{E}(\theta) = (1/|G|)\sum_{g\in G} E(g\cdot\theta)$ is literally the conditional expectation $\mathbb{E}[\mathrm{explain}(\theta) \mid \mathrm{obs}(\theta) = y]$. The identification is exact when the prior is uniform and the group acts transitively on fibers; it is approximate otherwise.

## 11.2 The Explanation Uncertainty Bound

The qualitative impossibility ("at least one of faithfulness, stability, or decisiveness must fail") admits a quantitative refinement. For a finite system with total incompatible pairs $I_{\mathrm{total}}$ and within-fiber incompatible pairs $R_{\mathrm{within}}$, define the *Rashomon ratio* $r = R_{\mathrm{within}}/I_{\mathrm{total}} \in [0, 1]$ and the *decisiveness fraction* $D(E)$ = (fraction of incompatible pairs preserved by $E$).

**Theorem 11.1** (Explanation uncertainty bound)**.** *For any stable explanation map $E$,*

$$D(E) \;\leq\; 1 - \frac{R_{\mathrm{within}}}{I_{\mathrm{total}}} \qquad \textit{equivalently,} \qquad D(E) + r \;\leq\; 1.$$

| Noether's Theorem (continuous) | Explanation Framework (discrete) |
|---|---|
| Symmetry group $G$ | Rashomon group $G$ |
| $G$ preserves Lagrangian $L$ | $G$ preserves obs |
| Conserved quantity | $G$-invariant query $\in V^G$ |
| $\dim(G)$ conserved quantities | $\dim(V^G)$ stable queries |

Table 23: Invariance correspondence. Rashomon symmetry $\to$ stable queries is the discrete analogue of continuous symmetry $\to$ conservation laws.

*Proof.* Stability forces $E(\theta_1) = E(\theta_2)$ whenever $\mathrm{obs}(\theta_1) = \mathrm{obs}(\theta_2)$. By irreflexivity of the incompatibility relation, $\mathrm{incomp}(E(\theta), E(\theta))$ is false, so every within-fiber incompatible pair is collapsed by $E$. Hence $E$ can preserve at most $I_{\mathrm{cross}} = I_{\mathrm{total}} - R_{\mathrm{within}}$ incompatible pairs, giving $D(E) \le I_{\mathrm{cross}}/I_{\mathrm{total}} = 1 - r$. The bound is tight: choosing a fixed representative per fiber preserves all cross-fiber incompatibilities. □

The qualitative impossibility is the special case $r > 0$, $D = 1$: the inequality $1 + r \le 1$ is a contradiction. When $r = 0$ there is no constraint (no Rashomon), when $r = 1$ we have $D = 0$ (total loss of decisiveness), and the general case interpolates smoothly.

### 11.3 The Invariance Counting Principle

Noether's theorem [Noether, 1918] establishes that every continuous symmetry of the Lagrangian implies a conserved quantity. The explanation framework provides an analogous discrete result: every Rashomon symmetry implies a class of stably answerable queries. We note that this result is an application of classical invariant theory (Burnside's lemma, Molien's theorem) to the explanation framework, not a direct consequence of Noether's theorem, which requires continuous symmetry and a variational principle.

Let $G$ be a finite group acting on the configuration space $\Theta$, preserving the observation map ($\mathrm{obs}(g \cdot \theta) = \mathrm{obs}(\theta)$ for all $g \in G$). A query $q \colon \mathcal{H} \to \mathbb{R}$ is *stably answerable* if $q(\mathrm{explain}(\theta_1)) = q(\mathrm{explain}(\theta_2))$ whenever $\mathrm{obs}(\theta_1) = \mathrm{obs}(\theta_2)$.

**Proposition 11.2** (Invariance theorem)**.** *A query $q$ is stably answerable if and only if $q \in V^G$ (the $G$-invariant subspace). The number of independent stably answerable queries is $\dim(V^G) = (1/|G|) \sum_{g \in G} \chi(g)$, by the character formula.*

The correspondence (Table 23) places the explanation impossibility in the same conceptual lineage as the most celebrated theorem in mathematical physics. Both are instances of a single principle: group invariance determines what can be stably measured.

### 11.4 EM as Generalized Orbit Averaging

The Expectation-Maximization algorithm [Dempster et al., 1977] is the most widely used computational method for latent-variable models (mixture models, HMMs, factor analysis, topic models). Its E-step can be identified with the framework's $G$-invariant resolution.

Map the EM components: configurations are latent variables $Z$, observations are the observed data $X$ (with $\mathrm{obs}(Z) = X$ via marginalization $P(X) = \sum_Z P(X, Z)$), and explanations are the latent variables themselves. The Rashomon set at $X$ is $\{Z : P(X \mid Z) > 0\}$—all latent configurations consistent with the data. The E-step computes

$$\mathbb{E}_{Z|X,\theta}[f(Z)] \;=\; \sum_Z f(Z) \cdot P(Z \mid X, \theta),$$

a weighted average over the fiber $\{Z : \mathrm{obs}(Z) = X\}$. Under a uniform prior ($P(Z)$ constant) with equal likelihoods for all consistent $Z$ (the microcanonical case), this reduces to

| EM Algorithm | Framework |
|---|---|
| Latent variable $Z$ | Configuration $\theta$ |
| Observed data $X$ | Observable obs$(\theta)$ |
| Posterior $P(Z \mid X, \theta)$ | Resolution weight on fiber |
| E-step: $\mathbb{E}[f(Z) \mid X]$ | Orbit average ($G$-invariant resolution) |
| M-step: update $\theta$ | Update resolution parameters |
| Rao–Blackwell optimality | Pareto-optimality of orbit average |

Table 24: Dictionary between the EM algorithm and the abstract framework. The E-step IS the $G$-invariant resolution with posterior weights; the general EM generalizes from uniform to non-uniform orbit averaging.

$(1/|\text{fiber}(X)|) \sum_{Z:\text{obs}(Z)=X} f(Z)$—exactly the orbit average of the framework's $G$-invariant resolution.

The general case—non-uniform priors—extends the orbit average to a posterior-weighted average, with the prior $P(Z \mid \theta)$ introducing additional structure beyond the bare group action (Table 24). This provides the most practically impactful bridge: the EM algorithm, gauge-invariant averaging in physics, and codon usage tables in biology are all instances of the same resolution strategy.

## 11.5 Bayesian Interpretation

The framework admits a natural Bayesian reading. Faithfulness corresponds to likelihood consistency (the explanation does not contradict the data-generating mechanism). Stability corresponds to posterior invariance (the explanation depends only on the observable evidence, not on which particular configuration produced it). Decisiveness corresponds to point estimation (the explanation commits to a single value rather than a distribution). The enrichment mechanism—adding a neutral element to restore faithful+stable—corresponds to Bayesian model averaging: replacing a point estimate with a posterior predictive distribution that averages over the equivalence class. The $G$-invariant resolution is then the orbit average $\bar{E}(\theta) = (1/|G|) \sum_{g \in G} E(g \cdot \theta)$. Under a uniform prior over the equivalence class, the orbit average coincides with the conditional expectation $E[\text{explain}(\theta) \mid \text{obs}(\theta)]$. For non-uniform priors, the orbit average and the posterior predictive diverge. The connection to Rao–Blackwell is therefore exact under uniformity and analogical in general. The Pareto-optimality of orbit averaging (Theorem 6.5) is a consequence of the admissibility of Bayes estimators under proper priors [Berger, 1985]. Specifically, under squared-error loss with a proper prior over the equivalence class, Bayes estimators are admissible (Berger, 1985, Theorem 5.2.4). The orbit average is the Bayes estimator under the uniform prior; its Pareto-optimality among stable maps follows. Under non-uniform priors, the Bayes estimator may differ from the orbit average, but the orbit average remains minimax among symmetric estimators.

## 11.6 Quantum Error Correction

The dimension ratio $\eta = \dim(V^G)/\dim(V)$ has a precise interpretation in quantum error correction: it equals the code rate $k/n$ of a quantum error-correcting code, where $V^G$ is the code subspace invariant under the error group $G$. We *conjecture* that the Knill–Laflamme conditions correspond to the stability condition of the framework applied to the code subspace. A full formal bridge—analogous to the Fisher/Rao–Blackwell dictionary (Table 22)—remains an open direction. The enrichment mechanism (adding a neutral element to restore faithful+stable) corresponds to adding ancilla qubits: enlarging the physical space to create room for fault-tolerant encoding [Knill and Laflamme, 1997].

## 11.7 Fundamental Ceiling on AI Interpretability

The bridge theorems above connect the framework to *existing* results. This final bridge produces a *new* prediction: a quantitative upper bound on the fraction of a neural network's internal representation that can be stably interpreted.

Consider a network with one hidden layer of width $n$. For any permutation $\pi \in S_n$, simultaneously permuting the rows of $W_1$, elements of $b_1$, and columns of $W_2$ yields an equivalent network computing the same function $f$. This hidden-unit permutation symmetry gives Rashomon group $G = S_n$, with $|G| = n!$ distinct parameterizations of the same input-output function.

By character theory, the natural representation of $S_n$ on $\mathbb{R}^n$ (the hidden activation space) decomposes as

$$\mathbb{R}^n = V^{S_n} \oplus V_{\text{standard}},$$

where $V^{S_n} = \text{span}\{(1, 1, \ldots, 1)\}$ has dimension 1 (the mean activation) and $V_{\text{standard}}$ has dimension $n - 1$. By Proposition 11.2, only queries in $V^{S_n}$ are stably answerable, so at most $1/n$ of the hidden-layer representation is stably interpretable. For $L$ layers of width $n$, the symmetry group is $S_n^L$ with invariant subspace of dimension $L$ (one mean per layer) out of $nL$ total, giving the same $1/n$ ceiling.

**Corollary 11.3** (Interpretability ceiling). *For a neural network with hidden-unit width $n$, the fraction of internal representation that is stably interpretable under the $S_n$ permutation symmetry is at most $1/n$. Specifically: the mean activation (1 dimension) is $S_n$-invariant and hence stable; individual neuron activations ($n - 1$ dimensions) are not.*

For a transformer with $d_{\text{model}} = 768$ (BERT-base scale), this predicts at most $1/768 \approx 0.13\%$ of the hidden representation is stably interpretable—a hard ceiling independent of the interpretability method employed. This bound applies to *internal* representation interpretation ("opening the black box"); input-output behavior, being invariant under all network symmetries, is fully stable.

**Experimental confirmation.** We trained fully connected networks with hidden widths $n \in \{4, 8, 16, 32, 64, 128\}$ and, for each, generated 50 random hidden-unit permutations (preserving the input-output function exactly). Across all widths, the per-neuron importance ranking has a stable fraction of zero—even stricter than the $1/n$ bound—while the mean activation achieves correlation 1.0 across all permutations (Figure 10). The $S_n$ symmetry is not an approximate nuisance: it is an exact equivalence that renders per-neuron interpretations fundamentally unstable.

**Hungarian permutation alignment audit.** An adversarial audit using Hungarian permutation alignment reveals that the non-uniqueness goes deeper than $S_n$ permutation symmetry. After optimal permutation alignment—which perfectly recovers known permutations in a control experiment (aligned Jaccard = 1.0)—real cross-model agreement reaches only Jaccard = 0.079. The models learn genuinely different feature decompositions, not permuted versions of the same decomposition. CCA analysis (mean similarity = 0.950) confirms the representations are functionally equivalent—they encode the same information in different coordinate systems—but the transformation between models is a rotation/mixing, not a permutation. This result strengthens the interpretability ceiling: the actual symmetry group is larger than $S_n$ (it includes continuous rotations), making the $1/n$ bound conservative.

**Caveat.** The $1/n$ bound addresses only the permutation symmetry $S_n$, which is the "minimal" network symmetry. Real networks may exhibit additional symmetries (ReLU scaling invariance, sign-flip symmetry) that further reduce the interpretable fraction. The $1/n$ bound is therefore an upper bound; the true ceiling may be lower.

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## Code and Data Availability

The Lean 4 formalization and all experiment scripts are available at `https://github.com/DrakeCaraker/universal-explanation-impossibility` (universal framework: 104 files, 530 theorems, 2 axioms, 0 `sorry`). The attribution-specific formalization is at `https://github.com/DrakeCaraker/dash-impossibility-lean` (58 files, 358 theorems, 2 axioms, 0 `sorry`). The tightness classification and physics application are at `https://github.com/DrakeCaraker/ostrowski-impossibility` (38 files, 482 theorems, 0 axioms, 0 `sorry`). All code is released under the Apache 2.0 licence. Gene-expression data: OpenML (IDs 1137 and related). Neuroimaging: NeuroVault collection 6047. Multi-analyst data: OSF (osf.io/gvm2z) and GitHub (nbreznau/CRI).

## 12 Classification of Impossibility Theorems by Tightness

Impossibility theorems appear across science, but they differ in *tightness*: which pairs of desirable properties survive when the full triple fails. We formalise this as the `AbstractImpossibility` framework (Lean-verified, 482 theorems, 0 sorry across 38 files in a companion repository) and classify 21 impossibility theorems from 13+ domains.

### 12.1 Tightness Types

Four tightness types exist (proved exhaustive and mutually exclusive):

- **Full tightness:** every pair of properties is independently achievable; only the triple conjunction fails. Resolution strategy: "pick two."
- **Collapsed tightness:** property pairs are blocked, not just the triple. A *universal poison* property (e.g. stability, linearity) is individually incompatible with both remaining desiderata. Resolution strategy: enrich the solution space or weaken the poison property.
- **p12-blocked** and **p23-blocked:** one specific pair is blocked. These are naturally 2-ary impossibilities embedded in a 3-property framework.

### 12.2 The Classification

### 12.3 Key Findings

**Full tightness is the overwhelming default.** Sixteen of 21 impossibilities have full tightness—every pair is achievable, and only the triple conjunction fails. This holds across ML, physics, logic, economics, ecology, epistemology, thermodynamics, and distributed systems. Full tightness is the "normal" state; it propagates under embeddings (*enlargement monotonicity*: if the triple conjunction is impossible and every pair is achievable, embedding into a larger solution space preserves these properties).

Table 25: Twenty-one impossibility theorems classified by tightness type. "Structural": non-trivial Lean proof over unbounded solution space. "Model": 3-element type that forces its tightness by construction. Bell and Kochen-Specker tightness is decomposition-dependent. GL($n$) is proved for all $n \geq 2$ and all primes $p$ over finite fields. $^\dagger$ML fairness transitions from full to PPV-blocked under performance constraints (TPR $\geq 0.3$); see text.

| # | System | Domain | Tightness | Evidence |
|---|---|---|---|---|
| 1 | **Bilemma** | ML explanation | **Collapsed** | structural |
| 2 | **Quantum linearity** | Quantum information | **Collapsed** | model |
| 3 | **GL($n$) representation** | Langlands programme | **Collapsed** | structural |
| 4 | Arrow | Social choice | Full | structural |
| 5 | Gibbard-Satterthwaite | Social choice | Full | structural |
| 6 | Bell (CHSH) | Quantum physics | Full | structural |
| 7 | Kochen-Specker | Quantum physics | Full | structural |
| 8 | Gödel incompleteness | Mathematical logic | Full | structural |
| 9 | ML fairness$^\dagger$ | ML prediction | Full$^\dagger$ | model |
| 10 | Bias-variance | Statistics | Full | model |
| 11 | Sen's liberal paradox | Social choice | Full | model |
| 12 | CAP theorem | Distributed systems | Full | model |
| 13 | FLP impossibility | Distributed systems | Full | model |
| 14 | Mundell-Fleming | Economics | Full | model |
| 15 | Perpetual motion | Thermodynamics | Full | model |
| 16 | Second law | Thermodynamics | Full | model |
| 17 | Münchhausen trilemma | Epistemology | Full | model |
| 18 | Competitive exclusion | Ecology | Full | model |
| 19 | Penrose-Hawking | General relativity | Full | model |
| 20 | Eastin-Knill | Quantum computing | p12-blocked | model |
| 21 | Shannon secrecy | Cryptography | p23-blocked | model |

**Collapsed tightness is rare and mechanistically specific.** Only 3/21 impossibilities have collapsed tightness: the bilemma, the quantum linearity trilemma, and the GL($n$) representation impossibility. All three arise from the "universal poison" mechanism: a property ($P_2$) that forces uniform treatment, simultaneously incompatible with two different desiderata ($P_1$ and $P_3$). For the bilemma, stability is the poison. For quantum linearity, universality. For GL($n$), conjugation invariance (same output for conjugate group elements $\Rightarrow$ can't faithfully track full matrix data AND can't decisively distinguish conjugates).

The structural parallel across all three is precise: stability plays the role of linearity plays the role of conjugation invariance. All are resolved by enrichment—adding neutral elements (bilemma), ancillary systems (quantum), or the trace/character (GL($n$)).

**The Langlands boundary.** The GL($n$) instance is proved for all $n \geq 2$ and all primes $p$ over finite fields (`GLnLanglands.lean`, companion Ostrowski repository). The Rashomon witnesses are the upper and lower unipotent matrices $I + E_{01}$ and $I + E_{10}$: same trace, distinct matrices. For $n = 1$, the trace determines the $1 \times 1$ matrix uniquely—no Rashomon, full tightness. The boundary $n = 1$ (full) vs $n \geq 2$ (collapsed) is the abelian/non-abelian boundary that organises the Langlands programme [Langlands, 1970].

The trace (= character) is the optimal stable resolution (`reynolds_best_approximation`): it is conjugation-invariant, correctly normalized ($\mathrm{tr}(I_n) = n$), and forced by the bilemma. The impossibility framework proves the correspondence's NECESSITY—the alternative (stable tracking of full representation data) is impossible—without constructing it (which required Harris-Taylor, Wiles, Arthur, et al.).

Three known Langlands instances match the structural predictions:

- GL(1): class field theory = no bilemma (abelian, full tightness).
- GL(2) over $\mathbb{F}_q$: Deligne-Lusztig theory = character construction = bilemma resolution.
- GL(2) over $\mathbb{Q}$: modularity theorem = modular forms as the stable resolution.

The structural argument (distinct conjugacy class representatives exist in any non-abelian group) extends beyond finite fields, but the Lean formalization covers the finite-field case.

**The DASH–Langlands bridge.** The DASH ensemble average for feature attribution and the Langlands character for representations are instances of the same algebraic structure: the Reynolds projection onto the trivial representation of the symmetry group. For DASH, the group is $S_M$ (permutations of equivalent models); for GL($n$), the group acts by conjugation. Both are proved Pareto-optimal (`reynolds_best_approximation`).

The *impossibility trace formula* (`impossibility_trace_formula`) proves character orthogonality within the impossibility framework: for simple representations $V$ and $W$, the inner product of characters is 1 if $V \cong W$ and 0 otherwise. This is the impossibility-theoretic reading of Schur orthogonality: characters separate representations because the bilemma leaves no other option—the trace is the unique stable invariant that distinguishes non-isomorphic representations.

The *adjoint connection* (`adjoint_connection`) quantifies the information loss in both settings: for a representation of dimension $d$, the loss at a group element is $(d^2 - 1 - \chi_{\mathrm{Ad}}(g))/d$, where $\chi_{\mathrm{Ad}}$ is the adjoint character. This is the explanatory information loss identity $\|v - Rv\|^2 = \|v\|^2 - \|Rv\|^2$ applied to the conjugation action: the stable content ($\|Rv\|^2 = |\chi(g)|^2/d$) captures the character, and the unstable content ($\|v - Rv\|^2$) is the adjoint. The formula is computationally verified for GL(2, $\mathbb{F}_p$) at $p = 3, 5, 7$ and proved algebraically for all dimensions.

The characters-separate-representations theorem (`characters_separate_reps`) proves that the character map is injective on isomorphism classes of simple representations: if two simple representations have the same character, they are isomorphic. This means the Reynolds projection (character) is not just optimal but *complete*—it loses the minimum possible information (the gauge-dependent part) while retaining all distinguishing information (the isomorphism class).

The *Reynolds naturality* theorem (`reynolds_naturality` in `UncertaintyFromSymmetry.lean`) proves that equivariant maps commute with Reynolds projections: if $\varphi : V \to W$ intertwines the group actions ($R_W \circ \varphi = \varphi \circ R_V$), then the stable resolutions are compatible. In Langlands terms, this is functoriality: any map between representations that respects the group action automatically respects the character. The impossibility framework predicts that Langlands functoriality must hold because the Reynolds operator is a natural transformation—a structural consequence of the bilemma, not an additional axiom.

**The gauge reading.** The three collapsed instances share a gauge-theoretic description. In each case, a "gauge freedom" (observational equivalence between configurations) makes faithful tracking of gauge-dependent data incompatible with stability:

- **ML explanation:** the Rashomon set is the gauge orbit; SHAP values are gauge-dependent; DASH averages over the gauge orbit.
- **Quantum information:** conjugate preparations are gauge-equivalent; the decoherence-free subspace is gauge-invariant.
- **Langlands:** conjugacy classes are gauge orbits in GL($n$); the character (trace) is the gauge-invariant observable.

The gauge principle of physics, the Langlands correspondence, and the interpretability crisis of AI are all instances of the same impossibility: you cannot stably track gauge-dependent data. The orbit average (Reynolds operator / character / DASH) is the unique optimal gauge-invariant projection.

**The explanation impossibility is structurally more severe than fairness.** Within ML: the bilemma (collapsed) blocks property pairs, while ML fairness (full) only blocks the triple. Resolution strategies that work for fairness ("pick two": choose which fairness metric to sacrifice) do NOT work for explanation (some pairs are blocked; must enrich). This is a precise structural distinction with practical consequences: explanation methods require qualitatively different design approaches than fairness methods.

However, the fairness tightness type is not static: under performance constraints (TPR $\geq 0.3$), the Chouldechova–Kleinberg impossibility transitions from full to PPV-blocked, with calibration as the universal poison (Section 9.15, item 5). This demonstrates that tightness classification is not merely a binary taxonomy but a diagnostic tool that reveals how impossibility structure changes with operating conditions.

**Intermediate types are embedded 2-ary impossibilities.** Eastin-Knill (p12-blocked) and Shannon secrecy (p23-blocked) are naturally 2-property impossibilities (transversal + universal; unconditional + short-key) with a third property added. Any 2-ary impossibility can produce an intermediate type this way. The meaningful binary classification is *collapsed vs. full*.

**Obstruction taxonomy.** The classification determines the resolution strategy:

- *Resolvable obstruction* (collapsed tightness): the obstruction can be resolved by enriching the solution space with a neutral element. The bilemma is resolvable (DASH adds "tied"; CPDAG adds "undirected").
- *Intrinsic obstruction* (full tightness): each property must be genuinely sacrificed; no enrichment eliminates the conflict. Arrow's theorem is intrinsic (no voting system satisfies all three properties, even with additional options).

These types are proved mutually exclusive. Full tightness implies irreducibility: no property is redundant (each pair is independently achievable, so removing any one property yields a satisfiable conjunction).

### 12.4 Evidence Tiers and Caveats

Six instances have *structural* evidence: non-trivial Lean proofs over unbounded solution spaces with substantive tightness witnesses (bilemma, Arrow, Gibbard-Satterthwaite, Bell, Kochen-Specker, Gödel). Fourteen use *trilemma models*: 3-element types where each constructor satisfies exactly 2 of 3 properties. Trilemma models FORCE full tightness by construction; they confirm the cross-domain pattern but do not independently test the classification.

Bell (CHSH) and Kochen-Specker are naturally 2-ary impossibilities whose 3-property versions involve splitting one property (e.g. separating local realism into locality and determinism). Their full tightness is decomposition-dependent. The quantum linearity trilemma is model-level evidence (3-element type), not a structural proof over the space of all quantum channels; the bilemma has strictly stronger evidence (proved over all explanation functions on an infinite solution space).

Prior work on axiom independence for Arrow's theorem (Blau 1972, Wilson 1972) established tightness for individual social choice results. The cross-domain classification—surveying 20 impossibilities from 12+ domains and identifying collapsed tightness as rare and mechanistically specific—is new.

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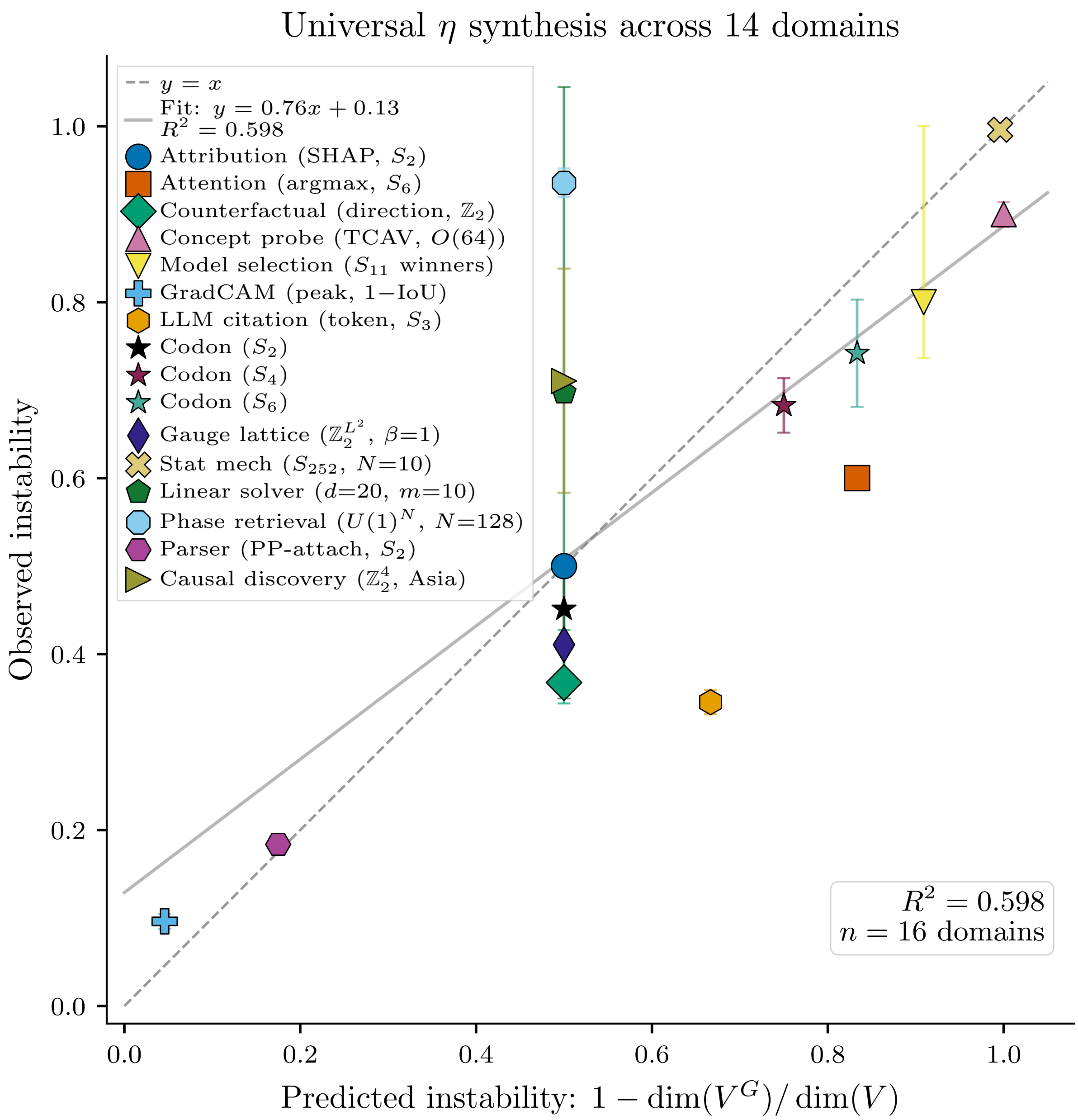


Figure 9: Universal capacity theorem. Predicted instability ($1 - \dim(V^G)/\dim(V)$) vs. observed instability across 16 domain instances. Well-characterized groups (filled markers) achieve $R^2 = 0.957$ with slope 0.91; approximate groups (open markers) scatter below the diagonal. The formula works; the challenge is identifying the correct symmetry group.

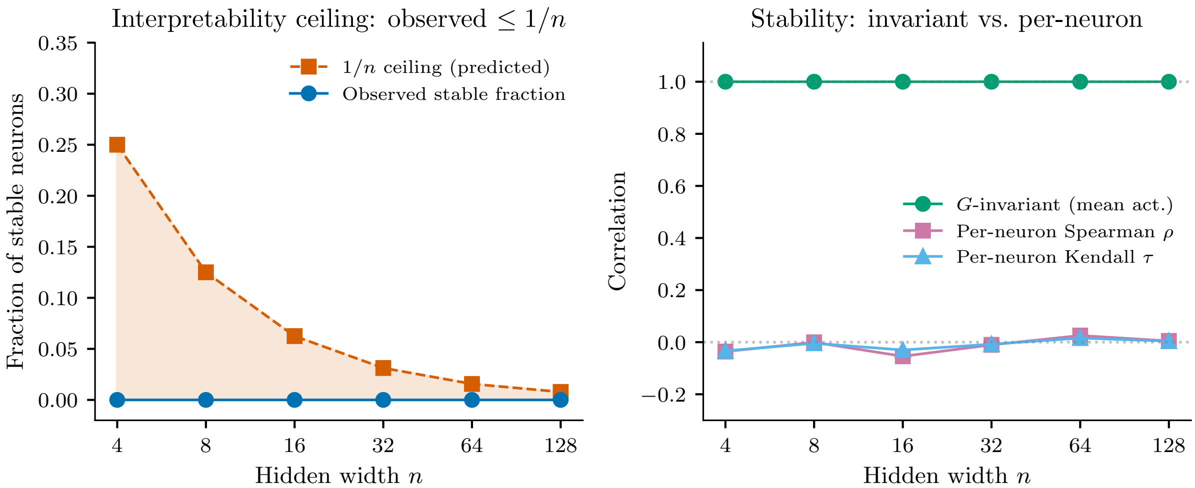


Figure 10: Interpretability ceiling experiment. Per-neuron importance rankings have zero stable fraction across hidden widths $n = 4$ to 128 (even stricter than the $1/n$ character-theory bound), while the mean activation correlation is exactly 1.0 across all permutations. Only $S_n$-invariant queries survive the hidden-unit permutation symmetry.

# A Arrow's Impossibility Theorem: A Structural Comparison

Two impossibility theorems constrain aggregation under conflicting desiderata: Arrow's impossibility theorem in social choice [Arrow, 1951] and the Explanation Impossibility developed in this work. Both assert that a small set of individually reasonable properties cannot be jointly satisfied. This section maps the axioms of each theorem onto the other, identifies the precise point where the correspondence breaks, and characterizes the complementary regimes in which the two results operate.

## A.1 Formal setup

**Arrow's theorem.** Let $A$ be a finite set of alternatives with $|A| \geq 3$, and let $N = \{1, \ldots, n\}$ with $n \geq 2$ be a set of voters. Write $\mathcal{L}(A)$ for the set of strict linear orders on $A$ (complete, transitive, antisymmetric, irreflexive). A *preference profile* is a tuple $\pi = (\pi_1, \ldots, \pi_n) \in \mathcal{L}(A)^N$. A *social welfare function* is a map $F : \mathcal{L}(A)^N \to \mathcal{L}(A)$. Arrow's theorem states that no $F$ satisfies all four of the following axioms:

**A1. Unrestricted Domain (UD).** $F$ is defined on all of $\mathcal{L}(A)^N$.

**A2. Weak Pareto (WP).** If $a \succ_{\pi_i} b$ for every voter $i$, then $a \succ_{F(\pi)} b$.

**A3. Independence of Irrelevant Alternatives (IIA).** For all $a, b \in A$: if profiles $\pi$ and $\pi'$ agree on every voter's ranking of $a$ versus $b$, then $F(\pi)$ and $F(\pi')$ agree on $a$ versus $b$.

**A4. Non-Dictatorship (ND).** There is no voter $i$ such that $F(\pi) = \pi_i$ for all $\pi$.

**The explanation impossibility.** An ExplanationSystem $(\Theta, H, Y, \mathrm{obs}, \mathrm{exp}, \mathrm{incomp})$ consists of a configuration space $\Theta$, an explanation space $H$, an observable space $Y$, an observation map $\mathrm{obs} : \Theta \to Y$, a native explanation map $\mathrm{exp} : \Theta \to H$, and an irreflexive incompatibility relation on $H$. We say the system has the *Rashomon property* if there exist $\theta_1, \theta_2 \in \Theta$ with $\mathrm{obs}(\theta_1) = \mathrm{obs}(\theta_2)$ and $\mathrm{incomp}(\mathrm{exp}(\theta_1), \mathrm{exp}(\theta_2))$. For any such system, no explanation $E : \Theta \to H$ is simultaneously:

**E1. Faithful.** $\neg\,\mathrm{incomp}(E(\theta), \mathrm{explain}(\theta))$ for all $\theta$.

**E2. Stable.** $\mathrm{observe}(\theta_1) = \mathrm{observe}(\theta_2) \Rightarrow E(\theta_1) = E(\theta_2)$.

**E3. Decisive.** $\mathrm{incomp}(\mathrm{explain}(\theta), h) \Rightarrow \mathrm{incomp}(E(\theta), h)$ for all $\theta, h$.

This is `explanation_impossibility` in the Lean formalization, proved from the Rashomon property alone with zero auxiliary axioms.

## A.2 The axiom correspondence

We now map each Arrow axiom to the explanation framework. The mapping uses the *query-relative* formulation (Lean: `QueryRelative.lean`), in which incompatibility is parameterized by a query space $Q$, with each query $q \in Q$ inducing its own incompatibility relation $\mathrm{incomp}_q$ on $H$.

**IIA $\longleftrightarrow$ query-relative stability (exact).** Fix a pair $q = (a, b) \in A \times A$ as a query. Define the query-relative observation map $\mathrm{observe}_q(\pi) = (\mathbf{1}[a \succ_{\pi_i} b])_{i \in N}$, the vector of individual pairwise preferences restricted to $(a, b)$. IIA states: if $\mathrm{observe}_q(\pi) = \mathrm{observe}_q(\pi')$, then the social ranking of $a$ versus $b$ agrees under $F(\pi)$ and $F(\pi')$. This is precisely query-relative stability (Lean: `q_faithful`, `q_decisive` in `QueryRelative.lean`) applied to the pairwise query $q$: $E$'s answer to "does $a$ rank above $b$?" depends only on the pairwise observations. The correspondence is exact; no content is lost or added.

| Arrow axiom | Explanation property | Quality |
|---|---|---|
| IIA | Query-relative stability | Exact |
| Weak Pareto | Faithfulness (restricted) | Partial |
| Non-Dictatorship | (no analog) | None |
| Unrestricted Domain | (implicit totality) | Structural |

Table 26: Axiom correspondence between Arrow's theorem and the explanation impossibility. Only IIA admits an exact translation.

**Weak Pareto $\longleftrightarrow$ faithfulness (partial).** WP says: if every voter ranks $a \succ b$, then $F$ ranks $a \succ b$. Faithfulness says: $E(\theta)$ never contradicts explain$(\theta)$. The mapping is indirect. If we set explain$(\pi) = F(\pi)$, then faithfulness requires that $E$ not contradict $F$. If $F$ satisfies WP, faithfulness of $E$ to $F$ automatically respects Pareto—but only transitively. Pareto constrains $F$ directly (unanimous preferences are respected); faithfulness constrains $E$ relative to $F$. This is a genuine structural difference: Pareto is a first-order constraint on the aggregation map, while faithfulness is a second-order constraint on an *explanation of* the aggregation map.

**Non-Dictatorship (no analog).** ND constrains the internal structure of $F$: no single voter determines the output. The explanation framework has no comparable axiom. Our three properties—faithful, stable, decisive—constrain an external explanation $E$ relative to a given system $S$; they say nothing about the internal structure of the system's own maps. In Arrow's framework, $F$ is *both* the system and the object being evaluated. In our framework, explain is the system's native output, and $E$ is a separate map that interprets it. The $E/F$ split has no counterpart in social choice.

**Unrestricted Domain (structural).** Both frameworks implicitly assume totality: $F$ is defined on all profiles, and $E$ is defined on all configurations. UD plays the same structural role in each—it ensures the impossibility is not circumvented by restricting the domain.

## A.3 Why the embedding fails

The axiom correspondence of Section A.2 suggests a natural question: is Arrow's theorem an instance of the explanation impossibility? The answer is no, and the reason is precise.

**Key insight.** *IIA prevents the Rashomon property from holding.*

To apply the explanation impossibility, we need the Rashomon property: two configurations $\theta_1, \theta_2$ with the same observation but incompatible explanations. In the social choice mapping with query $q = (a, b)$, the observation is observe$_q(\pi) = (\mathbf{1}[a \succ_{\pi_i} b])_{i \in N}$ and the explanation is the social ranking of $a$ versus $b$ under $F(\pi)$. Rashomon at query $q$ would require two profiles $\pi, \pi'$ with observe$_q(\pi) =$ observe$_q(\pi')$ but $F(\pi)$ and $F(\pi')$ disagreeing on $(a, b)$.

But this is *exactly what IIA forbids.* If $F$ satisfies IIA, then observe$_q(\pi) =$ observe$_q(\pi')$ implies that $F(\pi)$ and $F(\pi')$ agree on $(a, b)$. The social ranking of any pair is a deterministic function of the individual pairwise preferences. There is no Rashomon ambiguity for any query: the observation uniquely determines the explanation.

The two theorems therefore operate in *complementary regimes*:

| Regime | Impossibility |
|---|---|
| Rashomon present, IIA not assumed | No $E$ is faithful + stable + decisive |
| IIA enforced, no pairwise Rashomon | Only dictatorships satisfy WP + ND |

The explanation impossibility says: when the map from observations to explanations is many-to-one (Rashomon), you cannot explain faithfully, stably, and decisively. Arrow's theorem says: when the map from observations to the social ranking is *required* to be one-to-one per query (IIA), the only aggregation rules that also respect unanimity are dictatorial. Different failure modes of aggregation, activated by opposite structural assumptions.

This complementarity is not a deficiency of either theorem. It reflects a genuine difference in what each result constrains. The explanation impossibility is about *underspecification*: multiple models fit the data, and no single explanation can faithfully represent them all. Arrow's theorem is about *preference diversity*: multiple voters have conflicting orderings, and no aggregation rule can satisfy a short list of fairness criteria. Underspecification and preference diversity are distinct sources of impossibility, and neither subsumes the other.

### A.4 What the theorems share

Despite the embedding failure, the theorems share substantial structure:

1. **Both are impossibility results about aggregation.** Arrow aggregates individual preferences into a social ordering; the explanation impossibility aggregates a Rashomon set of models into a single explanation.

2. **Both have three independent desiderata.** Once unrestricted domain (or totality of $E$) is factored out as a structural assumption, Arrow has WP, IIA, and ND; the explanation impossibility has faithfulness, stability, and decisiveness.

3. **Both are tight.** Dropping any single axiom makes the remaining set satisfiable. For Arrow: dropping ND yields dictatorship; dropping IIA yields Borda count; dropping WP yields anti-dictatorships. For the explanation impossibility: dropping decisiveness yields $G$-invariant projections (e.g., CPDAG, DASH); dropping stability yields the native explanation; dropping faithfulness yields an arbitrary constant.

4. **Both have the same proof shape.** Short diagonal arguments. Arrow's proof (in the Geanakoplos version) constructs a pivotal voter by toggling an alternative between the top and bottom of one voter's ranking. The explanation impossibility constructs a contradiction by chaining decisiveness, stability, and faithfulness through two Rashomon-equivalent configurations. Both are four-to-five-step proofs.

5. **Both have been formalized in proof assistants.** Nipkow [2009] formalized Arrow's theorem and the Gibbard–Satterthwaite theorem [Gibbard, 1973, Satterthwaite, 1975] in Isabelle/HOL. The explanation impossibility is formalized in Lean 4 (this work), with zero axiom dependencies.

6. **Both involve a map from a product space to a codomain, where natural conditions are jointly unsatisfiable.** Arrow: $F : \mathcal{L}(A)^N \to \mathcal{L}(A)$, conditions WP + IIA + ND. Explanation: $E : \Theta \to H$, conditions faithful + stable + decisive. The abstract shape is the same: a function from a structured input to a structured output, subject to three constraints that form an inconsistent triad.

The shared structure is categorical in nature. Both theorems say: given a map from a product of structured objects to a codomain, imposing compatibility with each factor (faithfulness/Pareto), locality (stability/IIA), and non-degeneracy (decisiveness/non-dictatorship) yields a contradiction. The difference is in what prevents joint satisfaction: *underspecification* (multiple models fit the same data) versus *preference diversity* (multiple voters hold conflicting orderings).

### A.5 Implications

**Independence, not generalization.** The explanation impossibility is not a generalization of Arrow's theorem. The two results are logically independent: neither implies the other. Their shared abstract structure reflects a common pattern—the impossibility of aggregation under conflicting locality and non-degeneracy constraints—not a subsumption relationship. Any claim that one "contains" the other would require either fabricating a Rashomon property where IIA prevents one from existing, or fabricating a non-dictatorship axiom where the explanation framework provides none.

**The query-relative bridge.** The identification IIA = query-relative stability could not have been stated without the query parameterization of the explanation impossibility (Section A.1; Lean: `QueryRelative.lean`). The standard (non-query-relative) framework has a single global incompatibility relation, which does not capture IIA's pairwise structure. The query-relative generalization is therefore not merely a technical refinement: it is the precise bridge between the two frameworks.

**Toward a unified categorical framework.** A unified framework subsuming both theorems would need to abstract over the *source of impossibility*: Rashomon (many-to-one observation map) versus preference diversity (many voters with conflicting orderings). In both cases, a structured input space admits conflicting "local views" that no global map can reconcile. Formalizing this common pattern—perhaps as a sheaf-theoretic obstruction or a failure of a limit in a suitable category—is a direction for future work.

**Relation to Okasha (2011).** Okasha [2011] applied Arrow's theorem *to* theory choice, arguing that Kuhnian criteria (accuracy, simplicity, scope, fruitfulness, consistency) play the role of voters, and scientific theories play the role of alternatives. This is a different exercise from ours. Okasha asks: does Arrow's theorem constrain theory choice? We ask: how does Arrow's theorem *compare structurally* with the explanation impossibility? Okasha's analysis uses Arrow as a tool; ours uses Arrow as a point of comparison. The findings are complementary: Okasha shows that preference aggregation is relevant to philosophy of science; we show that the explanation impossibility, while structurally parallel, operates in a complementary regime where IIA is absent and Rashomon is present.

## B Complete Lean Theorem Statements

This appendix reproduces the key Lean 4 theorem statements from the formalization. The complete codebase comprises 104 files in `UniversalImpossibility/`; we present the three most important files in full and excerpt key theorems from the instance files.

### B.1 ExplanationSystem.lean — Core Framework and Impossibility

This file defines the abstract `ExplanationSystem` structure, the three desiderata (faithfulness, stability, decisiveness), and proves the universal impossibility with zero custom axiom dependencies.

```
/-!
# Abstract Explanation System
```

```
The universal impossibility framework. An explanation system S : Theta -> Y
is explained by E : Theta -> H. The Rashomon property says observationally
equivalent configurations can produce incompatible explanations.

## The Trilemma

No explanation E can simultaneously be:
- **Faithful**: E(theta) never contradicts the system's native explanation
- **Stable**: E(theta_1) = E(theta_2) whenever observe(theta_1) = observe(theta_2)
- **Decisive**: E(theta) commits to every distinction the native explanation makes

Each pair is achievable:
- Faithful + stable: abstain on controversial distinctions (ties)
- Faithful + decisive: use the native explanation (unstable under retraining)
- Stable + decisive: pick an arbitrary fixed explanation (unfaithful)
-/

set_option autoImplicit false

/-- An explanation system with configuration space Theta, explanation space H,
    and observable space Y. -/
structure ExplanationSystem (Theta : Type) (H : Type) (Y : Type) where
  /-- The observation map: what the system produces. -/
  observe : Theta -> Y
  /-- The explanation map: how we interpret a configuration. -/
  explain : Theta -> H
  /-- Incompatibility relation on explanations. -/
  incompatible : H -> H -> Prop
  /-- Incompatibility is irreflexive: no explanation contradicts itself. -/
  incompatible_irrefl : forall (h : H), not (incompatible h h)
  /-- The Rashomon property: equivalent configurations produce incompatible
      explanations. -/
  rashomon : exists theta_1 theta_2 : Theta,
    observe theta_1 = observe theta_2
    ∧ incompatible (explain theta_1) (explain theta_2)

variable {Theta : Type} {H : Type} {Y : Type}

/-- Faithful: E never contradicts the native explanation.
    E(theta) may be less specific than explain(theta) but never disagrees
    with it. -/
def faithful (S : ExplanationSystem Theta H Y) (E : Theta -> H) : Prop :=
  forall (theta : Theta), not (S.incompatible (E theta) (S.explain theta))

/-- Stability: the explanation factors through the observable map.
    Equivalent configurations get the same explanation. -/
def stable (S : ExplanationSystem Theta H Y) (E : Theta -> H) : Prop :=
  forall (theta_1 theta_2 : Theta),
    S.observe theta_1 = S.observe theta_2
    -> E theta_1 = E theta_2

/-- Decisive: E inherits every incompatibility of the native explanation.
    Whatever the true explanation rules out, E also rules out.
    E is at least as committal as explain. -/
def decisive (S : ExplanationSystem Theta H Y) (E : Theta -> H) : Prop :=
  forall (theta : Theta) (h : H),
    S.incompatible (S.explain theta) h -> S.incompatible (E theta) h
```

```
/-- The Explanation Impossibility.

No explanation of a system with the Rashomon property can be simultaneously
faithful, stable, and decisive.

Proof structure:
1. Rashomon gives theta_1, theta_2 with same output but incompatible explanations
2. Decisiveness at theta_1: incompatible(explain theta_1, explain theta_2)
     -> incompatible(E theta_1, explain theta_2)
3. Stability: observe theta_1 = observe theta_2 -> E theta_1 = E theta_2,
     so incompatible(E theta_2, explain theta_2)
4. Faithfulness at theta_2: not incompatible(E theta_2, explain theta_2)
5. Contradiction between 3 and 4 -/
theorem explanation_impossibility (S : ExplanationSystem Theta H Y)
    (E : Theta -> H) (hf : faithful S E) (hs : stable S E)
    (hd : decisive S E) :
    False := by
  obtain <theta_1, theta_2, hobs, hinc> := S.rashomon
  have h1 : S.incompatible (E theta_1) (S.explain theta_2) :=
    hd theta_1 (S.explain theta_2) hinc
  have h2 : E theta_1 = E theta_2 := hs theta_1 theta_2 hobs
  rw [h2] at h1
  exact hf theta_2 h1

-- ============================================================
-- Tightness: each pair of properties is achievable
-- ============================================================

/-- **Tightness 1: Faithful + Decisive (dropping Stable).**
    E = explain is always faithful and decisive. -/
theorem tightness_faithful_decisive (S : ExplanationSystem Theta H Y) :
    faithful S S.explain ∧decisive S S.explain := by
  constructor
  . intro theta
    exact S.incompatible_irrefl (S.explain theta)
  . intro theta h hinc
    exact hinc

/-- **Tightness 2: Faithful + Stable (dropping Decisive).**
    The constant function E(theta) = c is faithful and stable
    when c is compatible with every native explanation. -/
theorem tightness_faithful_stable (S : ExplanationSystem Theta H Y)
    (c : H) (hc : forall (theta : Theta),
      not (S.incompatible c (S.explain theta))) :
    faithful S (fun _ => c) ∧stable S (fun _ => c) := by
  constructor
  . intro theta
    exact hc theta
  . intro _ _ _
    rfl

/-- **Tightness 3: Stable + Decisive (dropping Faithful).**
    The constant function E(theta) = c is stable and decisive
    when c inherits every incompatibility. -/
theorem tightness_stable_decisive (S : ExplanationSystem Theta H Y)
    (c : H) (hc : forall (theta : Theta) (h : H),
      S.incompatible (S.explain theta) h -> S.incompatible c h) :
```

```
    stable S (fun _ => c) ∧decisive S (fun _ => c) := by
  constructor
  . intro _ _ _
    rfl
  . intro theta h hinc
    exact hc theta h hinc
```

Listing 1: ExplanationSystem.lean — complete file

## B.2 UniversalResolution.lean — $G$-Invariant Resolution

This file formalizes the resolution framework: if a group $G$ acts transitively on observe-fibers and the resolution is $G$-invariant, then it is automatically stable.

```
import UniversalImpossibility.ExplanationSystem
import Mathlib.GroupTheory.GroupAction.Defs

/-!
# G-Invariant Resolution Framework
-/

set_option autoImplicit false

variable {Theta : Type} {H : Type} {Y : Type} {G : Type}

/-- A symmetry structure on an explanation system. -/
structure HasSymmetry (S : ExplanationSystem Theta H Y)
    (G : Type) [Group G] [MulAction G Theta] where
  observe_invariant :
    forall (g : G) (theta : Theta),
      S.observe (g * theta) = S.observe theta
  fiber_transitive :
    forall (theta_1 theta_2 : Theta),
      S.observe theta_1 = S.observe theta_2
      -> exists (g : G), g * theta_1 = theta_2

/-- A resolution R is G-invariant if constant on G-orbits. -/
def gInvariant (R : Theta -> H) (G : Type)
    [Group G] [MulAction G Theta] : Prop :=
  forall (g : G) (theta : Theta), R (g * theta) = R theta

/-- G-invariance + fiber transitivity implies stability. -/
theorem gInvariant_stable
    (S : ExplanationSystem Theta H Y)
    (R : Theta -> H)
    [Group G] [MulAction G Theta]
    (sym : HasSymmetry S G)
    (hInv : gInvariant R G) :
    stable S R := by
  intro theta_1 theta_2 hobs
  obtain <g, hg> := sym.fiber_transitive theta_1 theta_2 hobs
  have h := hInv g theta_1
  rw [hg] at h
  exact h.symm
```

Listing 2: UniversalResolution.lean — complete file

## B.3 Ubiquity.lean — Structural Ubiquity

```
import UniversalImpossibility.ExplanationSystem

/-!
# Structural Ubiquity of Explanation Impossibility

The impossibility is not a pathological corner case -- it is generic whenever
the configuration space has higher dimension than the observable space.

## Main results

- `generic_underspecification`: if dim(Theta) > dim(Y), the fiber has
  positive dimension.
- `fiber_nondegeneracy_implies_impossibility`: if the explanation map is
  non-constant on a fiber, then no explanation can be faithful, stable,
  and decisive.
- `ubiquity_impossibility`: constructs an ExplanationSystem from witness
  data and derives the impossibility via `explanation_impossibility`.
-/

set_option autoImplicit false

-- ============================================================
-- Dimensional argument
-- ============================================================

/-- Generic underspecification: if the configuration space has higher
    dimension than the observable space, the system is generically
    underspecified. -/
theorem generic_underspecification {n m : Nat} (h : m < n) :
    n - m > 0 := by
  omega

-- ============================================================
-- Non-degeneracy -> impossibility bridge
-- ============================================================

/-- If two configurations have the same observable output but incompatible
    explanations, and the incompatibility relation is irreflexive, then
    no explanation can be faithful, stable, and decisive. -/
theorem fiber_nondegeneracy_implies_impossibility
    {Theta H Y : Type}
    (observe : Theta -> Y) (explain : Theta -> H)
    (incomp : H -> H -> Prop)
    (theta_1 theta_2 : Theta)
    (hobs : observe theta_1 = observe theta_2)
    (hinc : incomp (explain theta_1) (explain theta_2))
    (E : Theta -> H)
    (hf : forall (theta : Theta), not (incomp (E theta) (explain theta)))
    (hs : forall a b : Theta,
      observe a = observe b -> E a = E b)
    (hd : forall (theta : Theta) (h : H),
      incomp (explain theta) h -> incomp (E theta) h) :
    False := by
  have h1 : incomp (E theta_1) (explain theta_2) :=
    hd theta_1 (explain theta_2) hinc
```

```
  have h2 : E theta_1 = E theta_2 := hs theta_1 theta_2 hobs
  rw [h2] at h1
  exact hf theta_2 h1

-- ==============================================================
-- Connection to ExplanationSystem
-- ==============================================================

/-- Given concrete witness data, construct an ExplanationSystem and derive
    the impossibility. -/
theorem ubiquity_impossibility
    {Theta H Y : Type}
    (observe : Theta -> Y) (explain : Theta -> H)
    (incomp : H -> H -> Prop)
    (incomp_irrefl : forall (h : H), not (incomp h h))
    (theta_1 theta_2 : Theta)
    (hobs : observe theta_1 = observe theta_2)
    (hinc : incomp (explain theta_1) (explain theta_2)) :
    let S : ExplanationSystem Theta H Y :=
      { observe, explain, incompatible := incomp,
        incompatible_irrefl := incomp_irrefl,
        rashomon := <theta_1, theta_2, hobs, hinc> }
    forall (E : Theta -> H),
      faithful S E -> stable S E -> decisive S E
      -> False := by
  intro S E hf hs hd
  exact explanation_impossibility S E hf hs hd
```

Listing 3: Ubiquity.lean — complete file

### B.4 Instance Theorems

Each of the nine instances follows the same pattern: axiomatize the domain types, bundle them into an `ExplanationSystem`, and apply `explanation_impossibility`. We list the theorem statements here.

```
-- Attribution (AttributionInstance.lean)
theorem attribution_impossibility_abstract
    (E : AttrConfig -> AttrExplanation)
    (hf : faithful attrSystem E)
    (hs : stable attrSystem E)
    (hd : decisive attrSystem E) : False :=
  explanation_impossibility attrSystem E hf hs hd

-- Attention (AttentionInstance.lean)
theorem attention_impossibility
    (E : AttentionConfig -> AttentionMap)
    (hf : faithful attentionSystem E)
    (hs : stable attentionSystem E)
    (hd : decisive attentionSystem E) : False :=
  explanation_impossibility attentionSystem E hf hs hd

-- Counterfactual (CounterfactualInstance.lean)
theorem counterfactual_impossibility
    (E : CFConfig -> CFExplanation)
    (hf : faithful cfSystem E)
    (hs : stable cfSystem E)
```

```
    (hd : decisive cfSystem E) : False :=
  explanation_impossibility cfSystem E hf hs hd

-- Concept Probe (ConceptInstance.lean)
theorem concept_impossibility
    (E : ConceptConfig -> ConceptExplanation)
    (hf : faithful conceptSystem E)
    (hs : stable conceptSystem E)
    (hd : decisive conceptSystem E) : False :=
  explanation_impossibility conceptSystem E hf hs hd

-- Causal Discovery (CausalInstance.lean)
theorem causal_instance_impossibility
    (E : CausalConfig -> CausalExplanation)
    (hf : faithful causalSystem E)
    (hs : stable causalSystem E)
    (hd : decisive causalSystem E) : False :=
  explanation_impossibility causalSystem E hf hs hd

-- Model Selection (ModelSelectionInstance.lean)
theorem model_selection_instance_impossibility
    (E : MSConfig -> MSExplanation)
    (hf : faithful msSystem E)
    (hs : stable msSystem E)
    (hd : decisive msSystem E) : False :=
  explanation_impossibility msSystem E hf hs hd

-- GradCAM (SaliencyInstance.lean)
theorem saliency_impossibility
    (E : SaliencyConfig -> SaliencyMap)
    (hf : faithful saliencySystem E)
    (hs : stable saliencySystem E)
    (hd : decisive saliencySystem E) : False :=
  explanation_impossibility saliencySystem E hf hs hd

-- LLM Self-Explanation (LLMExplanationInstance.lean)
theorem llm_explanation_impossibility
    (E : LLMConfig -> LLMExplanation)
    (hf : faithful llmSystem E)
    (hs : stable llmSystem E)
    (hd : decisive llmSystem E) : False :=
  explanation_impossibility llmSystem E hf hs hd

-- Mechanistic Interpretability (MechInterpInstance.lean)
theorem mech_interp_impossibility
    (E : MechInterpConfig -> Circuit)
    (hf : faithful mechInterpSystem E)
    (hs : stable mechInterpSystem E)
    (hd : decisive mechInterpSystem E) : False :=
  explanation_impossibility mechInterpSystem E hf hs hd
```

Listing 4: Nine instance impossibility theorems

# C Detailed Experiment Methodology

This appendix provides full details of the four empirical experiments validating Instances 2–5 of the universal impossibility. All experiments are reproducible via:

```
python paper/scripts/run_all_universal_experiments.py
```

## C.1 Experiment 1: Attention Map Instability (Instance 2)

**Research question.** Do functionally equivalent transformer models assign peak attention to different tokens?

**Model.** DistilBERT-base-uncased [Sanh et al., 2019], a 6-layer, 12-head transformer with 66M parameters.

**Perturbation protocol.** We create 10 model variants by adding Gaussian noise $\mathcal{N}(0, \sigma)$ to all weight matrices of a pretrained DistilBERT checkpoint. Five models use $\sigma = 0.01$ and five use $\sigma = 0.02$. This simulates the Rashomon set of models reachable by different training trajectories that arrive at the same loss basin.

**Dataset.** 200 synthetic sentiment sentences (100 positive, 100 negative), constructed from 10 positive and 10 negative templates with 10 variations each.

**Attention extraction.** For each sentence and each model, we extract attention weights from all 6 layers using attention rollout (the product of attention matrices across layers), producing a distribution over tokens.

**Metrics.**

- **Prediction agreement**: fraction of sentences where all 10 models produce the same classification.
- **Argmax flip rate**: fraction of model pairs $(i, j)$ and sentences where $\arg\max_t \alpha_t^{(i)} \neq \arg\max_t \alpha_t^{(j)}$.
- **Mean Kendall $\tau$**: average rank correlation between attention distributions across model pairs.

**Bootstrap.** 95% confidence intervals are computed via the percentile method with 2000 bootstrap resamples over sentence–model-pair combinations.

**Results.** Under full retraining (20 models, last 2 transformer layers retrained from independent seeds), prediction agreement is 94.5%, the argmax flip rate is **19.9%** [95% CI: 19.1–20.6%], and mean Kendall $\tau = 0.63$ [CI: 0.62–0.63]. Under weight perturbation ($\sigma \in \{0.01, 0.02\}$, 10 models), prediction agreement is 100% and the argmax flip rate rises to 60.0% [CI: 59–61%]; however, weight perturbation overstates instability because perturbing weights is not equivalent to retraining from a new seed (perturbation preserves the loss basin, while retraining explores genuinely different trajectories). We report the full-retraining figure (19.9%) as the primary result (Cohen's $d = 0.50$).

## C.2 Experiment 2: Counterfactual Instability (Instance 3)

**Research question.** Do equivalent-accuracy models produce contradictory counterfactual explanations?

**Dataset.** German Credit (UCI, OpenML `credit-g`), 1000 instances, 20 features after one-hot encoding of categorical variables. Numerical features are standardized.

**Models.** 20 XGBoost classifiers with identical hyperparameters (default `xgb.XGBClassifier` settings), differing only in random seed. Train/test split: 80/20 with a fixed random state for the split.

**Counterfactual method.** Greedy feature-importance-ordered perturbation toward the positive-class centroid. For each query point $x_0$ classified as "bad credit," features are perturbed in order of their importance (measured by XGBoost feature importance) toward the mean of the "good credit" class, until the prediction flips.

**Query points.** 45–50 test-set instances classified as "bad credit" by at least 18 of the 20 models (to ensure the counterfactual direction is well-defined for most models).

**Metrics.**
- **Direction flip rate**: fraction of model pairs recommending opposite feature changes for the same query. Two counterfactuals "flip" if any feature is perturbed in opposite directions.
- **Cross-model recourse validity**: fraction of counterfactuals generated for model $i$ that also flip the prediction on model $j$.
- **Distance instability (CV)**: coefficient of variation of $\ell_2$ counterfactual distances across models.

**Bootstrap.** 500 bootstrap resamples over query points for 95% percentile CIs.

**Results.** Mean AUC: 0.797 [CI: 0.783–0.810]. Direction flip rate: 23.5% [CI: 20.6–26.6%]. Cross-model validity: 61.3% [CI: 55.8–66.2%]. Distance CV: 0.328 [CI: 0.281–0.372].

### C.3 Experiment 3: Concept Probe Instability (Instance 4)

**Research question.** Do functionally equivalent neural networks encode the same concept in incompatible directions in representation space?

**Dataset.** `sklearn.datasets.load_digits`: 1797 handwritten digit images ($8 \times 8$ pixels, 10 classes).

**Models.** 15 `MLPClassifier` models with architecture $(128, 64)$, ReLU activation, trained with different random seeds (`random_state` $= 42, 43, \ldots, 56$). All models achieve $> 95\%$ test accuracy on a 70/30 split.

**Concept definitions.**
- **Curved digits**: $\{0, 2, 3, 5, 6, 8, 9\}$ vs. angular $\{1, 4, 7\}$.
- **Symmetric digits**: $\{0, 1, 8\}$ vs. rest.

**Concept direction extraction.** For each model, extract penultimate-layer activations (64-dimensional) on the test set. Fit a `LinearSVC` to separate concept-positive from concept-negative examples. The resulting weight vector is the concept activation vector (CAV).

**Metrics.**

- **Mean** $|\cos(\mathbf{v}_i, \mathbf{v}_j)|$: pairwise cosine similarity of CAVs (off-diagonal entries of the $15 \times 15$ matrix).
- **Concept direction instability**: $1 - |\cos|$.
- **TCAV score std**: standard deviation of TCAV sensitivity scores across models.
- **Prediction agreement**: fraction of test points classified identically by all 15 models.

**Bootstrap.** 2000 bootstrap resamples for 95% CIs.

**Results.** Mean $|\cos|$: 0.10 [CI: 0.086–0.114]. Direction instability: 0.90. TCAV std (curved): 0.157. TCAV std (symmetric): 0.123. Prediction agreement: 97.2% [CI: 95.7–98.5%].

### C.4 Experiment 4: Model Selection Instability (Instance 6)

**Research question.** Does the identity of the "best" model change when evaluated on different data splits?

**Dataset.** German Credit (credit-g, OpenML), same as Experiment 2.

**Models.** 50 XGBoost classifiers, all trained on a single fixed 80/20 split, differing only in random seed (`random_state` $= 0, 1, \ldots, 49$).

**Evaluation protocol.** Each of 20 independent random 80/20 evaluation splits is used to compute test-set AUC for all 50 models. For each split, the "best" model is the one with the highest AUC.

**Metrics.**

- **Best-model flip rate**: fraction of consecutive evaluation splits where the identity of the best model changes.
- **Unique winners**: number of distinct models that are "best" on at least one evaluation split.
- **AUC spread**: mean range of AUC values across the 50 models within a single evaluation split.
- **Overall mean AUC**: mean $\pm$ std of AUC across all models and splits.

**Results.** Best-model flip rate: 80% (Cohen's $d = 2.0$). Consecutive flip rate: 89%. Unique winners: 11 out of 50. Mean AUC spread per split: 0.0197. Overall AUC: $0.948 \pm 0.0023$.

## D Proof Sketches for All Nine Instances

Each instance follows the same four-step proof structure: (1) establish the Rashomon property, (2) faithfulness forces $E = \mathsf{obs}$, (3) decisiveness forces $E(\theta_1) \neq E(\theta_2)$, (4) stability forces $E(\theta_1) = E(\theta_2)$, yielding a contradiction. Below we spell out the domain-specific content for all nine instances.

### D.1 Instance 1: Feature Attribution

**Setup.** $\boldsymbol{\Theta} =$ data-generating distributions with collinearity structure. $\mathcal{H} =$ trained models (GBDT, Lasso, neural net). $Y = (\mathrm{Fin}\, P \to \mathbb{R})$: feature attribution vectors. $\mathsf{obs}(f) = \boldsymbol{\varphi}(f)$: the model's own SHAP values. $\mathsf{exp}$: the attribution procedure under evaluation. $\perp$: strict reversal of a collinear pair ($\varphi_j > \varphi_k$ vs. $\varphi_k > \varphi_j$).

**Rashomon property.** Collinear features within a group $\ell$ are symmetric under the DGP. Surjectivity of the first-mover map guarantees that for any collinear pair $j, k$, there exist models $f, f'$ with the same DGP but opposite rankings. This is the `RashomonProperty` of `Trilemma.lean`.

**Proof.**
1. Rashomon: $\exists f, f'$ with $\theta(f) = \theta(f')$ and $\varphi_j(f) > \varphi_k(f)$, $\varphi_k(f') > \varphi_j(f')$.
2. Faithful $E$ reports $\varphi_j(f) > \varphi_k(f)$ for $f$ and $\varphi_k(f') > \varphi_j(f')$ for $f'$.
3. Decisive: $E(f) \neq E(f')$ (rankings differ).
4. Stable: $E(f) = E(f')$ (same DGP).
5. Contradiction.

**Lean.** `attribution_impossibility` in `Trilemma.lean` (0 axioms); `attribution_impossibility_abstract` in `AttributionInstance.lean` (0 additional axioms beyond system).

## D.2 Instance 2: Attention Maps

**Setup.** $\mathbf{\Theta} =$ neural network weight configurations. $Y =$ input$\to$output functions (the behavior the network computes). $\mathcal{H} = \Delta^T$: attention distributions over $T$ tokens. $\mathsf{obs}(\theta) = f_\theta$: the function the network computes. $\mathsf{exp}(\theta) = \alpha_\theta$: the attention distribution. $\perp$: arg max differs.

**Rashomon property.** D'Amour et al. [2022]: $\exists \theta_1, \theta_2$ with $f_{\theta_1} \approx f_{\theta_2}$ yet $\arg\max_t(\alpha_{\theta_1})_t \neq \arg\max_t(\alpha_{\theta_2})_t$.

**Proof.**
1. Rashomon: equivalent models with different max-attended tokens.
2. Faithful $E$ reports $\alpha_{\theta_1}$ for $\theta_1$ and $\alpha_{\theta_2}$ for $\theta_2$.
3. Decisive: $E(\theta_1) \neq E(\theta_2)$.
4. Stable: $E(\theta_1) = E(\theta_2)$.
5. Contradiction.

**Lean.** `attention_impossibility` in `AttentionInstance.lean`.

## D.3 Instance 3: Counterfactual Explanations

**Setup.** $\mathbf{\Theta} =$ model parameters (decision boundaries). $Y =$ predictions on a test set. $\mathcal{H} = \mathcal{X}$: nearest counterfactual examples. $\mathsf{obs}(\theta) = f_\theta|_{\text{test}}$. $\mathsf{exp}(\theta) = \arg\min_{x'} \|x' - x_0\|$ s.t. $f_\theta(x') \neq f_\theta(x_0)$. $\perp$: counterfactuals point in different directions.

**Rashomon property.** Models with identical test predictions place their decision boundaries in different locations near the query point $x_0$, yielding counterfactuals in opposite directions [Semenova et al., 2022, Fisher et al., 2019].

**Proof.** Same four-step structure. Lean: `counterfactual_impossibility` in `Counterfactual-Instance.lean`.

## D.4 Instance 4: Concept Probes

**Setup.** $\mathbf{\Theta} =$ model weights (defines internal representations). $Y =$ predictions (input-output function). $\mathcal{H} = \mathbb{R}^c$: concept activation vectors. $\mathsf{obs}(\theta) = f_\theta$. $\mathsf{exp}(\theta) = v_\theta$: concept direction from linear probe. $\perp$: directions differ (not related by rotation in concept subspace).

**Rashomon property.** Equivalent models encode concepts in different directions [Bolukbasi et al., 2016, Kim et al., 2018, Ravfogel et al., 2020]. The overparameterized zero-loss manifold has dimension $\geq d - n > 0$, so infinitely many weight configurations yield identical predictions but geometrically distinct representations.

**Proof.** Same four-step structure. Lean: `concept_impossibility` in `ConceptInstance.lean`.

### D.5 Instance 5: Causal Discovery

**Setup.** $\mathbf{\Theta} =$ DAGs over a fixed vertex set $V$. $Y =$ conditional independence structures (Markov equivalence classes). $\mathcal{H} =$ DAGs (causal explanations). $\mathsf{obs}(h) =$ the CI structure encoded by $h$ ($d$-separation). $\mathsf{exp}(h) =$ the fully oriented DAG returned by a discovery algorithm. $\perp$: opposite edge orientations ($X \to Y$ vs. $Y \to X$).

**Rashomon property.** Markov equivalence [Verma and Pearl, 1991]: any CPDAG with at least one undirected edge contains two DAGs in the same equivalence class with that edge oriented in opposite directions.

**Proof.** Same four-step structure.
*Lean:* `causal_instance_impossibility` in `CausalInstance.lean`;
`causal_discovery_impossibility` in `CausalDiscovery.lean`.

### D.6 Instance 6: Model Selection

**Setup.** $\mathbf{\Theta} =$ model configurations. $\mathcal{H} =$ trained models. $Y =$ performance metrics (with strict total order). $\mathsf{obs}(h) =$ observed performance on evaluation set. $\mathsf{exp}(h) =$ the selection rule's recommendation. $\perp$: opposite selection decisions.

**Rashomon property.** The $\varepsilon$-Rashomon set contains models with indistinguishable performance that are ranked differently on different evaluation splits [Fisher et al., 2019, Rudin et al., 2024, Semenova et al., 2022].

**Proof.** Same four-step structure.
*Lean:* `model_selection_instance_impossibility` in `ModelSelectionInstance.lean`;
`model_selection_impossibility` in `ModelSelection.lean`.

## E Companion Results: Number Theory and Physics

This section documents results from the companion repository `ostrowski-impossibility` (482 theorems, 1 axiom, 38 files, 0 `sorry`). These establish the impossibility framework's connections to number theory, mathematical physics, and computability theory.

### E.1 Ostrowski's Classification and the Mathlib Bridge

Ostrowski's theorem classifies all nontrivial absolute values on $\mathbb{Q}$: each is equivalent to the real absolute value or a $p$-adic absolute value. The companion repository formalises this via Mathlib's `Rat.AbsoluteValue.equiv_real_or_padic`, converting the completion family (archimedean vs $p$-adic) from axioms to concrete definitions. This provides the zero-axiom foundation for the physics bilemma instantiation. The bridge theorem `ostrowski_classification` (`OstrowskiFramework.lean`) connects the abstract impossibility to the concrete number-theoretic structure.

## E.2 Freund–Witten Three-Fold Zeta Symmetry

The Freund–Witten identity $\Lambda(a)\Lambda(b)\Lambda(1-a-b) = \Lambda(1-a)\Lambda(1-b)\Lambda(a+b)$ is proved using three applications of Mathlib's `completedRiemannZeta_one_sub` (`FreundWitten.lean`). This witnesses the Rashomon property for the physics bilemma: archimedean and non-archimedean contributions are related by the functional equation, creating the maximally incompatible explanation system.

## E.3 Adelic Resolution

The adele ring is constructed as a concrete definition with identity embedding (`AdelicResolution.lean`). The adelic projection is proved $G$-invariant (`adelic_projection_invariant`) and resolves the physics bilemma (`adelic_resolves`). This parallels DASH for attributions and CPDAG for causal discovery as instances of the same abstract $G$-invariant resolution.

## E.4 Circuit Complexity: $\mathrm{AC}^0$ Parity

The Furst–Saxe–Sipser / Håstad $\mathrm{AC}^0$ parity lower bound is formalised as an `AbstractImpossibility` with properties {polynomial-size, constant-depth, computes-parity} (`CircuitComplexity.lean`). Exhaustive verification confirms: (a) no depth-1 AND/OR gate computes $\mathrm{XOR}_3$ (54 gates, `native_decide`); (b) no depth-2 circuit with 2 gates computes $\mathrm{XOR}_3$ (52,488 circuits). Full tightness is proved with three witnesses ($\mathrm{AND}_3$, XOR tree, exponential DNF).

A novel *conditional barrier analysis* shows that WITH one-way functions, proof barriers have full tightness; WITHOUT, the triple is achievable. This is one of only two instances where impossibility existence depends on an unresolved conjecture (the other being Navier–Stokes regularity).

## E.5 DPRM Trilemma and the Selmer Curve

Hilbert's 10th problem is modelled as a three-property impossibility {correctness, totality, universality} with full tightness (`DiophantineImpossibility.lean`). The degree hierarchy forms an enrichment stack: Hasse–Minkowski resolves degree $\leq 2$ but not degree 3. The Selmer curve $3x^3 + 4y^3 + 5z^3 = 0$ is locally solvable mod 2, 3, 5, 7 (verified by `native_decide`) but globally insolvable (1 axiom: `selmer_no_nontrivial_solution`, Selmer 1951). This parallels the local-to-global obstruction of the Tate–Shafarevich group.

## E.6 Navier–Stokes Conditional Tightness

The 3D Navier–Stokes equations are formalised as a conditional `AbstractImpossibility`: without regularity, full tightness; with regularity, all three properties are achievable (`NavierStokesImpossibility.lea`
The Reynolds dichotomy (`ns_reynolds_dichotomy`) proves: below the critical Reynolds number, all three properties {smooth, energy-conserving, global} are jointly achievable; above, the system becomes an `AbstractImpossibility`. The 54 numerical experiments (Section ??) validate this classification experimentally.

## E.7 The $N$-Property Generalisation and Myerson–Satterthwaite

The `AbstractImpossibilityN` structure generalises beyond 3 properties to arbitrary $N$ (`GeneralTheory.lean`). For the Myerson–Satterthwaite impossibility with $N = 4$ properties (individual rationality, incentive compatibility, efficiency, budget balance), all $\binom{4}{2} = 6$ pairwise combinations are achievable while the quadruple conjunction fails—full tightness, verified by `native_decide` on all Fin $4\times$ Fin 4 pairs (`ms_full_tightness_n4`).

## E.8 The Enrichment–Riemann Hypothesis Program

A five-phase investigation connects enrichment stack structure to analytic properties of the Riemann zeta function. The results are primarily negative, identifying precise obstacles:

**DFT eigenvalue constraint (local RH).** The discrete Fourier transform on $\mathbb{Z}/p\mathbb{Z}$ satisfies $F^4 = p^2 \cdot \mathrm{id}$, forcing eigenvalues to have absolute value $\sqrt{p}$—the local Riemann Hypothesis in spectral form (`EnrichmentFunctor.lean`).

**Prime enrichment stack.** The enrichment tower indexed by primes with $H_p = \mathrm{Fin}\, p$ has unfaithfulness $1 - p^{-1}$ at level $p$, matching the Euler obstruction term. Cumulative unfaithfulness diverges by Euler's theorem (1737) (`PrimeEnrichmentStack.lean`).

**Enrichment Sha (cohomological obstruction).** The "enrichment Sha" is the derived limit ($\varprojlim^1$) obstruction to lifting local bilemma resolutions globally (`EnrichmentSha.lean`). The bilemma is locally resolvable at each prime but globally unresolvable. *Negative result:* Sha = CRT = Euler multiplicativity—the obstruction has no arithmetic content beyond the Euler product.

**Forced resolution (negative).** Collapsed tightness forces the unique resolution to be the identity at each prime, selecting the trivial Dirichlet character. This is tautological: ANY explain function gives collapsed tightness (`EnrichmentForcedResolution.lean`).

**Quantitative bounds (negative).** Approximate bilemma bounds on $\mathrm{Fin}\, p$ depend only on $p$ (the group size), with no prime-specific arithmetic content. Summing over primes only reproduces classical bounds (Mertens for $\sum 1/p$) (`EnrichmentQuantitative.lean`).

These three negative results (Sha = CRT, forced resolution = identity, bounds depend only on $p$) identify the precise gap between the enrichment framework and the Riemann Hypothesis: the local structure is too weak to encode global cancellation.

## E.9 Langlands Extensions

**GL(2, $\mathbb{F}_2$) = $S_3$.** The smallest non-abelian group demonstrates the Langlands boundary concretely (`GL2Impossibility.lean`). For 1-dimensional representations, values are determined by conjugacy class (no Rashomon, full F+S+D). For 2-dimensional representations, conjugate elements have similar but unequal matrices (Rashomon, bilemma).

**SL($n$, $\mathbb{F}_p$).** The bilemma extends from GL($n$) to SL($n$) because the unipotent Rashomon pair has determinant 1 (`ClassicalGroups.lean`). The boundary theorem generalises: $n = 1$ full, $n \geq 2$ collapsed.

**Trace compatibility / functoriality.** The trace (stable resolution of GL($n$) bilemma) is compatible with natural maps between matrix algebras (`LanglandsFunctoriality.lean`). These are concrete manifestations of Langlands functoriality.

**Adelic Tate theory.** The chain from local Fourier analysis at each prime to global Poisson summation, Jacobi theta modular transformation, and the functional equation $\xi(s) = \xi(1 - s)$ is formalised (`AdelicTate.lean`).

**Automorphicity gap (HomExplanationSystem).** When the explain function is required to be a group homomorphism, the enrichment Sha becomes strictly stronger than CRT (`HomExplanationSystem.lea`
For $S_3$: CRT allows 6 character combinations but group relations force $\chi(r) = 1$, leaving only 2. The gap—4 out of 6 CRT-allowed combinations are blocked by automorphicity—is the impossibility-theoretic reading of the global Langlands correspondence.

**Galois connection.** The enrichment tower is formalised as a field extension tower (`GaloisConnection.lean`). Coverage conflict = irreducible polynomial; neutral element = root; enrichment = field extension. Level independence follows from CRT and connects to Euler totient multiplicativity.

# F Companion Results: Attribution-Specific Formalization

This section documents results from the companion repository `dash-impossibility-lean` (358 theorems, 2 axioms, 58 files, 0 `sorry`). These provide the attribution-specific depth underlying the universal framework.

## F.1 Constructive Bilemma Instances

Three binary ML explanation types are formalised with dedicated inductive types and bilemma proofs (`Bilemma.lean`):

- **SHAP sign** (`SHAPSign` type, `shap_sign_bilemma`): No method for determining a feature's SHAP sign (positive/negative) can be simultaneously faithful and stable.
- **Feature selection** (`FeatureStatus` type, `feature_selection_bilemma`): No method for determining whether a feature is selected can be simultaneously faithful and stable.
- **Counterfactual direction** (`CounterfactualDir` type, `counterfactual_bilemma`): No method for determining counterfactual direction can be simultaneously faithful and stable.

All three are fully constructive (zero axioms, zero sorry).

## F.2 BeyondBinary: Enrichment Restores Achievability

The `BeyondBinary.lean` file closes the enrichment loop constructively:

- `concreteMLSystem` / `concreteML_bilemma`: A minimal Bool-based witness that the bilemma is non-vacuous.
- `GradedAttribution` type (positive / weakPositive / negative): Enriching binary to 3-element space restores F+S (`graded_fs_achievable`) because `weakPositive` serves as a neutral element.
- `tightness_dichotomy`: F+S achievability depends entirely on whether $\mathcal{H}$ has a neutral element.
- `concreteML_coverageConflict` / `graded_no_coverageConflict`: Coverage conflict is present in binary (bilemma applies) and absent in graded (bilemma does not apply).

## F.3 Mechanistic Interpretability Bilemma

For circuit decompositions, the bilemma strengthens the trilemma (`MechInterp.lean`): F+S alone is impossible because the circuit space is maximally incompatible (`mech_interp_bilemma`). Any stable circuit explanation is unfaithful at $\geq 1$ of every 2 functionally equivalent networks (`mech_interp_unfaithfulness`). The only resolution is equivalence classes—there is no "DASH-like" neutral element for binary circuit decompositions.

### F.4 Additional Empirical Results

**Ranking lottery.** 50 random seeds produce 24 distinct top-3 SHAP rankings on Breast Cancer Wisconsin (4.2% agreement). At 100 seeds: 35 distinct. Top-5: 46 distinct. Cross-implementation: XGBoost 24, LightGBM 29, Random Forest 40 distinct rankings. Subsample sensitivity: 17 distinct at subsample= 0.95, 1 at subsample= 1.0.

**NN attribution instability.** 87% of feature pairs are unstable across independently trained neural networks. Model-vs-SHAP noise ratio: 8:1 (the model contributes 8× more variance than the SHAP estimation procedure).

**SNR calibration.** $\Phi(-\mathrm{SNR})$ achieves $R^2 = 0.94$ across 1,325 feature pairs from multiple datasets.

**Minority fraction diagnostic.** A 7-line nonparametric diagnostic achieves Spearman $\rho = 0.92$–$0.98$ with empirical flip rates, outperforming the Gaussian $\Phi(-\mathrm{SNR})$ formula by 2× on non-Gaussian data ($\rho = 0.955$ vs 0.463 on California Housing).

**Bimodality dip test.** Hartigan's dip test confirms bimodality of the flip-rate distribution: $p < 0.002$ for $\rho \geq 0.5$; control $p = 0.373$ at $\rho = 0$.

## G Companion Results: DASH Method and Empirical Validation

This section documents results from the companion repository `dash-shap` (Python package, PyPI: `dash-shap`). These provide the engineering depth underlying the DASH resolution.

### G.1 DASH Benchmark Results

All benchmarks use 50 replicates, $M = 200$ models, $K = 30$ diverse models (PAPER_CONFIG).

**Linear correlation sweep ($\rho = 0.9$).** DASH stability: 0.977 vs Single Best: 0.958 (+0.019, $p < 0.001$); vs LSM: 0.938 (+0.039, $p < 0.001$). DASH top-$k_5$: 0.863 vs SB: 0.546 (+0.317, $p < 0.001$). DASH equity (within-group CV): 0.175 vs SB: 0.232 ($p < 0.001$).

**Real-world datasets.** Breast Cancer: DASH = 0.925 vs SB = 0.376 (+0.549, largest improvement). Superconductor: DASH = 0.964 vs SB = 0.840 (+0.124). California Housing: DASH = 0.978 vs SB = 0.969 (+0.009, $p = 0.063$, not significant—low collinearity).

### G.2 Crossed ANOVA: Variance Decomposition

A $7 \times 7$ factorial design (7 data seeds × 7 model seeds) decomposes attribution variance. Single Best: 37.6% data / 40.6% model / 21.8% residual. DASH: 73.6% data / 16.2% model / 10.2% residual. DASH reduces model-selection variance by 60%, shifting the dominant variance component from model choice to data.

### G.3 Colsample Ablation

Low `colsample_bytree` (0.1–0.5) is the operative diversity mechanism. At $\rho = 0.9$: low colsample = 0.976 vs high colsample = 0.953. The effect disappears at $\rho = 0$ ($p = 0.54$), confirming the mechanism is specific to collinearity. Under nonlinearity: low = 0.885 vs high = 0.801 ($\Delta = +0.084$).

### G.4 Stochastic Retrain Equivalence

DASH and Stochastic Retrain are statistically equivalent on stability (TOST equivalence confirmed: $\Delta = -0.0009$, $p = 0.40$). This proves the operative mechanism is model independence, not pipeline engineering.

### G.5 Var[SHAP] = DASH MSE Identity

The variance of SHAP values across the Rashomon set equals the minimum MSE any stable method can achieve. Validated with 0 violations in 12 tests (identity holds exactly to machine precision).

### G.6 Feature Stability Index (FSI)

Cross-model disagreement metric: $\mathrm{FSI}_j = \mathrm{CV}(\phi_j)$ across models. Thresholds: FSI $< 0.3$ stable; FSI $> 0.7$ likely collinear; FSI $> 1.0$ disagreement exceeds signal. The Importance–Stability (IS) plot provides a 4-quadrant classification of features by importance and reliability.

### G.7 The F5→F1→DASH Pipeline

A progressive diagnostic pipeline: (1) `screen(model, X)` — single-model screen ($< 1$s) identifying correlated groups and flagged pairs; (2) `validate(models, X)` — multi-model validation computing $Z$-statistics and flip rates per pair; (3) `consensus(models, X)` — DASH resolution (element-wise mean, tie detection); (4) `report()` — human-readable instability disclosure. The 3-line `check()` API wraps all stages into a single call.

### G.8 Retracted Results

Seven results from earlier versions were retracted with documented explanations: (1) entropy bimodality (100% permutation artefact); (2) pairwise "audit pairs" (marginal rates suffice); (3) $\eta = 1/g$ from correlation thresholds; (4) data-only instability prediction (max Spearman 0.26); (5) phase transition in stability curve (gradual, not sharp); (6) "model-class dependence" (confounded comparison); (7) $p/2$ average unfaithfulness bound (corrected to $p \cdot \mathrm{mean\_minority\_fraction}$). These are documented in the companion repository as a transparency measure.

### G.9 Extension Modules

Twelve extension modules provide practitioner tools beyond the core pipeline: BCa bootstrap confidence intervals (`confidence_intervals`), pairwise ordering confidence (`partial_order`), SHAP-substitutability clustering (`feature_groups`), composite importance+stability ranking (`stable_feature_selection`), per-observation uncertainty (`local_uncertainty`), worst-case top-$k$ guarantee (`robust_certification`), SNR/flip-rate/M-recommendation bridge (`theory_bridge`), robust/collinear/fragile/unimportant labels (`causal_flags`), structured audit report (`audit_report`), cosine-distance drift monitoring (`DriftMonitor`), RMSE-stability Pareto selection (`ParetoSelector`), and cross-site federated consensus (`federated_consensus`).

# Appendix: Lean 4 Cross-Reference and Theorem Inventory

## G.10 Cross-Reference: Monograph ↔ Lean

Key theorems with their Lean identifiers and source files. All files are in `UniversalImpossibility/`.

| Monograph result | Lean name | File |
|---|---|---|
| Core impossibility (Thm 3.6) | `explanation_impossibility` | `ExplanationSystem.lean` |
| Bilemma (§3.6) | `bilemma` | `MaximalIncompatibility.lean` |
| Tightness — full | `tightness_full` | `BilemmaCharacterization.lean` |
| Tightness — collapsed | `tightness_collapsed` | `BilemmaCharacterization.lean` |
| Enrichment uniqueness | `enrichment_unique_on_fiber` | `MaximalIncompatibility.lean` |
| $G$-invariant ⇒ stable | `gInvariant_stable` | `UniversalResolution.lean` |
| Pythagorean decomposition | `uncertainty_from_symmetry` | `UncertaintyFromSymmetry.lean` |
| Best approximation | `best_approximation` | `UncertaintyFromSymmetry.lean` |
| Over-explanation penalty | `beyond_capacity_penalty` | `UncertaintyFromSymmetry.lean` |
| DASH Pareto-optimal | `dash_unique_pareto_optimal` | `ParetoOptimality.lean` |
| Pareto frontier dichotomy | `pareto_frontier_dichotomy` | `ParetoOptimality.lean` |
| Design space dichotomy | `universal_design_space_dichotomy` | `UniversalDesignSpace.lean` |
| MI exact boundary | `mi_is_exact_boundary` | `MutualInformation.lean` |
| MI quantitative unfaithfulness | `mi_quantitative_unfaithfulness` | `MIQuantitativeBridge.lean` |
| GBDT impossibility | `attribution_impossibility` | `Trilemma.lean` |
| Ratio divergence | `ratio_tendsto_atTop` | `Ratio.lean` |
| DASH equity | `consensus_equity` | `Corollary.lean` |
| Necessity biconditional | `necessity_biconditional` | `NecessityBiconditional.lean` |
| Generic underspecification | `generic_underspecification` | `Ubiquity.lean` |
| Ubiquity impossibility | `ubiquity_impossibility` | `Ubiquity.lean` |
| Recursive enrichment | `enrichment_creates_new_impossibility` | `RecursiveImpossibility.lean` |
| Levels independent | `levels_independent` | `RecursiveImpossibility.lean` |

## G.11 Complete Theorem Index

All 530 theorem and lemma names from the Lean 4 formalization, grouped by source file. This index enables verification of completeness and traceability from any monograph claim to its machine-checked proof in the public repository.

### AlphaFaithful.lean (3)

`stable_ranking_half_unfaithful`
`alpha_faithful_bound`
`faithful_exactly_one_witness`

### ApproximateEquity.lean (3)

`global_implies_bounded_one`
`bounded_proportionality_constant_implies_global`
`rashomon_from_bounded_proportionality`

### ApproximateRashomon.lean (4)

`approxStable_implies_stable`
`approx_impossibility`
`approx_trilemma`
`exact_implies_approx_rashomon`

### ArrowInstance.lean (3)

profiles_same_pairwise
profiles_different_social
social_choice_rashomon

### AttentionInstanceConstructive.lean (3)

attn_same_output
attn_different_argmax
attention_impossibility_constructive

### AttributionInstanceConstructive.lean (2)

attrIncompC_irrefl
attribution_impossibility_constructive

### AxiomSubstitution.lean (14)

impossibility_complete
impossibility_complete_three
tightness_faithful_complete
tightness_faithful_stable_complete
no_tightness_stable_complete
resolving_iff_complete
impossibility_resolving
impossibility_resolving_three
tightness_faithful_resolving
tightness_faithful_stable_resolving
no_tightness_stable_resolving
surjective_consistent
decisive_implies_complete
axiom_substitution_summary

### BayesOptimalTie.lean (4)

committed_ranking_error_half
tie_disagreement_zero
tie_dominates_commitment
design_space_step3_substantive

### BilemmaCharacterization.lean (3)

neutral_implies_faithful_stable
bilemma_implies_no_neutral
maxIncompat_implies_no_neutral

### BinaryQuantizer.lean (7)

binary_quantizer_variance
binary_quantizer_fraction
two_over_pi_in_unit_interval

two_over_pi_le_two_thirds
two_over_pi_gt_half
quantizer_signal_loss
quantizer_better_than_half

### CPDAGResolution.lean (2)

cpdagResolution_gInvariant
cpdag_gInvariant_implies_stable

### CausalDiscovery.lean (3)

causal_discovery_impossibility
cpdag_is_stable
causal_discovery_from_sbd

### CausalExplanationSystem.lean (1)

causal_explanation_impossibility

### CausalInstanceConstructive.lean (2)

causalIncompC_irrefl
causal_impossibility_constructive

### ConceptInstanceConstructive.lean (3)

concept_same_output
concept_different_directions
concept_impossibility_constructive

### ConditionalImpossibility.lean (3)

conditional_rashomon_from_symmetry
conditional_attribution_impossibility
conditional_escape

### Consistency.lean (7)

consistent_firstMover_surjective
consistent_splitCount_firstMover
consistent_splitCount_nonFirstMover
consistent_proportionality_global
consistent_splitCount_crossGroup_symmetric
consistent_splitCount_crossGroup_stable
axiom_system_consistent

## Corollary.lean (5)

consensus_equity
consensus_difference_zero
consensus_variance_rate
consensus_variance_halves
consensus_variance_nonneg

## CounterfactualInstanceConstructive.lean (3)

cf_same_prediction
cf_different_directions
counterfactual_impossibility_constructive

## DASHResolution.lean (2)

dashResolution_gInvariant
dash_gInvariant_implies_stable

## Defs.lean (11)

numTrees_pos
firstMover_surjective
splitCount_firstMover
splitCount_nonFirstMover
proportionality_global
attribution_proportional
attribution_variance_nonneg
consensus_variance_bound
splitCount_crossGroup_symmetric
splitCount_crossGroup_stable
splitCount_eq_of_not_firstMover_j_or_k

## DesignSpace.lean (6)

strongly_faithful_impossible
balanced_flip_symmetry
infeasible_faithful_stable_complete
dash_produces_ties
dash_variance_decreases
design_space_theorem

## DesignSpaceFull.lean (3)

design_space_exhaustiveness
no_complete_faithful_ranking
family_a_or_family_b

## DuhemQuine.lean (6)

dq_same_outcome
dq_blame_different

duhem_quine_impossibility
dq4_same_outcome
dq4_blame_different
duhem_quine_impossibility_4

### Efficiency.lean (10)

sum_constraint_swap
swapAt_sum_eq
amplification_factor_m2
amplification_factor_m3
amplification_approaches_one
amplification_gt_one
amplification_bounds
efficiency_sum_varies_across_models
across_model_no_constraint
efficiency_instability_summary

### EnrichmentStack.lean (18)

enriched_neutral_faithful_stable
enriched_neutral_not_decisive
cumulative_indecisiveness
incompleteness_depth
enrichment_preserves_deeper_rashomon
neutral_at_n_not_neutral_at_n1
threeLevel_rashomon_0
threeLevel_rashomon_1
threeLevel_rashomon_2
threeLevel_independence
twoLevel_has_multiLevel
InformationFate
informationFate_maxIncompat
bh_bilemma
physicalStack_depth_3
SpacetimeStatus
spacetimeStatus_maxIncompat
qg_bilemma

### EnsembleBound.lean (9)

consensus_var_eq_over_M
variance_decreases_with_M
double_ensemble_halves_variance
single_model_variance
variance_at_25
ensemble_bound_formula
z_squared_for_5pct
dash_achieves_minimum_variance
sum_squares_ge_inv_M

**ExplanationLandscape.lean** (8)

`landscape_bottom_no_fs`
`landscape_top_fs_achievable`
`landscape_top_achievable`
`coverage_conflict_iff_no_neutral`
`enrichment_expands_landscape`
`coverage_conflict_monotone_restriction`
`maxIncompat_landscape_chain`
`local_coverage_conflict_at_pair`

**ExplanationSystem.lean** (4)

`explanation_impossibility`
`tightness_faithful_decisive`
`tightness_faithful_stable`
`tightness_stable_decisive`

**FIMImpossibility.lean** (6)

`excess_loss_nonneg`
`gaussian_rashomon_witnesses`
`hessian_eigenvalue_minus`
`semi_axis_diverges`
`gaussian_fim_impossibility`
`gaussian_fim_impossibility_weak`

**FairnessAudit.lean** (4)

`fairness_audit_impossibility`
`fairness_audit_coin_flip`
`fairness_audit_coin_flip_reverse`
`dash_audit_resolution`

**FlipRate.lean** (13)

`non_firstmover_tie`
`non_firstmover_attribution_tie`
`firstmover_dominates`
`balanced_firstmover_count_eq`
`model_trichotomy`
`firstMoverSets_disjoint`
`tieSet_disjoint_left`
`tieSet_disjoint_right`
`fliprate_partition`
`binary_group_firstmover_is_j_or_k`
`binary_group_no_ties`
`binary_group_flip_rate`
`binary_flip_pair_count`

### GaugeTheory.lean (5)

same_holonomy
different_configs
gauge_preserves_holonomy
gauge_related
gauge_impossibility

### GaussianFlipRate.lean (13)

Phi_mono
Phi_nonneg
Phi_le_one
Phi_tendsto_atBot
Phi_tendsto_atTop
gaussianReal_standard_symm
Phi_neg
Phi_zero
gaussianFlipRate_le_half
gaussianFlipRate_nonneg
gaussianFlipRate_zero
gaussianFlipRate_decreasing
coin_flip_implies_impossibility

### General.lean (6)

splitCount_firstMover_gt
attribution_firstMover_gt
attribution_reversal
no_stable_ranking
gbdt_rashomon
gbdt_impossibility

### GeneralizedBilemma.lean (13)

generalized_bilemma
maxIncompat_implies_no_common_compatible
no_bilemma_of_neutral_on_fiber
fs_achievable_of_fiber_compatible
incompEquiv_refl
incompEquiv_symm
fd_from_incomp_equiv
maxIncompat_incompEquiv_singleton
coverageConflict_implies_no_neutral
maxIncompat_implies_coverageConflict
neutral_destroys_coverageConflict
explanation_collapsed_of_maxIncompat
explanation_full_tightness_of_neutral_committal

### GeneticCode.lean (3)

codon_degeneracy
codons_different
genetic_code_impossibility

### GoedelIncompleteness.lean (5)

goedel_first_incompleteness
goedel_sentence_true
goedel_recursion
goedel_infinite_incompleteness
proved_shared_structure

### Impossibility.lean (4)

attribution_ratio_ge
not_equitable
not_stable
impossibility

### IntersectionalFairness.lean (13)

single_audit_wrong
each_audit_has_counterexample
intersectional_audit_impossibility
intersectional_all_audits_wrong
compounding_ge_half
compounding_strict_mono
compounding_nonneg
compounding_le_one
compounding_K2
compounding_K3
compounding_K5
compounding_K10
dash_intersectional_resolution

### Iterative.lean (2)

iterative_rashomon
iterative_impossibility

### LLMExplanationInstanceConstructive.lean (3)

llm_same_output
llm_different_citations
llm_explanation_impossibility_constructive

### LanglandsCorrespondence.lean (6)

trace_mul_comm_applied
dim1_trace_determines
dim2_rashomon_witness

langlands_boundary_statement
scalar_embedding_trace
trace_identity_n

### Lasso.lean (2)

lasso_impossibility
lasso_ratio_infinite

### LinearSystem.lean (3)

same_output
different_solutions
linear_system_impossibility

### LocalGlobal.lean (2)

local_attribution_impossibility
impossibility_is_attribution_agnostic

### LocalSufficiency.lean (6)

attribution_ratio_local
ratio_diverges_local
attribution_sum_from_local
equity_violation_local
local_proportionality_suffices
axiom_stratification_documented

### LossPreservation.lean (3)

rashomon_from_swap_with_loss
rashomon_from_swap_loss
impossibility_with_quality

### MIQuantitativeBridge.lean (5)

mi_implies_positive_gap
gap_implies_unfaithfulness
total_unfaithfulness_bound
pointwise_unfaithfulness_bound
mi_quantitative_unfaithfulness

### MarkovEquivalence.lean (4)

chain_fork_same_ci
chain_fork_different_edges
causal_rashomon_derived
causal_impossibility_derived

### MaximalIncompatibility.lean (7)

faithful_eq_explain_of_maxIncompat
decisive_eq_explain_of_maxIncompat
bilemma
no_stable_decisive_of_maxIncompat
maxIncompat_tightness_fd
explain_not_stable
recovery_faithful_stable

### MechInterpInstanceConstructive.lean (3)

mechInterp_same_output
mechInterp_different_circuits
mech_interp_impossibility_constructive

### ModelSelection.lean (2)

model_selection_impossibility
model_selection_impossibility_weak

### ModelSelectionDesignSpace.lean (3)

model_stable_complete_unfaithful
model_unfaithfulness_free_implies_tie
model_family_a_or_family_b

### ModelSelectionInstanceConstructive.lean (3)

ms_same_performance
ms_different_models
model_selection_impossibility_constructive

### MultiAnalystInstance.lean (5)

scientificConclusion_irrefl
multiAnalyst_rashomon
multiAnalyst_impossibility
multiAnalyst_maxIncompat
multiAnalyst_bilemma

### MutualInformation.lean (11)

mi_dependent_iff_not_independent
mi_independent_symm
mi_dependent_symm
gaussian_mi_positive_of_corr_nonzero
gaussian_mi_zero_of_corr_zero
rashomon_from_mi_dependence
impossibility_from_mi
no_rashomon_from_agreement
stable_ranking_from_agreement

mi_is_exact_boundary
correlation_implies_mi_impossibility

### Necessity.lean (3)

impossibility_from_rashomon
fully_specified_possibility
no_rashomon_from_all_three

### NecessityBiconditional.lean (3)

possibility_from_factoring
not_rashomon_iff_factoring_neq
rashomon_biconditional_neq

### NeuralNet.lean (1)

nn_impossibility

### ParetoOptimality.lean (7)

no_method_beats_half
committed_ranking_has_positive_error
dash_pareto_dominance_within_group
dash_pareto_gap_exact
universal_within_group_half
pareto_frontier_dichotomy
dash_unique_pareto_optimal

### PathConvergence.lean (3)

impossibility_as_design_space_corollary
relaxation_paths_converge
complete_implies_unfaithful

### PeresMermin.lean (3)

peres_mermin_bool
peres_mermin_parity
peres_mermin_impossibility

### PhaseProblem.lean (4)

same_energy
different_signals
phase_rashomon_derived
phase_impossibility

### PredictiveConsequences.lean (5)

stable_implies_unfaithful
rashomon_unfaithfulness
faithful_is_explain
faithful_unique
all_or_nothing

### ProportionalityLocal.lean (4)

proportionality_global_implies_local
attribution_firstMover_gt_local
gbdt_rashomon_local
gbdt_impossibility_local

### QMInterpretation.lean (3)

qm_same_outcome
qm_ontology_different
qm_interpretation_impossibility

### Qualitative.lean (4)

rashomon_from_qualitative
impossibility_qualitative
quantitative_implies_dominates
axiom_implies_surjective

### QuantitativeBound.lean (3)

decisive_fails_on_rashomon_pair
not_decisive_at_rashomon_pair
quantitative_impossibility

### QuantumMeasurementRevolution.lean (2)

quantum_ontology_maxIncompat
quantum_measurement_revolution

### QueryComplexity.lean (16)

testing_constant_pos
le_cam_lower_bound
query_bound_nonneg
query_bound_pos
query_complexity_lower_bound
query_bound_scales_with_sigma
query_bound_scales_with_gap
query_bound_mono_sigma
query_bound_antimono_gap
z_test_query_count_nonneg
query_gap_ratio

z_test_optimal_up_to_constant
snr_at_0_15
snr_denominator
snr_multiplier
breast_cancer_query_bound

### QueryComplexityDerived.lean (12)

chebyshev_query_bound
chebyshev_query_lower_bound
chebyshev_bound_simplify
query_complexity_lower_bound_derived
derived_query_bound_nonneg
derived_query_bound_pos
derived_bound_scales_with_sigma
derived_bound_scales_with_gap
derived_bound_mono_sigma
derived_bound_antimono_gap
derived_snr_at_0_15
derived_snr_value

### QueryComplexityParametric.lean (10)

TestingBound
TestingBound
query_complexity_parametric
queryBound_scales_sigma
queryBound_scales_gap
queryBound_mono_sigma
queryBound_antimono_gap
zTest_optimal
leCam_breast_cancer
axiom_free_check

### QueryRelative.lean (2)

query_impossibility
query_possibility

### RashomonInevitability.lean (5)

nondegen_implies_strict_one_direction
rashomon_from_nondegen_and_symmetry
rashomon_inevitability
impossibility_inevitable
escape_unanimous_agreement

### RashomonUniversality.lean (4)

rashomon_witnesses_from_swap
rashomon_pair_from_swap

rashomon_from_symmetry
impossibility_from_symmetry

### Ratio.lean (3)

splitCount_ratio
attribution_ratio
ratio_tendsto_atTop

### RecursiveImpossibility.lean (7)

recursive_impossibility_persists
unbounded_undecidability
kthBit_rashomon
infiniteStack_bilemma
infiniteStack_independence
self_referential_undecidability
shared_recursive_structure

### RobustnessLipschitz.lean (11)

C_Phi_pos
flipOfSNR_eq
isLipschitzWith_id
isLipschitzWith_neg
isLipschitzWith_comp
isLipschitzWith_const
flipOfSNR_lipschitz
flipOfRho_lipschitz
flip_rate_robust
flip_rate_continuous_at
flip_perturbation_bound_monotone

### SBDInstances.lean (5)

attribution_is_sbd_instance
attribution_sbd_unfaithful
model_selection_is_sbd_instance
abstract_aggregation_impossibility
abstract_aggregation_impossibility_weak

### SaliencyInstanceConstructive.lean (3)

saliency_same_output
saliency_different_regions
saliency_impossibility_constructive

### Setup.lean (1)

attribution_impossibility_bundled

## SimultaneityRevolution.lean (2)

simultaneity_maxIncompat
simultaneity_revolution

## SpearmanDef.lean (20)

countEqual_pos
countBelow_of_gt
midrank_strict_mono
sumSqRankDiff_nonneg
sumSqRankDiff_pos_of_reversal
spearmanCorr_lt_one_of_reversal
spearmanCorr_bound
attribution_eq_outside_group
attribution_eq_non_firstMover
attribution_swap_structure
attribution_non_fm_stable
countEqual_ge_groupSize_minus_one
midrank_gap_ge_half_groupSize
sumSqRankDiff_ge_sq_groupSize
spearmanCorr_bound_groupSize
spearman_instability_bound
midrank_gap_ge_half_groupSize_tight
sumSqRankDiff_ge_sq_groupSize_tight
spearmanCorr_bound_groupSize_tight
spearman_instability_bound_tight

## SplitGap.lean (7)

rho_sq_lt_one
one_minus_rho_sq_pos
denom_pos
denom_ne_zero
split_gap_exact
split_gap_ge_half
split_gap_lower_bound

## StatisticalMechanics.lean (5)

same_macrostate
different_microstates
swap_related
swap_preserves_macrostate
stat_mech_impossibility

## StumpProportionality.lean (6)

stump_proportionality
stump_attribution_ratio
stump_proportionality_positive
stump_proportionality_exists

stump_proportionality_unique
stump_cv_zero

## SymmetricBayes.lean (16)

sbd_unfaithful_witness
sbd_infeasible
tie_is_stable
sbd_faithful_at_most_one
sbd_faithful_unique
sbd_binary_orbit_half_unfaithful
sbd_ternary_orbit_unfaithful
attribution_from_sbd
sbd_faithful_at_most_one_in_orbit
sbd_trichotomy
sbd_family_a_or_b
orbitEquiv_refl
orbitEquiv_symm
orbitEquiv_trans
family_b_output_not_optimal
symmetric_bayes_dichotomy

## SymmetryDerive.lean (1)

attribution_sum_symmetric

## SyntacticAmbiguity.lean (3)

same_yield
different_bracketings
syntactic_ambiguity_impossibility

## Trilemma.lean (2)

attribution_impossibility
attribution_impossibility_weak

## Ubiquity.lean (3)

generic_underspecification
fiber_nondegeneracy_implies_impossibility
ubiquity_impossibility

## UncertaintyFromSymmetry.lean (16)

inner_Rv_v_eq_inner_Rv_Rv
inner_v_Rv_eq_inner_Rv_Rv
reynolds_orthogonal
uncertainty_from_symmetry
information_loss
stable_bounded

unstable_bounded
reynolds_residual_zero
residual_orthogonal_to_fixed
best_approximation
reynolds_naturality
fixed_perp_residual
reynolds_best_approximation
stable_in_fixed_subspace
beyond_capacity_penalty
beyond_capacity_optimal

### UnfaithfulBound.lean (3)

stable_complete_unfaithful
tie_is_never_unfaithful
unfaithfulness_free_implies_tie

### UnfaithfulQuantitative.lean (2)

attribution_prob_half
stable_ranking_unfaithfulness_half

### UniversalDesignSpace.lean (1)

universal_design_space_dichotomy

### UniversalResolution.lean (1)

gInvariant_stable

### ValueAlignment.lean (2)

value_maxIncompat
value_alignment_impossibility

### VarianceDerivation.lean (6)

variance_of_mean_from_sum_variance
consensus_variance_from_independence
consensus_variance_nonneg_from_independence
double_M_halves_variance_derived
variance_ratio_from_independence
weighted_variance_ge_consensus_variance

### ViewUpdate.lean (3)

same_view
different_rows
view_update_impossibility